

Chapter 1

Some Results on Linear Algebra, Matrix Theory and Distributions

We need some basic knowledge to understand the topics in the analysis of variance.

Vectors:

A vector Y is an ordered n -tuple of real numbers. A vector can be expressed as a row vector or a column vector as

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

is a column vector of order $n \times 1$ and

$$Y' = (y_1, y_2, \dots, y_n)$$

is a row vector of order $1 \times n$.

If all $y_i = 0$ for all $i = 1, 2, \dots, n$ then $Y' = (0, 0, \dots, 0)$ is called the **null vector**.

If

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, Z = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}$$

then

$$X + Y = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad kY = \begin{pmatrix} ky_1 \\ ky_2 \\ \vdots \\ ky_n \end{pmatrix}$$

$$X + (Y + Z) = (X + Y) + Z$$

$$X'(Y + Z) = X'Y + X'Z$$

$$k(X'Y) = (kX)'Y = X'(kY)$$

$$k(X + Y) = kX + kY$$

$$X'Y = x_1y_1 + x_2y_2 + \dots + x_ny_n$$

where k is a scalar.

Orthogonal vectors:

Two vectors X and Y are said to be orthogonal if $X'Y = Y'X = 0$.

The null vector is orthogonal to every vector X and is the only such vector.

Linear combination:

If $x_1, x_2, \dots, x_m$ are m vectors and $k_1, k_2, \dots, k_m$ are m scalars, then

$$t = \sum_{i=1}^m k_i x_i$$

is called the linear combination of $x_1, x_2, \dots, x_m$.

Linear independence

If $X_1, X_2, \dots, X_m$ are m vectors then they are said to be linearly independent if there exist scalars $k_1, k_2, \dots, k_m$ such that

$$\sum_{i=1}^m k_i X_i = 0 \Rightarrow k_i = 0 \text{ for all } i = 1, 2, \dots, m.$$

If there exist $k_1, k_2, \dots, k_m$ with at least one k_i to be nonzero, such that $\sum_{i=1}^m k_i x_i = 0$ then

$x_1, x_2, \dots, x_m$ are said to be **linearly dependent**.

- Any set of vectors containing the null vector is linearly dependent.
- Any set of non-null pair-wise orthogonal vectors is linearly independent.
- If $m > 1$ vectors are linearly dependent, it is always possible to express at least one of them as a linear combination of the others.

Linear function:

Let $K = (k_1, k_2, \dots, k_m)'$ be a $m \times 1$ vector of scalars and $X = (x_1, x_2, \dots, x_m)$ be a $m \times 1$ vector of variables,

then $K'Y = \sum_{i=1}^m k_i y_i$ is called a linear function or linear form. The vector K is called the **coefficient**

vector. For example, the mean of $x_1, x_2, \dots, x_m$ can be expressed as

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i = \frac{1}{m} (1, 1, \dots, 1) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \frac{1}{m} 1'_m X$$

where $1'_m$ is a $m \times 1$ vector of all elements unity.

Contrast:

The linear function $K'X = \sum_{i=1}^m k_i x_i$ is called a contrast in $x_1, x_2, \dots, x_m$ if $\sum_{i=1}^m k_i = 0$.

For example, the linear functions

$$x_1 - x_2, 2x_1 - 3x_2 + x_3, \frac{x_1}{2} - x_2 + \frac{x_3}{3}$$

are contrasts.

- A linear function $K'X$ is a contrast if and only if it is orthogonal to a linear function $\sum_{i=1}^m x_i$ or to the linear function $\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$.
- Contrasts $x_1 - x_2, x_1 - x_3, \dots, x_1 - x_j$ are linearly independent for all $j = 2, 3, \dots, m$.
- Every contrast in $x_1, x_2, \dots, x_n$ can be written as a linear combination of $(m - 1)$ contrasts $x_1 - x_2, x_1 - x_3, \dots, x_1 - x_m$.

Matrix:

A matrix is a rectangular array of real numbers. For example

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

is a matrix of order $m \times n$ with m rows and n columns.

- If $m = n$, then A is called a square matrix.
- If $a_{ij} = 0, i \neq j, m = n$, then A is a diagonal matrix and is denoted as

$$A = \text{diag}(a_{11}, a_{22}, \dots, a_{mm}).$$

- If $m = n$ (square matrix) and $a_{ij} = 0$ for $i > j$, then A is called an **upper triangular matrix**. On the other hand if $m = n$ and $a_{ij} = 0$ for $i < j$ then A is called a **lower triangular matrix**.
- If A is a $m \times n$ matrix, then the matrix obtained by writing the rows of A and columns of A as columns of A and rows of A respectively is called the **transpose of a matrix** A and is denoted as A' .
- If $A = A'$ then A is a **symmetric matrix**.
- If $A = -A'$ then A is **skew-symmetric matrix**.
- A matrix whose all elements are equal to zero is called a **null matrix**.

- An **identity matrix** is a square matrix of order p whose diagonal elements are unity (ones) and all the off diagonal elements are zero. It is denoted as I_p .
- If A and B are matrices of order $m \times n$ then $(A+B)' = A' + B'$.
- If A and B are the matrices of order $m \times n$ and $n \times p$ respectively and k is any scalar, then

$$(AB)' = B' A'$$

$$(kA)B = A(kB) = k(AB) = kAB.$$
- If the orders of matrices A is $m \times n$, B is $n \times p$ and C is $n \times p$ then $A(B+C) = AB + AC$.
- If the orders of matrices A is $m \times n$, B is $n \times p$ and C is $p \times q$ then $(AB)C = A(BC)$.
- If A is the matrix of order $m \times n$ then $I_m A = A I_n = A$.

Trace of a matrix:

The trace of $n \times n$ matrix A , denoted as $tr(A)$ or $trace(A)$ is defined to be the sum of all the diagonal

elements of A , i.e. $tr(A) = \sum_{i=1}^n a_{ii}$.

- If A is of order $m \times n$ and B is of order $n \times m$, then

$$tr(AB) = tr(BA).$$
- If A is $n \times n$ matrix and P is any nonsingular $n \times n$ matrix then

$$tr(A) = tr(P^{-1}AP).$$

If P is an orthogonal matrix then $tr(A) = tr(P'AP)$.

- * If A and B are $n \times n$ matrices, a and b are scalars then

$$tr(aA + bB) = a tr(A) + b tr(B).$$

- If A is an $m \times n$ matrix, then

$$tr(A'A) = tr(AA') = \sum_{j=1}^n \sum_{i=1}^n a_{ij}^2$$

and

$$tr(A'A) = tr(AA') = 0 \text{ if and only if } A = 0.$$

- If A is $n \times n$ matrix then

$$tr(A') = trA.$$

Rank of matrices

The rank of a matrix A of $m \times n$ is the number of linearly independent rows in A .

Let B be any other matrix of order $n \times q$.

- A square matrix of order m is called **non-singular** if it has full rank.
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
- $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$
- Rank A is equal to the maximum order of all nonsingular square sub-matrices of A .
- $\text{rank}(AA') = \text{rank}(A'A) = \text{rank}(A) = \text{rank}(A')$.
- A is of full row rank if $\text{rank}(A) = m < n$.
- A is of full column rank if $\text{rank}(A) = n < m$.

Inverse of a matrix

The inverse of a square matrix A of order m , is a square matrix of order m , denoted as A^{-1} , such that $A^{-1}A = AA^{-1} = I_m$.

The inverse of A exists if and only if A is non-singular.

- $(A^{-1})^{-1} = A$.
- If A is non-singular, then $(A')^{-1} = (A^{-1})'$
- If A and B are non-singular matrices of the same order, then their product, if defined, is also nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.

Idempotent matrix:

A square matrix A is called idempotent if $A^2 = AA = A$.

If A is an $n \times n$ idempotent matrix with $\text{rank}(A) = r \leq n$. Then

- eigenvalues of A are 1 or 0.
- $\text{trace}(A) = \text{rank}(A) = r$.
- If A is of full rank n , then $A = I_n$.
- If A and B are idempotent and $AB = BA$, then AB is also idempotent.
- If A is idempotent then $(I - A)$ is also idempotent and $A(I - A) = (I - A)A = 0$.

Quadratic forms:

If A is a given matrix of order $m \times n$ and X and Y are two given vectors of order $m \times 1$ and $n \times 1$ respectively

$$X'AY = \sum_{i=1}^m \sum_{j=1}^n a_{ij}x_iy_j$$

where a_{ij} are the nonstochastic elements of A .

If A is a square matrix of order m and $X = Y$, then

$$X'AX = a_{11}x_1^2 + \dots + a_{mm}x_m^2 + (a_{12} + a_{21})x_1x_2 + \dots + (a_{m-1,m} + a_{m,m-1})x_{m-1}x_m.$$

If A is symmetric also, then

$$\begin{aligned} X'AX &= a_{11}x_1^2 + \dots + a_{mm}x_m^2 + 2a_{12}x_1x_2 + \dots + 2a_{m-1,m}x_{m-1}x_m \\ &= \sum_{i=1}^m \sum_{j=1}^n a_{ij}x_ix_j \end{aligned}$$

is called a quadratic form in m variables $x_1, x_2, \dots, x_m$ or a quadratic form in X .

- To every quadratic form corresponds a symmetric matrix and vice versa.
- The matrix A is called the matrix of the quadratic form.
- The quadratic form $X'AX$ and the matrix A of the form is called.
 - ❖ Positive definite if $X'AX > 0$ for all $x \neq 0$.
 - ❖ Positive semidefinite if $X'AX \geq 0$ for all $x \neq 0$.
 - ❖ Negative definite if $X'AX < 0$ for all $x \neq 0$.
 - ❖ Negative semidefinite if $X'AX \leq 0$ for all $x \neq 0$.
- If A is positive semi-definite matrix then $a_{ii} \geq 0$ and if $a_{ii} = 0$ then $a_{ij} = 0$ for all j , and $a_{ji} = 0$ for all j .
- If P is any nonsingular matrix and A is any positive definite matrix (or positive semi-definite matrix) then $P'AP$ is also a positive definite matrix (or positive semi-definite matrix).
- A matrix A is positive definite if and only if there exists a non-singular matrix P such that $A = P'P$.
- A positive definite matrix is a nonsingular matrix.
- If A is $m \times n$ matrix and $\text{rank}(A) = m < n$ then AA' is positive definite and $A'A$ is positive semidefinite.
- If A $m \times n$ matrix and $\text{rank}(A) = k < m < n$, then both $A'A$ and AA' are positive semidefinite.

Simultaneous linear equations

The set of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ and scalars a_{ij} and b_i , $i = 1, 2, \dots, m, j = 1, 2, \dots, n$ of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

can be formulated as $AX = b$

where A is a real matrix of known scalars of order $m \times n$ called as a coefficient matrix, X is $n \times 1$ real vector and b is $m \times 1$ real vector of known scalars given by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \text{ is an } m \times n \text{ real matrix called as coefficient matrix,}$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \text{ is an } n \times 1 \text{ vector of variables and } b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \text{ is an } m \times 1 \text{ real vector.}$$

- If A is a $n \times n$ nonsingular matrix, then $AX = b$ has a unique solution.
- Let $B = [A, b]$ is an augmented matrix. A solution to $AX = b$ exist if and only if $\text{rank}(A) = \text{rank}(B)$.
- If A is a $m \times n$ matrix of rank m , then $AX = b$ has a solution.
- Linear homogeneous system $AX = 0$ has a solution other than $X = 0$ if and only if $\text{rank}(A) < n$.
- If $AX = b$ is consistent then $AX = b$ has a unique solution if and only if $\text{rank}(A) = n$
- If a_{ii} is the i^{th} diagonal element of an orthogonal matrix, then $-1 \leq a_{ii} \leq 1$.
- Let the $n \times n$ matrix be partitioned as $A = [a_1, a_2, \dots, a_n]$ where a_i is an $n \times 1$ vector of the elements of i^{th} column of A . A necessary and sufficient condition that A is an orthogonal matrix is given by the following:
 - $a_i' a_i = 1$ for $i = 1, 2, \dots, n$
 - $a_i' a_j = 0$ for $i \neq j = 1, 2, \dots, n$.

Orthogonal matrix

A square matrix A is called an orthogonal matrix if $A'A = AA' = I$ or equivalently if $A^{-1} = A'$.

- An orthogonal matrix is non-singular.
- If A is orthogonal, then AA' is also orthogonal.
- If A is an $n \times n$ matrix and let P is an $n \times n$ orthogonal matrix, then the determinants of A and $P'AP$ are the same.

Random vectors:

Let $Y_1, Y_2, \dots, Y_n$ be n random variables then $Y = (Y_1, Y_2, \dots, Y_n)'$ is called a random vector.

- The **mean vector** Y is
$$E(Y) = (E(Y_1), E(Y_2), \dots, E(Y_n))'$$
- The **covariance matrix** or **dispersion matrix** of Y is

$$Var(Y) = \begin{pmatrix} Var(Y_1) & Cov(Y_1, Y_2) & \dots & Cov(Y_1, Y_n) \\ Cov(Y_2, Y_1) & Var(Y_2) & \dots & Cov(Y_2, Y_n) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(Y_n, Y_1) & Cov(Y_n, Y_2) & \dots & Var(Y_n) \end{pmatrix}$$

which is a symmetric matrix

- If $Y_1, Y_2, \dots, Y_n$ are independently distributed, then the covariance matrix is a diagonal matrix.
- If $Var(Y_i) = \sigma^2$ for all $i = 1, 2, \dots, n$ then $Var(Y) = \sigma^2 I_n$.

Linear function of random variable :

If $Y_1, Y_2, \dots, Y_n$ are n random variables, and $k_1, k_2, \dots, k_n$ are scalars, then $\sum_{i=1}^n k_i Y_i$ is called a linear function of random variables $Y_1, Y_2, \dots, Y_n$.

If $Y = (Y_1, Y_2, \dots, Y_n)'$, $K = (k_1, k_2, \dots, k_n)'$ then $K'Y = \sum_{i=1}^n k_i Y_i$.

- the mean $K'Y$ is $E(K'Y) = K'E(Y) = \sum_{i=1}^n k_i E(Y_i)$ and
- the variance of $K'Y$ is $Var(K'Y) = K'Var(Y)K$.

Multivariate normal distribution

A random vector $Y = (Y_1, Y_2, \dots, Y_n)'$ has a multivariate normal distribution with mean vector $\mu = (\mu_1, \mu_2, \dots, \mu_n)$ and dispersion matrix Σ if its probability density function is

$$f(Y / \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (Y - \mu)' \Sigma^{-1} (Y - \mu) \right]$$

assuming Σ is a nonsingular matrix.

Chi-square distribution

- If $Y_1, Y_2, \dots, Y_k$ are independently distributed following the normal distribution random variables with common mean 0 and common variance 1, then the distribution of $\sum_{i=1}^k Y_i^2$ is called the χ^2 -distribution with k degrees of freedom.
- The probability density function of χ^2 -distribution with k degrees of freedom is given as

$$f_{\chi^2}(x) = \frac{1}{\Gamma(k/2) 2^{k/2}} x^{\frac{k}{2}-1} \exp\left(-\frac{x}{2}\right); \quad 0 < x < \infty$$

- If $Y_1, Y_2, \dots, Y_k$ are independently distributed following the normal distribution with common mean 0 and common variance σ^2 , then $\frac{1}{\sigma^2} \sum_{i=1}^k Y_i^2$ has χ^2 distribution with k degrees of freedom.
- If the random variables $Y_1, Y_2, \dots, Y_k$ are normally distributed with non-null means $\mu_1, \mu_2, \dots, \mu_k$ but common variance 1, then the distribution of $\sum_{i=1}^k Y_i^2$ has noncentral χ^2 distribution with k degrees of freedom and noncentrality parameter $\lambda = \sum_{i=1}^k \mu_i^2$.
- If $Y_1, Y_2, \dots, Y_k$ are independently and normally distributed following the normal distribution with means $\mu_1, \mu_2, \dots, \mu_k$ but common variance σ^2 then $\frac{1}{\sigma^2} \sum_{i=1}^k Y_i^2$ has non-central χ^2 distribution with k degrees of freedom and noncentrality parameter $\lambda = \frac{1}{\sigma^2} \sum_{i=1}^k \mu_i^2$.
- If U has a Chi-square distribution with k degrees of freedom then $E(U) = k$ and $Var(U) = 2k$.
- If U has a noncentral Chi-square distribution with k degrees of freedom and

noncentrality parameter λ then $E(U) = k + \lambda$ and $Var(U) = 2k + 4\lambda$.

- If $U_1, U_2, \dots, U_k$ are independently distributed random variables with each U_i having a noncentral Chi-square distribution with n_i degrees of freedom and non-centrality parameter $\lambda_i, i = 1, 2, \dots, k$ then $\sum_{i=1}^k U_i$ has noncentral Chi-square distribution with $\sum_{i=1}^k n_i$ degrees of freedom and noncentrality parameter $\sum_{i=1}^k \lambda_i$.
- Let $X = (X_1, X_2, \dots, X_n)'$ has a multivariate distribution with mean vector μ and positive definite covariance matrix Σ . Then $X'AX$ is distributed as noncentral χ^2 with k degrees of freedom if and only if ΣA is an idempotent matrix of rank k .
- Let $X = (X_1, X_2, \dots, X_n)$ has a multivariate normal distribution with mean vector μ and positive definite covariance matrix Σ . Let the two quadratic forms-
 - $X'A_1X$ is distributed as χ^2 with n_1 degrees of freedom and noncentrality parameter $\mu'A_1\mu$ and
 - $X'A_2X$ is distributed as χ^2 with n_2 degrees of freedom and noncentrality parameter $\mu'A_2\mu$.

Then $X'A_1X$ and $X'A_2X$ are independently distributed if $A_1\Sigma A_2 = 0$

t-distribution

- If
 - X has a normal distribution with mean 0 and variance 1,
 - Y has a χ^2 distribution with n degrees of freedom, and
 - X and Y are independent random variables,

then the distribution of the statistic $T = \frac{X}{\sqrt{Y/n}}$ is called the t -distribution with n degrees of freedom. The probability density function of t is

$$f_T(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\sqrt{n\pi}} \left(1 + \frac{t^2}{n}\right)^{-\left(\frac{n+1}{2}\right)}; \quad -\infty < t < \infty$$

- If the mean of X is nonzero then the distribution of $\frac{X}{\sqrt{Y/n}}$ is called the noncentral t -distribution with n degrees of freedom and noncentrality parameter μ .

F-distribution

- If X and Y are independent random variables with χ^2 -distribution with m and n degrees of freedom respectively, then the distribution of the statistic $F = \frac{X/m}{Y/n}$ is called the F -distribution with m and n degrees of freedom. The probability density function of F is

$$f_F(f) = \frac{\Gamma\left(\frac{m+n}{2}\right)\left(\frac{m}{n}\right)^{m/2}}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} f^{\left(\frac{m-2}{2}\right)} \left(1 + \left(\frac{m}{n}\right)f\right)^{-\left(\frac{m+n}{2}\right)}; \quad 0 < f < \infty$$

- If X has a noncentral Chi-square distribution with m degrees of freedom and noncentrality parameter λ ; Y has a χ^2 distribution with n degrees of freedom, and X and Y are independent random variables, then the distribution of $F = \frac{X/m}{Y/n}$ is the noncentral F distribution with m and n degrees of freedom and noncentrality parameter λ .

Linear model:

Suppose there are n observations. In the linear model, we assume that these observations are the values taken by n random variables $Y_1, Y_2, \dots, Y_n$ satisfying the following conditions:

- $E(Y_i)$ is a linear combination of p unknown parameters

$$\beta_1, \beta_2, \dots, \beta_p,$$

$$E(Y_i) = x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ip}\beta_p, \quad i = 1, 2, \dots, n$$

where x_{ij} 's are known constants.

- $Y_1, Y_2, \dots, Y_n$ are uncorrelated and normality distributed with variance $\text{Var}(Y_i) = \sigma^2$.

The linear model can be rewritten by introducing independent normal random variables following $N(0, \sigma^2)$, as

$$Y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ip}\beta_p + \varepsilon_i, \quad i = 1, 2, \dots, n.$$

These equations can be written using the matrix notations as

$$Y = X\beta + \varepsilon$$

where Y is a $n \times 1$ vector of observation, X is a $n \times p$ matrix of n observations on each of $X_1, X_2, \dots, X_p$ variables, β is a $p \times 1$ vector of parameters and ε is a $n \times 1$ vector of random error components with

$\varepsilon \sim N(0, \sigma^2 I_n)$. Here Y is called **study or dependent variable**, $X_1, X_2, \dots, X_p$ are called **explanatory or independent variables** and $\beta_1, \beta_2, \dots, \beta_p$ are called as **regression coefficients**.

Alternatively since $Y \sim N(X\beta, \sigma^2 I)$ so the linear model can also be expressed in the expectation form as a normal random variable Y with

$$E(y) = X\beta$$

$$Var(y) = \sigma^2 I.$$

Note that β and σ^2 are unknown but X is known.

Estimable functions:

A linear parametric function $\lambda'\beta$ of the parameter is said to be an estimable parametric function or estimable if there exists a linear function of random variables $\ell'y$ of Y where $Y = (Y_1, Y_2, \dots, Y_n)'$ such that

$$E(\ell'y) = \lambda'\beta$$

with $\ell = (\ell_1, \ell_2, \dots, \ell_n)'$ and $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)'$ being vectors of known scalars.

Best linear unbiased estimates (BLUE)

The unbiased minimum variance linear estimate $\ell'Y$ of an estimable function $\lambda'\beta$ is called the best linear unbiased estimate of $\lambda'\beta$.

- Suppose $\ell_1'Y$ and $\ell_2'Y$ are the BLUE of $\lambda_1'\beta$ and $\lambda_2'\beta$ respectively. Then $(a_1\ell_1 + a_2\ell_2)'Y$ is the BLUE of $(a_1\lambda_1 + a_2\lambda_2)'\beta$.
- If $\lambda'\beta$ is estimable, its best estimate is $\lambda'\hat{\beta}$ where $\hat{\beta}$ is any solution of the equations $X'X\beta = X'Y$.

Least squares estimation

The least-squares estimate of β is $Y = X\beta + \varepsilon$ is the value of β which minimizes the error sum of squares $\varepsilon'\varepsilon$.

Let

$$S = \varepsilon'\varepsilon = (Y - X\beta)'(Y - X\beta)$$

$$= Y'Y - 2\beta'X'Y + \beta'X'X\beta.$$

Minimizing S with respect to β involves

$$\frac{\partial S}{\partial \beta} = 0$$

$$\Rightarrow X'X\beta = X'Y$$

which is termed as **normal equation**. This normal equation has a unique solution given by

$$\hat{\beta} = (X'X)^{-1}X'Y$$

assuming $\text{rank}(X) = p$. Note that $\frac{\partial^2 S}{\partial \beta \partial \beta'} = X'X$ is a positive definite matrix. So $\hat{\beta} = (X'X)^{-1}X'Y$ is the value of β which minimizes $\varepsilon'\varepsilon$ and is termed as **ordinary least squares estimator** of β .

- In this case, $\beta_1, \beta_2, \dots, \beta_p$ are estimable and consequently, all the linear parametric function are estimable.
- $E(\hat{\beta}) = (X'X)^{-1}X'E(Y) = (X'X)^{-1}X'X\beta = \beta$
- $\text{Var}(\hat{\beta}) = (X'X)^{-1}X'\text{Var}(Y)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$
- If $\lambda'\hat{\beta}$ and $\mu'\hat{\beta}$ are the estimates of $\lambda'\beta$ and $\mu'\beta$ respectively, then
 - ❖ $\text{Var}(\lambda'\hat{\beta}) = \lambda'\text{Var}(\hat{\beta})\lambda = \sigma^2[\lambda'(X'X)^{-1}\lambda]$
 - ❖ $\text{Cov}(\lambda'\hat{\beta}, \mu'\hat{\beta}) = \sigma^2[\mu'(X'X)^{-1}\lambda]$
- $Y - X\hat{\beta}$ is called the **residual vector**
- $E(Y - X\hat{\beta}) = 0$.

Linear model with correlated observations:

In the linear model

$$Y = X\beta + \varepsilon$$

with $E(\varepsilon) = 0$, $\text{Var}(\varepsilon) = \Sigma$ and ε is normally distributed, we find

$$E(Y) = X\beta, \text{Var}(Y) = \Sigma$$

Assuming Σ to be positive definite, so we can write

$$\Sigma = P'P$$

where P is a nonsingular matrix. Premultiplying $Y = X\beta + \varepsilon$ by P , we get

$$PY = PX\beta + P\varepsilon$$

$$\text{or } Y^* = X^*\beta + \varepsilon^*$$

$$\text{where } Y^* = PY, X^* = PX \text{ and } \varepsilon^* = P\varepsilon.$$

Note that in this model $E(\varepsilon^*) = 0$ and $\text{Var}(\varepsilon^*) = \sigma^2 I$.

Distribution of $\ell'Y$:

In the linear model $Y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$ consider a linear function $\ell'Y$ which is normally distributed with

$$E(\ell'Y) = \ell'X\beta,$$

$$\text{Var}(\ell'Y) = \sigma^2(\ell'\ell).$$

Then

$$\frac{\ell'Y}{\sigma\sqrt{\ell'\ell}} \sim N\left(\frac{\ell'X\beta}{\sigma\sqrt{\ell'\ell}}, 1\right).$$

Further, $\frac{(\ell'Y)^2}{\sigma^2\ell'\ell}$ has a noncentral Chi-square distribution with one degree of freedom and noncentrality parameter $\frac{(\ell'X\beta)^2}{\sigma^2\ell'\ell}$.

Degrees of freedom:

A linear function $\ell'Y$ of the observations ($\ell \neq 0$) is said to carry one degrees of freedom. A set of linear functions $L'Y$ where L is $r \times n$ matrix, is said to have M degrees of freedom if there exist M linearly independent functions in the set and no more. Alternatively, the degrees of freedom carried by the set $L'Y$ equals $\text{rank}(L)$. When the set $L'Y$ are the estimates of $\Lambda'\beta$, the degrees of freedom of the set $L'Y$ will also be called the degrees of freedom for the estimates of $\Lambda'\beta$.

Sum of squares:

If $\ell'Y$ is a linear function of observations, then the projection of Y on ℓ is the vector $\frac{Y'\ell}{\ell'\ell}\ell$. The square of this projection is called the sum of squares (SS) due to $\ell'y$ is given by $\frac{(\ell'Y)^2}{\ell'\ell}$. Since $\ell'Y$ has one degree of freedom, so the SS due $\ell'Y$ to has one degree of freedom.

The sum of squares and the degrees of freedom arising out of the mutually orthogonal sets of functions can be added together to give the sum of squares and degrees of freedom for the set of all the function together and vice versa.

Let $X = (X_1, X_2, \dots, X_n)$ has a multivariate normal distribution with mean vector μ and positive definite covariance matrix Σ . Let the two quadratic forms.

- $X' A_1 X$ is distribution χ^2 with n_1 degrees of freedom and noncentrality parameter $\mu' A_1 \mu$ and
- $X' A_2 X$ is distributed as χ^2 with n_2 degrees of freedom and noncentrality parameter $\mu' A_2 \mu$.

Then $X' A_1 X$ and $X' A_2 X$ are independently distributed if $A_1 \Sigma A_2 = 0$.

Fisher-Cochran theorem

If $X = (X_1, X_2, \dots, X_n)$ has a multivariate normal distribution with mean vector μ and positive definite covariance matrix Σ and let

$$X' \Sigma^{-1} X = Q_1 + Q_2 + \dots + Q_k$$

where $Q_i = X' A_i X$ with $\text{rank}(A_i) = N_i, i = 1, 2, \dots, k$. Then Q_i 's are independently distributed noncentral Chi-square distribution with N_i degrees of freedom and noncentrality parameter $\mu' A_i \mu$ if and only if $\sum_{i=1}^k N_i = N$ is which case

$$\mu' \Sigma^{-1} \mu = \sum_{i=1}^k \mu' A_i \mu.$$

Derivatives of quadratic and linear forms:

Let $X = (x_1, x_2, \dots, x_n)'$ and $f(X)$ be any function of n independent variables $x_1, x_2, \dots, x_n$, then

$$\frac{\partial f(X)}{\partial X} = \begin{pmatrix} \frac{\partial f(X)}{\partial x_1} \\ \frac{\partial f(X)}{\partial x_2} \\ \vdots \\ \frac{\partial f(X)}{\partial x_n} \end{pmatrix}.$$

If $K = (k_1, k_2, \dots, k_n)'$ is a vector of constants, then

$$\frac{\partial K' X}{\partial X} = K$$

If A is a $m \times n$ matrix, then

$$\frac{\partial X' A X}{\partial X} = 2(A + A')X.$$

Independence of linear and quadratic forms:

- Let Y be an $n \times 1$ vector having multivariate normal distribution $N(\mu, I)$ and B be a $m \times n$ matrix. Then the $m \times 1$ vector linear form BY is independent of the quadratic form $Y'AY$ if $BA = 0$ where A is a symmetric matrix of known elements.
- Let Y be a $n \times 1$ vector having multivariate normal distribution $N(\mu, \Sigma)$ with $\text{rank}(\Sigma) = n$. If $B\Sigma A = 0$, then the quadratic form $Y'AY$ is independent of linear form BY where B is a $m \times n$ matrix.

Chapter 10

Partial Confounding

The objective of confounding is to mix the less important treatment combinations with the block effect differences so that higher accuracy can be provided to the other important treatment comparisons. When such mixing of treatment contrasts and block differences is done in all the replicates, then it is termed as **total confounding**. On the other hand, when the treatment contrast is not confounded in all the replicates but only in some of the replicates, then it is said to be partially confounded with the blocks. It is also possible that one treatment combination is confounded in some of the replicates and another treatment combination is confounded in other replicates which are different from the earlier replicates. So the treatment combinations confounded in some of the replicates and unconfounded in other replicates. So the treatment combinations are said to be partially confounded. The partially confounded contrasts are estimated only from those replicates where it is not confounded. Since the variance of the contrast estimator is inversely proportional to the number of replicates in which they are estimable, so some factors on which information is available from all the replicates are more accurately determined.

Balanced and unbalanced partially confounded design

If all the effects of a certain order are confounded with incomplete block differences in equal number of replicates in a design, then the design is said to be balanced partially confounded design. If all the effects of a certain order are confounded an unequal number of times in a design, then the design is said to be unbalanced partially confounded design.

We discuss only the analysis of variance in the case of balanced partially confounded design through examples on 2^2 and 2^3 factorial experiments.

Example 1:

Consider the case of 2^2 factorial as in following table in the set up of a randomized block design.

Factorial effects	Treatment combinations				Divisor
	(1)	(a)	(b)	(ab)	
<i>M</i>	+	+	+	+	4
<i>A</i>	-	+	-	+	2
<i>B</i>	-	-	+	+	2
<i>AB</i>	+	-	-	+	2

where $y_{*i} = ((1), a, b, ab)'$ denotes the vector of total responses in the i^{th} replication and each treatment is replicated r times, $i = 1, 2, \dots, r$. If no factor is confounded then the factorial effects are estimated using all the replicates as

$$\begin{aligned} A &= \frac{1}{2r} \sum_{i=1}^r \ell'_A y_{*i}, \\ B &= \frac{1}{2r} \sum_{i=1}^r \ell'_B y_{*i}, \\ AB &= \frac{1}{2r} \sum_{i=1}^r \ell'_{AB} y_{*i}, \end{aligned}$$

where the vectors of contrasts $\ell_A, \ell_B, \ell_{AB}$ are given by

$$\begin{aligned} \ell_A &= (-1 \quad 1 \quad -1 \quad 1)' \\ \ell_B &= (-1 \quad -1 \quad 1 \quad 1)' \\ \ell_{AB} &= (1 \quad -1 \quad -1 \quad 1)'. \end{aligned}$$

We have in this case

$$\ell'_A \ell_A = \ell'_B \ell_B = \ell'_{AB} \ell_{AB} = 4.$$

The sum of squares due to A, B and AB can be accordingly modified and expressed as

$$\begin{aligned} SS_A &= \frac{(\sum_{i=1}^r \ell'_A y_{*i})^2}{r \ell'_A \ell_A} = \frac{(ab + a - b - (1))^2}{4r} \\ SS_B &= \frac{(\sum_{i=1}^r \ell'_B y_{*i})^2}{r \ell'_B \ell_B} = \frac{(ab + b - a - (1))^2}{4r} \end{aligned}$$

and

$$S_{AB} = \frac{(\sum_{i=1}^r \ell'_{AB} y_{*i})^2}{r \ell'_{AB} \ell_{AB}} = \frac{(ab + (1) - a - b)^2}{4r},$$

respectively.

Now consider a situation with 3 replicates with each consisting of 2 incomplete blocks as in the following figure:

Replicate 1 (A confounded)		Replicate 2 (B confounded)		Replicate 3 (AB confounded)	
Block 1	Block 2	Block 1	Block 2	Block 1	Block 2
$\begin{array}{ c } \hline ab \\ \hline (1) \\ \hline \end{array}$	$\begin{array}{ c } \hline a \\ \hline b \\ \hline \end{array}$	$\begin{array}{ c } \hline ab \\ \hline a \\ \hline \end{array}$	$\begin{array}{ c } \hline b \\ \hline (1) \\ \hline \end{array}$	$\begin{array}{ c } \hline ab \\ \hline b \\ \hline \end{array}$	$\begin{array}{ c } \hline a \\ \hline (1) \\ \hline \end{array}$

There are three factors A, B and AB . In case of total confounding, a factor is confounded in all the replicates. We consider here the situation of partial confounding in which a factor is not confounded in all the replicates.

Rather, the factor A is confounded in replicate 1, the factor B is confounded in replicate 2 and the interaction AB is confounded in replicate 3. Suppose each of the three replicate is repeated r times. So the observations are now available on r repetitions of each of the blocks in the three replicates. The partitions of replications, the blocks within replicates and plots within blocks being randomized. Now from the setup of figure,

- the factor A can be estimated from replicates 2 and 3 as it is confounded in replicate 1.
- the factor B can be estimated from replicates 1 and 3 as it is confounded in replicate 2 and
- the interaction AB can be estimated from replicates 1 and 2 as it is confounded in replicate 3.

When A is estimated from the replicate 2 only, then its estimate is given by

$$A_{rep2} = \frac{(\sum_{i=1}^r \ell'_{A2} y_{*i})_{rep2}}{2r}$$

and when A is estimated from the replicate 3 only, then its estimate is given by

$$A_{rep3} = \frac{(\sum_{i=1}^r \ell'_{A3} y_{*i})_{rep3}}{2r}$$

where ℓ'_{A2} and ℓ'_{A3} are the suitable vectors of +1 and -1 for being the linear function to be contrasts under replicates 2 and 3, respectively. Note that ℓ'_{A2} and ℓ'_{A3} each is a (4×1) vector having 4 elements in it. Now since A is estimated from both the replicates 2 and 3, so to combine them and to obtain a single estimator of A , we consider the arithmetic mean of A_{rep2} and A_{rep3} as an estimator of A given by

$$\begin{aligned}
A_{pc} &= \frac{A_{rep2} + A_{rep3}}{2} \\
&= \frac{\left(\sum_{i=1}^r \ell'_{A2} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell'_{A3} y_{*i} \right)_{rep3}}{4r} \\
&= \frac{\sum_{i=1}^r \ell'_{A} y_{*i}}{4r}
\end{aligned}$$

where (8×1) the vector

$$\ell_A^{*'} = (\ell_{A2}, \ell_{A3})$$

has 8 elements in it and subscript pc in A_{pc} denotes the estimate of A under “partial confounding” (pc). The sum of squares under partial confounding in this case is obtained as

$$SS_{A_{pc}} = \frac{\left(\sum_{i=1}^r \ell_A^{*'} y_{*i} \right)^2}{r \ell_A^{*'} \ell_A^{*'}} = \frac{\left(\sum_{i=1}^r \ell_A^{*'} y_{*i} \right)^2}{8r}.$$

Assuming that y_{ij} 's are independent and $Var(y_{ij}) = \sigma^2$ for all i and j , the variance of A_{pc} is given by

$$\begin{aligned}
Var(A_{pc}) &= \left(\frac{1}{4r} \right)^2 Var \left(\sum_{i=1}^r \ell_A^{*'} y_{*i} \right) \\
&= \left(\frac{1}{4r} \right)^2 Var \left(\left(\sum_{i=1}^r \ell'_{A2} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell'_{A3} y_{*i} \right)_{rep3} \right) \\
&= \left(\frac{1}{4r} \right)^2 (4r\sigma^2 + 4r\sigma^2) \\
&= \frac{\sigma^2}{2r}.
\end{aligned}$$

Now suppose A is not confounded in any of the blocks in the three replicates in this example. Then A can be estimated from all the three replicates, each repeated r times. Under such a condition, an estimate of A can be obtained using the same approach as the arithmetic mean of the estimates obtained from each of the three replicates as

$$\begin{aligned}
A_{pc}^* &= \frac{A_{rep1} + A_{rep2} + A_{rep3}}{3} \\
&= \frac{\left(\sum_{i=1}^r \ell'_{A1} y_{*i} \right)_{rep1} + \left(\sum_{i=1}^r \ell'_{A2} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell'_{A3} y_{*i} \right)_{rep3}}{6r} \\
&= \frac{\sum_{i=1}^r \ell_A^{*'} y_{*i}}{6r}
\end{aligned}$$

where (12×1) the vector

$$\ell_A^{**} = (\ell_{A1}, \ell_{A2}, \ell_{A3})$$

has 12 elements in it. The variance of A assuming that y_{ij} 's are independent and $\text{Var}(y_{ij}) = \sigma^{*2}$ for all i and j in this case is obtained as

$$\begin{aligned} \text{Var}(A_{pc}^*) &= \left(\frac{1}{6r}\right)^2 \text{Var} \left[\left(\sum_{i=1}^r \ell'_{A1} y_{*i} \right)_{rep1} + \left(\sum_{i=1}^r \ell'_{A2} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell'_{A3} y_{*i} \right)_{rep3} \right] \\ &= \left(\frac{1}{6r}\right)^2 (4r\sigma^{*2} + 4r\sigma^{*2} + 4r\sigma^{*2}) \\ &= \frac{\sigma^{*2}}{3r}. \end{aligned}$$

If we compare this estimator with the earlier estimator for the situation where A is unconfounded in all the r replicates and was estimated by

$$A = \frac{\sum_{i=1}^r \ell'_A y_{*i}}{2r}$$

and in the present situation of partial confounding, the corresponding estimator of A is given by

$$A_{pc}^* = \frac{A_{rep1} + A_{rep2} + A_{rep3}}{3} = \frac{\sum_{i=1}^r \ell_A^{**} y_{*i}}{6r}.$$

Both the estimators, viz., A and A_{pc}^* are same because A is based on r replications whereas A_{pc}^* is based on $3r$ replications. If we denote $r^* = 3r$ then A_{pc}^* becomes same as A . The expressions of variance of A and A_{pc}^* then also are same if we use $r^* = 3r$. Comparing them, we see that the information on A in the partially confounded scheme relative to that in unconfounded scheme is

$$\frac{2r/\sigma^2}{3r/\sigma^{*2}} = \frac{2}{3} \frac{\sigma^{*2}}{\sigma^2}.$$

If $\sigma^{*2} > \frac{3}{2} \sigma^2$, then the information in partially confounded design is more than the information in unconfounded design.

In total confounding case, the confounded effect is completely lost but in the case of partial confounding, some information about the confounded effect can be recovered. For example, two third of the total information can be recovered .his case for A .

Similarly, when B is estimated from the replicates 1 and 3 separately, then the individual estimates of B are given by

$$B_{rep1} = \frac{\left(\sum_{i=1}^r \ell'_{B1} y_{*i} \right)_{rep1}}{2r}$$

and

$$B_{rep3} = \frac{\left(\sum_{i=1}^r \ell'_{B3} y_{*i} \right)_{rep3}}{2r}.$$

Both the estimators are combined as arithmetic mean and the estimator of B based on partial confounding is

$$\begin{aligned} B_{pc} &= \frac{B_{rep1} + B_{rep3}}{2} \\ &= \frac{\left(\sum_{i=1}^r \ell'_{B1} y_{*i} \right)_{rep1} + \left(\sum_{i=1}^r \ell'_{B3} y_{*i} \right)_{rep3}}{4r} \\ &= \frac{\left(\sum_{i=1}^r \ell_{B}^{*'} y_{*i} \right)}{4r} \end{aligned}$$

where the (8×1) vector

$$\ell_B^{*'} = (\ell_{B1}, \ell_{B3})$$

has 8 elements. The sum of squares due to B_{pc} is obtained as

$$SS_{B_{pc}} = \frac{\left(\sum_{i=1}^r \ell_B^{*'} y_{*i} \right)^2}{r \ell_B^{*'} \ell_B^{*'}} = \frac{\left(\sum_{i=1}^r \ell_B^{*'} y_{*i} \right)^2}{8r}.$$

Assuming that y_{ij} 's are independent and $Var(y_{ij}) = \sigma^2$, the variance of B_{pc} is

$$\begin{aligned} Var(B_{pc}) &= \left(\frac{1}{4r} \right)^2 Var \left(\sum_{i=1}^r \ell_B^{*'} y_{*i} \right) \\ &= \frac{\sigma^2}{2r}. \end{aligned}$$

When AB is estimated from the replicates 1 and 2 separately, then its estimators based on the observations available from replicates 1 and 2 are

$$AB_{rep1} = \frac{\left(\sum_{i=1}^r \ell'_{AB} y_{*i} \right)_{rep1}}{2r}$$

and

$$AB_{rep2} = \frac{\left(\sum_{i=1}^r \ell'_{AB} y_{*i} \right)_{rep2}}{2r}$$

respectively. Both the estimators are combined as arithmetic mean and the estimator of AB is obtained as

$$\begin{aligned} AB_{pc} &= \frac{AB_{rep1} + AB_{rep2}}{2} \\ &= \frac{\left(\sum_{i=1}^r \ell'_{AB1} y_{*i} \right)_{rep1} + \left(\sum_{i=1}^r \ell'_{AB2} y_{*i} \right)_{rep2}}{4r} \\ &= \frac{\sum_{i=1}^r \ell_{AB}^{*'} y_{*i}}{4r} \end{aligned}$$

Where (8×1) the vector

$$\ell_{AB}^{*'} = (\ell_{AB1}, \ell_{AB2})$$

consists of 8 elements. The sum of squares due to AB_{pc} is

$$\begin{aligned} SS_{AB_{pc}} &= \frac{\left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right)^2}{r \ell_{AB}^* \ell_{AB}^*} \\ &= \frac{\left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right)^2}{8r} \end{aligned}$$

and the variance of AB_{pc} under the assumption that y_{ij} 's are independent and $Var(y_{ij}) = \sigma^2$ is given by

$$\begin{aligned} Var(AB_{pc}) &= \left(\frac{1}{4r} \right)^2 Var \left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right) \\ &= \frac{\sigma^2}{2r}. \end{aligned}$$

Block sum of squares:

Note that in case of partial confounding, the block sum of squares will have two components – due to replicates and within replicates. So the usual sum of squares due to blocks need to be divided into two components based on these two variants.. Now we illustrate how the sum of squares due to blocks are adjusted under partial confounding. We consider the setup as in the earlier example. There are 6 blocks (2 blocks under each replicate 1, 2 and 3), each repeated r times. So there are total $(6r - 1)$ degrees of freedom associated with the sum of squares due to blocks. The sum of squares due to blocks is divided into two parts-

- the sum of squares due to replicates with $(3r - 1)$ degrees of freedom and
- the sum of squares due to within replicates with $3r$ degrees of freedom.

Now, denoting

- B_i to be the total of i^{th} block and
- R_i to be the total due to i^{th} replicate,

the sum of squares due to blocks is

$$\begin{aligned}
 SS_{Block(pc)} &= \frac{1}{\text{Total number of treatment}} \sum_{i=1}^{\text{Total number of blocks}} B_i^2 - \frac{G^2}{N} \\
 &= \frac{1}{2^2} \sum_{i=1}^{3r} B_i^2 - \frac{G^2}{12r}; \quad (N = 12r) \\
 &= \frac{1}{2^2} \sum_{i=1}^{3r} (B_i^2 - R_i^2 + R_i^2) - \frac{G^2}{12r} \\
 &= \frac{1}{2^2} \sum_{i=1}^{3r} (B_i^2 - R_i^2) + \left(\frac{1}{2^2} \sum_{i=1}^{3r} R_i^2 - \frac{G^2}{12r} \right) \\
 &= \frac{1}{2^2} \sum_{i=1}^{3r} \left(\frac{B_{1i}^2 + B_{2i}^2}{2} - R_i^2 \right) + \left(\frac{1}{2^2} \sum_{i=1}^{3r} R_i^2 - \frac{G^2}{12r} \right)
 \end{aligned}$$

where B_{ji} denotes the total of j^{th} block in i^{th} replicate ($j = 1, 2$). The sum of squares due to blocks within replications (wr) is

$$SS_{Block(wr)} = \frac{1}{2^2} \sum_{i=1}^{3r} \left(\frac{B_{1i}^2 + B_{2i}^2}{2} - R_i^2 \right).$$

The sum of squares due to replications is

$$SS_{Block(r)} = \frac{1}{2^2} \sum_{i=1}^{3r} R_i^2 - \frac{G^2}{12r}.$$

So we have in case of partial confounding

$$SS_{\text{Block}} = SS_{\text{Block}(wr)} + SS_{\text{Block}(r)}.$$

The total sum of squares remains same as usual and is given by

$$SS_{\text{Total}(pc)} = \sum_i \sum_j \sum_k y_{ijk}^2 - \frac{G^2}{N}; \quad (N = 12r).$$

The analysis of variance table in this case of partial confounding is given in the following table. The test of hypothesis can be carried out in a usual way as in the case of factorial experiments.

Source	Sum of squares	Degrees of freedom	Mean squares
Replicates	$SS_{\text{Block}(r)}$	$3r(=r^*)$	$MS_{\text{Block}(r)}$
Blocks within replicates	$SS_{\text{Block}(wr)}$	$3r-1(=r^*-1)$	$MS_{\text{Block}(wr)}$
Factor A	$SS_{A_{pc}}$	1	$MS_{A(pc)}$
Factor B	$SS_{B_{pc}}$	1	$MS_{B(pc)}$
AB	$SS_{AB_{pc}}$	1	$MS_{AB(pc)}$
Error	By subtraction	$6r-3(=2r^*-3)$	$MS_{E(pc)}$
Total	$SS_{\text{Total}(pc)}$	$12r-1(=4r^*-1)$	

Example 2:

Consider the setup of 2^3 factorial experiment. The block size is 2^2 and 4 replications are made as in the following figure.

Replicate 1 (AB confounded)		Replicate 2 (AC confounded)		Replicate 3 (BC confounded)		Replicate 4 (ABC confounded)	
Block 1	Block 2	Block 1	Block 2	Block 1	Block 2	Block 1	Block 2
(1) ab c abc	a b ac bc	(1) b ac abc	a ab c bc	(1) a bc abc	b c ab ac	(1) ab ac bc	a b c abc

The arrangement of the treatments in different blocks in various replicates is based on the fact that different interaction effects are confounded in the different replicates. The interaction effect AB is confounded in replicate 1, AC is confounded in replicate 2, BC is confounded in replicate 3 and ABC is confounded in replicate 4. Then the r replications of each block are obtained. There are total eight factors involved in this case including (1). Out of them, three factors, viz., A, B and C are unconfounded whereas AB, BC, AC and ABC are partially confounded. Our objective is to estimate all these factors. The unconfounded factors can be estimated from all the four replicates whereas partially confounded factors can be estimated from the following replicates:

- AB from the replicates 2, 3 and 4,
- AC from the replicates 1, 3 and 4,
- BC from the replicates 1, 2 and 4 and
- ABC from the replicates 1, 2 and 3.

We first consider the estimation of unconfounded factors A, B and C which are estimated from all the four replicates.

The estimation of factor A from ℓ^{th} replicate ($\ell = 1, 2, 3, 4$) is as follows:

$$\begin{aligned}
 A_{rep\ j} &= \frac{\sum_{i=1}^r \ell'_{Aj} y_{*i}}{4r} \\
 A &= \frac{\sum_{j=1}^4 \ell'_{Aj} A_{rep\ j}}{4} = \frac{\sum_{j=1}^4 \sum_{i=1}^r \ell'_{Aj} y_{*i}}{16r} \\
 &= \frac{\sum_{i=1}^r \ell_{A*}^* y_{*i}}{16r}
 \end{aligned}$$

where (32×1) the vector

$$\ell_A^* = (\ell_{A1}, \ell_{A2}, \ell_{A3}, \ell_{A4})$$

has 32 elements and each ℓ_{Aj} ($j = 1, 2, 3, 4$) is having 8 elements in it. The sum of square due to A is now based on $32r$ elements as

$$SS_A = \frac{\left(\sum_{i=1}^r \ell_{A*}^* y_{*i} \right)^2}{r \ell_A^* \ell_A^*} = \frac{\left(\sum_{i=1}^r \ell_{A*}^* y_{*i} \right)^2}{32r}.$$

Assuming that y_{ij} 's are independent and $Var(y_{ij}) = \sigma^2$ for all i and j , the variance of A is obtained as

$$\begin{aligned} Var(A) &= \left(\frac{1}{16r}\right)^2 Var\left(\sum_{i=1}^r \ell_{A y_{*i}}^{**}\right) \\ &= \left(\frac{1}{16r}\right)^2 \times 32r\sigma^2 \\ &= \frac{\sigma^2}{8r}, \end{aligned}$$

Similarly, the factor B is estimated as an arithmetic mean of the estimates of B from each replicate as

$$B = \frac{\sum_{i=1}^r \ell_{B y_{*i}}^{**}}{16r}$$

where (32×1) the vector

$$\ell_B^{**} = (\ell_{B1}, \ell_{B2}, \ell_{B3}, \ell_{B4})$$

consists of 32 elements.

The sum of squares due to B is obtained on the similar line as in case of A as

$$SS_B = \frac{\left(\sum_{i=1}^r \ell_{B y_{*i}}^{**}\right)^2}{32r}.$$

The variance of B is obtained on the similar lines as in the case of A as

$$Var(B) = \frac{\sigma^2}{8r}.$$

The unconfounded factor C is also estimated as the average of estimates of C from all the replicates as

$$C = \frac{\sum_{i=1}^r \ell_{C y_{*i}}^{**}}{16r}$$

where (32×1) the vector

$$\ell_C^{**} = (\ell_{C1}, \ell_{C2}, \ell_{C3}, \ell_{C4})$$

consists of 32 elements.

The sum of square due to C in this case is obtained as

$$SS_C = \frac{\left(\sum_{i=1}^r \ell_{C y_{*i}}^{**}\right)^2}{32r}.$$

The variance of C is obtained as

$$Var(C) = \frac{\sigma^2}{8r}.$$

Next we consider the estimation of the confounded factor AB . This factor AB can be estimated from each of the replicates 2, 3 and 4 and the final estimate of AB can be obtained as the arithmetic mean of those three estimates as

$$\begin{aligned} AB_{pc} &= \frac{AB_{rep2} + AB_{rep3} + AB_{rep4}}{3} \\ &= \frac{1}{12r} \left(\left(\sum_{i=1}^r \ell'_{AB2} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell'_{AB3} y_{*i} \right)_{rep3} + \left(\sum_{i=1}^r \ell'_{AB4} y_{*i} \right)_{rep4} \right) \\ &= \frac{\sum_{i=1}^r \ell'_{AB} y_{*i}}{12r} \end{aligned}$$

where (24×1) the vector

$$\ell_{AB}^{*'} = (\ell_{AB2}, \ell_{AB3}, \ell_{AB4})$$

consists of 24 elements and each of the (8×1) vectors ℓ_{AB2} , ℓ_{AB3} and ℓ_{AB4} is having 8 elements in it. The sum of squares due to AB_{pc} is then based on $24r$ elements given as

$$SS_{AB_{pc}} = \frac{\left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right)^2}{r \ell_{AB}^{*'} \ell_{AB}^{*'}} = \frac{\left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right)^2}{24r}.$$

The variance AB_{pc} in this case is obtained under the assumption that y_{ij} 's are independent and each has variance σ^2 as

$$\begin{aligned} Var(AB_{pc}) &= \left(\frac{1}{12r} \right)^2 Var \left(\sum_{i=1}^r \ell_{AB}^{*'} y_{*i} \right) \\ &= \left(\frac{1}{12r} \right)^2 Var \left(\left(\sum_{i=1}^r \ell_{AB2}^{*'} y_{*i} \right)_{rep2} + \left(\sum_{i=1}^r \ell_{AB3}^{*'} y_{*i} \right)_{rep3} + \left(\sum_{i=1}^r \ell_{AB4}^{*'} y_{*i} \right)_{rep4} \right) \\ &= \left(\frac{1}{12r} \right)^2 (8r\sigma^2 + 8r\sigma^2 + 8r\sigma^2) \\ &= \frac{\sigma^2}{6r}. \end{aligned}$$

The confounded effects AC is obtained as the average of estimates of AC obtained from the replicates 1, 3 and 4 as

$$AC_{pc} = \frac{AC_{rep1} + AC_{rep3} + AC_{rep4}}{3}$$

$$= \frac{\sum_{i=1}^r \ell_{AC}^* y_{*i}}{12r}$$

where (24×1) the vector

$$\ell_{AC}^* = (\ell_{AB1}, \ell_{AC3}, \ell_{AC4})$$

consists of 24 elements,

The sum of squares due to AC in this case is given by

$$SS_{AC_{pc}} = \frac{\left(\sum_{i=1}^r \ell_{AC}^* y_{*i} \right)^2}{24r}.$$

The variance of AC in this case under the assumption that y_{ij} 's are independent and each has variance σ^2 is given by

$$Var(AC_{pc}) = \frac{\sigma^2}{6r}.$$

Similarly, the confounded effect BC is estimated as the average of the estimates of BC obtained from the replicates 1, 2 and 4 as

$$BC_{pc} = \frac{BC_{rep1} + BC_{rep2} + BC_{rep4}}{3}$$

$$= \frac{\sum_{i=1}^r \ell_{BC}^* y_{*i}}{12r}$$

where the vector

$$\ell_{BC}^* = (\ell_{BC1}, \ell_{BC3}, \ell_{BC4})$$

consists of 24 elements.

The sum of squares due to BC in this case is based on $24r$ elements and is given as

$$SS_{BC_{pc}} = \frac{\left(\sum_{i=1}^r \ell_{BC}^* y_{*i} \right)^2}{24r}.$$

The variance of BC in this case is obtained under the assumption that y_{ij} 's are independent and each has variance σ^2 as

$$\text{Var}(BC_{pc}) = \frac{\sigma^2}{6r}.$$

Lastly, the confounded effect ABC can be estimated first from the replicates 1,2 and 3 and then the estimate of ABC is obtained as an average of these three individual estimates as

$$ABC_{pc} = \frac{ABC_{rep1} + ABC_{rep2} + ABC_{rep3}}{3}$$

where (24×1) the vector

$$\ell_{ABC}^* = (\ell_{ABC1}, \ell_{ABC2}, \ell_{ABC3})$$

consists of 24 elements.

The sum of squares due to ABC in this case is based on $24r$ elements and is given by

$$SS_{ABC_{pc}} = \frac{\left(\sum_{i=1}^r \ell_{ABC}^* y_{*i} \right)^2}{24r}.$$

The variance of ABC in this case assuming that y_{ij} 's are independent and each has variance σ^2 is given by

$$\text{Var}(ABC_{pc}) = \frac{\sigma^2}{6r}.$$

If an unconfounded design with $4r$ replication was used then the variance of each of the factors A, B, C, AB, BC, AC and ABC is $\sigma^{*2}/8r$ where σ^{*2} is the error variance on blocks of size 8. So the relative efficiency of a confounded effect in the partially confounded design with respect to that of an unconfounded one in a comparable design is

$$\frac{6r/\sigma^2}{8r/\sigma^{*2}} = \frac{3}{4} \frac{\sigma^{*2}}{\sigma^2}.$$

So the information on a partially confounded effect relative to an unconfounded effect is $\frac{3}{4}$. If $\sigma^{*2} > 4\sigma^2/3$, then partially confounded design gives more information than the unconfounded design.

Further, the sum of squares due to block can be divided into two components – within replicates and due to replications. So we can write

$$SS_{\text{Block}} = SS_{\text{Block}(wr)} + SS_{\text{Block}(r)}$$

where the sum of squares due to blocks within replications (wr) is

$$SS_{\text{Block}(wr)} = \frac{1}{2^3} \sum_{i=1}^{4r} \left(\frac{B_{1i}^2 + B_{2i}^2}{2} - R_i^2 \right)$$

which carries $4r$ degrees of freedom and the sum of squares due to replication is

$$SS_{\text{Block}(r)} = \frac{1}{2^3} \sum_{i=1}^{4r} R_i^2 - \frac{G^2}{32r}$$

which carries $(4r - 1)$ degrees of freedom. The total sum of squares is

$$SS_{\text{Total}(pc)} = \sum_i \sum_j \sum_k y_{ijk}^2 - \frac{G^2}{32r}.$$

The analysis of variance table in this case of 2^3 factorial under partial confounding is given as

Source	Sum of squares	Degrees of freedom	Mean squares
Replicates	$SS_{\text{Block}(r)}$	$4r - 1$	$MS_{\text{Block}(r)}$
Blocks within replicate	$SS_{\text{Block}(wr)}$	$4r$	$MS_{\text{Block}(wr)}$
Factor A	SS_A	1	MS_A
Factor B	SS_B	1	MS_B
Factor C	SS_C	1	MS_C
AB	$SS_{AB(pc)}$	1	$MS_{AB(pc)}$
AC	$SS_{AC(pc)}$	1	$MS_{AC(pc)}$
BC	$SS_{BC(pc)}$	1	$MS_{BC(pc)}$
ABC	$SS_{ABC(pc)}$	1	$MS_{ABC(pc)}$
Error	by subtraction	$24r - 7$	$MS_{E(pc)}$
Total	$SS_{\text{Total}(pc)}$	$32r - 1$	

Test of hypothesis can be carried out in the usual way as in the case of factorial experiment.

Chapter 11

Fractional Replications

Consider the set up of complete factorial experiment, say 2^k . If there are four factors, then the total number of plots needed to conduct the experiment is $2^4 = 16$. When the number of factors increases to six, then the required number of plots to conduct the experiment becomes $2^6 = 64$ and so on. Moreover, the number of treatment combinations also become large when the number of factors increases. Sometimes, it is so large that it becomes practically difficult to organize such a huge experiment. Also, the quantity of experimental material needed, time, manpower etc. also increase and sometimes even it may not be possible to have so many resources to conduct a complete factorial experiment. The non-experimental type of errors also enters in the planning and conduct of the experiment. For example, there can be a slip in numbering the treatments or plots or they may be wrongly reported if they are too large in numbers.

About the degree of freedoms, in the 2^6 factorial experiment there are $2^6 - 1 = 63$ degrees of freedom which are divided as 6 for main effects, 15 for two-factor interactions and rest 42 for three or higher-order interactions. In case, the higher-order interactions are not of much use or much importance, then they can possibly be ignored. The information on main and lower-order interaction effects can then be obtained by conducting a fraction of complete factorial experiments. Such experiments are called as **fractional factorial experiments**. The utility of such experiments becomes more when the experimental process is more influenced and governed by the main and lower-order interaction effects rather than the higher-order interaction effects. The fractional factorial experiments need less number of plots and lesser experimental material than required in the complete factorial experiments. Hence it involves less cost, less manpower, less time etc.

It is possible to combine the runs of two or more fractional factorials to assemble sequentially a larger experiment to estimate the factor and interaction effects of interest.

To explain the fractional factorial experiment and its related concepts, we consider here examples in the set up of 2^k factorial experiments.

One-half fraction of 2^3 factorial experiment

First, we consider the set up of 2^3 factorial experiment and consider its one-half fraction. This is a very simple set up to understand the basics, definitions, terminologies and concepts related to the fractional factorials.

Consider the setup of 2^3 factorial experiment consisting of three factors, each at two levels. There is a total of 8 treatment combinations involved. So 8 plots are needed to run the complete factorial experiment.

Suppose the material needed to conduct the complete factorial experiment in 8 plots is not available or the cost of total experimental material is too high. The experimenter has material or money which is sufficient only for four plots. So the experimenter decides to have only four runs, i.e., $\frac{1}{2}$ fraction of 2^3 factorial experiment. Such an experiment contains a one-half fraction of a 2^3 experiment and is called 2^{3-1} factorial experiment. Similarly, $\frac{1}{2^2}$ fraction of 2^3 factorial experiment requires only 2 runs and contains $\frac{1}{2^2}$ fraction of 2^3 factorial experiment and is called as 2^{3-2} factorial experiment. In general, $\frac{1}{2^p}$ fraction of a 2^k factorial experiment requires only 2^{k-p} runs and is denoted as 2^{k-p} factorial experiment.

We consider the case of $\frac{1}{2}$ fraction of 2^3 factorial experiment to describe the various issues involved and to develop the concepts. The first question is how to choose four out of eight treatment combinations for conductive the experiment. In order to decide this, first we have to choose an interaction factor which the experimenter feels can be ignored. Generally, this can be a higher-order interaction which is usually difficult to interpret. We choose ABC in this case. Now we create the table of treatment combinations as in the following table.

Arrangement of treatment combinations for one-half fraction of 2^3 factorial experiment

Factors	<i>I</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>AB</i>	<i>AC</i>	<i>BC</i>	<i>ABC</i>
Treatment combinations								
<i>a</i>	+	+	-	-	-	-	+	+
<i>b</i>	+	-	+	-	-	+	-	+
<i>c</i>	+	-	-	+	+	-	-	+
<i>abc</i>	+	+	+	+	+	+	+	+
*****	*****	*****	*****	*****	*****	*****	*****	*****
<i>ab</i>	+	+	+	-	+	-	-	-
<i>ac</i>	+	+	-	+	-	+	-	-
<i>bc</i>	+	+	+	+	-	-	+	-
(1)	+	+	-	-	+	+	+	-

This table is obtained by the following steps.

- Write down the factor to be ignored which is ABC in our case. We can express ABC as $ABC = (a + b + c + abc) - (ab + ac + bc + (1))$.
- Collect the treatment combinations with plus (+) and minus (–) signs together; divide the eight treatment combinations into two groups with respect to the + and – signs. This is done in the last column corresponding to ABC .
- Write down the symbols + or – of the other factors A, B, C, AB, AC and BC corresponding to (a, b, c, abc) and $(ab, ac, bc, (1))$.

This provides the arrangement of treatments as given in the table. Now consider the column of ABC . The treatment combinations corresponding to + signs of treatment combinations in ABC provide one-half fraction of 2^3 factorial experiment. The remaining treatment combinations corresponding to the – signs in ABC will constitute another one-half fractions of 2^3 factorial experiment. Here one of the one-half fractions corresponding to + signs contains the treatment combinations a, b, c and abc . Another one-half fraction corresponding to - signs contains the treatment combinations ab, ac, bc and (1) . Both the one-half fractions are separated by a starred line in the Table.

Generator:

The factor which is used to generate the one-half fractions is called as the **generator**. For example, ABC is the generator of a fraction in the present case and we have two one-half fractions.

Defining relation:

The defining relation for a fractional factorial is the set of all columns that are equal to the identity column I . The identity column I always contains all the + signs. So in our case, $I = ABC$ is called the **defining relation** of this fractional factorial experiment.

The number of degrees of freedom associated with a one-half fraction of 2^3 factorial experiment, i.e., 2^{3-1} factorial experiment is 3 which is essentially used to estimate the main effects.

Now consider the one-half fraction containing the treatment combinations a, b, c and abc (corresponding to + signs in the column of ABC).

The factors A , B , C , AB , AC and BC are now estimated from this block as follows:

$$A = a - b - c + abc,$$

$$B = -a + b - c + abc,$$

$$C = -a - b + c + abc,$$

$$AB = -a - b + c + abc,$$

$$AC = -a + b - c + abc,$$

$$BC = a - b - c + abc.$$

Aliases:

We notice that the estimates of A and BC are same. So it is not possible to differentiate between whether A is being estimated or BC is being estimated. As such, $A = BC$. Similarly, the estimates of B and of AC as well as the estimates of C and of AB are also the same. We write this as $B = AC$ and $C = AB$. So it is not possible to differentiate between B and AC as well as between C and AB in the sense that which one is being estimated. Two or more effects that have this property are called **aliases**. Thus

- A and BC are aliases,
- B and AC are aliases and
- C and AB are aliases.

Note that the estimates of A, B, C, AB, BC, AC and ABC are obtained are one-half fraction set up. These estimates can also be obtained from the complete factorial set up. A question arises that how the estimate of an effect in the two different setups are related? The answer is as follows:

In fact, when we estimate A , B and C in 2^{3-1} factorial experiment, then we are essentially estimating $A + BC$, $B + AC$ and $C + AB$, respectively in a complete 2^3 factorial experiment. To understand this, consider the setup of a complete 2^3 factorial experiment in which A and BC are estimated by

$$A = -(1) + a - b + ab - c + ac - bc + abc,$$

$$BC = (1) + a - b - ab - c - ac + bc + abc.$$

Adding A and BC and ignoring the common multiplier, we have

$$A + BC = a - b - c + abc$$

which is the same as A or BC is a one-half fraction with $I = ABC$.

Similarly, considering the estimates of B and AC in 2^3 factorial experiment, adding them together and ignoring the common multiplier, we have

$$B = -(1) - a + b + ab - c - ac + bc + abc,$$

$$AC = (1) - a + b - ab + ac - bc + abc,$$

$$B + AC = -a + b - c + abc,$$

which is same B or AC in one-half fraction with $I = ABC$

The estimates of C and AB in 2^3 factorial experiment and their sum is as follows:

$$C = -(1) - a - b - ab + c + ac + bc + abc,$$

$$AB = (1) - a - b + ab + c - ac - bc + abc,$$

$$C + AB = -a - b - c + abc,$$

which is same as C or AB in one-half fraction with $I = ABC$.

Determination of alias structure:

The alias structure is determined by using the defining relation. Multiplying any column (or effect) by the defining relation yields the aliases for that column (or effect). For example, in this case, the defining relation is $I = ABC$. Now multiply the factors on both sides of $I = ABC$ yields

$$A \times I = (A) \times (ABC) = A^2 BC = BC,$$

$$B \times I = (B) \times (ABC) = AB^2 C = AC,$$

$$C \times I = (C) \times (ABC) = ABC^2 = AB.$$

The systematic rule to find aliases is to write down all the effects of a $2^{3-1} = 2^2$ factorial in a standard order and multiply each factor by the defining contrast.

Alternate or complementary one-half fraction:

We have considered up to now the one-half fraction corresponding to + signs of treatment combinations in ABC column in the table. Now suppose we choose other one-half fraction, i.e., treatment combinations with – signs in ABC column in the table. This is called an **alternate or complementary one-half fraction**. In this case, the effects are estimated as

$$A = ab + ac - bc - (1),$$

$$B = ab - ac + bc - (1),$$

$$C = -ab + ac + bc - (1),$$

$$AB = ab - ac - bc + (1),$$

$$AC = -ab + ac - bc + (1),$$

$$BC = -ab - ac + bc + (1),$$

In this case, we notice that $A = -BC$, $B = -AC$, $C = -AB$, so the same factors remain aliases again which are aliases in the one-half fraction with + sign in ABC . If we consider the setup of the complete 2^3 factorial experiment, then in case of complete fractional

$$A = -(1) + a - b + ab - c + ac - bc + abc$$

$$BC = (1) + a - b - ab - c - ac + bc + abc,$$

we observe that $A - BC$ in the complete 2^3 factorial experiment the same as A or BC in the one-half fractional with $I = -ABC$ (ignoring the common multiplier). In order to find the relationship between the estimates under this one-half fraction and a complete factorial, we find that what we estimate in the one-half fraction with - sign in ABC is the same as of estimating $A - BC$ from a complete 2^3 factorial experiment. Similarly, using $B - AC$ in the complete 2^3 factorial, is the same as using B or AC in a one-half fraction with $I = ABC$. Using $C - AB$ in a complete 2^3 factorial experiment is the same using as C or AB in one-half fraction with $I = ABC$ (ignoring the common multiplier).

Now there are two one-half fractions corresponding to + and - signs of treatment combinations in ABC . Based on that, there are now two sets of treatment combinations. A question arises that which one to use?

In practice, it does not matter which fraction is actually used. Both the one-half fractions belong to the same family of 2^3 factorial experiment. Moreover, the difference of negative signs in aliases of both the halves becomes positive while obtaining the sum of squares in the analysis of variance.

Use of more than one defining relations:

Further, suppose we want to have $1/2^2$ fraction of 2^3 factorial experiment with one more defining relation, say $I = BC$ along with $I = ABC$. So the one-half with + signs of ABC can further be divided into two halves. In this case, each half fraction will contain two treatments corresponding to

- + sign of BC , (viz., a and abc) and
- - sign of BC , (viz., b and c).

These two halves will constitute the one-fourth fraction of 2^3 factorial experiment. Similarly, we can consider the other one-half fraction corresponding to - sign of ABC . Now we look for + and - sign corresponding to $I = BC$ which constitute the two one-half fractions consisting of the treatments

- (1), bc and
- ab , ac .

This will again constitute the one-fourth fraction of 2^3 factorial experiment.

Example in 2^6 factorial experiment:

In order to have more understanding of the fractional factorial, we consider the setup of 2^6 factorial experiment. Since the highest order interaction, in this case, is $ABCDEF$, so we construct the one-half fraction using $I = ABCDEF$ as defining relation. Then we write all the factors of $2^{6-1} = 2^5$ factorial experiment in the standard order. Then multiply all the factors with the defining relation.

For example,

$$\begin{aligned} I \times A &= ABCDEF \times A \\ &= A^2 BCDEF \\ \text{or } A &= BCDEF \end{aligned}$$

Similarly,

$$\begin{aligned} I \times ABC &= ABCDEF \times ABC \\ &= A^2 B^2 CDEF \\ \text{or } ABC &= CDEF \text{ etc.} \end{aligned}$$

All such operations are illustrated in the following table.

One-half fraction of 2^6 factorial experiment using $I = ABCDEF$ as defining relation

$I = ABCDEF$	$D = ABCEF$	$E = ABCDF$	$DE = ABCF$
$A = BCDEF$	$AD = BCEF$	$AE = BCDF$	$ADE = BCF$
$B = ACDEF$	$BD = ACEF$	$BE = ACDF$	$BDE = ACF$
$AB = CDEF$	$ABD = CEF$	$ABE = CDF$	$ABDE = CF$
$C = ABDEF$	$CD = ABEF$	$CE = ABDF$	$CDE = ABF$
$AC = BDEF$	$ACD = BEF$	$ACE = BDF$	$ACDE = BF$
$BC = ADEF$	$BCD = AEF$	$BCE = ADF$	$BCDE = AF$
$ABC = DEF$	$ABCD = EF$	$ABCE = DF$	$ABCDE = F$

In this case, we observe that

- all the main effects have 5 factor interactions as aliases,
- all the 2 factor interactions have 4 factor interactions as aliases and
- all the 3 factor interactions have 3 factor interactions as aliases.

Suppose a completely randomized design is adopted with blocks of size 16. There are 32 treatments and $abcdef$ is chosen as the defining contrast for half replicate. Now all the 32 treatments cannot be accommodated. Only 16 treatments can be accommodated. So the treatments are to be divided and allocated into two blocks of size 16 each. This is equivalent to saying that one factorial effect (and its alias) are confounded with blocks. Suppose we decide that the three-factor interactions and their aliases (which are also three factors interactions in this case) are to be used as an error. So choose one of the three-factor interaction, say ABC (and its alias DEF) to be confounded. Now one of the blocks contains all the treatment combinations having an even number of letters a, b or c . These blocks are constructed in the following table.

One-half replicate of 2^6 factorial experiment in the blocks of size 16

Block 1	Block 2
(1)	ab
de	ae
df	af
ef	bd
ab	be
ac	bf
bc	cd
$abde$	ce
$abdf$	cf
$abef$	$adef$
$acde$	$bdef$
$acdf$	$cdef$
$acef$	$abcd$
$bcde$	$abce$
$bcd f$	$abcf$
$bcef$	$abcdef$

There are total 31 degrees of freedom, out of which 6 degrees of freedom are used by the main effects, 15 degrees of freedom are used by the two-factor interactions and 9 degrees of freedom are used by the error (from three-factor interactions). Additionally, one more division of degree of freedom arises in this case which is due to blocks. So the degree of freedom carried by blocks is 1. That is why the error degrees of freedom are 9 (and not 10) because one degree of freedom goes to block.

- Suppose the block size is to be reduced further and we want to have blocks of size 8 in the same setup. This can be achieved by $1/2^2$ fraction of 2^6 factorial experiment. In terms of confounding setup, this is equivalent to saying that the two factorial effects are to be confounded. Suppose we choose ABD (and its alias CEF) in addition to ABC (and its alias DEF). When we confound two effects, then their generalized interaction also gets confounded. So the interaction $ABC \times ABD = A^2B^2CD = CD$ (or $DEF \times CEF = CDE^2F^2 = CD$) and its alias $ABEF$ also get confounded. One may note that a two-factor interaction is getting confounded in this case which is not a good strategy. A good strategy in such cases where an important factor is getting confounded is to choose the least important two-factor interaction. The blocks arising with this plan are described in the following table. These blocks are derived by dividing each block of the earlier table of a one-half replicate of 2^6 factorial experiment in the blocks of size 16 into two halves. These halves contain respectively an odd and even number of the letters c and d .

Block 1	Block 2	Block 3	Block 4
(1)	<i>de</i>	<i>ae</i>	<i>ad</i>
<i>ef</i>	<i>df</i>	<i>af</i>	<i>bd</i>
<i>ab</i>	<i>ac</i>	<i>be</i>	<i>ce</i>
<i>abef</i>	<i>bc</i>	<i>bf</i>	<i>cf</i>
<i>acde</i>	<i>abde</i>	<i>cd</i>	<i>abce</i>
<i>acdf</i>	<i>abdf</i>	<i>abcd</i>	<i>abcf</i>
<i>bcde</i>	<i>acef</i>	<i>cdef</i>	<i>adef</i>
<i>bcdf</i>	<i>bcef</i>	<i>abcdef</i>	<i>bdef</i>

The total degrees of freedom, in this case, is 31 which are divided as follows:

- the blocks carry 3 degrees of freedom,
- the main effect carries 6 degrees of freedom.
- the two-factor interactions carry 14 degrees of freedom and

the error carries 8 degrees of freedom.

The analysis of variance in the case of fractional factorial experiments is conducted in the usual way, as in the case of any factorial experiment. The sums of squares for blocks, main effects and two-factor interactions are computed in the usual way.

Resolution:

The criterion of resolution is used to compare the fractional factorial designs for overall quality of the statistical inferences which can be drawn. It is defined as the length of the shortest word (or order of the lowest-order effect) aliased with “ I ” in the generating relationship.

A fractional factorial design with the greater resolution is considered to be better than a design with lower resolution. An important objective in the designs is to find a fractional factorial design that has the greatest resolution for a given number of runs and numbers of factors. The resolution of design is generally denoted by a subscripted roman numeral. For example, a fractional factorial design constructed by using “ $I = ABCD = \pm ABEF (= \pm CDEF)$ ” is denoted as 2_{IV}^{6-2} fractional factorial plan. In practice, the designs with resolution III, IV and V are used in practice.

When the design is of **resolution II**, it means that, e.g., “ $I = +AB$ ”. It means that in this case “ $A = +B$ ” which indicates that at least some pairs of main effects are aliased.

When the design is of **resolution III**, the generating relation is like e.g., “ $I = +ABC$ ”. In this case “ $A = +BC = \dots$ ” This means that the main effects will not be aliased with each other but some of them will be aliased with two-factor interaction. Thus such design can estimate all main effects if all interactions are absent.

When the design is of **resolution IV**, then the generating relationship is like “ $I = +ABCD$ ”. Then the main effects will not be aliased with each other or with two-factor interactions but some will get aliased with three-factor interaction, e.g., “ $A = +BCD$ ”. Some pairs of two-factor interactions will also get aliased, e.g., “ $AB = +CD = \dots$ ”. So this type of design unbiasedly estimates all the main effects even when two-factor interactions are present.

Similarly, the generating relations like “ $I = +ABCDE$ ” are used in resolution V designs. In this case, all main effects can be estimated unbiasedly in the absence of all interactions of the order less than five. The two-factor interactions can be estimated if no effects of higher-order are present. So resolution V design provides a complete estimation of the second-order model.

The designs of resolution II or higher than resolution V are not used in practice. Reason being that resolution II design cannot separate the influence of main effects and design of resolution VI or higher require a large number of units which may not be feasible all the times.

Chapter 12

Analysis of Covariance

Any scientific experiment is performed to know something that is unknown about a group of treatments and to test certain hypothesis about the corresponding treatment effect.

When variability of experimental units is small relative to the treatment differences and the experimenter does not wish to use experimental design, then just take large number of observations on each treatment effect and compute its mean. The variation around mean can be made as small as desired by taking more observations.

When there is considerable variation among observations on the same treatment and it is not possible to take an unlimited number of observations, the techniques used for reducing the variation are

- (i) use of proper experimental design and
- (ii) use of concomitant variables.

The use of concomitant variables is accomplished through the technique of analysis of covariance. If both the techniques fail to control the experimental variability then the number of replications of different treatments (in other words, the number of experimental units) are needed to be increased to a point where adequate control of variability is attained.

Introduction to analysis of covariance model

In the linear model

$$Y = X_1\beta_1 + X_2\beta_2 + \dots + X_p\beta_p + \varepsilon,$$

if the explanatory variables are quantitative variables as well as indicator variables, i.e., some of them are qualitative and some are quantitative, then the linear model is termed as analysis of covariance (ANCOVA) model.

Note that the indicator variables do not provide as much information as the quantitative variables. For example, the quantitative observations on age can be converted into indicator variable. Let an indicator variable be

$$D = \begin{cases} 1 & \text{if age} \geq 17 \text{ years} \\ 0 & \text{if age} < 17 \text{ years.} \end{cases}$$

Now the following quantitative values of age can be changed into indicator variables.

Ages (years)	Ages
14	0
15	0
16	0
17	1
20	1
21	1
22	1

In many real application, some variables may be quantitative and others may be qualitative. In such cases, ANCOVA provides a way out.

It helps in reducing the sum of squares due to error which in turn reflects the better model adequacy diagnostics.

See how does this work:

In one way model : $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, we have $TSS_1 = SSA_1 + SSE_1$

In two way model : $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$, we have $TSS_2 = SSA_2 + SSB_2 + SSE_2$

In three way model : $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \varepsilon_{ijk}$, we have $TSS_3 = SSA_3 + SSB_3 + SS\gamma_3 + SSE_3$

If we have a given data set, then ideally

$$TSS_1 = TSS_2 = TSS_3$$

$$SSA_1 = SSA_2 = SSA_3;$$

$$SSB_2 = SSB_3$$

So $SSE_1 \geq SSE_2 \geq SSE_3$.

Note that in the construction of F -statistics, we use $\frac{SS(\text{effects}) / df}{SSE / df}$.

So F -statistic essentially depends on the SSE s.

Smaller $SSE \Rightarrow$ large $F \Rightarrow$ more chance of rejection.

Since SSA , SSB etc. here are based on dummy variables, so obviously if SSA, SSB , etc. are based on quantitative variables, they will provide more information. Such ideas are used in ANCOVA models and we construct the model by incorporating the quantitative explanatory variables in ANOVA models.

In another example, suppose our interest is to compare several different kinds of feed for their ability to put weight on animals. If we use ANOVA, then we use the final weights at the end of experiment. However, final weights of the animals depend upon the initial weight of the animals at the beginning of the experiment as well as upon the difference in feeds.

Use of ANCOVA models enables us to adjust or correct these initial differences.

ANCOVA is useful for improving the precision of an experiment. Suppose response Y is linearly related to covariate X (or **concomitant variable**). Suppose experimenter cannot control X but can observe it. ANCOVA involves adjusting Y for the effect of X . If such an adjustment is not made, then the X can inflate the error mean square and makes the true differences in Y due to treatment harder to detect.

If, for a given experimental material, the use of proper experimental design cannot control the experimental variation, the use of concomitant variables (which are related to experimental material) may be effective in reducing the variability.

Consider the one way classification model as

$$\begin{aligned} E(Y_{ij}) &= \beta_i & i = 1, \dots, p, j = 1, \dots, N_i, \\ \text{Var}(Y_{ij}) &= \sigma^2. \end{aligned}$$

If usual analysis of variance for testing the hypothesis of **equality of treatment effects** shows a highly significant difference in the treatment effects due to some factors affecting the experiment, then consider the model which takes into account this effect

$$\begin{aligned} E(Y_{ij}) &= \beta_i + \gamma t_{ij} & i = 1, \dots, p, j = 1, \dots, N_i, \\ \text{Var}(Y_{ij}) &= \sigma^2 \end{aligned}$$

where t_{ij} are the observations on concomitant variables (which are related to X_{ij}) and γ is the regression coefficient associated with t_{ij} . With this model, the variability of treatment effects can be considerably reduced.

For example, in any agricultural experimental, if the experimental units are plots of land then, t_{ij} can be measure of fertility characteristic of the j^{th} plot receiving i^{th} treatment and X_{ij} can be yield.

In another example, if experimental units are animals and suppose the objective is to compare the growth rates of groups of animals receiving different diets. Note that the observed differences in growth rates can be attributed to diet only if all the animals are similar in some observable characteristics like weight, age etc. which influence the growth rates.

In the absence of similarity, use t_{ij} which is the weight or age of j^{th} animal receiving i^{th} treatment.

If we consider the quadratic regression in t_{ij} then in

$$E(Y_{ij}) = \beta_i + \gamma_1 t_{ij} + \gamma_2 t_{ij}^2, \quad i = 1, \dots, p, j = 1, \dots, n_i,$$

$$Var(Y_{ij}) = \sigma^2.$$

ANCOVA in this case is the same as ANCOVA with two concomitant variables t_{ij} and t_{ij}^2 .

In two way classification with one observation per cell,

$$E(Y_{ij}) = \mu + \alpha_i + \beta_j + \gamma t_{ij}, \quad i = 1, \dots, I, j = 1, \dots, J$$

or

$$E(Y_{ij}) = \mu + \alpha_i + \beta_j + \gamma_1 t_{ij} + \gamma_2 w_{ij}$$

$$\text{with } \sum_i \alpha_i = 0, \sum_j \beta_j = 0,$$

then (y_{ij}, t_{ij}) or (y_{ij}, t_{ij}, w_{ij}) are the observations in $(i, j)^{th}$ cell and t_{ij}, w_{ij} are the concomitant variables.

The concomitant variables can be fixed or random.

We consider the case of fixed concomitant variables only.

One way classification

Let Y_{ij} ($j = 1 \dots n_i, i = 1 \dots p$) be a random sample of size n_i from i^{th} normal populations with mean

$$\mu_{ij} = E(Y_{ij}) = \beta_i + \gamma t_{ij}$$

$$Var(Y_{ij}) = \sigma^2$$

where β_i, γ and σ^2 are the unknown parameters, t_{ij} are known constants which are the observations on a concomitant variable.

The null hypothesis is

$$H_0 : \beta_1 = \dots = \beta_p.$$

Let

$$\bar{y}_{io} = \frac{1}{n_i} \sum_j y_{ij}, \quad \bar{y}_{oj} = \frac{1}{p} \sum_i y_{ij}, \quad \bar{y}_{oo} = \frac{1}{n} \sum_i \sum_j y_{ij}$$

$$\bar{t}_{io} = \frac{1}{n_i} \sum_j t_{ij}, \quad \bar{t}_{oj} = \frac{1}{p} \sum_i t_{ij}, \quad \bar{t}_{oo} = \frac{1}{n} \sum_i \sum_j t_{ij}$$

$$n = \sum_i n_i.$$

Under the whole parametric space (π_Ω) , use likelihood ratio test for which we obtain the $\hat{\beta}_i$'s and $\hat{\gamma}$ using the least squares principle or maximum likelihood estimation as follows:

$$\begin{aligned} \text{Minimize } S &= \sum_i \sum_j (y_{ij} - \mu_{ij})^2 \\ &= \sum_i \sum_j (y_{ij} - \beta_i - \gamma t_{ij})^2 \end{aligned}$$

$$\frac{\partial S}{\partial \beta_i} = 0 \text{ for fixed } \gamma$$

$$\Rightarrow \beta_i = \bar{y}_{io} - \gamma \bar{t}_{io}$$

$$\text{Put } \beta_i \text{ in } S \text{ and minimize the function by } \frac{\partial S}{\partial \gamma} = 0,$$

$$\text{i.e., minimize } \sum_i \sum_j [y_{ij} - \bar{y}_{io} - \gamma(t_{ij} - \bar{t}_{io})]^2 \text{ with respect to } \gamma \text{ gives}$$

$$\hat{\gamma} = \frac{\sum_i \sum_j (y_{ij} - \bar{y}_{io})(t_{ij} - \bar{t}_{io})}{\sum_i \sum_j (t_{ij} - \bar{t}_{io})^2}.$$

$$\text{Thus } \hat{\beta}_i = \bar{y}_{io} - \hat{\gamma} \bar{t}_{io}$$

$$\hat{\mu}_{ij} = \hat{\beta}_i + \hat{\gamma} t_{ij}.$$

$$\begin{aligned}\text{Since } y_{ij} - \hat{\mu}_{ij} &= y_{ij} - \hat{\beta}_i - \hat{\gamma}t_{ij} \\ &= y_{ij} - \bar{y}_{io} - \hat{\gamma}(t_{ij} - \bar{t}_{io}),\end{aligned}$$

we have

$$\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 = \sum_i \sum_j (y_{ij} - \bar{y}_{io})^2 - \frac{\left[\sum_i \sum_j (y_{ij} - \bar{y}_{io})(t_{ij} - \bar{t}_{io}) \right]}{\sum_i \sum_j (t_{ij} - \bar{t}_{io})^2}.$$

Under $H_0 : \beta_1 = \dots = \beta_p = \beta$ (say), consider $S_w = \sum_i \sum_j [y_{ij} - \beta - \gamma t_{ij}]^2$ and minimize S_w under sample space (π_w) ,

$$\frac{\partial S_w}{\partial \beta} = 0,$$

$$\frac{\partial S_w}{\partial \gamma} = 0$$

$$\Rightarrow \hat{\beta} = \bar{y}_{oo} - \hat{\gamma}\bar{t}_{oo}$$

$$\hat{\gamma} = \frac{\sum_i \sum_j (y_{ij} - \bar{y}_{oo})(t_{ij} - \bar{t}_{oo})}{\sum_i \sum_j (t_{ij} - \bar{t}_{oo})^2}$$

$$\hat{\mu}_{ij} = \hat{\beta} + \hat{\gamma}t_{ij}.$$

Hence

$$\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 = \sum_i \sum_j (y_{ij} - \bar{y}_{oo})^2 - \frac{\left[\sum_i \sum_j (y_{ij} - \bar{y}_{oo})(t_{ij} - \bar{t}_{oo}) \right]^2}{\sum_i \sum_j (t_{ij} - \bar{t}_{oo})^2}$$

and

$$\sum_i \sum_j (\hat{\mu}_{ij} - \hat{\mu}_{ij})^2 = \sum_i \sum_j \left[(\bar{y}_i - \bar{y}_{oo}) + \hat{\gamma}(t_{ij} - \bar{t}_{io}) - \hat{\gamma}(t_{ij} - \bar{t}_{oo}) \right]^2.$$

The likelihood ratio test statistic in this case is given by

$$\lambda = \frac{\max_w L(\beta, \gamma, \sigma^2)}{\max_{\Omega} L(\beta, \gamma, \sigma^2)} = \frac{\sum_i \sum_j (\hat{\mu}_{ij} - \hat{\mu}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2}.$$

Now we use the following theorems:

Theorem 1: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follow a multivariate normal distribution $N(\mu, \Sigma)$ with mean vector μ and positive definite covariance matrix Σ . Then $Y'AY$ follows a noncentral chi-square distribution with p degrees of freedom and noncentrality parameter $\mu'A\mu$, i.e., $\chi^2(p, \mu'A\mu)$ if and only if ΣA is an idempotent matrix of rank p .

Theorem 2: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follows a multivariate normal distribution $N(\mu, \Sigma)$ with mean vector μ and positive definite covariance matrix Σ . Let $Y'A_1Y$ follows $\chi^2(p_1, \mu'A_1\mu)$ and $Y'A_2Y$ follows $\chi^2(p_2, \mu'A_2\mu)$. Then $Y'A_1Y$ and $Y'A_2Y$ are independently distributed if $A_1\Sigma A_2 = 0$.

Theorem 3: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follows a multivariate normal distribution $N(\mu, \sigma^2 I)$, then the maximum likelihood (or least squares) estimator $L'\hat{\beta}$ of estimable linear parametric function is independently distributed of $\hat{\sigma}^2$; $L'\hat{\beta}$ follow $N[L'\beta, L'(XX)^{-1}L]$ and $\frac{n\hat{\sigma}^2}{\sigma^2}$ follows $\chi^2(n-p)$ where $\text{rank}(X) = p$.

Using these theorems on the independence of quadratic forms and dividing the numerator and denominator by respective degrees of freedom, we have

$$F = \frac{n-p-1}{p-1} \frac{\sum_i \sum_j (\hat{\mu}_{ij} - \hat{\mu}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2} \sim F(p-1, n-p) \text{ under } H_0$$

So reject H_0 whenever $F \geq F_{1-\alpha}(p-1, n-p)$ at α level of significance.

The terms involved in λ can be simplified for computational convenience follows:

We can write

$$\begin{aligned}
& \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 \\
&= \sum_i \sum_j \left[y_{ij} - \hat{\beta} - \hat{\gamma} t_{ij} \right]^2 \\
&= \sum_i \sum_j \left[(y_{ij} - \bar{y}_{oo}) - \hat{\gamma} (t_{ij} - \bar{t}_{oo}) \right]^2 \\
&= \sum_i \sum_j \left[(y_{ij} - \bar{y}_{oo}) - \hat{\gamma} (t_{ij} - \bar{t}_{oo}) + \hat{\gamma} (t_{ij} - \bar{t}_{io}) - \hat{\gamma} (t_{ij} - \bar{t}_{io}) \right]^2 \\
&= \sum_i \sum_j \left[(y_{ij} - \bar{y}_{io}) - \hat{\gamma} (t_{ij} - \bar{t}_{io}) \right]^2 \\
&= \sum_i \sum_j \left[(y_{ij} - \bar{y}_{oo}) + \hat{\gamma} (t_{ij} - \bar{t}_{io}) - \hat{\gamma} (t_{ij} - \bar{t}_{oo}) \right]^2 \\
&= \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 + \sum_i \sum_j (\hat{\mu}_{ij} - \hat{\hat{\mu}}_{ij})^2.
\end{aligned}$$

For computational convenience

$$\lambda = \frac{\sum_i \sum_j (\hat{\mu}_{ij} - \hat{\hat{\mu}}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2} = \frac{\left(T_{yy} - \frac{T_{yt}^2}{T_{tt}} \right) - \left(E_{yy} - \frac{E_{yt}^2}{E_{tt}} \right)}{\left(E_{yy} - \frac{E_{yt}^2}{E_{yy}} \right)}$$

where

$$\begin{aligned}
T_{yy} &= \sum_i \sum_j (y_{ij} - \bar{y}_{oo})^2 \\
T_{tt} &= \sum_i \sum_j (t_{ij} - \bar{t}_{oo})^2 \\
T_{yt} &= \sum_i \sum_j (y_{ij} - \bar{y}_{oo})(t_{ij} - \bar{t}_{oo}) \\
E_{yy} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io})^2 \\
E_{tt} &= \sum_i \sum_j (t_{ij} - \bar{t}_{io})^2 \\
E_{yt} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io})(t_{ij} - \bar{t}_{io}).
\end{aligned}$$

Analysis of covariance table for one way classification is as follows:

Source of variation	Degrees of freedom	Sum of products			Adjusted sum of squares		F
		yy	yt	tt	Degress of freedom	Sum of squares	
Population	$p-1$	$P_{yy}(=T_{yy}-E_{yy})$	$P_{yt}(=T_{yt}-E_{yt})$	$P_{tt}(=T_{tt}-E_{tt})$	$p-1$	$q_1 = q_0 - q_2$	$\frac{n-p-1}{p-1} \frac{q_1}{q_2}$
Error	$n-p$	E_{yy}	E_{yt}	E_{tt}	$n-p-1$	$q_2 = E_{yy} - \frac{E_{yt}^2}{E_{yy}}$	
Total	$n-1$	T_{yy}	T_{yt}	T_{tt}	$n-2$	$q_0 = T_{yy} - \frac{T_{yt}^2}{T_{tt}}$	

If H_0 is rejected, employ multiple comprises methods to determine which of the contrasts in β_i are responsible for this.

For any estimable linear parametric contrast

$$\varphi = \sum_{i=1}^p C_i \beta_i \text{ with } \sum_{i=1}^p C_i = 0,$$

$$\hat{\varphi} = \sum_{i=1}^p C_i \hat{\beta}_i = \sum_{i=1}^p C_i \bar{y}_i - \hat{\gamma} \sum_{i=1}^p C_i \bar{t}_i$$

$$Var(\hat{\gamma}) = \frac{\sigma^2}{\sum_i \sum_j (t_{ij} - \bar{t}_i)^2}$$

$$\Rightarrow Var(\hat{\varphi}) = \sigma^2 \left[\sum_i \frac{C_i^2}{n_i} + \frac{\left(\sum_i C_i \bar{t}_i \right)^2}{\sum_i \sum_j (t_{ij} - \bar{t}_i)^2} \right].$$

Two way classification (with one observations per cell)

Consider the case of two way classification with one observation per cell.

Let $y_{ij} \sim N(\mu_{ij}, \sigma^2)$ be independently distributed with

$$E(y_{ij}) = \mu + \alpha_i + \beta_j + \gamma t_{ij}, \quad i = 1 \dots I, j = 1 \dots J$$

$$V(y_{ij}) = \sigma^2$$

where

μ : Grand mean

α_i : Effect of i^{th} level of A satisfying $\sum_i \alpha_i = 0$

β_j : Effect of j^{th} level of B satisfying $\sum_j \beta_j = 0$

t_{ij} : observation (known) on concomitant variable.

The null hypothesis under consideration are

$$H_{0\alpha} : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$$

$$H_{0\beta} : \beta_1 = \beta_2 = \dots = \beta_J = 0$$

Dimension of whole parametric space $(\pi_\Omega) : I + J$

Dimension of sample space $(\pi_{w\alpha}) : J + 1$ under $H_{0\alpha}$

Dimension of sample space $(\pi_{w\beta}) : I + 1$ under $H_{0\beta}$

with respective alternative hypotheses as

$H_{1\alpha}$: At least one pair of α 's is not equal

$H_{1\beta}$: At least one pair of β 's is not equal.

Consider the estimation of parameters under the whole parametric space (π_Ω) .

Find minimum value of $\sum_i \sum_j (y_{ij} - \mu_{ij})^2$ under π_Ω .

To do this, minimize

$$\sum_i \sum_j (y_{ij} - \mu - \alpha_i - \beta_j - \gamma t_{ij})^2.$$

For fixed γ , which gives on solving the least squares estimates (or the maximum likelihood estimates) of the respective parameters as

$$\begin{aligned}
\mu &= \bar{y}_{oo} - \gamma \bar{t}_o \\
\alpha_i &= \bar{y}_i - \bar{y}_{oo} - \gamma(\bar{t}_{io} - \bar{t}_{oo}) \\
\beta_j &= \bar{y}_{oj} - \bar{y}_{oo} - \gamma(\bar{t}_{oj} - \bar{t}_{oo}).
\end{aligned} \tag{1}$$

Under these values of μ, α_i and β_j , the sum of squares $\sum_i \sum_j (y_{ij} - \mu - \alpha_i - \beta_j - \gamma t_{ij})^2$ reduces to

$$\sum_i \sum_j \left[y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo} + \gamma(t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo}) \right]^2. \tag{2}$$

Now minimization of (2) with respect to γ gives

$$\hat{\gamma} = \frac{\sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})(t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo})}{\sum_{i=1}^I \sum_{j=1}^J (t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo})^2}.$$

Using $\hat{\gamma}$, we get from (1)

$$\begin{aligned}
\hat{\mu} &= \bar{y}_{oo} - \hat{\gamma} \bar{t}_{oo} \\
\hat{\alpha}_i &= (\bar{y}_{io} - \bar{y}_{oo}) - \hat{\gamma}(\bar{t}_{io} - \bar{t}_{oo}) \\
\hat{\beta}_j &= (\bar{y}_{oj} - \bar{y}_{oo}) - \hat{\gamma}(\bar{t}_{oj} - \bar{t}_{oo}).
\end{aligned}$$

Hence

$$\begin{aligned}
& \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 \\
&= \sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2 - \frac{\left[\sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})(t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo}) \right]^2}{\sum_i \sum_j (t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo})^2} \\
&= E_{yy} - \frac{E_{yt}^2}{E_{tt}}
\end{aligned}$$

where

$$\begin{aligned}
E_{yy} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2 \\
E_{yt} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})(t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo}) \\
E_{tt} &= \sum_i \sum_j (t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo})^2.
\end{aligned}$$

Case (i) : Test of $H_{0\alpha}$

Minimize $\sum_i \sum_j (y_{ij} - \mu - \beta_j - \gamma t_{ij})^2$ with respect to μ, β_j and γ gives the least squares estimates) (or the maximum likelihood estimates) of respective parameters as

$$\begin{aligned}\Rightarrow \hat{\mu} &= \bar{y}_{oo} - \hat{\gamma} \bar{t}_{oo} \\ \hat{\beta}_j &= \bar{y}_{oj} - \bar{y}_{oo} - \hat{\gamma}(\bar{t}_{oj} - \bar{t}_{oo}) \\ \hat{\gamma} &= \frac{\sum_i \sum_j (y_{ij} - \bar{y}_{oj})(t_{ij} - \bar{t}_{oj})}{\sum_i \sum_j (t_{ij} - \bar{t}_{oj})^2} \\ \hat{\mu} &= \hat{\mu} + \hat{\beta}_j + \hat{\gamma} t_{ij}.\end{aligned}\tag{3}$$

Substituting these estimates in (3) we get

$$\begin{aligned}\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 &= \sum_i \sum_j (y_{ij} - \bar{y}_j)^2 - \frac{\left[\sum_i \sum_j (y_{ij} - \bar{y}_{oj})(t_{ij} - \bar{t}_{oj}) \right]}{\sum_i \sum_j (t_{ij} - \bar{t}_{oj})^2} \\ &= E_{yy} + A_{yy} - \frac{[E_{yt} + A_{yt}]}{E_{tt} + A_{tt}}\end{aligned}$$

where

$$\begin{aligned}A_{yy} &= \sum_i J(\bar{y}_{io} - \bar{y}_{oo})^2 \\ A_{tt} &= \sum_i J(\bar{t}_{io} - \bar{t}_{oo})^2 \\ A_{yt} &= \sum_i J(\bar{y}_{io} - \bar{y}_{oo})(\bar{t}_{io} - \bar{t}_{oo})^2 \\ E_{yy} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2 \\ E_{tt} &= \sum_i \sum_j (t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo})^2 \\ E_{yt} &= \sum_i \sum_j (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})(t_{ij} - \bar{t}_{io} - \bar{t}_{oj} + \bar{t}_{oo}).\end{aligned}$$

Thus the likelihood ratio test statistic for testing $H_{0\alpha}$ is

$$\lambda_1 = \frac{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 - \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2}.$$

Adjusting with degrees of freedom and using the earlier result for the independence of two quadratic forms and their distribution

$$F_1 = \frac{(IJ - I - J)}{(I - 1)} \left[\frac{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2 - \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2} \right] \sim F(I - 1, IJ - I - J) \text{ under } H_{0\alpha}.$$

So the decision rule is to reject $H_{0\alpha}$ whenever $F_1 > F_{1-\alpha}(I - 1, IJ - I - J)$.

Case b: Test of $H_{0\beta}$

Minimize $\sum_i \sum_j (y_{ij} - \mu - \alpha_i - \gamma t_{ij})^2$ with respect to μ, α_i and γ gives the least squares estimates (or maximum likelihood estimates) of respective parameters as

$$\begin{aligned} \hat{\mu} &= \bar{y}_{oo} - \tilde{\gamma} \bar{t}_{oo} \\ \tilde{\alpha}_j &= \bar{y}_{io} - \bar{y}_{oo} - \tilde{\gamma}(\bar{t}_{io} - \bar{t}_{oo}) \\ \tilde{\gamma} &= \frac{\sum_i \sum_j (y_{ij} - \bar{y}_{io})(t_{ij} - \bar{t}_{io})}{\sum_i \sum_j (t_{ij} - \bar{t}_{io})^2} \\ \tilde{\mu}_{ij} &= \tilde{\mu} + \tilde{\alpha}_i + \tilde{\gamma}_{ij}. \end{aligned} \quad (4)$$

From (4), we get

$$\begin{aligned} \sum_i \sum_j (y_{ij} - \tilde{\mu}_{ij})^2 &= \sum_i \sum_j (y_{ij} - \bar{y}_{io})^2 - \frac{\left[\sum_i \sum_j (y_{ij} - \bar{y}_{io})(t_{ij} - \bar{t}_{io}) \right]^2}{\sum_i \sum_j (t_{ij} - \bar{t}_{io})^2} \\ &= E_{yy} + B_{yy} - \frac{[E_{yt} + B_{yt}]^2}{B_{tt}} \end{aligned}$$

$$B_{yy} = \sum_j I(\bar{y}_{oj} - \bar{y}_{oo})^2$$

$$\text{where } B_{tt} = \sum_j I(\bar{t}_{oj} - \bar{t}_{oo})^2$$

$$B_{yt} = \sum_j I(\bar{y}_{io} - \bar{y}_{oo})(\bar{t}_{oj} - \bar{t}_{oo}).$$

Thus the likelihood ratio test statistic for testing $H_{0\beta}$ is

$$F_2 = \frac{(IJ - I - J)}{(J - 1)} \left[\frac{\sum_i \sum_j (y_{ij} - \tilde{\mu}_{ij})^2 - \sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2}{\sum_i \sum_j (y_{ij} - \hat{\mu}_{ij})^2} \right] \sim F(J - 1, IJ - I - J) \text{ under } H_{0\beta}.$$

So the decision rule is to reject $H_{0\beta}$ whenever $F_2 \geq F_{1-\alpha}(J - 1, IJ - I - J)$.

If $H_{\alpha\alpha}$ is rejected, use multiple comparison methods to determine which of the contrasts α_i are responsible for this rejection. The same is true for $H_{\alpha\beta}$.

The analysis of covariance table for two way classification is as follows:

Source of variation	Degrees of freedom	Sum of products				F
	yy	yt	tt			
Between evels of A	$I - 1$	A_{yy}	A_{yt}	A_{tt}	$I - 1 \quad q_0 = q_3 - q_2$	$F_1 = \frac{IJ - I - J}{I - 1} \frac{q_0}{q_2}$
Between levels of B	$J - 1$	B_{yy}	B_{yt}	B_{tt}	$J - 1 \quad q_1 = q_4 - q_2$	$F_2 = \frac{IJ - I - J}{J - 1} \frac{q_1}{q_2}$
Error	$(I - 1)(J - 1)$	E_{yy}	E_{yt}	E_{tt}	$IJ - I - J \quad q_2 = E_{yy} - \frac{E_{yt}^2}{E_{tt}}$	
Total	$IJ - 1$	T_{yy}	T_{yt}	T_{tt}	$IJ - 2$	
Error + levels of A	$IJ - J$				$q_3 = (A_{yy} + E_{yy}) - \frac{(A_{yt} + E_{yt})^2}{A_{tt} + E_{tt}}$	
Error + levels of B	$IJ - I$				$q_4 = (B_{yy} + E_{yt}) - \frac{(B_{yt} + E_{yt})^2}{B_{tt} + E_{tt}}$	

Chapter 2

General Linear Hypothesis and Analysis of Variance

Regression model for the general linear hypothesis

Let $Y_1, Y_2, \dots, Y_n$ be a sequence of n independent random variables associated with responses. Then we can write it as

$$E(Y_i) = \sum_{j=1}^p \beta_j x_{ij}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, p$$
$$\text{Var}(Y_i) = \sigma^2.$$

This is the linear model in the expectation form where $\beta_1, \beta_2, \dots, \beta_p$ are the unknown parameters and x_{ij} 's are the known values of independent covariates $X_1, X_2, \dots, X_p$.

Alternatively, the linear model can be expressed as

$$Y_i = \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i, \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, p$$

where ε_i 's are identically and independently distributed random error component with mean 0 and variance σ^2 , i.e., $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 (i \neq j)$.

In matrix notations, the linear model can be expressed as

$$Y = X\beta + \varepsilon$$

where

- $Y = (Y_1, Y_2, \dots, Y_n)'$ is a $n \times 1$ vector of observations on the response variable,
- the matrix $X = \begin{pmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{pmatrix}$ is a $n \times p$ matrix of n observations on p independent covariates $X_1, X_2, \dots, X_p$,
- $\beta = (\beta_1, \beta_2, \dots, \beta_p)'$ is a $p \times 1$ vector of unknown regression parameters (or regression coefficients) $\beta_1, \beta_2, \dots, \beta_p$ associated with $X_1, X_2, \dots, X_p$, respectively and
- $\varepsilon = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)'$ is a $n \times 1$ vector of random errors or disturbances.
- We assume that $E(\varepsilon) = 0$, the covariance matrix $V(\varepsilon) = E(\varepsilon\varepsilon') = \sigma^2 I_p$, $\text{rank}(X) = p$.

In the context of analysis of variance and design of experiments,

- ♦ the matrix X is termed as the **design matrix**;
- ♦ unknown $\beta_1, \beta_2, \dots, \beta_p$ are termed as **effects**,
- ♦ the covariates $X_1, X_2, \dots, X_p$, are **counter variables** or **indicator variables** where x_{ij} counts the number of times the effect β_j occurs in the i^{th} observation x_i .
- ♦ x_{ij} mostly takes the values 1 or 0 but not always.
- ♦ The value $x_{ij} = 1$ indicates the presence of effect β_j in x_i and $x_{ij} = 0$ indicates the absence of effect β_j in x_i .

Note that in the **linear regression model**, the covariates are usually continuous variables.

When some of the covariates are counter variables, and rest are continuous variables, then the model is called a **mixed model** and is used in the analysis of covariance.

Relationship between the regression model and analysis of variance model

The same linear model is used in the linear regression analysis as well as in the analysis of variance. So it is important to understand the role of a linear model in the context of linear regression analysis and analysis of variance.

Consider the multiple linear model

$$Y = \beta_0 + X_1\beta_1 + X_2\beta_2 + \dots + X_p\beta_p + \varepsilon.$$

In the case of analysis of variance model,

- the one-way classification considers only one covariate,
- two-way classification model considers two covariates,
- three-way classification model considers three covariates and so on.

If β, γ and δ denote the effects associated with the covariates X, Z and W which are the counter variables, then in

$$\text{One-way model: } Y = \alpha + X\beta + \varepsilon$$

$$\text{Two-way model: } Y = \alpha + X\beta + Z\gamma + \varepsilon$$

$$\text{Three-way model: } Y = \alpha + X\beta + Z\gamma + W\delta + \varepsilon \text{ and so on.}$$

Consider an example of agricultural yield. The study variable Y denotes the yield which depends on various covariates $X_1, X_2, \dots, X_p$. In the case of regression analysis, the covariates $X_1, X_2, \dots, X_p$ are the different variables like temperature, the quantity of fertilizer, amount of irrigation etc.

Now consider the case of one-way model and try to understand its interpretation in terms of the multiple regression model. The covariate X is now measured at different levels, e.g., if X is the quantity of fertilizer then suppose there are p possible values, say 1 Kg., 2 Kg., ..., p Kg. then $X_1, X_2, \dots, X_p$ denotes these p values in the following way.

The linear model now can be expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

by defining

$$\begin{aligned} X_1 &= \begin{cases} 1 & \text{if effect of 1 Kg.fertilizer is present} \\ 0 & \text{if effect of 1 Kg.fertilizer is absent} \end{cases} \\ X_2 &= \begin{cases} 1 & \text{if effect of 2 Kg.fertilizer is present} \\ 0 & \text{if effect of 2 Kg.fertilizer is absent} \end{cases} \\ &\vdots \\ X_p &= \begin{cases} 1 & \text{if effect of } p \text{ Kg.fertilizer is present} \\ 0 & \text{if effect of } p \text{ Kg.fertilizer is absent.} \end{cases} \end{aligned}$$

If the effect of 1 Kg. of fertilizer is present, then other effects will obviously be absent and the linear model is expressible as

$$\begin{aligned} Y &= \beta_0 + \beta_1 (X_1 = 1) + \beta_2 (X_2 = 0) + \dots + \beta_p (X_p = 0) + \varepsilon \\ &= \beta_0 + \beta_1 + \varepsilon \end{aligned}$$

If the effect of 2 Kg. of fertilizer is present then

$$\begin{aligned} Y &= \beta_0 + \beta_1 (X_1 = 0) + \beta_2 (X_2 = 1) + \dots + \beta_p (X_p = 0) + \varepsilon \\ &= \beta_0 + \beta_2 + \varepsilon \end{aligned}$$

If the effect of p Kg. of fertilizer is present then

$$\begin{aligned} Y &= \beta_0 + \beta_1 (X_1 = 0) + \beta_2 (X_2 = 0) + \dots + \beta_p (X_p = 1) + \varepsilon \\ &= \beta_0 + \beta_p + \varepsilon \end{aligned}$$

and so on.

If the experiment with 1 Kg. of fertilizer is repeated n_1 number of times then n_1 observation on response variables are recorded which can be represented as

$$\begin{aligned} Y_{11} &= \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 0 + \varepsilon_{11} \\ Y_{12} &= \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 0 + \varepsilon_{12} \\ &\vdots \\ Y_{1n_1} &= \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 0 + \varepsilon_{1n_1} \end{aligned}$$

If $X_2=1$ is repeated n_2 times, then on the same lines n_2 number of times then n_2 observation on response variables are recorded which can be represented as

$$\begin{aligned} Y_{21} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \dots + \beta_p \cdot 0 + \varepsilon_{21} \\ Y_{22} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \dots + \beta_p \cdot 0 + \varepsilon_{22} \\ &\vdots \\ Y_{2n_2} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \dots + \beta_p \cdot 0 + \varepsilon_{2n_2} \end{aligned}$$

The experiment is continued and if $X_p = 1$ is repeated n_p times, then on the same lines

$$\begin{aligned} Y_{p1} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 1 + \varepsilon_{p1} \\ Y_{p2} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 1 + \varepsilon_{p2} \\ &\vdots \\ Y_{pn_p} &= \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 1 + \varepsilon_{pn_p} \end{aligned}$$

All these $n_1, n_2, \dots, n_p$ observations can be represented as

$$\begin{pmatrix} Y_{11} \\ Y_{12} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ Y_{22} \\ \vdots \\ Y_{2n_2} \\ \vdots \\ Y_{p1} \\ Y_{p2} \\ \vdots \\ Y_{pn_p} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 \dots 0 & 0 \\ 1 & 1 & 0 & 0 \dots 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 0 & 0 \dots 0 & 0 \\ 1 & 0 & 1 & 0 \dots 0 & 0 \\ 1 & 0 & 1 & 0 \dots 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 1 & 0 \dots 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 \dots 0 & 1 \\ 1 & 0 & 0 & 0 \dots 0 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 \dots 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \vdots \\ \varepsilon_{1n_1} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \vdots \\ \varepsilon_{2n_2} \\ \vdots \\ \varepsilon_{p1} \\ \varepsilon_{p2} \\ \vdots \\ \varepsilon_{pn_p} \end{pmatrix}$$

or

$$Y = X\beta + \varepsilon.$$

In the two-way analysis of variance model, there are two covariates and the linear model is expressible as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \gamma_1 Z_1 + \gamma_2 Z_2 + \dots + \gamma_q Z_q + \varepsilon$$

where $X_1, X_2, \dots, X_p$ denotes, e.g., the p levels of the quantity of fertilizer, say 1 Kg., 2 Kg., ..., p Kg. and $Z_1, Z_2, \dots, Z_q$ denotes, e.g., the q levels of level of irrigation, say 10 Cms., 20 Cms., ..., $10q$ Cms. etc. The levels $X_1, X_2, \dots, X_p, Z_1, Z_2, \dots, Z_q$ are the counter variable indicating the presence or absence of the effect as in the earlier case. If the effect of X_1 and Z_1 are present, i.e., 1 Kg of fertilizer and 10 Cms. of irrigation is used then the linear model is written as

$$\begin{aligned} Y &= \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 0 + \dots + \beta_p \cdot 0 + \gamma_1 \cdot 1 + \gamma_2 \cdot 0 + \dots + \gamma_q \cdot 0 + \varepsilon \\ &= \beta_0 + \beta_1 + \gamma_1 + \varepsilon. \end{aligned}$$

If $X_2 = 1$ and $Z_2 = 1$ is used, then the model is

$$Y = \beta_0 + \beta_2 + \gamma_2 + \varepsilon.$$

The design matrix can be written accordingly as in the one-way analysis of variance case.

In the three-way analysis of variance model

$$Y = \alpha + \beta_1 X_1 + \dots + \beta_p X_p + \gamma_1 Z_1 + \dots + \gamma_q Z_q + \delta_1 W_1 + \dots + \delta_r W_r + \varepsilon$$

The regression parameters β 's can be fixed or random.

- If all β 's are unknown constants, they are called as **parameters** of the model and the model is called as a **fixed effect model** or **model I**. The objective, in this case, is to make inferences about the parameters and the error variance σ^2 .
- If for some j , $x_{ij} = 1$ for all $i = 1, 2, \dots, n$ then β_j is termed an **additive constant**. In this case, β_j occurs with every observation and so it is also called a **general mean effect**.
- If all β 's are observable random variables except the additive constant, then the linear model is termed as **random effect model, model II** or **variance components model**. The objective, in this case, is to make inferences about the variances of β 's, i.e., $\sigma_{\beta_1}^2, \sigma_{\beta_2}^2, \dots, \sigma_{\beta_p}^2$ and error variance σ^2 and/or certain functions of them..
- If some parameters are fixed and some are random variables, then the model is called a **mixed effect model** or **model III**. In the mixed effect model, at least one β_j is constant and at least one β_j is a random variable. The objective is to make inference about the fixed effect parameters, variance of random effects and error variance σ^2 .

Analysis of variance

Analysis of variance is a body of statistical methods of analyzing the measurements assumed to be structured as

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

where x_{ij} are integers, generally 0 or 1 indicating usually the absence or presence of effects β_j ; and ε_i 's are assumed to be identically and independently distributed with mean 0 and variance σ^2 . It may be noted that the ε_i 's can be assumed additionally to follow a normal distribution $N(0, \sigma^2)$. It is needed for the maximum likelihood estimation of parameters from the beginning of the analysis, but in the least-squares estimation, it is needed only when conducting the tests of hypothesis and the confidence interval estimation of parameters. The least-squares method does not require any knowledge of distribution like normal up to the stage of estimation of parameters.

We need some basic concepts to develop tools.

Least squares estimate of β :

Let $y_1, y_2, \dots, y_n$ be a sample of observations on $Y_1, Y_2, \dots, Y_n$. The least-squares estimate of β is the values $\hat{\beta}$ of β for which the sum of squares due to errors, i.e.,

$$\begin{aligned} S^2 &= \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (y - X\beta)'(y - X\beta) \\ &= y'y - 2X'y + \beta'X'X\beta \end{aligned}$$

is minimum where $y = (y_1, y_2, \dots, y_n)'$. Differentiating S^2 with respect to β and substituting it to be zero, the normal equations are obtained as

$$\frac{dS^2}{d\beta} = 2X'X\beta - 2X'y = 0$$

or $X'X\beta = X'y$.

If X has full rank p , then $(X'X)$ has a unique inverse and the unique least squares estimate of β is

$$\hat{\beta} = (X'X)^{-1} X'y$$

which is the best linear unbiased estimator of β in the sense of having minimum variance in the class of linear and unbiased estimator. If the rank of X is not full, then generalized inverse is used for finding the inverse of $(X'X)$.

If $L'\beta$ is a linear parametric function where $L = (\ell_1, \ell_2, \dots, \ell_p)'$ is a non-null vector, then the least-squares estimate of $L'\beta$ is $L'\hat{\beta}$.

A question arises that what are the conditions under which a linear parametric function $L'\beta$ admits a unique least-squares estimate in the general case.

The concept of estimable function is needed to find such conditions.

Estimable functions:

A linear function $\lambda'\beta$ of the parameters with known λ is said to be an estimable parametric function (or estimable) if there exists a linear function $L'Y$ of Y such that

$$E(L'Y) = \lambda'\beta \text{ for all } \beta \in R^p.$$

Note that not all parametric functions are estimable.

Following results will be useful in understanding further topics.

Theorem 1: A linear parametric function $L'\beta$ admits a unique least squares estimate if and only if $L'\beta$ is estimable.

Theorem 2 (Gauss Markoff theorem):

If the linear parametric function $L'\beta$ is estimable then the linear estimator $L'\hat{\beta}$ where $\hat{\beta}$ is a solution of

$$X'X\hat{\beta} = X'Y$$

is the best linear unbiased estimator of $L'\beta$ in the sense of having minimum variance in the class of all linear and unbiased estimators of $L'\beta$.

Theorem 3: If the linear parametric function $\phi_1 = l_1'\beta, \phi_2 = l_2'\beta, \dots, \phi_k = l_k'\beta$ are estimable, then any linear combination of $\phi_1, \phi_2, \dots, \phi_k$ is also estimable.

Theorem 4: All linear parametric functions in β are estimable if and only if X has full rank.

If X is not of full rank, then some linear parametric functions do not admit the unbiased linear estimators and nothing can be inferred about them. The linear parametric functions which are not estimable are said to be **confounded**. A possible solution to this problem is to add linear restrictions on β so as to reduce the linear model to full rank.

Theorem 5: Let $L_1'\beta$ and $L_2'\beta$ be two estimable parametric functions and let $L_1'\hat{\beta}$ and $L_2'\hat{\beta}$ be their least squares estimators. Then

$$\begin{aligned} \text{Var}(L_1'\hat{\beta}) &= \sigma^2 L_1'(X'X)^{-1}L_1 \\ \text{Cov}(L_1'\hat{\beta}, L_2'\hat{\beta}) &= \sigma^2 L_1'(X'X)^{-1}L_2 \end{aligned}$$

assuming that X is a full rank matrix. If not, the generalized inverse of $X'X$ can be used in place of the unique inverse.

Estimator of σ^2 based on least squares estimation:

Consider an estimator of σ^2 as,

$$\begin{aligned} \hat{\sigma}^2 &= \frac{1}{n-p} (y - X\hat{\beta})'(y - X\hat{\beta}) \\ &= \frac{1}{n-p} [y - X(X'X)^{-1}X'y]'[y - X(X'X)^{-1}X'y] \\ &= \frac{1}{n-p} y'[I - X(X'X)^{-1}X'] [I - X(X'X)^{-1}X']y \\ &= \frac{1}{n-p} y'[I - X(X'X)^{-1}X']y \end{aligned}$$

where the hat matrix $[I - X(X'X)^{-1}X']$ is an idempotent matrix with its trace as

$$\begin{aligned} \text{tr}[I - X(X'X)^{-1}X'] &= \text{tr}I - \text{tr}X(X'X)^{-1}X' \\ &= n - \text{tr}(X'X)^{-1}X'X \text{ (using the result } \text{tr}(AB) = \text{tr}(BA)) \\ &= n - \text{tr}I_p \\ &= n - p. \end{aligned}$$

Note that using $E(y'Ay) = \text{tr}(A\Sigma)$, with $\text{Cov}(y) = \Sigma$, we have

$$\begin{aligned} E(\hat{\sigma}^2) &= \frac{\sigma^2}{n-p} \text{tr}[I - X(X'X)^{-1}X'] \\ &= \sigma^2 \end{aligned}$$

and so $\hat{\sigma}^2$ is an unbiased estimator of σ^2 .

Maximum Likelihood Estimation

The least-squares method does not use any distribution of the random variables in case of the estimation of parameters. We need the distributional assumption in case of least squares only while constructing the tests for hypothesis and the confidence intervals. For maximum likelihood estimation, we need the distributional assumption from the beginning.

Suppose $y_1, y_2, \dots, y_n$ are independently and identically distributed following a normal distribution with mean $E(y_i) = \sum_{j=1}^p \beta_j x_{ij}$ and variance $Var(y_i) = \sigma^2$ ($i = 1, 2, \dots, n$). Then the likelihood function of $y_1, y_2, \dots, y_n$ is

$$L(y|\beta, \sigma^2) = \frac{1}{(2\pi)^{\frac{n}{2}} (\sigma^2)^{\frac{n}{2}}} \exp \left[-\frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta) \right]$$

where $y = (y_1, y_2, \dots, y_n)'$. Then

$$L = \ln L(y|\beta, \sigma^2) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta).$$

Differentiating the log-likelihood with respect to β and σ^2 , we have

$$\begin{aligned} \frac{\partial L}{\partial \beta} &= 0 \Rightarrow X'X\tilde{\beta} = X'y \\ \frac{\partial L}{\partial \sigma^2} &= 0 \Rightarrow \tilde{\sigma}^2 = \frac{1}{n} (y - X\tilde{\beta})'(y - X\tilde{\beta}) \end{aligned}$$

Assuming the full rank of X , the normal equations are solved and the maximum likelihood estimators are obtained as

$$\begin{aligned} \tilde{\beta} &= (X'X)^{-1} X'y \\ \tilde{\sigma}^2 &= \frac{1}{n} (y - X\tilde{\beta})'(y - X\tilde{\beta}) \\ &= \frac{1}{n} y' [I - X(X'X)^{-1} X'] y. \end{aligned}$$

The second-order differentiation conditions can be checked and they are satisfied for $\hat{\beta}$ and $\hat{\sigma}^2$ to be the maximum likelihood estimators.

Note that in the maximum likelihood estimator $\tilde{\beta}$ is the same as the least-squares estimator $\tilde{\beta}$ and

- $\tilde{\beta}$ is an unbiased estimator of β , i.e., $E(\tilde{\beta}) = \beta$ unlike the least-squares estimator but
- $\tilde{\sigma}^2$ is not an unbiased estimator of σ^2 , i.e., $E(\tilde{\sigma}^2) = \frac{n-p}{n}\sigma^2 \neq \sigma^2$ like the least-squares estimator.

Now we use the following theorems for developing the test of hypothesis.

Theorem 6: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follow a multivariate normal distribution $N(\mu, \Sigma)$ with mean vector μ and positive definite covariance matrix Σ . Then $Y'AY$ follows a noncentral chi-square distribution with p degrees of freedom and noncentrality parameter $\mu'A\mu$, i.e., $\chi^2(p, \mu'A\mu)$ if and only if ΣA is an idempotent matrix of rank p .

Theorem 7: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follows a multivariate normal distribution $N(\mu, \Sigma)$ with mean vector μ and positive definite covariance matrix Σ . Let $Y'A_1Y$ follows $\chi^2(p_1, \mu'A_1\mu)$ and $Y'A_2Y$ follows $\chi^2(p_2, \mu'A_2\mu)$. Then $Y'A_1Y$ and $Y'A_2Y$ are independently distributed if $A_1\Sigma A_2 = 0$.

Theorem 8: Let $Y = (Y_1, Y_2, \dots, Y_n)'$ follows a multivariate normal distribution $N(\mu, \sigma^2 I)$, then the maximum likelihood (or least squares) estimator $L'\hat{\beta}$ of estimable linear parametric function is independently distributed of $\hat{\sigma}^2$; $L'\hat{\beta}$ follows $N[L'\beta, L'(X'X)^{-1}L]$ and $\frac{n\hat{\sigma}^2}{\sigma^2}$ follows $\chi^2(n-p)$ where $\text{rank}(X) = p$.

Proof: Consider $\hat{\beta} = (X'X)^{-1}X'Y$, then

$$\begin{aligned} E(L'\hat{\beta}) &= L'(X'X)^{-1}X'E(Y) \\ &= L'(X'X)^{-1}X'X\beta \\ &= L\beta \\ \text{Var}(L'\hat{\beta}) &= L'\text{Var}(\hat{\beta})L \\ &= L'E(\hat{\beta} - \beta)(\hat{\beta} - \beta)L \\ &= \sigma^2 L'(X'X)^{-1}L. \end{aligned}$$

Since $\hat{\beta}$ is a linear function of y and $L'\hat{\beta}$ is a linear function of $\hat{\beta}$, so $L'\hat{\beta}$ follows a normal distribution $N[L\beta, \sigma^2 L'(X'X)^{-1}L]$. Let $A = I - X(X'X)^{-1}X'$ and $B = L(X'X)^{-1}X'$, then

$$L'\hat{\beta} = L'(X'X)^{-1}X'Y = BY$$

$$\text{and } n\hat{\sigma}^2 = (Y - X\beta)'[I - X(X'X)^{-1}X'](Y - X\beta) = Y'AY.$$

So, using Theorem 6 with $\text{rank}(A) = n - p$, $\frac{n\hat{\sigma}^2}{\sigma^2}$ follows a $\chi^2(n - p)$. Also

$$\begin{aligned} BA &= L'(X'X)^{-1}X' - L'(X'X)^{-1}XX(X'X)^{-1}X' \\ &= 0. \end{aligned}$$

So using Theorem 7, $Y'AY$ and BY are independently distributed.

Tests of Hypothesis in the Linear Regression Model

First, we discuss the development of the tests of hypothesis concerning the parameters of a linear regression model. These tests of the hypothesis will be used later in the development of tests based on the analysis of variance.

Analysis of Variance

The technique in the analysis of variance involves the breaking down of total variation into orthogonal components. Each orthogonal factor represents the variation due to a particular factor contributing in the total variation.

Model

Let $Y_1, Y_2, \dots, Y_n$ be independently distributed following a normal distribution with mean

$$E(Y_i) = \sum_{j=1}^p \beta_j x_{ij} \text{ and variance } \sigma^2. \text{ Denoting } Y = (Y_1, Y_2, \dots, Y_n)' \text{ a } n \times 1 \text{ column vector, such}$$

assumption can be expressed in the form of a linear regression model

$$Y = X\beta + \varepsilon$$

where X is a $n \times p$ matrix, β is a $p \times 1$ vector and ε is a $n \times 1$ vector of disturbances with $E(\varepsilon) = 0$

$\text{Cov}(\varepsilon) = \sigma^2 I$ and ε follows a normal distribution.

This implies that

$$E(Y) = X\beta$$

$$E(Y - X\beta)(Y - X\beta)' = \sigma^2 I.$$

Now we consider four different types of tests of hypothesis.

In the first two cases, we develop the likelihood ratio test for the null hypothesis related to the analysis of variance. Note that, later we will derive the same test on the basis of least squares principle also. An important idea behind the development of this test is to demonstrate that the test used in the analysis of variance can be derived using the least-squares principle as well as the likelihood ratio test.

Case 1: Consider the null hypothesis for testing $H_0 : \beta = \beta^0$ where $\beta = (\beta_1, \beta_2, \dots, \beta_p)'$, $\beta^0 = (\beta_1^0, \beta_2^0, \dots, \beta_p^0)'$ is specified and σ^2 is unknown. This null hypothesis is equivalent to $H_0 : \beta_1 = \beta_1^0, \beta_2 = \beta_2^0, \dots, \beta_p = \beta_p^0$.

Assume that all β_i 's are estimable, i.e., $\text{rank}(X) = p$ (full column rank). We now develop the likelihood ratio test.

The $(p+1) \times 1$ dimensional parametric space Ω is a collection of points such that

$$\Omega = \{(\beta, \sigma^2); -\infty < \beta_i < \infty, \sigma^2 > 0, i = 1, 2, \dots, p\}.$$

Under H_0 , all β 's are known and equal, say β^0 all are known and the Ω reduces to one-dimensional space given by

$$\omega = \{(\beta^0, \sigma^2); \sigma^2 > 0\}.$$

The likelihood function of $y_1, y_2, \dots, y_n$ is

$$L(y|\beta, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2}(y - X\beta^0)'(y - X\beta^0)\right]$$

The likelihood function is maximum over Ω when β and σ^2 are substituted with their maximum likelihood estimators, i.e.,

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ \hat{\sigma}^2 &= \frac{1}{n}(y - X\hat{\beta})'(y - X\hat{\beta}).\end{aligned}$$

Substituting $\hat{\beta}$ and $\hat{\sigma}^2$ in $L(y|\beta, \sigma^2)$ gives

$$\begin{aligned}M_{\Omega} L(y|\beta, \sigma^2) &= \left(\frac{1}{2\pi\hat{\sigma}^2}\right)^{\frac{n}{2}} \exp\left(-\frac{1}{2\hat{\sigma}^2}(y - X\hat{\beta})'(y - X\hat{\beta})\right) \\ &= \left(\frac{n}{2\pi(y - X\hat{\beta})'(y - X\hat{\beta})}\right)^{\frac{n}{2}} \exp\left(-\frac{n}{2}\right).\end{aligned}$$

Under H_0 , the maximum likelihood estimator of σ^2 is $\hat{\sigma}^2 = \frac{1}{n}(y - X\beta^0)'(y - X\beta^0)$.

The maximum value of the likelihood function under H_0 is

$$\begin{aligned} \text{Max}_{\omega} L(y|\beta, \sigma^2) &= \left(\frac{1}{2\pi\hat{\sigma}^2} \right)^{\frac{n}{2}} \exp \left(-\frac{1}{2\hat{\sigma}^2} (y - X\beta^0)'(y - X\beta^0) \right) \\ &= \left(\frac{n}{2\pi(y - X\beta^0)'(y - X\beta^0)} \right)^{\frac{n}{2}} \exp \left(-\frac{n}{2} \right) \end{aligned}$$

The likelihood ratio test statistic is

$$\begin{aligned} \lambda &= \frac{\text{Max}_{\omega} L(y|\beta, \sigma^2)}{\text{Max}_{\Omega} L(y|\beta, \sigma^2)} \\ &= \left[\frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{(y - X\beta^0)'(y - X\beta^0)} \right]^{\frac{n}{2}} \\ &= \left[\frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{[(y - X\hat{\beta}) + (X\hat{\beta} - X\beta^0)]'[(y - X\hat{\beta}) + (X\hat{\beta} - X\beta^0)]} \right]^{\frac{n}{2}} \\ &= \left[1 + \frac{(\hat{\beta} - \beta^0)'X'X(\hat{\beta} - \beta^0)}{(y - X\hat{\beta})'(y - X\hat{\beta})} \right]^{-\frac{n}{2}} \\ &= \left[1 + \frac{q_1}{q_2} \right]^{-\frac{n}{2}} \end{aligned}$$

where $q_2 = (y - X\hat{\beta})'(y - X\hat{\beta})$

and $q_1 = (\hat{\beta} - \beta^0)'X'X(\hat{\beta} - \beta^0)$.

The expression of q_1 and q_2 can be further simplified as follows.

Consider

$$\begin{aligned} q_1 &= (\hat{\beta} - \beta^0)'X'X(\hat{\beta} - \beta^0) \\ &= [(X'X)^{-1}X'y - \beta^0]'X'X[(X'X)^{-1}X'y - \beta^0] \\ &= [(X'X)^{-1}X'(y - X\beta^0)]'X'X[(X'X)^{-1}X'(y - X\beta^0)] \\ &= (y - X\beta^0)'X(X'X)^{-1}X'X(X'X)^{-1}X'(y - X\beta^0) \\ &= (y - X\beta^0)'X(X'X)^{-1}X'(y - X\beta^0) \end{aligned}$$

$$\begin{aligned}
q_2 &= (y - X\hat{\beta})'(y - X\hat{\beta}) \\
&= [y - X(X'X)^{-1}X'y]'[y - X(X'X)^{-1}X'y] \\
&= y'[I - X(X'X)^{-1}X']y \\
&= [(y - X\beta^0) + X\beta^0]'[I - X(X'X)^{-1}X'][(y - X\beta^0)' + X\beta^0] \\
&= (y - X\beta^0)'[I - X(X'X)^{-1}X'](y - X\beta^0)
\end{aligned}$$

Other two terms become zero using

$$[I - X(X'X)^{-1}X']X = 0$$

In order to find out the decision rule for H_0 based on λ , first, we need to find if λ is a monotonic increasing or decreasing function of $\frac{q_1}{q_2}$. So we proceed as follows:

Let $g = \frac{q_1}{q_2}$, so that

$$\begin{aligned}
\lambda &= \left(1 + \frac{q_1}{q_2}\right)^{-\frac{n}{2}} \\
&= (1 + g)^{-\frac{n}{2}}
\end{aligned}$$

then

$$\frac{d\lambda}{dg} = -\frac{n}{2} \frac{1}{(1+g)^{\frac{n}{2}+1}}$$

So as g increases, λ decreases.

Thus λ is a monotonic decreasing function of $\frac{q_1}{q_2}$.

The decision rule is to reject H_0 if $\lambda \leq \lambda_0$ where λ_0 is a constant to be determined on the basis of size of the test α . Let us simplify this in our context.

$$\lambda \leq \lambda_0$$

$$\text{or} \quad \left(1 + \frac{q_1}{q_2}\right)^{-\frac{n}{2}} \leq \lambda_0$$

$$\text{or} \quad \frac{1}{(1+g)^{\frac{n}{2}}} \leq \lambda_0$$

$$\text{or} \quad (1+g) \geq \lambda_0^{-\frac{2}{n}}$$

$$\text{or} \quad g \geq \lambda_0^{-\frac{2}{n}} - 1$$

$$\text{or} \quad g \geq C$$

where C is a constant to be determined by the size α condition of the test.

So reject H_0 whenever $\frac{q_1}{q_2} \geq C$.

Note that the statistic $\frac{q_1}{q_2}$ can also be obtained by the least-squares method as follows. The least-

squares methodology will also be discussed in further lectures.

$$\begin{array}{ccccc}
 q_1 = (\hat{\beta} - \beta^0)' X' X (\hat{\beta} - \beta^0) & & & & \\
 q_1 = \underset{\omega}{\text{Min}}(y - X\beta)'(y - X\beta) & - & \underset{\Omega}{\text{Min}}(y - X\beta)'(y - X\beta) & & \\
 \downarrow & & \downarrow & & \downarrow \\
 \begin{array}{|l|} \hline \text{sum} \\ \text{of} \\ \text{squares} \\ \text{due to} \\ \text{deviation} \\ \text{from } H_0 \\ \text{OR} \\ \text{sum of} \\ \text{squares due} \\ \text{to } \beta \\ \hline \end{array} & & \begin{array}{|l|} \hline \text{sum} \\ \text{of} \\ \text{squares} \\ \text{due to } H_0 \\ \text{OR} \\ \text{Total sum of} \\ \text{squares} \\ \hline \end{array} & & \begin{array}{|l|} \hline \text{sum} \\ \text{of} \\ \text{squares} \\ \text{due to error} \\ \hline \end{array}
 \end{array}$$

It will be seen later that the test statistic will be based on the ratio $\frac{q_1}{q_2}$. In order to find an appropriate

distribution of $\frac{q_1}{q_2}$, we use the following theorem:

Theorem 9: Let

$$Z = Y - X\beta^0$$

$$Q_1 = Z'X(X'X)^{-1}X'Z$$

$$Q_2 = Z'[I - X(X'X)^{-1}X']Z.$$

Then $\frac{Q_1}{\sigma^2}$ and $\frac{Q_2}{\sigma^2}$ are independently distributed. Further, when H_0 is true, then $\frac{Q_1}{\sigma^2} \sim \chi^2(p)$

and $\frac{Q_2}{\sigma^2} \sim \chi^2(n-p)$ where $\chi^2(m)$ denotes the χ^2 distribution with 'm' degrees of freedom.

Proof: Under H_0 ,

$$E(Z) = X\beta^0 - X\beta^0 = 0$$

$$\text{Var}(Z) = \text{Var}(Y) = \sigma^2 I.$$

Further Z is a linear function of Y and Y follows a normal distribution. So

$$Z \sim N(0, \sigma^2 I)$$

The matrices $X(X'X)^{-1}X'$ and $[I - X(X'X)^{-1}X']$ are idempotent matrices. So

$$\text{tr}[X(X'X)^{-1}X'] = \text{tr}[(X'X)^{-1}X'X] = \text{tr}(I_p) = p$$

$$\text{tr}[I - X(X'X)^{-1}X'] = \text{tr} I_n - \text{tr}[X(X'X)^{-1}X'] = n - p$$

So using theorem 6, we can write that under H_0

$$\frac{Q_1}{\sigma^2} \sim \chi^2(p) \quad \text{and} \quad \frac{Q_2}{\sigma^2} \sim \chi^2(n-p)$$

where the degrees of freedom p and $(n-p)$ are obtained by the trace of $[X(X'X)^{-1}X']$ and trace of $[I - X(X'X)^{-1}X']$, respectively.

Since

$$[I - X(X'X)^{-1}X'] [X(X'X)^{-1}X'] = 0,$$

So using theorem 7, the quadratic forms Q_1 and Q_2 are independent under H_0 .

Hence the theorem is proved.

Since Q_1 and Q_2 are independently distributed, so under H_0

$\frac{Q_1/p}{Q_2/(n-p)}$ follows a central F distribution, i.e.

$$\left(\frac{n-p}{p} \right) \frac{Q_1}{Q_2} \sim F(p, n-p).$$

Hence the constant C in the likelihood ratio test statistic λ is given by

$$C = F_{1-\alpha}(p, n-p)$$

where $F_{1-\alpha}(n_1, n_2)$ denotes the upper $100\alpha\%$ points of F -distribution with n_1 and n_2 degrees of freedom.

The computations of this test of hypothesis can be represented in the form of an analysis of variance table.

ANOVA for testing $H_0 : \beta = \beta^0$

Source of variation	Degrees of freedom	Sum of squares	Mean squares	F-value
Due to β	p	q_1	$\frac{q_1}{p}$	$\left(\frac{n-p}{p} \right) \frac{q_1}{q_2}$
Error	$n-p$	q_2	$\frac{q_2}{(n-p)}$	
Total	n	$(y - X\beta^0)'(y - X\beta^0)$		

Case 2: Test of a subset of parameters $H_0 : \beta_k = \beta_k^0, k = 1, 2, \dots, r < p$ when $\beta_{r+1}, \beta_{r+2}, \dots, \beta_p$ and σ^2 are unknown.

In case 1, the test of hypothesis was developed when all β 's are considered in the sense that we test for each $\beta_i = \beta_i^0, i = 1, 2, \dots, p$. Now consider another situation, in which the interest is to test only a subset of $\beta_1, \beta_2, \dots, \beta_p$, i.e., not all but only a few parameters. This type of test of hypothesis can be used, e.g., in the following situation. Suppose five levels of voltage are applied to check the rotations per minute (rpm) of a fan at 160 volts, 180 volts, 200 volts, 220 volts and 240 volts. It can be realized in practice that when the voltage is low, then the rpm at 160, 180 and 200 volts can be observed easily. At 220 and 240 volts, the fan rotates at the full speed and there is not much difference in the rotations per minute at these voltages. So the interest of the experimenter lies in testing the hypothesis related to only first three effects, viz., β_1 , for 160 volts, β_2 for 180 volts and β_3 for 200 volts. The null hypothesis in this case can be written as:

$$H_0 : \beta_1 = \beta_1^0, \beta_2 = \beta_2^0, \beta_3 = \beta_3^0$$

when β_4, β_5 and σ^2 are unknown.

Note that under case 1, the null hypothesis will be

$$H_0 : \beta_1 = \beta_1^0, \beta_2 = \beta_2^0, \beta_3 = \beta_3^0, \beta_4 = \beta_4^0, \beta_5 = \beta_5^0.$$

Let $\beta_1, \beta_2, \dots, \beta_p$ be the p parameters.

We can divide them into two parts such that out of $\beta_1, \beta_2, \dots, \beta_r, \beta_{r+1}, \dots, \beta_p$ and we are interested in testing a hypothesis of a subset of it.

Suppose, We want to test the null hypothesis

$$H_0 : \beta_k = \beta_k^0, k = 1, 2, \dots, r < p \text{ when } \beta_{r+1}, \beta_{r+2}, \dots, \beta_p \text{ and } \sigma^2 \text{ are unknown.}$$

The alternative hypothesis under consideration is $H_1 : \beta_k \neq \beta_k^0$ for at least one $k = 1, 2, \dots, r < p$.

In order to develop a test for such a hypothesis, the linear model

$$Y = X\beta + \varepsilon$$

under the usual assumptions can be rewritten as follows:

$$\text{Partition } X = (X_1 \ X_2), \ \beta = \begin{pmatrix} \beta_{(1)} \\ \beta_{(2)} \end{pmatrix}$$

$$\text{where } \beta_{(1)} = (\beta_1, \beta_2, \dots, \beta_r)', \ \beta_{(2)} = (\beta_{r+1}, \beta_{r+2}, \dots, \beta_p)'$$

$$\text{with the order as } X_1 : n \times r, \ X_2 : n \times (p-r), \ \beta_{(1)} : r \times 1 \text{ and } \beta_{(2)} : (p-r) \times 1.$$

The model can be rewritten as

$$\begin{aligned} Y &= X\beta + \varepsilon \\ &= (X_1 \ X_2) \begin{pmatrix} \beta_{(1)} \\ \beta_{(2)} \end{pmatrix} + \varepsilon \\ &= X_1\beta_{(1)} + X_2\beta_{(2)} + \varepsilon \end{aligned}$$

The null hypothesis of interest is now

$$H_0 : \beta_{(1)} = \beta_{(1)}^0 = (\beta_1^0, \beta_2^0, \dots, \beta_r^0) \text{ where } \beta_{(2)} \text{ and } \sigma^2 \text{ are unknown.}$$

The complete parametric space is

$$\Omega = \{(\beta, \sigma^2); -\infty < \beta_i < \infty, \sigma^2 > 0, i = 1, 2, \dots, p\}$$

and sample space under H_0 is

$$\omega = \{(\beta_{(1)}, \beta_{(2)}, \sigma^2); -\infty < \beta_i < \infty, \sigma^2 > 0, i = r+1, r+2, \dots, p\}.$$

The likelihood function is

$$L(y|\beta, \sigma^2) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\sigma^2} (y - X\beta)'(y - X\beta) \right].$$

The maximum value of likelihood function under Ω is obtained by substituting the maximum likelihood estimates of β and σ^2 , i.e.,

$$\hat{\beta} = (X'X)^{-1} X'y$$

$$\hat{\sigma}^2 = \frac{1}{n} (y - X\hat{\beta})'(y - X\hat{\beta})$$

as

$$Max_{\Omega} L(y|\beta, \sigma^2) = \left(\frac{1}{2\pi\hat{\sigma}^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\hat{\sigma}^2} (y - X\hat{\beta})'(y - X\hat{\beta}) \right]$$

$$= \left(\frac{n}{2\pi(y - X\hat{\beta})'(y - X\hat{\beta})} \right)^{\frac{n}{2}} \exp \left(-\frac{n}{2} \right).$$

Now we find the maximum value of likelihood function under H_0 . The model under H_0 becomes

$Y = X_1\beta_{(1)}^0 + X_2\beta_2 + \varepsilon$. The likelihood function under H_0 is

$$L(y|\beta, \sigma^2) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\sigma^2} (y - X_1\beta_{(1)}^0 - X_2\beta_2)'(y - X_1\beta_{(1)}^0 - X_2\beta_2) \right]$$

$$= \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\sigma^2} (y^* - X_2\beta_2)'(y^* - X_2\beta_2) \right]$$

where $y^* = y - X_1\beta_{(1)}^0$. Note that $\beta_{(2)}$ and σ^2 are the unknown parameters. This likelihood function looks like as if it is written for $y^* \sim N(X_2\beta_{(2)}, \sigma^2)$.

This helps in writing the maximum likelihood estimators of $\beta_{(2)}$ and σ^2 directly as

$$\hat{\beta}_{(2)} = (X_2'X_2)^{-1} X_2'y^*$$

$$\hat{\sigma}^2 = \frac{1}{n} (y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}).$$

Note that $X_2'X_2$ is a principal minor of $X'X$. Since $X'X$ is a positive definite matrix, so $X_2'X_2$ is also positive definite. Thus $(X_2'X_2)^{-1}$ exists and is unique.

Thus the maximum value of likelihood function under H_0 is obtained as

$$Max_{\omega} L(y^*|\hat{\beta}, \sigma^2) = \left(\frac{1}{2\pi\hat{\sigma}^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\hat{\sigma}^2} (y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}) \right]$$

$$= \left(\frac{n}{2\pi(y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)})} \right)^{\frac{n}{2}} \exp \left(-\frac{n}{2} \right)$$

The likelihood ratio test statistic for $H_0 : \beta_{(1)} = \beta_{(1)}^0$ is

$$\begin{aligned}
\lambda &= \frac{\max_{\omega} L(y|\beta, \sigma^2)}{\max_{\Omega} L(y|\beta, \sigma^2)} \\
&= \left[\frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{(y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)})} \right]^{\frac{n}{2}} \\
&= \left[\frac{(y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}) - (y - X\hat{\beta})'(y - X\hat{\beta}) + (y - X\hat{\beta})'(y - X\hat{\beta})}{(y - X\hat{\beta})'(y - X\hat{\beta})} \right]^{-\frac{n}{2}} \\
&= \left[1 + \frac{(y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}) - (y - X\hat{\beta})'(y - X\hat{\beta})}{(y - X\hat{\beta})'(y - X\hat{\beta})} \right]^{-\frac{n}{2}} \\
&= \left[1 + \frac{q_1}{q_2} \right]^{-\frac{n}{2}}
\end{aligned}$$

where $q_1 = (y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}) - (y - X\hat{\beta})'(y - X\hat{\beta})$ and $q_2 = (y - X\hat{\beta})'(y - X\hat{\beta})$.

Now we simplify q_1 and q_2 .

Consider

$$\begin{aligned}
(y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)}) &= \\
&= (y^* - X_2(X_2'X_2)^{-1}X_2'y^*)'(y^* - X_2(X_2'X_2)^{-1}X_2'y^*) \\
&= y^{*'}[I - X_2(X_2'X_2)^{-1}X_2']y^* \\
&= [(y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) + X_2\beta_{(2)}]'[I - X_2(X_2'X_2)^{-1}X_2'][(y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) + X_2\beta_{(2)}] \\
&= (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})'[I - X_2(X_2'X_2)^{-1}X_2'](y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}).
\end{aligned}$$

The other terms become zero using the result $X_2'[I - X_2(X_2'X_2)^{-1}X_2'] = 0$.

Note that under H_0 , $X_1\beta_{(1)}^0 + X_2\beta_{(2)}$ can be expressed as $(X_1 \ X_2)(\beta_{(1)}^0 \ \beta_{(2)})'$,

Consider

$$\begin{aligned}
(y - X\hat{\beta})'(y - X\hat{\beta})' &= \\
&= (y - X(X'X)^{-1}X'y)'(y - X(X'X)^{-1}X'y) \\
&= y'[I - X(X'X)^{-1}X']y \\
&= \left[(y - X_1\beta_{(1)}^0 - X_2\hat{\beta}_{(2)}) + X_1\beta_{(1)}^0 + X_2\beta_{(2)} \right]' \\
&\quad \times \left[I - X(X'X)^{-1}X' \right] \left[(y - X_1\beta_{(1)}^0 - X_2\hat{\beta}_{(2)}) + X_1\beta_{(1)}^0 + X_2\beta_{(2)} \right] \\
&\quad - (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) \left[I - X(X'X)^{-1}X' \right] (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) \\
&= (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})' \left[I - X(X'X)^{-1}X' \right] (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})
\end{aligned}$$

and another term becomes zero using the result $X'[I - X(X'X)^{-1}X'] = 0$.

Note that under H_0 , the term $X_1\beta_{(1)}^0 + X_2\beta_{(2)}$ can be expressed as $(X_1 \ X_2)(\beta_{(1)}^0 \ \beta_{(2)})'$. Thus

$$\begin{aligned}
q_1 &= (y^* - X_2\hat{\beta}_{(2)})'(y^* - X_2\hat{\beta}_{(2)})' - (y - X\hat{\beta})'(y - X\hat{\beta})' \\
&= y^*'[I - X_2(X_2'X_2)^{-1}X_2']y^* - y'[I - X(X'X)^{-1}X']y \\
&= (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})'[I - X_2(X_2'X_2)^{-1}X_2'](y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) \\
&\quad - (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})'[I - X(X'X)^{-1}X'](y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) \\
&= (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})'[X(X'X)^{-1}X' - X_2(X_2'X_2)^{-1}X_2'](y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})
\end{aligned}$$

$$\begin{aligned}
q_2 &= (y - X\hat{\beta})'(y - X\hat{\beta}) \\
&= y'[I - X(X'X)^{-1}X']y \\
&= \left[(y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) + (X_1\beta_{(1)}^0 + X_2\beta_{(2)}) \right]' \left[I - X(X'X)^{-1}X' \right] \\
&\quad \times \left[(y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}) + (X_1\beta_{(1)}^0 + X_2\beta_{(2)}) \right] \\
&= (y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})'[I - X(X'X)^{-1}X'](y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}).
\end{aligned}$$

Other terms become zero. Note that in simplifying the terms q_1 and q_2 , we tried to write them in the quadratic form with the same variable $(y - X_1\beta_{(1)}^0 - X_2\beta_{(2)})$.

Using the same argument as in case 1, we can say that since λ is a monotonic decreasing function of $\frac{q_1}{q_2}$, so the likelihood ratio test rejects H_0 whenever

$$\frac{q_1}{q_2} > C$$

where C is a constant to be determined by the size α of the test.

The likelihood ratio test statistic can also be obtained through least-squares method as follows:

$(q_1 + q_2)$: Minimum value of $(y - X\beta)'(y - X\beta)$ when $H_0 : \beta_{(1)} = \beta_{(1)}^0$ holds true.

q_1 : Sum of squares due to H_0

q_2 : Sum of squares due to error.

q_1 : Sum of squares due to the deviation from H_0 or sum of squares due to $\beta_{(1)}$ adjusted for $\beta_{(2)}$.

If $\beta_{(1)}^0 = 0$ then

$$\begin{aligned} q_1 &= (y - X_2\hat{\beta}_{(2)})'(y - X_2\hat{\beta}_{(2)}) - (y - X\hat{\beta})'(y - X\hat{\beta}) \\ &= (y'y - \hat{\beta}_{(2)}'X_2'y) - (y'y - \hat{\beta}'X'y) \\ &= \hat{\beta}'X'y - \hat{\beta}_{(2)}'X_2'y. \end{aligned}$$

↓

↓

Reduction sum of squares or sum of squares due to β	sum of squares due to $\beta_{(2)}$ ignoring $\beta_{(1)}$
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Now we have the following theorem based on Theorems 6 and 7.

Theorem 10: Let $Z = Y - X_1\beta_{(1)}^0 - X_2\beta_{(2)}$; $Q_1 = Z'AZ$ and $Q_2 = Z'BZ$ where

$$A = X(X'X)^{-1}X' - X_2(X_2'X_2)^{-1}X_2' \text{ and } B = I - X(X'X)^{-1}X'.$$

Then $\frac{Q_1}{\sigma^2}$ and $\frac{Q_2}{\sigma^2}$ are independently distributed. Further $\frac{Q_1}{\sigma^2} \sim \chi^2(r)$ and $\frac{Q_2}{\sigma^2} \sim \chi^2(n-p)$.

Thus under H_0 , $\frac{Q_1/r}{Q_2/(n-p)} = \frac{n-p}{r} \frac{Q_1}{Q_2}$ follows an F -distribution $F(r, n-p)$.

Hence the constant C in λ is

$$C = F_{1-\alpha}(r, n-p),$$

where $F_{1-\alpha}(r, n-p)$ denotes the upper $100\alpha\%$ points on F -distribution with r and $(n-p)$ degrees of freedom.

The analysis of variance table for this null hypothesis is as follows:

ANOVA for testing $H_0 : \beta_{(1)} = \beta_{(1)}^0$

Source of Variation	Degrees of Freedom	Sum of squares	Mean squares	F-value
Due to $\beta_{(1)}$	r	q_1	$\frac{q_1}{r}$	$\left(\frac{n-p}{r} \right) \frac{q_1}{q_2}$
Error	$n-p$	q_2	$\frac{q_2}{(n-p)}$	
Total	$n-(p-r)$	$q_1 + q_2$		

Case 3: Test of $H_0 : L'\beta = \delta$

Let us consider the test of hypothesis related to a linear parametric function. Assuming that the linear parameter function $L'\beta$ is estimable where $L = (\ell_1, \ell_2, \dots, \ell_p)'$ is a $p \times 1$ vector of known constants and $\beta = (\beta_1, \beta_2, \dots, \beta_p)'$. The null hypothesis of interest is

$$H_0 : L'\beta = \delta.$$

where δ is some specified constant.

Consider the set up of linear model $Y = X\beta + \varepsilon$ where $Y = (Y_1, Y_2, \dots, Y_n)'$ follows $N(X\beta, \sigma^2 I)$. The maximum likelihood estimators of β and σ^2 are

$$\hat{\beta} = (X'X)^{-1} X'y \text{ and } \hat{\sigma}^2 = \frac{1}{n} (y - X\hat{\beta})'(y - X\hat{\beta})$$

respectively. The maximum likelihood estimate of estimable $L'\beta$ is $L'\hat{\beta}$, with

$$E(L'\hat{\beta}) = L'\beta$$

$$\text{Cov}(L'\hat{\beta}) = \sigma^2 L'(X'X)^{-1} L$$

$$L'\hat{\beta} \sim N[L'\beta, \sigma^2 L'(X'X)^{-1} L]$$

and

$$\frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-p)$$

assuming X to be the full column rank matrix. Further, $L'\hat{\beta}$ and $\frac{n\hat{\sigma}^2}{\sigma^2}$ are also independently distributed.

Under $H_0 : L'\beta = \delta$, the statistic

$$t = \frac{\sqrt{(n-p)}(L'\hat{\beta} - \delta)}{\sqrt{n\hat{\sigma}^2 L'(X'X)^{-1}L}}$$

follows a t -distribution with $(n-p)$ degrees of freedom. So the test for $H_0 : L'\beta = \delta$ against $H_1 : L'\beta \neq \delta$ rejects H_0 whenever

$$|t| \geq t_{1-\frac{\alpha}{2}}(n-p)$$

where $t_{1-\alpha}(n_1)$ denotes the upper $100\alpha\%$ points on t -distribution with n_1 degrees of freedom.

Case 4: Test of $H_0 : \phi_1 = \delta_1, \phi_2 = \delta_2, \dots, \phi_k = \delta_k$

Now we develop the test of hypothesis related to more than one linear parametric functions. Let the i^{th} estimable linear parametric function is

$\phi_i = L_i'\beta$ and there are k such functions with L_i and β both being $p \times 1$ vectors as in the Case 3.

Our interest is to test the hypothesis

$$H_0 : \phi_1 = \delta_1, \phi_2 = \delta_2, \dots, \phi_k = \delta_k$$

where $\delta_1, \delta_2, \dots, \delta_k$ are the known constants.

Let $\phi = (\phi_1, \phi_2, \dots, \phi_k)'$ and $\delta = (\delta_1, \delta_2, \dots, \delta_k)'$.

Then H_0 is expressible as $H_0 : \phi = L'\beta = \delta$

where L' is a $k \times p$ matrix of constants associated with $L_1, L_2, \dots, L_k$. The maximum likelihood estimator of ϕ_i is : $\hat{\phi}_i = L_i'\hat{\beta}$ where $\hat{\beta} = (X'X)^{-1}X'y$.

Then $\hat{\phi} = (\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_k) = L\hat{\beta}$.

Also $E(\hat{\phi}) = \phi$

$$\text{Cov}(\hat{\phi}) = \sigma^2 V$$

where $V = ((L_i'(X'X)^{-1}L_j))$ where $(L_i'(X'X)^{-1}L_j)$ is the $(i, j)^{\text{th}}$ element of V . Thus

$$\frac{(\hat{\phi} - \phi)' V^{-1} (\hat{\phi} - \phi)}{\sigma^2}$$

follows a χ^2 -distribution with k degrees of freedom and

$\frac{n\hat{\sigma}^2}{\sigma^2}$ follows a χ^2 -distribution with $(n-p)$ degrees of freedom where

$\hat{\sigma}^2 = \frac{1}{n}(y - X\hat{\beta})'(y - X\hat{\beta})$ is the maximum likelihood estimator of σ^2 .

Further $\frac{(\hat{\phi} - \phi)'V^{-1}(\hat{\phi} - \phi)}{\sigma^2}$ and $\frac{n\hat{\sigma}^2}{\sigma^2}$ are also independently distributed.

Thus under $H_0 : \phi = \delta$

$$\frac{\left(\frac{(\hat{\phi} - \delta)'V^{-1}(\hat{\phi} - \delta)}{\sigma^2} \right)}{\left(\frac{\frac{n\hat{\sigma}^2}{\sigma^2}}{(n-p)} \right)}$$

or $\left(\frac{n-p}{k} \right) \frac{(\hat{\phi} - \delta)'V^{-1}(\hat{\phi} - \delta)}{n\hat{\sigma}^2}$

follows F -distribution with k and $(n-p)$ degrees of freedom. So the hypothesis $H_0 : \phi = \delta$ is rejected against H_1 : At least one $\phi_i \neq \delta_i$ for $i = 1, 2, \dots, k$ whenever $F \geq F_{1-\alpha}(k, n-p)$ where $F_{1-\alpha}(k, n-p)$ denotes the $100\alpha\%$ points on F -distribution with k and $(n-p)$ degrees of freedom.

One-way classification with fixed effect linear models of full rank:

The objective in the one-way classification is to test the hypothesis about the equality of means on the basis of several samples which have been drawn from univariate normal populations with different means but the same variances.

Let there be p univariate normal populations and samples of different sizes are drawn from each of the population. Let y_{ij} ($j = 1, 2, \dots, n_i$) be a random sample from the i^{th} normal population with mean β_i and variance σ^2 , $i = 1, 2, \dots, p$, i.e.,

$$Y_{ij} \sim N(\beta_i, \sigma^2), \quad j = 1, 2, \dots, n_i; \quad i = 1, 2, \dots, p.$$

The random samples from different populations are assumed to be independent of each other.

These observations follow the set up of linear model

$$Y = X\beta + \varepsilon$$

where

$$\begin{aligned}
Y &= (Y_{11}, Y_{12}, \dots, Y_{1n_1}, Y_{21}, \dots, Y_{2n_2}, \dots, Y_{p1}, Y_{p2}, \dots, Y_{pn_p})' \\
y &= (y_{11}, y_{12}, \dots, y_{1n_1}, y_{21}, \dots, y_{2n_2}, \dots, y_{p1}, y_{p2}, \dots, y_{pn_p})' \\
\beta &= (\beta_1, \beta_2, \dots, \beta_p)' \\
\varepsilon &= (\varepsilon_{11}, \varepsilon_{12}, \dots, \varepsilon_{1n_1}, \varepsilon_{21}, \dots, \varepsilon_{2n_2}, \dots, \varepsilon_{p1}, \varepsilon_{p2}, \dots, \varepsilon_{pn_p})'
\end{aligned}$$

$$X = \begin{pmatrix} \left. \begin{matrix} 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 1 & 0 & & 0 \end{matrix} \right\} n_1 \text{ values} \\ \left. \begin{matrix} 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \end{matrix} \right\} n_2 \text{ values} \\ \vdots \\ \left. \begin{matrix} 0 & 0 & \dots & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{matrix} \right\} n_p \text{ values} \end{pmatrix}$$

$$x_{ij} = \begin{cases} 1 & \text{if } \beta_i \text{ occurs in the } j^{\text{th}} \text{ observation } x_j \\ & \text{or if effect } \beta_i \text{ is present in } x_j \\ 0 & \text{if effect } \beta_i \text{ is absent in } x_j \end{cases}$$

$$n = \sum_{i=1}^p n_i.$$

So X is a matrix of order $n \times p$, β is fixed and

- first n_1 rows of ε are $\varepsilon'_1 = (1, 0, 0, \dots, 0)$,
- next n_2 rows of ε are $\varepsilon'_2 = (0, 1, 0, \dots, 0)$
- and similarly, the last n_p rows of ε are $\varepsilon'_p = (0, 0, \dots, 0, 1)$.

Obviously, $\text{rank}(X) = p$, $E(Y) = X\beta$ and $\text{Cov}(Y) = \sigma^2 I$.

This completes the representation of a **fixed effect linear model of full rank**.

The null hypothesis of interest is $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = \beta$ (say)

and $H_1 : \text{At least one } \beta_i \neq \beta_j (i \neq j)$

where β and σ^2 are unknown.

We would develop here the likelihood ratio test. It may be noted that the same test can also be derived through the least-squares method. This will be demonstrated in the next module. This way the readers will understand both the methods.

We already have developed the likelihood ratio for the hypothesis $H_0 : \beta_1 = \beta_2 = \dots = \beta_p$ in case 1.

The whole parametric space Ω is a $(p+1)$ dimensional space

$$\Omega = \{(\beta, \sigma^2) : -\infty < \beta < \infty, \sigma^2 > 0, i=1, 2, \dots, p\}.$$

Note that there are $(p+1)$ parameters are $\beta_1, \beta_2, \dots, \beta_p$ and σ^2 .

Under H_0 , Ω reduces to two dimensional space ω as

$$\omega = \{(\beta, \sigma^2) : -\infty < \beta < \infty, \sigma^2 > 0\}..$$

The likelihood function under Ω is

$$\begin{aligned} L(y|\beta, \sigma^2) &= \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \beta_i)^2\right] \\ L = \ln L(y|\beta, \sigma^2) &= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \beta_i)^2 \\ \frac{\partial L}{\partial \beta_i} &= 0 \Rightarrow \hat{\beta}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} = \bar{y}_{io} \\ \frac{\partial L}{\partial \sigma^2} &= 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2. \end{aligned}$$

The dot sign ($\circ$) in $\bar{y}_{io}$ indicates that the average has been taken over the second subscript j . The Hessian matrix of second-order partial derivation of $\ln L$ with respect to β_i and σ^2 is negative definite at $\beta = \bar{y}_{io}$ and $\sigma^2 = \hat{\sigma}^2$ which ensures that the likelihood function is maximized at these values.

Thus the maximum value of $L(y|\beta, \sigma^2)$ over Ω is

$$\begin{aligned} \text{Max}_{\Omega} L(y|\beta, \sigma^2) &= \left(\frac{1}{2\pi\hat{\sigma}^2}\right)^{\frac{n}{2}} \exp\left[-\frac{1}{2\hat{\sigma}^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \hat{\beta}_i)^2\right] \\ &= \left[\frac{n}{2\pi \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2}\right]^{n/2} \exp\left(-\frac{n}{2}\right). \end{aligned}$$

The likelihood function under ω is

$$L(y|\beta, \sigma^2) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \beta)^2 \right]$$

$$\ln L(y|\beta, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \beta)^2$$

The normal equations and the least-squares are obtained as follows:

$$\frac{\partial \ln L(y|\beta, \sigma^2)}{\partial \beta} = 0 \Rightarrow \hat{\beta} = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij} = \bar{y}_{oo}$$

$$\frac{\partial \ln L(y|\beta, \sigma^2)}{\partial \sigma^2} = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2.$$

The maximum value of the likelihood function over ω under H_0 is

$$\begin{aligned} \text{Max}_{\omega} L(y|\beta, \sigma^2) &= \left(\frac{1}{2\pi\hat{\sigma}^2} \right)^{\frac{n}{2}} \exp \left[-\frac{1}{2\hat{\sigma}^2} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \hat{\beta})^2 \right] \\ &= \left[\frac{n}{2\pi \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2} \right]^{n/2} \exp \left(-\frac{n}{2} \right). \end{aligned}$$

The likelihood ratio test statistic is

$$\begin{aligned} \lambda &= \frac{\text{Max}_{\omega} L(y|\beta, \sigma^2)}{\text{Max}_{\Omega} L(y|\beta, \sigma^2)} \\ &= \left[\frac{\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2}{\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2} \right]^{n/2} \end{aligned}$$

We have that

$$\begin{aligned} \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2 &= \sum_{i=1}^p \sum_{j=1}^{n_i} \left[(y_{ij} - \bar{y}_{io}) + (\bar{y}_{io} - \bar{y}_{oo}) \right]^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2 + \sum_{i=1}^p n_i (\bar{y}_{io} - \bar{y}_{oo})^2 \end{aligned}$$

Thus

$$\lambda = \left[\frac{\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 + \sum_{i=1}^p n_i (\bar{y}_{io} - \bar{y}_{oo})^2}{\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2} \right]^{-\frac{n}{2}}$$

$$= \left[1 + \frac{q_1}{q_2} \right]^{-\frac{n}{2}}$$

where

$$q_1 = \sum_{i=1}^p n_i (\bar{y}_{io} - \bar{y}_{oo})^2, \text{ and } q_2 = \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2.$$

Note that if the least-squares principal is used, then

q_1 : sum of squares due to deviations from H_0 or the between population sum of squares,

q_2 : sum of squares due to error or the within-population sum of squares,

$q_1 + q_2$: sum of squares due to H_0 or the total sum of squares.

Using theorems 6 and 7, let

$$Q_1 = \sum_{i=1}^p n_i (\bar{Y}_{io} - \bar{Y}_{oo})^2, \quad Q_2 = \sum_{i=1}^p S_i^2$$

where

$$S_i^2 = \sum_{j=1}^{n_i} (\bar{Y}_{ij} - \bar{Y}_{io})^2$$

$$\bar{Y}_{oo} = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} Y_{ij}$$

$$\bar{Y}_{io} = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$$

then under H_0

$$\frac{Q_1}{\sigma^2} \sim \chi^2(p-1)$$

$$\frac{Q_2}{\sigma^2} \sim \chi^2(n-p)$$

and $\frac{Q_1}{\sigma^2}$ and $\frac{Q_2}{\sigma^2}$ are independently distributed.

Thus under H_0

$$\frac{\left(\frac{Q_1}{\sigma^2} \right)}{\left(\frac{Q_2}{\sigma^2} \right)} \sim F(p-1, n-p).$$

The likelihood ratio test reject H_0 whenever

$$\frac{q_1}{q_2} > C$$

where the constant $C = F_{1-\alpha}(p-1, n-p)$.

The analysis of variance table for the one-way classification in fixed effect model is

Source of Variation	Degrees of freedom	Sum of squares	Mean sum of squares	F
Between Population	$p-1$	q_1	$\frac{q_1}{p-1}$	$\left(\frac{n-p}{p-1} \right) \cdot \frac{q_1}{q_2}$
Within Population	$n-p$	q_2	$\frac{q_2}{n-p}$	
Total	$n-1$	$q_1 + q_2$		

Note that

$$E\left[\frac{Q_2}{n-p} \right] = \sigma^2$$

$$E\left[\frac{Q_1}{p-1} \right] = \sigma^2 + \frac{\sum_{i=1}^p (\beta_i - \bar{\beta})^2}{p-1}; \bar{\beta} = \frac{1}{p} \sum_{i=1}^p \beta_i$$

Case of rejection of H_0

If $F > F_{1-\alpha}(p-1, n-p)$, then $H_0 : \beta_1 = \beta_2 = \dots = \beta_p$ is rejected. This means that at least one β_i is different from others which is responsible for the rejection. So the objective is to investigate and find out such β_i and divide the population into groups such that the means of populations within the groups are same. This can be done by pairwise testing of β 's.

Test $H_0 : \beta_i = \beta_k$ ($i \neq k$) against $H_1 : \beta_i \neq \beta_k$.

This can be tested using following t -statistic

$$t = \frac{\bar{Y}_{io} - \bar{Y}_{ko}}{\sqrt{s^2 \left(\frac{1}{n_i} + \frac{1}{n_k} \right)}}$$

which follows the t distribution with $(n - p)$ degrees of freedom under H_0 and $s^2 = \frac{q_2}{n - p}$. Thus

the decision rule is to reject H_0 at the level α if the observed difference

$$|(\bar{Y}_{io} - \bar{Y}_{ko})| > t_{1-\frac{\alpha}{2}, n-p} \sqrt{s^2 \left(\frac{1}{n_i} + \frac{1}{n_k} \right)}$$

The quantity $t_{1-\frac{\alpha}{2}, n-p} \sqrt{s^2 \left(\frac{1}{n_i} + \frac{1}{n_k} \right)}$ is called the **critical difference**.

Thus following steps are followed :

1. Compute all possible critical differences arising out of all possible pair (β_i, β_k) , $i \neq k = 1, 2, \dots, p$.
2. Compare them with their observed differences
3. Divide the p populations into different groups such that the populations in the same group have the same means.

The computation are simplified if $n_i = n$ for all i . In such a case, the **common critical difference (CCD)** is

$$CCD = t_{1-\frac{\alpha}{2}, n-p} \sqrt{\frac{2s^2}{n}}$$

and the observed difference $(\bar{y}_{io} - \bar{y}_{ko}), i \neq k$ are compared with CCD.

If $|\bar{y}_{io} - \bar{y}_{ko}| > CCD$

then the corresponding effects/means $\bar{y}_{io}$ and $\bar{y}_{ko}$ are coming from populations with different means.

Note: In general we say that if there are three effects $\beta_1, \beta_2, \beta_3$ then

if $H_{01} : \beta_1 = \beta_2$ ($\equiv$ denote as event A) is accepted

and if $H_{02} : \beta_2 = \beta_3$ ($\equiv$ denote as event B) is accepted

then $H_{03} : \beta_1 = \beta_3$ ($\equiv$ denote as event C) will be accepted.

The question arises here that in what sense do we conclude such a statement about the acceptance of H_{03} . The reason is as follows:

Since event $A \cap B \subset C$, so

$$P(A \cap B) \leq P(C)$$

In this sense if the probability of an event is higher than the intersection of the events, i.e., the probability that H_{03} is accepted is higher than the probability of acceptance of H_{01} and H_{02} both, so we conclude, in general, that the acceptance of H_{01} and H_{02} implies the acceptance of H_{03} .

Multiple comparison test:

One interest in the analysis of variance is to decide whether population means are equal or not. If the hypothesis of equal means is rejected then one would like to divide the populations into subgroups such that all populations with the same means come to the same subgroup. This can be achieved by the multiple comparison tests.

A multiple comparison test procedure conducts the test of hypothesis for all the pairs of effects and compares them at a significance level α i.e., it works on per comparison basis.

This is based mainly on the t -statistic. If we want to ensure that the significance level α simultaneously for all group comparison of interest, the approximate multiple test procedure is one that controls the error rate per experiment basis.

There are various available multiple comparison tests. We will discuss some of them in the context of one-way classification. In two-way or higher classification, they can be used on similar lines.

1. Studentized range test:

It is assumed in the Studentized range test that the p samples, each of size n , have been drawn from p normal populations. Let their sample means be $\bar{y}_{1o}, \bar{y}_{2o}, \dots, \bar{y}_{po}$. These means are ranked and arranged in ascending order as $\bar{y}_1^*, \bar{y}_2^*, \dots, \bar{y}_p^*$ where $\bar{y}_1^* = \text{Min}_i \bar{y}_{io}$ and $\bar{y}_p^* = \text{Max}_i \bar{y}_{io}$, $i = 1, 2, \dots, p$.

Find the range as $R = \bar{y}_p^* - \bar{y}_1^*$.

The Studentized range is defined as

$$q_{p, n-p} = \frac{R\sqrt{n}}{s}$$

where $q_{\alpha,p,\gamma}$ is the upper $100\alpha\%$ point of Studentized range when $\gamma = n - p$. The tables for $q_{\alpha,p,\gamma}$ are available.

The testing procedure involves the comparison of $q_{p,\gamma}$ with $q_{\alpha,p,\gamma}$ in the usual way as-

- if $q_{p,n-p} < q_{\alpha,p,n-p}$ then conclude that $\beta_1 = \beta_2 = \dots = \beta_p$.
- if $q_{p,n-p} > q_{\alpha,p,n-p}$ then all β 's in the group are not the same.

2. Student-Newman-Keuls test:

The Student-Newman-Keuls test is similar to Studentized range test in the sense that the range is compared with $100\alpha\%$ points on critical Studentized range W_p given by

$$W_p = q_{\alpha,p,\gamma} \sqrt{\frac{S^2}{n}}.$$

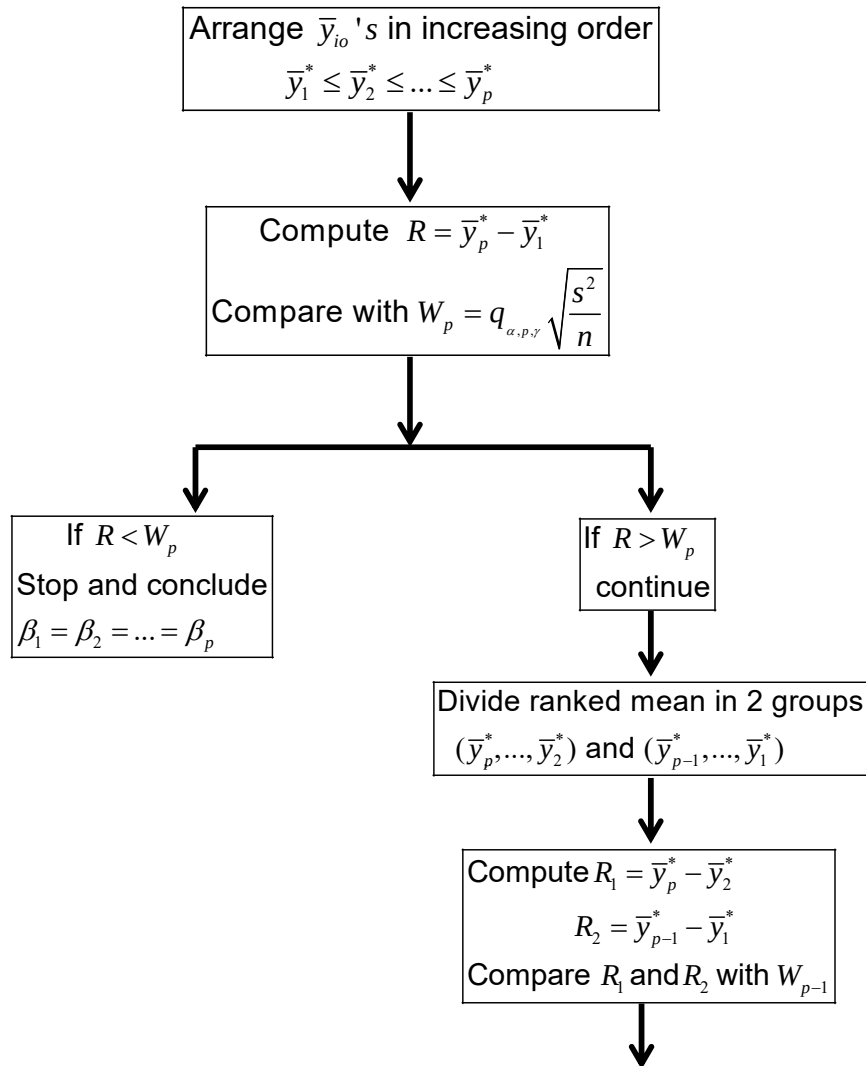
The observed range $R = \bar{y}_p^* - \bar{y}_1^*$ is now compared with W_p . Let the effects $\beta_1, \beta_2, \dots, \beta_p$ be denoted as $\beta_1^*, \beta_2^*, \dots, \beta_p^*$ corresponding to $\bar{y}_1^*, \bar{y}_2^*, \dots, \bar{y}_p^*$ respectively in the context of Student-Newman-Keuls test. For example, the largest mean $\bar{y}_p^*$ maybe $\bar{y}_3$ and so $\beta_p^* = \beta_3$.

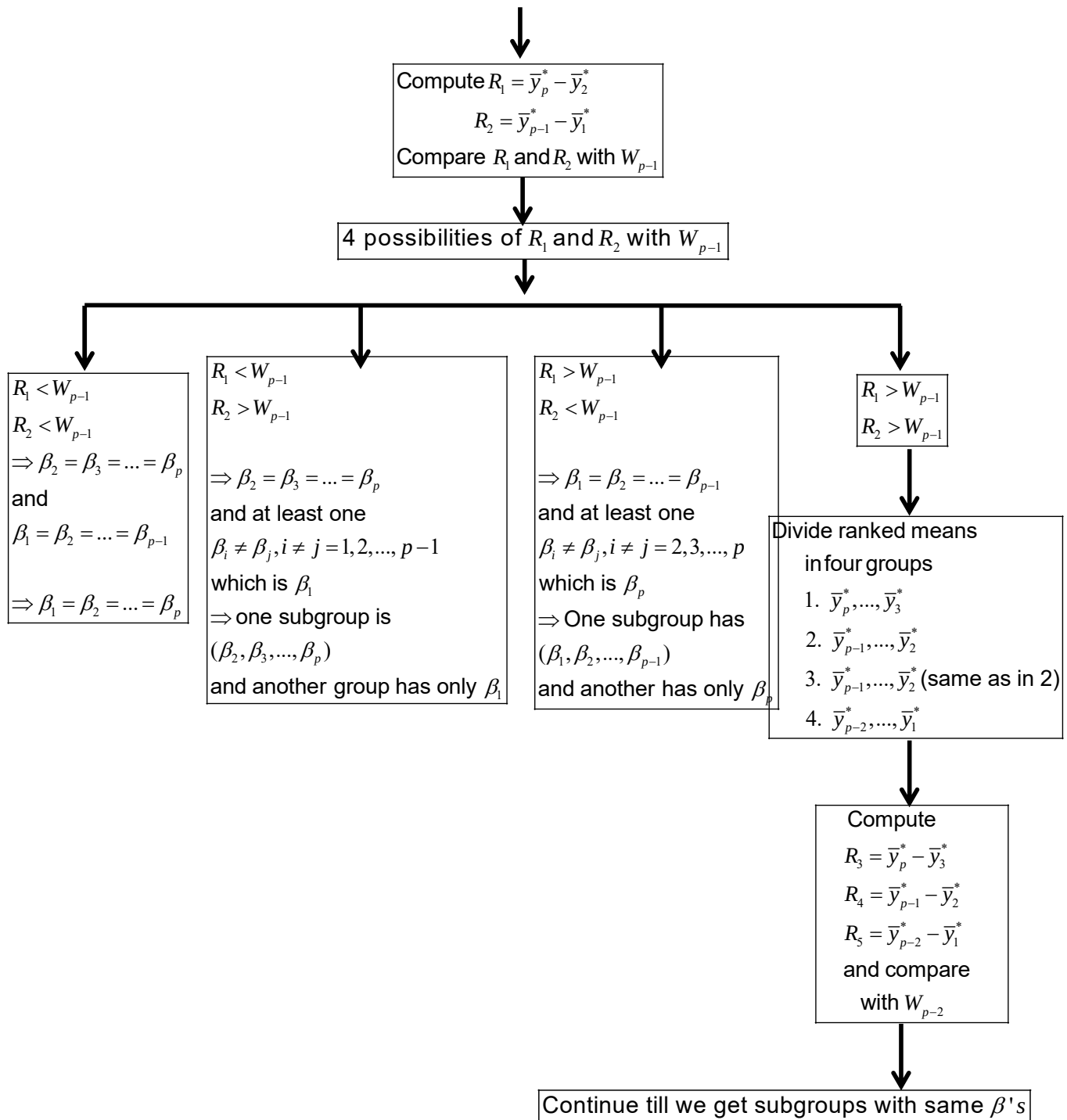
- If $R < W_p$ then stop the process of comparison and conclude that $\beta_1^* = \beta_2^* = \dots = \beta_p^*$.
- If $R > W_p$ then
 - divide the ranked means $\bar{y}_1^*, \bar{y}_2^*, \dots, \bar{y}_p^*$ into two subgroups containing - $(\bar{y}_p^*, \bar{y}_{p-1}^*, \dots, \bar{y}_2^*)$ and $(\bar{y}_{p-1}^*, \bar{y}_{p-2}^*, \dots, \bar{y}_1^*)$
 - Compute the ranges $R_1 = \bar{y}_p^* - \bar{y}_2^*$ and $R_2 = \bar{y}_{p-1}^* - \bar{y}_1^*$. Then compare the ranges R_1 and R_2 with W_{p-1} .
 - If either range (R_1 or R_2) is smaller than W_{p-1} , then the means (or β_i 's) in each of the groups are equal.
 - If R_1 and/or R_2 are greater than W_{p-1} , then the $(p-1)$ means (or β_i 's) in the group concerned are divided into two groups of $(p-2)$ means (or β_i 's) each and compare the range of the two groups with W_{p-2} .

Continue with this procedure until a group of i means (or β_i 's) is found whose range does not exceed W_i .

By this method, the difference between any two means under test is significant when the range of the observed means of each and every subgroup containing the two means under test is significant according to the studentized critical range.

This procedure can be easily understood by the following flow chart.





3. Duncan's multiple comparison test:

The test procedure in Duncan's multiple comparison test is the same as in the Student-Newman-Keuls test except the observed ranges are compared with Duncan's 100 α % critical range

$$D_p = q_{\alpha_p, p, \gamma}^* \sqrt{\frac{s^2}{n}}$$

where $\alpha_p = 1 - (1 - \alpha)^{p-1}$, $q_{\alpha_p, p, \gamma}^*$ denotes the upper 100 α % points of the Studentized range based on Duncan's range.

Tables for Duncan's range are available.

Duncan felt that this test is better than the Student-Newman-Keuls test for comparing the differences between any two ranked means. Duncan regarded that the Student-Newman-Keuls method is too stringent in the sense that the true differences between the means will tend to be missed too often. Duncan notes that in testing the equality of a subset $k, (2 \leq k \leq p)$ means through null hypothesis, we are in fact testing whether $(p-1)$ orthogonal contrasts between the β 's differ from zero or not. If these contrasts were tested in separate independent experiments, each at level α , the probability of incorrectly rejecting the null hypothesis would be $[1 - (1 - \alpha)^{p-1}]$. So Duncan proposed to use $[1 - (1 - \alpha)^{p-1}]$ in place of α in the Student-Newman-Keuls test.

[Reference: Contributions to order statistics, Wiley 1962, Chapter 9 (Multiple decision and multiple comparisons, H.A. David, pages 147-148)].

Case of unequal sample sizes:

When sample means are not based on the same number of observations, the procedures based on Studentized range, Student-Newman-Keuls test and Duncan's test are not applicable. Kramer proposed that in Duncan's method, if a set of p means is to be tested for equality, then replace

$$q_{\alpha_p, p, \gamma}^* \frac{s}{\sqrt{n}} \text{ by } q_{\alpha_p, p, \gamma}^* s \sqrt{\frac{1}{2} \left(\frac{1}{n_U} + \frac{1}{n_L} \right)}$$

where n_U and n_L are the number of observations corresponding to the largest and smallest means in the data. This procedure is only an approximate procedure but will tend to be conservative since means based on a small number of observations will tend to be overrepresented in the extreme groups of means.

Another option is to replace n by the harmonic mean of $n_1, n_2, \dots, n_p$, i.e., $\frac{p}{\sum_{i=1}^p \left(\frac{1}{n_i} \right)}$.

4. The “Least Significant Difference” (LSD):

In the usual testing of $H_0 : \beta_i = \beta_k$ against $H_1 : \beta_i \neq \beta_k$, the t -statistic

$$t = \frac{\bar{y}_{io} - \bar{y}_{ko}}{\sqrt{\widehat{Var}(\bar{y}_{io} - \bar{y}_{ko})}}$$

is used which follows a t -distribution, say with degrees of freedom ' df '. Thus H_0 is rejected whenever

$$|t| > t_{df, 1-\frac{\alpha}{2}}$$

and it is concluded that β_i and β_k are significantly different. The inequality $|t| > t_{df, 1-\frac{\alpha}{2}}$ can be equivalently written as

$$|\bar{y}_{io} - \bar{y}_{ko}| > t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{io} - \bar{y}_{ko})}.$$

If every pair of sample for which

$$|\bar{y}_{io} - \bar{y}_{ko}| \text{ exceeds } t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{io} - \bar{y}_{ko})}$$

then this will indicate that the difference between β_i and β_k is significantly different. So according to this, the quantity $t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{io} - \bar{y}_{ko})}$ would be the least difference of $\bar{y}_{io}$ and $\bar{y}_{ko}$ for which it will be declared that the difference between β_i and β_k is significant. Based on this idea, we use the pooled variance of the two samples $\widehat{Var}(\bar{y}_{io} - \bar{y}_{ko})$ as s^2 and the Least Significant Difference (LSD) is defined as

$$LSD = t_{df, 1-\frac{\alpha}{2}} \sqrt{s^2 \left(\frac{1}{n_i} + \frac{1}{n_k} \right)}.$$

If $n_1 = n_2 = n$, then

$$LSD = t_{df, 1-\frac{\alpha}{2}} \sqrt{\frac{2s^2}{n}}.$$

Now all $\frac{p(p-1)}{2}$ pairs of $\bar{y}_{io}$ and $\bar{y}_{ko}$, ($i \neq k = 1, 2, \dots, p$) are compared with LSD . Use of LSD criterion may not lead to good results if it is used for comparisons suggested by the data (largest/smallest sample mean) or if all pairwise comparisons are done without correction of the test level. If LSD is used for all the pairwise comparisons then these tests are not independent. Such correction for test levels was incorporated in Duncan's test.

5. Tukey's "Honestly significant Difference" (HSD)

In this procedure, the Studentized rank values $q_{\alpha, n, \gamma}$ are used in place of t -quantiles and the standard error of the difference of pooled mean is used in place of standard error of mean in common critical difference for testing $H_0: \beta_i = \beta_k$ against $H_0: \beta_i \neq \beta_k$ and Tukey's Honestly Significant Difference is computed as

$$HSD = q_{1-\frac{\alpha}{2}, p, \gamma} \sqrt{\frac{MS_{error}}{n}}$$

assuming all samples are of the same size n . All $\frac{p(p-1)}{2}$ pairs $|\bar{y}_{io} - \bar{y}_{ko}|$ are compared with HSD .

If $|\bar{y}_{io} - \bar{y}_{ko}| > HSD$ then β_i and β_k are significantly different.

We notice that all the multiple comparison test procedure discussed up to now are based on the testing of hypothesis. There is one-to-one relationship between the testing of hypothesis and the confidence interval estimation. So the confidence interval can also be used for such comparisons. Since $H_0: \beta_i = \beta_k$ is same as $H_0: \beta_i - \beta_k = 0$ so first we establish the relationship and then describe the Tukey's and Scheffe's procedures for multiple comparison test which are based on the confidence interval. We need the following concepts.

Contrast:

A linear parametric function $L = l' \beta = \sum_{i=1}^p \ell_i \beta_i$ where $\beta = (\beta_1, \beta_2, \dots, \beta_p)$ and $\ell = (\ell_1, \ell_2, \dots, \ell_p)$ are the $p \times 1$ vectors of parameters and constants respectively is said to be a contrast when $\sum_{i=1}^p \ell_i = 0$.

For example, $\beta_1 - \beta_2 = 0$, $\beta_1 + \beta_2 - \beta_3 - \beta_1 = 0$, $\beta_1 + 2\beta_2 - 3\beta_3 = 0$ etc. are contrast whereas $\beta_1 + \beta_2 = 0$, $\beta_1 + \beta_2 + \beta_3 + \beta_4 = 0$, $\beta_1 - 2\beta_2 - 3\beta_3 = 0$ etc. are not contrasts.

Orthogonal contrast:

If $L_1 = \ell' \beta = \sum_{i=1}^p \ell_i \beta_i$ and $L_2 = m' \beta = \sum_{i=1}^p m_i \beta_i$ are contrasts such that $\ell' m = 0$ or $\sum_{i=1}^p \ell_i m_i = 0$ then L_1 and L_2 are called orthogonal contrasts.

For example, $L_1 = \beta_1 + \beta_2 - \beta_3 - \beta_4$ and $L_2 = \beta_1 - \beta_2 + \beta_3 - \beta_4$ are contrasts. They are also the orthogonal contrasts.

The condition $\sum_{i=1}^p \ell_i m_i = 0$ ensures that L_1 and L_2 are independent in the sense that

$$\text{Cov}(L_1, L_2) = \sigma^2 \sum_{i=1}^p \ell_i m_i = 0.$$

Mutually orthogonal contrasts:

If there are more than two contrasts then they are said to be mutually orthogonal, if they are pair-wise orthogonal.

It may be noted that the number of mutually orthogonal contrasts is the number of degrees of freedom.

Coming back to the multiple comparison test, if the null hypothesis of equality of all effects is rejected then it is reasonable to look for the contrasts which are responsible for the rejection. In terms of contrasts, it is desirable to have a procedure

- (i) that permits the selection of the contrasts after the data is available.
- (ii) with which a known level of significance is associated.

Such procedures are Tukey's and Scheffe's procedures. Before discussing these procedures, let us consider the following example which illustrates the relationship between the testing of hypothesis and confidence intervals.

Example: Consider the test of hypothesis for

$$H_0 : \beta_i = \beta_j \quad (i \neq j = 1, 2, \dots, p)$$

$$\text{or } H_0 : \beta_i - \beta_j = 0$$

$$\text{or } H_0 : \text{contrast} = 0$$

$$\text{or } H_0 : L = 0.$$

The test statistic for $H_0 : \beta_i = \beta_j$ is

$$t = \frac{(\hat{\beta}_i - \hat{\beta}_j) - (\beta_i - \beta_j)}{\sqrt{\widehat{\text{Var}}(\bar{y}_{i0} - \bar{y}_{j0})}} = \frac{\hat{L} - L}{\sqrt{\widehat{\text{Var}}(\hat{L})}}$$

where $\hat{\beta}$ denotes the maximum likelihood (or least-squares) estimator of β and t follows a t -distribution with df degrees of freedom. This statistic, in fact, can be extended to any linear contrast, say e.g., $L = \beta_1 + \beta_2 - \beta_3 - \beta_4$, $\hat{L} = \hat{\beta}_1 + \hat{\beta}_2 - \hat{\beta}_3 - \hat{\beta}_4$.

The decision rule is

reject $H_0 : L = 0$ against $H_1 : L \neq 0$.

If $|\hat{L}| > t_{df} \sqrt{\widehat{Var}(\hat{L})}$.

The $100(1 - \alpha)\%$ confidence interval of L is obtained as

$$P \left[-t_{df} \leq \frac{\hat{L} - L}{\sqrt{\widehat{Var}(\hat{L})}} \leq t_{df} \right] = 1 - \alpha$$

$$\text{or } P \left[\hat{L} - t_{df} \sqrt{\widehat{Var}(\hat{L})} \leq L \leq \hat{L} + t_{df} \sqrt{\widehat{Var}(\hat{L})} \right] = 1 - \alpha$$

so that the $100(1 - \alpha)\%$ confidence interval of L is

$$\left[\hat{L} - t_{df} \sqrt{\widehat{Var}(\hat{L})}, \hat{L} + t_{df} \sqrt{\widehat{Var}(\hat{L})} \right]$$

and

$$\hat{L} - t_{df} \sqrt{\widehat{Var}(\hat{L})} \leq L \leq \hat{L} + t_{df} \sqrt{\widehat{Var}(\hat{L})}$$

If this interval includes $L = 0$ between lower and upper confidence limits, then $H_0 : L = 0$ is accepted. Our objective is to know if the confidence interval contains zero or not.

Suppose for some given data the confidence intervals for $\beta_1 - \beta_2$ and $\beta_1 - \beta_3$ are obtained as

$$-3 \leq \beta_1 - \beta_2 \leq 2 \quad \text{and} \quad 2 \leq \beta_1 - \beta_3 \leq 4.$$

Thus we find that the interval of $\beta_1 - \beta_2$ includes zero which implies that $H_0 : \beta_1 - \beta_2 = 0$ is accepted. Thus $\beta_1 = \beta_2$. On the other hand interval of $\beta_1 - \beta_3$ does not include zero and so $H_0 : \beta_1 - \beta_3 = 0$ is not accepted. Thus $\beta_1 \neq \beta_3$.

If the interval of $\beta_1 - \beta_3$ is $-1 \leq \beta_1 - \beta_3 \leq 1$ then $H_0 : \beta_1 = \beta_3$ is accepted. If both $H_0 : \beta_1 = \beta_2$ and $H_0 : \beta_1 = \beta_3$ are accepted then we can conclude that $\beta_1 = \beta_2 = \beta_3$.

Tukey's procedure for multiple comparisons (*T*-method)

The *T*-method uses the distribution of the studentized range statistic. (The *S*-method (discussed next) utilizes the *F*-distribution). The *T*-method can be used to make the simultaneous confidence statements about contrasts $(\beta_i - \beta_j)$ among a set of parameters $\{\beta_1, \beta_2, \dots, \beta_p\}$ and an estimate s^2 of error variance if certain restrictions are satisfied.

These restrictions have to be viewed according to the given conditions. For example, one of the restrictions is that all $\hat{\beta}_i$'s have equal variances. In the setup of one-way classification, $\hat{\beta}_i$ has its mean $\bar{Y}_i$ and its variance is $\frac{\sigma^2}{n_i}$. This reduces to a simple condition that all n_i 's are same, i.e., $n_i = n$ for all i . so that all the variances are the same.

Another assumption is to assume that $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p$ are statistically independent and the only contrasts considered are the $\frac{p(p-1)}{2}$ differences $\{\beta_i - \beta_j, i \neq j = 1, 2, \dots, p\}$.

We make the following assumptions:

- (i) The $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p$ are statistically independent
- (ii) $\hat{\beta}_i \sim N(\beta_i, a^2 \sigma^2), i = 1, 2, \dots, p, a > 0$ is a known constant.
- (iii) s^2 is an independent estimate of σ^2 with γ degrees of freedom (here $\gamma = n - p$), i.e.,

$$\frac{\gamma s^2}{\sigma^2} \sim \chi^2(\gamma)$$

and

- (iv) s^2 is statistically independent of $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p$.

The statement of *T*-method is as follows:

Under the assumptions (i)-(iv), the probability is $(1 - \alpha)$ that the values of contrasts

$$L = \sum_{i=1}^p C_i \beta_i \quad \left(\sum_{i=1}^p C_i = 0 \right) \text{ simultaneously satisfy}$$

$$\hat{L} - Ts \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right) \leq L \leq \hat{L} + Ts \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right)$$

where $\hat{L} = \sum_{i=0}^p C_i \hat{\beta}_i$, $\hat{\beta}_i$ is the maximum likelihood (or least squares) estimate of β_i , $T = a q_{\alpha, p, \gamma}$,

with $q_{\alpha, p, \gamma}$ being the upper α point of the distribution of studentized range.

Note that if L is a contrast like $\beta_i - \beta_j (i \neq j)$ then $\frac{1}{2} \sum_{i=1}^p |C_i| = 1$ and the variance is σ^2 so that

$a = 1$ and the interval simplifies to

$$(\hat{\beta}_i - \hat{\beta}_j) - Ts \leq \beta_i - \beta_j \leq (\hat{\beta}_i - \hat{\beta}_j) + Ts$$

where $T = q_{\alpha, p, \gamma}$. Thus the maximum likelihood (or least squares) estimate $\hat{L} = \hat{\beta}_i - \hat{\beta}_j$ of

$L = \beta_i - \beta_j$ is said to be significantly different from zero according to T -criterion if the interval

$(\hat{\beta}_i - \hat{\beta}_j - Ts, \hat{\beta}_i - \hat{\beta}_j + Ts)$ does not cover $\beta_i - \beta_j = 0$, i.e.,

if $|\hat{\beta}_i - \hat{\beta}_j| > Ts$

or more general if $|\hat{L}| > Ts \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right)$.

The steps involved in the testing now involve the following steps:

- Compute $\hat{L}$ or $(\hat{\beta}_i - \hat{\beta}_j)$.
- Compute all possible pairwise differences.
- Compare all the differences with

$$q_{\alpha, p, \gamma} \cdot \frac{s}{\sqrt{n}} \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right)$$

- If $|\hat{L}|$ or $(\hat{\beta}_i - \hat{\beta}_j) > Ts \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right)$

then $\hat{\beta}_i$ and $\hat{\beta}_j$ are significantly different where $T = \frac{q_{\alpha, p, \gamma}}{\sqrt{n}}$.

Tables for T are available.

When sample sizes are not equal, then **Tukey-Kramer Procedure** suggests to compare $|\hat{L}|$ with

$$q_{\alpha, p, \gamma} s \sqrt{\frac{1}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right)$$

or

$$T \sqrt{\frac{1}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \left(\frac{1}{2} \sum_{i=1}^p |C_i| \right).$$

The Scheffe's method (S-method) of multiple comparisons

S-method generally gives shorter confidence intervals than T -method. It can be used in a number of situations where T -method is not applicable, e.g., when the sample sizes are not equal.

A set L of estimable functions $\{\psi\}$ is called a p -dimensional space of estimable functions if there exists p linearly independent estimable functions $(\psi_1, \psi_2, \dots, \psi_p)$ such that every ψ in L is of the form $\psi = \sum_{i=1}^p C_i y_i$ where $C_1, C_2, \dots, C_p$ are known constants. In other words, L is the set of all linear combinations of $\psi_1, \psi_2, \dots, \psi_p$.

Under the assumption that the parametric space Ω is $Y \sim N(X\beta, \sigma^2 I)$ with $\text{rank}(X) = p$, $\beta = (\beta_1, \dots, \beta_p)$, X is $n \times p$ matrix, consider a p -dimensional space L of estimable functions generated by a set of p linearly independent estimable functions $\{\psi_1, \psi_2, \dots, \psi_p\}$.

For any $\psi \in L$, Let $\hat{\psi} = \sum_{i=1}^n C_i y_i$ be its least squares (or maximum likelihood) estimator,

$$\begin{aligned} \text{Var}(\hat{\psi}) &= \sigma^2 \sum_{i=1}^n C_i^2 \\ &= \sigma_{\psi}^2 \text{ (say)} \end{aligned}$$

$$\text{and } \hat{\sigma}_{\psi}^2 = s^2 \sum_{i=1}^n C_i^2$$

where s^2 is the mean square due to error with $(n - p)$ degrees of freedom.

The statement of S-method is as follows:

Under the parametric space Ω , the probability is $(1 - \alpha)$ that simultaneously for all $\psi \in L$, $\hat{\psi} - S \hat{\sigma}_{\hat{\psi}} \leq \psi \leq \hat{\psi} + S \hat{\sigma}_{\hat{\psi}}$ where the constant $S = \sqrt{p F_{1-\alpha}(p, n - p)}$.

Method: For a given space L of estimable functions and confidence coefficient $(1 - \alpha)$, the least square (or maximum likelihood) estimate $\hat{\psi}$ of $\psi \in L$ will be said to be significantly different from zero according to S-criterion if the confidence interval

$$(\hat{\psi} - S \hat{\sigma}_{\hat{\psi}} \leq \psi \leq \hat{\psi} + S \hat{\sigma}_{\hat{\psi}})$$

does not cover $\psi = 0$, i.e., if $|\hat{\psi}| > S \hat{\sigma}_{\hat{\psi}}$.

The S-method is less sensitive to the violation of assumptions of normality and homogeneity of variances.

Comparison of Tukey's and Scheffe's methods:

1. Tukey's method can be used only with equal sample size for all factor level but S-method is applicable whether the sample sizes are equal or not.
2. Although, Tukey's method is applicable for any general contrast, the procedure is more powerful when comparing simple pairwise differences and not making more complex comparisons.
3. If only pairwise comparisons are of interest, and all factor levels have equal sample sizes, Tukey's method gives shorter confidence interval and thus is more powerful.
4. In the case of comparisons involving general contrasts, Scheffe's method tends to give narrower confidence interval and provides a more powerful test.
5. Scheffe's method is less sensitive to the violations of assumptions of normal distribution and homogeneity of variances.

Chapter 3

Experimental Design Models

We consider the models which are used in designing an experiment. The experimental conditions, experimental setup and the objective of the study primarily determine that what type of design is to be used and hence which type of design model can be used for the further statistical analysis to conclude about the decisions. These models are based on one-way classification, two-way classifications (with or without interactions), etc. We discuss them now in detail in a few setups which can be extended further to any order of classification. We discuss them now under the set up of one-way and two-way classifications.

It may be noted that it has already been described how to develop the likelihood ratio tests for the testing the hypothesis of equality of more than two means from normal distributions and now we will concentrate more on deriving the same tests through the least-squares principle under the setup of the linear regression model. The design matrix is assumed to be not necessarily of full rank and consists of 0's and 1's only.

One way classification:

Let p random samples from p normal populations with the same variances but different means and different sample sizes have been independently drawn.

Let the observations Y_{ij} follow the linear regression model setup and

Y_{ij} denotes the j^{th} observation of dependent variable Y when the effect of i^{th} level of the factor is present.

Then Y_{ij} are independently normally distributed with

$$E(Y_{ij}) = \mu + \alpha_i, i = 1, 2, \dots, p, j = 1, 2, \dots, n_i$$

$$V(Y_{ij}) = \sigma^2$$

where

μ – is the general mean effect.

- is fixed.

- gives an idea about the general conditions of the experimental units and treatments.

α_i – is the effect of i^{th} level of the factor.

- can be fixed or random.

Example: Consider a medicine experiment in which there are three different dosages of drugs - 2 mg., 5 mg., 10 mg. which are given to patients for controlling the fever. These are the 3 levels of drugs, and so denote $\alpha_1 = 2$ mg., $\alpha_2 = 5$ mg., $\alpha_3 = 10$ mg. Let Y denotes the time taken by the medicine to reduce the body temperature from high to normal. Suppose two patients have been given 2 mg. of dosage, so Y_{11} and Y_{12} will denote their responses. So we can write that when $\alpha_1 = 2$ mg is given to the two patients, then

$$E(Y_{1j}) = \mu + \alpha_1; j = 1, 2.$$

Similarly, if $\alpha_2 = 5$ mg. and $\alpha_3 = 10$ mg. of dosages are given to 4 and 7 patients respectively then the responses follow the model

$$E(Y_{2j}) = \mu + \alpha_2; j = 1, 2, 3, 4$$

$$E(Y_{3j}) = \mu + \alpha_3; j = 1, 2, 3, 4, 5, 6, 7.$$

Here μ denotes the general mean effect which may be thought as follows: The human body has a tendency to fight against the fever, so the time taken by the medicine to bring down the temperature depends on many factors like body weight, height, general health condition etc. of the patient. So μ denotes the general effect of all these factors which is present in all the observations.

In the terminology of the linear regression model, μ denotes the intercept term which is the value of the response variable when all the independent variables are set to take value zero. In experimental designs, the models with intercept term are more commonly used and so generally we consider these types of models.

Also, we can express

$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}; i = 1, 2, \dots, p, j = 1, 2, \dots, n_i$ where ε_{ij} is the random error component in Y_{ij} . It indicates the variations due to uncontrolled causes which can influence the observations. We assume that ε_{ij} 's are identically and independently distributed as $N(0, \sigma^2)$ with $E(\varepsilon_{ij}) = 0, \text{Var}(\varepsilon_{ij}) = \sigma^2$.

Note that the general linear model considered is

$$E(Y) = X\beta$$

for which Y_{ij} can be written as

$$E(Y_{ij}) = \beta_i.$$

When all the entries in X are 0's or 1's, then this model can also be re-expressed in the form of

$$E(Y_{ij}) = \mu + \alpha_i.$$

This gives rise to some more issues.

Consider and rewrite

$$\begin{aligned} E(Y_{ij}) &= \beta_i \\ &= \bar{\beta} + (\beta_i - \bar{\beta}) \\ &= \mu + \alpha_i \end{aligned}$$

where

$$\begin{aligned} \mu &\equiv \bar{\beta} = \frac{1}{p} \sum_{i=1}^p \beta_i \\ \alpha_i &\equiv \beta_i - \bar{\beta}. \end{aligned}$$

Now let us see the changes in the structure of the design matrix and the vector of regression coefficients.

The model $E(Y_{ij}) = \beta_i = \mu + \alpha_i$ can now be rewritten as

$$\begin{aligned} E(Y) &= X^* \beta^* \\ \text{Cov}(Y) &= \sigma^2 I \end{aligned}$$

where $\beta^* = (\mu, \alpha_1, \alpha_2, \dots, \alpha_p)'$ is a $p \times 1$ vector and

$$X^* = \begin{bmatrix} 1 & & & \\ 1 & & & X \\ \vdots & & & \\ 1 & & & \end{bmatrix}$$

is a $n \times (p+1)$ matrix, and X denotes the earlier defined design matrix in which

- first n_1 rows as $(1, 0, 0, \dots, 0)$,
- second n_2 rows as $(0, 1, 0, \dots, 0)$
- ..., and
- last n_p rows as $(0, 0, 0, \dots, 1)$.

We earlier assumed that $\text{rank}(X) = p$ but can we also say that $\text{rank}(X^*)$ is also p in the present case?

Since the first column of X^* is the vector sum of all its remaining p columns, so

$$\text{rank}(X^*) = p.$$

It is thus apparent that all the linear parametric functions of $\alpha_1, \alpha_2, \dots, \alpha_p$ are not estimable. The question now arises is what kind of linear parametric functions are estimable?

Consider any linear estimator

$$L = \sum_{i=1}^p \sum_{j=1}^{n_i} a_{ij} Y_{ij}$$

with

$$C_i = \sum_{j=1}^{n_i} a_{ij}$$

Now

$$\begin{aligned} E(L) &= \sum_{i=1}^p \sum_{j=1}^{n_i} a_{ij} E(Y_{ij}) \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} a_{ij} (\mu + \alpha_i) \\ &= \mu \sum_{i=1}^p \sum_{j=1}^{n_i} a_{ij} + \sum_{i=1}^p \sum_{j=1}^{n_i} a_{ij} \alpha_i \\ &= \mu \left(\sum_{i=1}^p C_i \right) + \sum_{i=1}^p C_i \alpha_i. \end{aligned}$$

Thus $\sum_{i=1}^p C_i \alpha_i$ is estimable if and only if

$$\sum_{i=1}^p C_i = 0,$$

i.e., $\sum_{i=1}^p C_i \alpha_i$ is a contrast.

Thus, in general, neither $\sum_{i=1}^p \alpha_i$ nor any $\mu, \alpha_1, \alpha_2, \dots, \alpha_p$ is estimable. If it is a contrast, then it is estimable.

This effect and outcome can also be seen from the following explanation based on the estimation of parameters $\mu, \alpha_1, \alpha_2, \dots, \alpha_p$.

Consider the least-squares estimation $\hat{\mu}, \hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_p$ of $\mu, \alpha_1, \alpha_2, \dots, \alpha_p$ respectively.

Minimize the sum of squares due to ε_{ij} 's

$$S = \sum_{i=1}^p \sum_{j=1}^{n_i} \varepsilon_{ij}^2 = \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i)^2$$

to obtain $\hat{\mu}, \hat{\alpha}_1, \dots, \hat{\alpha}_p$.

$$(a) \quad \frac{\partial S}{\partial \mu} = 0 \Rightarrow \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i) = 0$$

$$(b) \quad \frac{\partial S}{\partial \alpha_i} = 0 \Rightarrow \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i) = 0, \quad i = 1, 2, \dots, p.$$

Note that (a) can be obtained from (b) or vice versa. So (a) and (b) are linearly dependent in the sense that there are $(p + 1)$ unknowns and p linearly independent equations. Consequently $\hat{\mu}, \hat{\alpha}_1, \dots, \hat{\alpha}_p$ do not have a unique solution. Same applies to the maximum likelihood estimation of $\mu, \alpha_1, \dots, \alpha_p$.

If a side condition that

$$\sum_{i=1}^p n_i \hat{\alpha}_i = 0 \quad \text{or} \quad \sum_{i=1}^p n_i \alpha_i = 0$$

is imposed then (a) and (b) have a unique solution as

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij} = \bar{y}_{oo},$$

$$\begin{aligned} \hat{\alpha}_i &= \frac{1}{n_p} \sum_{j=1}^{n_i} y_{ij} - \hat{\mu} \\ &= \bar{y}_{io} - \bar{y}_{oo} \end{aligned}$$

$$\text{where } n = \sum_{i=1}^p n_i.$$

In case, all the sample sizes are the same, then the condition $\sum_{i=1}^p n_i \hat{\alpha}_i = 0$ or $\sum_{i=1}^p n_i \alpha_i = 0$ reduces to

$$\sum_{i=1}^p \hat{\alpha}_i = 0 \quad \text{or} \quad \sum_{i=1}^p \alpha_i = 0.$$

So the model $y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$ needs to be rewritten so that all the parameters can be uniquely estimated. Thus

$$\begin{aligned} Y_{ij} &= \mu + \alpha_i + \varepsilon_{ij} \\ &= (\mu + \bar{\alpha}) + (\alpha_i - \bar{\alpha}) + \varepsilon_{ij} \\ &= \mu^* + \alpha_i^* + \varepsilon_{ij} \end{aligned}$$

where

$$\mu^* = \mu + \bar{\alpha}$$

$$\alpha_i^* = \alpha_i - \bar{\alpha}$$

$$\bar{\alpha} = \frac{1}{p} \sum_{i=1}^p \alpha_i$$

and

$$\sum_{i=1}^p \alpha_i^* = 0$$

This is a **reparameterized form** of the linear model.

Thus in a linear model when X is not of full rank, then the parameters do not have unique estimates.

In such conditions, a restriction $\sum_{i=1}^p \alpha_i = 0$ (or equivalently $\sum_{i=1}^p n_i \alpha_i = 0$ in case all n_i 's are not the same) can be added and then the least squares (or maximum likelihood) estimators obtained are unique.

The model

$$E(Y_{ij}) = \mu^* + \alpha_i^*; \quad \sum_{i=1}^p \alpha_i^* = 0$$

is called a **reparametrization of the original linear model**.

Let us now consider the analysis of variance with an additional constraint. Let

$$\begin{aligned} Y_{ij} &= \beta_i + \varepsilon_{ij}, \quad i = 1, 2, \dots, p; \quad j = 1, 2, \dots, n_i \\ &= \bar{\beta} + (\beta_i - \bar{\beta}) + \varepsilon_{ij} \\ &= \mu + \alpha_i + \varepsilon_{ij} \end{aligned}$$

with

$$\mu = \bar{\beta} = \frac{1}{p} \sum_{i=1}^p \beta_i, \quad \alpha_i = \beta_i - \bar{\beta},$$

$$\sum_{i=1}^p n_i \alpha_i = 0, \quad n = \sum_{i=1}^p n_i.$$

and ε_{ij} 's are identically and independently distributed with mean 0 and variance σ^2 .

The null hypothesis is

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0$$

and the alternative hypothesis is

$$H_1 : \text{atleast one } \alpha_i \neq \alpha_j \text{ for all } i, j.$$

This model is a one-way layout in the sense that the observations y_{ij} 's are assumed to be affected by only one treatment effect α_i . So the null hypothesis is equivalent to testing the equality of p population means or equivalently the equality of p treatment effects.

We use the principle of least squares to estimate the parameters $\mu, \alpha_1, \alpha_2, \dots, \alpha_p$.

Minimize the error sum of squares

$$E = \sum_{i=1}^p \sum_{j=1}^{n_i} \varepsilon_{ij}^2 = \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i)^2$$

with respect to $\mu, \alpha_1, \alpha_2, \dots, \alpha_p$. The normal equations are obtained as

$$\frac{\partial E}{\partial \mu} = 0 \Rightarrow -2 \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i) = 0$$

or

$$n\mu + \sum_{i=1}^p n_i \alpha_i = \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij} \quad (1)$$

$$\frac{\partial E}{\partial \alpha_i} = 0 \Rightarrow -2 \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i) = 0$$

or

$$n_i \mu + n_i \alpha_i = \sum_{j=1}^{n_i} y_{ij} \quad (i = 1, 2, \dots, p). \quad (2)$$

Using $\sum_{i=1}^p n_i \alpha_i = 0$ in (1) gives

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij} = \frac{G}{n} = \bar{y}_{oo}$$

where $G = \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij}$ is the grand total of all the observations.

Substituting $\hat{\mu}$ in (2) gives

$$\begin{aligned}\hat{\alpha}_i &= \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} - \hat{\mu} \\ &= \frac{T_i}{n_i} - \hat{\mu} \\ &= \bar{y}_{io} - \bar{y}_{oo}\end{aligned}$$

where $T_i = \sum_{j=1}^{n_i} y_{ij}$ is the treatment total due to i^{th} effect α_i , i.e., a total of all the observations

receiving the i^{th} treatment and $\bar{y}_{io} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$.

Now the fitted model is $y_{ij} = \hat{\mu} + \hat{\alpha}_i$ and the error sum of squares after substituting $\hat{\mu}$ and $\hat{\alpha}_i$ in E becomes

$$\begin{aligned}E &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \hat{\mu} - \hat{\alpha}_i)^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} [(y_{ij} - \bar{y}_{oo}) - (\bar{y}_{io} - \bar{y}_{oo})]^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2 - \sum_{i=1}^p \sum_{j=1}^{n_i} (\bar{y}_{io} - \bar{y}_{oo})^2 \\ &= \left(\sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij}^2 - \frac{G^2}{n} \right) - \left(\sum_{i=1}^p \frac{T_i^2}{n_i} - \frac{G^2}{n} \right)\end{aligned}$$

where the total sum of squares (TSS)

$$\begin{aligned}TSS &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij}^2 - \frac{G^2}{n},\end{aligned}$$

and $\frac{G^2}{n}$ is called the **correction factor (CF)**.

To obtain a measure of variation due to treatments, let

$$H_0 = \alpha_1 = \alpha_2 = \dots = \alpha_p = 0$$

be true. Then the model becomes

$$Y_{ij} = \mu + \varepsilon_{ij}, \quad i = 1, 2, \dots, p; \quad j = 1, 2, \dots, n_i.$$

Minimizing the error sum of squares

$$E_1 = \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu)^2$$

with respect to μ , the normal equation is obtained as

$$\frac{\partial E_1}{\partial \mu} = 0 \Rightarrow -2 \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \mu) = 0$$

or

$$\hat{\mu} = \frac{G}{n} = \bar{y}_{oo}.$$

Substituting $\hat{\mu}$ in E_1 , the error sum of squares becomes

$$\begin{aligned} E_1 &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \hat{\mu})^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^p \sum_{j=1}^{n_i} y_{ij}^2 - \frac{G^2}{n}. \end{aligned}$$

Note that

E_1 : Contains variation due to treatment and error both

E : Contains variation due to error only

So $E_1 - E$: contain variation due to treatment only.

The **sum of squares due to treatment** ($SSTr$) is given by

$$\begin{aligned} SSTr &= E_1 - E \\ SSTr &= \sum_{i=1}^p \sum_{j=1}^{n_i} (\bar{y}_{io} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^p \frac{T_i^2}{n_i} - \frac{G^2}{n}. \end{aligned}$$

The following quantity is called the **error sum of squares** or **sum of squares due to error** (SSE)

$$SSE = \sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2.$$

These sum of squares forms the basis for the development of tools in the analysis of variance. We can write

$$TSS = SStr + SSE.$$

The distribution of degrees of freedom among these sum of squares is as follows:

- The total sum of squares is based on n quantities subject to the constraint that

$$\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo}) = 0 \text{ so } TSS \text{ carries } (n-1) \text{ degrees of freedom.}$$

- The sum of squares due to the treatments is based on p quantities subject to the constraint

$$\sum_{i=1}^p n_i (\bar{y}_{io} - \bar{y}_{oo}) = 0 \text{ so } SStr \text{ has } (p-1) \text{ degrees of freedom.}$$

- The sum of squares due to errors is based on n quantities subject to p constraints

$$\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io}) = 0, i = 1, 2, \dots, p$$

so SSE carries $(n-p)$ degrees of freedom.

Also, note that

$$TSS = SStr + SSE,$$

the TSS has been divided into two orthogonal components - $SSTr$ and SSE . Moreover, all TSS , $SSTr$ and SSE can be expressed in a quadratic form. Since ε_{ij} are assumed to be identically and independently distributed following $N(0, \sigma^2)$, so y_{ij} are also independently distributed following $N(\mu + \alpha_i, \sigma^2)$.

Now using the theorems 7 and 8 with $q_1 = SStr$, $q_2 = SSE$, we have under H_0 ,

$$\frac{SSTr}{\sigma^2} \sim \chi^2(p-1)$$

and

$$\frac{SSE}{\sigma^2} \sim \chi^2(n-p).$$

Moreover, $SSTr$ and SSE are independently distributed.

The mean squares is defined as the sum of squares divided by the degrees of freedom. So the mean square due to treatment is

$$MSTr = \frac{SSTr}{p-1}$$

and the mean square due to error is

$$MSE = \frac{SSE}{n-p}.$$

Thus, under H_0 ,

$$F = \frac{\left(\frac{MSTr}{\sigma^2} \right)}{\left(\frac{MSE}{\sigma^2} \right)} \sim F(p-1, n-p).$$

The decision rule is that reject H_0 if

$$F > F_{1-\alpha, p-1, n-p}$$

at α % level of significance.

If H_0 does not hold true, then

$$\frac{MSTr}{MSE} \sim \text{noncentral } F(p-1, n-p, \delta)$$

where $\delta = \sum_{i=1}^p \frac{n_i \alpha_i^2}{\sigma^2}$ is the non-centrality parameter.

Note that the test statistic $\frac{MSTr}{MSE}$ can also be obtained from the likelihood ratio test.

If H_0 is rejected, then we go for multiple comparison tests and try to divide the population into several groups having the same effects.

The analysis of variance table is as follows:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F-value
Treatment	$p-1$	$SSTr$	$MSTr$	$\frac{MSTr}{MSE}$
Error	$n-p$	SSE	MSE	
Total	$n-1$	TSS		

Now we find the expectations of $SSTr$ and SSE .

$$\begin{aligned} E(SSTr) &= E \left[\sum_{i=1}^p n_i (\bar{y}_{io} - \bar{y}_{oo})^2 \right] \\ &= E \left[\sum_{i=1}^p n_i \{(\mu + \alpha_i + \bar{\varepsilon}_{io}) - (\mu + \bar{\varepsilon}_{oo})\}^2 \right] \end{aligned}$$

where

$$\bar{\varepsilon}_{io} = \frac{1}{n_i} \sum_{j=1}^{n_i} \varepsilon_{ij}, \quad \bar{\varepsilon}_{oo} = \frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} \varepsilon_{ij} \quad \text{and} \quad \sum_{i=1}^p \frac{n_i \alpha_i}{n} = 0.$$

$$\begin{aligned} E(SSTr) &= E \left[\sum_{i=1}^p n_i \{ \alpha_i + (\bar{\varepsilon}_{io} - \bar{\varepsilon}_{oo}) \}^2 \right] \\ &= \sum_{i=1}^p n_i E(\alpha_i^2) + \sum_{i=1}^p n_i E(\bar{\varepsilon}_{io} - \bar{\varepsilon}_{oo})^2 + 0. \end{aligned}$$

Since

$$\begin{aligned} E(SSTr) &= \sum_{i=1}^p n_i \alpha_i^2 + \sigma^2 \sum_{i=1}^p n_i \left(\frac{1}{n_i} - \frac{1}{n} \right) \\ &= \sum_{i=1}^p n_i \alpha_i^2 + (p-1) \sigma^2 \end{aligned}$$

$$\text{or } E \left(\frac{SSTr}{p-1} \right) = \sigma^2 + \frac{\sum_{i=1}^p n_i \alpha_i^2}{p-1}$$

$$\text{or } E(MSTr) = \sigma^2 + \frac{\sum_{i=1}^p n_i \alpha_i^2}{p-1}.$$

$$E(\bar{\varepsilon}_{io}^2) = \text{Var}(\bar{\varepsilon}_{io}) = \text{Var} \left(\frac{1}{n_i} \sum_{j=1}^{n_i} \varepsilon_{ij} \right) = \frac{1}{n_i^2} n_i \sigma^2 = \frac{\sigma^2}{n_i}$$

$$E(\bar{\varepsilon}_{oo}^2) = \text{Var}(\bar{\varepsilon}_{oo}) = \text{Var} \left(\frac{1}{n} \sum_{i=1}^p \sum_{j=1}^{n_i} \varepsilon_{ij} \right) = \frac{1}{n^2} n \sigma^2 = \frac{\sigma^2}{n}$$

$$\begin{aligned} E(\bar{\varepsilon}_{io} \bar{\varepsilon}_{oo}) &= \text{Cov}(\bar{\varepsilon}_{io}, \bar{\varepsilon}_{oo}) \\ &= \frac{1}{n_i n} \text{Cov} \left(\sum_{j=1}^{n_i} \varepsilon_{ij}, \sum_{i=1}^p \sum_{j=1}^{n_i} \varepsilon_{ij} \right) \\ &= \frac{n_i \sigma^2}{n_i n} = \frac{\sigma^2}{n}. \end{aligned}$$

Next

$$\begin{aligned}
E(SSE) &= E \left[\sum_{i=1}^p \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2 \right] \\
&= E \left[\sum_{i=1}^p \sum_{j=1}^{n_i} \{ (\mu + \alpha_i + \varepsilon_{ij}) - (\mu + \alpha_i + \bar{\varepsilon}_{io}) \}^2 \right] \\
&= E \left[\sum_{i=1}^p \sum_{j=1}^{n_i} (\varepsilon_{ij} - \bar{\varepsilon}_{io})^2 \right] \\
&= \sum_{i=1}^p \sum_{j=1}^{n_i} E(\varepsilon_{ij}^2 + \bar{\varepsilon}_{io}^2 - 2\varepsilon_{ij}\bar{\varepsilon}_{io}) \\
&= \sum_{i=1}^p \sum_{j=1}^{n_i} \left(\sigma^2 + \frac{\sigma^2}{n_i} - \frac{2\sigma^2}{n_i} \right) \\
&= \sigma^2 \sum_{i=1}^p \sum_{j=1}^{n_i} \left(\frac{n_i - 1}{n_i} \right) \\
&= \sigma^2 \sum_{i=1}^p \frac{n_i(n_i - 1)}{n_i} \\
&= \sigma^2 \sum_{i=1}^p (n_i - 1) \\
&= (n - p)\sigma^2
\end{aligned}$$

$$\text{or } E \left(\frac{SSE}{n - p} \right) = \sigma^2$$

$$\text{or } E(MSE) = \sigma^2.$$

Thus MSE is an unbiased estimator of σ^2 .

Two-way classification under fixed effects model

Suppose the response of an outcome is affected by the two factors – A and B . For example, suppose I varieties of mangoes are grown on I different plots of the same size in each of the J different locations. All plots are given the same treatment like an equal amount of water, an equal amount of fertilizer etc. So there are two factors in the experiment which affect the yield of mangoes.

- Location (A)

- Variety of mangoes (B)

Such an experiment is called a **two – factor experiment**. The different locations correspond to the different levels of A and the different varieties correspond to the different levels of factor B . The observations are collected on the basis of per plot.

The combined effect of the two factors (A and B in our case) is called the **interaction effect** (of A and B).

Mathematically, let a and b be the levels of factors A and B respectively then a function $f(a, b)$ is called a function of no interaction if and only if there exists functions $g(a)$ and $h(b)$ such that

$$f(a, b) = g(a) + h(b).$$

Otherwise, the factors are said to interact.

For a function $f(a, b)$ of no interaction,

$$f(a_1, b) = g(a_1) + h(b)$$

$$f(a_2, b) = g(a_2) + h(b)$$

$$\Rightarrow f(a_1, b) - f(a_2, b) = g(a_1) - g(a_2)$$

and so it is independent of b . Such no interaction functions are called **additive functions**.

Now there are two options:

- Only one observation per plot is collected.
- More than one observations per plot are collected.

If there is only one observation per plot then there cannot be any interaction effect among the observations and we assume it to be zero.

If there are more than one observations per plot then the interaction effect among the observations can be considered.

We consider here two cases

1. One observation per plot in which the interaction effect is zero.
2. More than one observations per plot in which the interaction effect is present.

Two-way classification without interaction

Let y_{ij} be the response of observation from i^{th} level of the first factor, say A and j^{th} level of the second factor, say B . So assume Y_{ij} are independently distributed as

$$N(\mu_{ij}, \sigma^2) \quad i=1, 2, \dots, I, j=1, 2, \dots, J.$$

This can be represented in the form of a linear model as

$$\begin{aligned} E(Y_{ij}) &= \mu_{ij} \\ &= \mu_{oo} + (\mu_{io} - \mu_{oo}) + (\mu_{oj} - \mu_{oo}) + (\mu_{ij} - \mu_{io} - \mu_{oj} + \mu_{oo}) \\ &= \mu + \alpha_i + \beta_j + \gamma_{ij} \end{aligned}$$

where

$$\mu = \mu_{oo}$$

$$\alpha_i = \mu_{io} - \mu_{oo}$$

$$\beta_j = \mu_{oj} - \mu_{oo}$$

$$\gamma_{ij} = \mu_{ij} - \mu_{io} - \mu_{oj} + \mu_{oo}$$

with

$$\sum_{i=1}^I \alpha_i = \sum_{i=1}^I (\mu_{io} - \mu_{oo}) = 0$$

$$\sum_{j=1}^J \beta_j = \sum_{j=1}^J (\mu_{oj} - \mu_{oo}) = 0$$

Here

α_i : effect of i^{th} level of factor A

or excess of mean of i^{th} level of A over the general mean.

β_j : effect of j^{th} level of B

or excess of mean of j^{th} level of B over the general mean.

γ_{ij} : Interaction effect of i^{th} level of A and j^{th} level of B .

Here we assume $\gamma_{ij} = 0$ as we have only one observation per plot.

We also assume that the model $E(Y_{ij}) = \mu_{ij}$ is a full rank model so that μ_{ij} and all linear parametric functions of μ_{ij} are estimable.

The total number of observations are $I \times J$ which can be arranged in a two-way classified $I \times J$ table where the rows correspond to the different levels of A and the column corresponds to the different levels of B .

The observations on Y and the design matrix X in this case are

$$\begin{array}{c|cccc|cccc}
 Y & \mu & \alpha_1 & \alpha_2 & \cdots & \alpha_I & \beta_1 & \beta_2 & \cdots & \beta_J \\
 y_{11} & 1 & 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\
 y_{12} & 1 & 1 & 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\
 \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots & & & \\
 y_{1J} & 1 & 1 & 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \\
 \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & & \vdots \\
 y_{I1} & 1 & 0 & 0 & \cdots & 1 & 1 & 0 & & 0 \\
 y_{I2} & 1 & 0 & 0 & & 1 & 0 & 1 & \cdots & 0 \\
 \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \ddots & & \vdots \\
 y_{IJ} & 1 & 0 & 0 & & 1 & 0 & 0 & & 1
 \end{array}$$

If the design matrix is not of full rank, then the model can be reparameterized. In such a case, we can start the analysis by assuming that the model $E(Y_{ij}) = \mu + \alpha_i + \beta_j$ is obtained after reparameterization.

There are two null hypotheses of interest:

$$H_{0\alpha} : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$$

$$H_{0\beta} : \beta_1 = \beta_2 = \dots = \beta_J = 0$$

against

$$H_{1\alpha} : \text{at least one } \alpha_i (i=1, 2, \dots, I) \text{ is different from others}$$

$$H_{1\beta} : \text{at least one } \beta_j (j=1, 2, \dots, J) \text{ is different from others.}$$

Now we derive the least-squares estimators (or equivalently the maximum likelihood estimator) of μ, α_i and β_j , $i=1, 2, \dots, I, j=1, 2, \dots, J$ by minimizing the error sum of squares

$$E = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \mu - \alpha_i - \beta_j)^2.$$

The normal equations are obtained as

$$\frac{\partial E}{\partial \mu} = 0 \Rightarrow -2 \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \mu - \alpha_i - \beta_j) = 0$$

$$\frac{\partial E}{\partial \alpha_i} = 0 \Rightarrow -2 \sum_{j=1}^J (y_{ij} - \mu - \alpha_i - \beta_j) = 0, \quad i = 1, 2, \dots, I,$$

$$\frac{\partial E}{\partial \beta_j} = 0 \Rightarrow -2 \sum_{i=1}^I (y_{ij} - \mu - \alpha_i - \beta_j) = 0, \quad j = 1, 2, \dots, J.$$

Solving the normal equations and using $\sum_{i=1}^I \alpha_i = 0$ and $\sum_{j=1}^J \beta_j = 0$, the least-squares estimators are obtained as

$$\hat{\mu} = \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J y_{ij} = \frac{G}{IJ} = \bar{y}_{oo}$$

$$\hat{\alpha}_i = \frac{1}{J} \sum_{j=1}^J y_{ij} - \bar{y}_{oo} = \frac{T_i}{J} - \bar{y}_{oo} = \bar{y}_{io} - \bar{y}_{oo} \quad i = 1, 2, \dots, I$$

$$\hat{\beta}_j = \frac{1}{I} \sum_{i=1}^I y_{ij} - \bar{y}_{oo} = \frac{B_j}{I} - \bar{y}_{oo} = \bar{y}_{oj} - \bar{y}_{oo}, \quad j = 1, 2, \dots, J$$

where

T_i : treatment totals due to i^{th} α effect, i.e., the sum of all the observations receiving the i^{th} treatment.

B_j : block totals due to j^{th} β effect, i.e., sum of all the observations in the j^{th} block.

Thus the error sum of squares is

$$\begin{aligned} SSE &= \text{Min}_{\mu, \alpha_i, \beta_j} E \\ &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J [(y_{ij} - \bar{y}_{oo}) - (\bar{y}_{io} - \bar{y}_{oo}) - (\bar{y}_{oj} - \bar{y}_{oo})]^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{oo})^2 - J \sum_{i=1}^I (\bar{y}_{io} - \bar{y}_{oo})^2 - I \sum_{j=1}^J (\bar{y}_{oj} - \bar{y}_{oo})^2 \end{aligned}$$

which carries

$$IJ - (I - 1) - (J - 1) - 1 = (I - 1)(J - 1) \text{ degrees of freedom.}$$

Next, we consider the estimation of μ and β_j under the null hypothesis $H_{0\alpha} : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$

by minimizing the error sum of squares

$$E_1 = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \mu - \beta_j)^2.$$

The normal equations are obtained by

$$\frac{\partial E_1}{\partial \mu} = 0 \text{ and } \frac{\partial E_1}{\partial \beta_j} = 0, \quad j = 1, 2, \dots, J$$

which on solving gives the least square estimates

$$\hat{\mu} = \bar{y}_{oo}$$

$$\hat{\beta}_j = \bar{y}_{oj} - \bar{y}_{oo}.$$

The sum of squares due to $H_{0\alpha}$ is

$$\begin{aligned}
 \text{Min}_{\mu, \beta_j} E_1 &= \text{Min}_{\mu, \beta_j} \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \mu - \beta_j)^2 \\
 &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \hat{\mu} - \hat{\beta}_j)^2 \\
 &= J \sum_{i=1}^I (\bar{y}_{io} - \bar{y}_{oo})^2 + \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2.
 \end{aligned}$$

$\downarrow$

Sum of squares due to factor A

$\downarrow$

Error sum of squares

Thus the sum of squares due to deviation from $H_{0\alpha}$ (or sum of squares due to rows or sum of squares are to factor A)

$$SSA = J \sum_{i=1}^I (\bar{y}_{io} - \bar{y}_{oo})^2 = J \sum_{i=1}^I \bar{y}_{io}^2 - IJ\bar{y}_{oo}^2$$

and carries $(IJ - J) - (I - 1)(J - 1) = I - 1$ degrees of freedom.

Now we find the estimates of μ and α_i under $H_{0\beta} : \beta_1 = \beta_2 = \dots = \beta_J = 0$ by minimizing

$$E_2 = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \mu - \alpha_i)^2.$$

The normal equations are

$$\frac{\partial E_2}{\partial \mu} = 0 \text{ and } \frac{\partial E_2}{\partial \alpha_i} = 0, i = 1, 2, \dots, I$$

which on solving give the estimators as

$$\begin{aligned}
 \hat{\mu} &= \bar{y}_{oo} \\
 \hat{\alpha}_i &= \bar{y}_{io} - \bar{y}_{oo}.
 \end{aligned}$$

The minimum value of the error sum of squares is obtained by

$$\begin{aligned}
 \text{Min}_{\mu, \alpha_j} E_2 &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \hat{\mu} - \hat{\alpha}_i)^2 \\
 &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io})^2 \\
 &= I \sum_{j=1}^J (\bar{y}_{oj} - \bar{y}_{oo})^2 + \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2
 \end{aligned}$$

$\downarrow$

Sum of squares due to factor B

$\downarrow$

Error sum of squares

The sum of squares due to deviation from $H_{0\beta}$ (or the sum of squares due to columns or sum of squares due to factor B) is

$$SSB = I \sum_{j=1}^J (\bar{y}_{oj} - \bar{y}_{oo})^2 = I \sum_j \bar{y}_{oj}^2 - IJ \bar{y}_{oo}^2$$

and its degrees of freedom are

$$(IJ - I) - (I - 1)(J - 1) = J - 1.$$

Note that the total sum of squares is

$$\begin{aligned} TSS &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J [(\bar{y}_{io} - \bar{y}_{oo}) + (\bar{y}_{oj} - \bar{y}_{oo}) + (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})]^2 \\ &= J \sum_{i=1}^I (\bar{y}_{io} - \bar{y}_{oo})^2 + I \sum_{j=1}^J (\bar{y}_{oj} - \bar{y}_{oo})^2 + \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2 \\ &= SSA + SSB + SSE. \end{aligned}$$

The partitioning of degrees of freedom into the corresponding groups is

$$IJ - 1 = (I - 1) + (J - 1) + (I - 1)(J - 1).$$

Note that SSA , SSB and SSE are mutually orthogonal and that is why the degrees of freedom can be divided like this.

Now using the theory explained while discussing the likelihood ratio test or assuming y_{ij} 's to be independently distributed as $N(\mu + \alpha_i + \beta_j, \sigma^2)$, $i = 1, 2, \dots, I$; $j = 1, 2, \dots, J$, and using the Theorems 6 and 7, we can write

$$\frac{SSA}{\sigma^2} \sim \chi^2(I - 1), \quad \frac{SSB}{\sigma^2} \sim \chi^2(J - 1), \quad \text{and} \quad \frac{SSE}{\sigma^2} \sim \chi^2((I - 1)(J - 1)).$$

So the test statistic for $H_{0\alpha}$ is obtained as

$$\begin{aligned} F_1 &= \frac{\left(\frac{SSA / \sigma^2}{I - 1} \right)}{\left(\frac{SSE / \sigma^2}{(I - 1)(J - 1)} \right)} \\ &= \frac{(I - 1)(J - 1)}{(I - 1)} \cdot \frac{SSA}{SSE} \\ &= \frac{MSA}{MSE} \sim F((I - 1), (I - 1)(J - 1)) \text{ under } H_{0\alpha} \end{aligned}$$

where

$$MSA = \frac{SSA}{I-1}$$

$$MSE = \frac{SSE}{(I-1)(J-1)}.$$

The same statistic is also obtained using the likelihood ratio test for $H_{0\alpha}$.

The decision rule is

Reject $H_{0\alpha}$ if $F_1 > F_{1-\alpha}[(I-1), (I-1)(J-1)]$.

Under $H_{1\alpha}$, F_1 follows a noncentral F distribution $F(\delta, (J-1), (I-1)(J-1))$ where $\delta = \frac{J \sum_{i=1}^I \alpha_i^2}{\sigma^2}$ is

the associated non-centrality parameter.

Similarly, the test statistic for $H_{0\beta}$ is obtained as

$$F_2 = \frac{\left(\frac{SSB / \sigma^2}{J-1} \right)}{\left(\frac{SSE / \sigma^2}{(I-1)(J-1)} \right)} = \frac{(I-1)(J-1)}{(J-1)} \frac{SSB}{SSE} = \frac{MSB}{MSE} \sim F((J-1), (I-1)(J-1)) \text{ under } H_{0\beta}$$

where $MSB = \frac{SSB}{J-1}$.

The decision rule is

Reject $H_{0\beta}$ if $F_2 > F_{1-\alpha}((J-1), (I-1)(J-1))$.

The same test statistic can also be obtained from the likelihood ratio test.

The analysis of variance table is as follows:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F -value
Factor A (or rows)	$(I-1)$	SSA	MSA	$F_1 = \frac{MSA}{MSE}$
Factor B (or column)	$(J-1)$	SSB	MSB	$F_2 = \frac{MSB}{MSE}$
Error	$(I-1)(J-1)$	SSE	MSE	
		(by subtraction)		
Total	$IJ-1$	TSS		

It can be found on similar lines as in the case of one way classification that

$$E(MSA) = \sigma^2 + \frac{J}{I-1} \sum_{i=1}^I \alpha_i^2$$

$$E(MSB) = \sigma^2 + \frac{I}{J-1} \sum_{j=1}^J \beta_j^2$$

$$E(MSE) = \sigma^2.$$

If the null hypothesis is rejected, then we use the multiple comparison tests to divide the α_i 's (or β_j 's) into groups such that α_i 's (or β_j 's) belonging to the same group are equal and those belonging to different groups are different. Generally, in practice, the interest of experimenter is more in using the multiple comparison test for treatment effects rather on the block effects. So the multiple comparison test are used generally for the treatment effects only.

Two-way classification with interactions:

Consider the two-way classification with an equal number, say K observations per cell. Let y_{ijk} : k^{th} observation in $(i, j)^{\text{th}}$ cell, i.e., receiving the treatments i^{th} level of factor A and j^{th} level of factor B , $i = 1, 2, \dots, I$; $j = 1, 2, \dots, J$; $k = 1, 2, \dots, K$ and

y_{ijk} are independently drawn from $N(\mu_{ij}, \sigma^2)$ so that the linear model under consideration is

$$y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$$

where ε_{ijk} are identically and independently distributed following $N(0, \sigma^2)$. Thus

$$\begin{aligned} E(y_{ijk}) &= \mu_{ij} \\ &= \mu_{oo} + (\mu_{io} - \mu_{oo}) + (\mu_{oj} - \mu_{oo}) + (\mu_{ij} - \mu_{io} - \mu_{oj} + \mu_{oo}) \\ &= \mu + \alpha_i + \beta_j + \gamma_{ij} \end{aligned}$$

where

$$\mu = \mu_{oo}$$

$$\alpha_i = \mu_{io} - \mu_{oo}$$

$$\beta_j = \mu_{oj} - \mu_{oo}$$

$$\gamma_{ij} = \mu_{ij} - \mu_{io} - \mu_{oj} + \mu_{oo}$$

with

$$\sum_{i=1}^I \alpha_i = 0, \sum_{j=1}^J \beta_j = 0, \sum_{i=1}^I \gamma_{ij} = 0, \sum_{j=1}^J \gamma_{ij} = 0.$$

Assume that the design matrix X is of full rank so that all the parametric functions of μ_{ij} are estimable.

The null and the corresponding alternative hypotheses are

$$H_{0\alpha} : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$$

$$H_{1\alpha} : \text{At least one } \alpha_i \neq \alpha_j, \text{ for } i \neq j,$$

$$H_{0\beta} : \beta_1 = \beta_2 = \dots = \beta_J = 0$$

$$H_{1\beta} : \text{At least one } \beta_i \neq \beta_j, \text{ for } i \neq j$$

and

$$H_{0\gamma} : \text{All } \gamma_{ij} = 0 \text{ for all } i, j.$$

$$H_{1\gamma} : \text{At least one } \gamma_{ij} \neq \gamma_{ik}, \text{ for } j \neq k.$$

Minimizing the error sum of squares

$$E = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \alpha_i - \beta_j - \gamma_{ij})^2,$$

The normal equations are obtained as

$$\frac{\partial E}{\partial \mu} = 0, \quad \frac{\partial E}{\partial \alpha_i} = 0 \text{ for all } i, \quad \frac{\partial E}{\partial \beta_j} = 0 \text{ for all } j \text{ and } \frac{\partial E}{\partial \gamma_{ij}} = 0 \text{ for all } i \text{ and } j$$

The least-squares estimates are obtained as

$$\hat{\mu} = \bar{y}_{ooo} = \frac{1}{IJK} \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K y_{ijk}$$

$$\hat{\alpha}_i = \bar{y}_{ioo} - \bar{y}_{ooo} = \frac{1}{JK} \sum_{j=1}^J \sum_{k=1}^K y_{ijk} - \bar{y}_{ooo}$$

$$\hat{\beta}_j = \bar{y}_{ojo} - \bar{y}_{ooo} = \frac{1}{IK} \sum_{i=1}^I \sum_{k=1}^K y_{ijk} - \bar{y}_{ooo}$$

$$\hat{\gamma}_{ij} = \bar{y}_{ijo} - \bar{y}_{ioo} - \bar{y}_{ojo} + \bar{y}_{ooo} = \frac{1}{K} \sum_{k=1}^K y_{ijk} - \bar{y}_{ioo} - \bar{y}_{ojo} + \bar{y}_{ooo}.$$

The error sum of squares is

$$\begin{aligned} SSE &= \min_{\hat{\mu}, \hat{\alpha}_i, \hat{\beta}_j, \hat{\gamma}_{ij}} \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \alpha_i - \beta_j - \gamma_{ij})^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j - \hat{\gamma}_{ij})^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \bar{y}_{ijo})^2 \end{aligned}$$

$$\text{with } \frac{SSE}{\sigma^2} \sim \chi^2(IJ(K-1)).$$

Now minimizing the error sum of squares under $H_{0\alpha} = \alpha_1 = \alpha_2 = \dots = \alpha_I = 0$, i.e., minimizing

$$E_1 = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \beta_j - \gamma_{ij})^2 \text{ with respect to } \mu, \beta_j \text{ and } \gamma_{ij} \text{ and solving the normal equations}$$

$$\frac{\partial E_1}{\partial \mu} = 0, \quad \frac{\partial E_1}{\partial \beta_j} = 0 \text{ for all } j \text{ and } \frac{\partial E_1}{\partial \gamma_{ij}} = 0 \text{ for all } i \text{ and } j$$

gives the least squares estimates as

$$\begin{aligned}\hat{\mu} &= \hat{y}_{ooo} \\ \hat{\beta}_j &= \bar{y}_{oj0} - \bar{y}_{ooo} \\ \hat{\gamma}_{ij} &= \bar{y}_{ijo} - \bar{y}_{ooo} - \bar{y}_{oj0} + \bar{y}_{ooo} = \bar{y}_{ijo} - \bar{y}_{oj0}.\end{aligned}$$

The sum of squares due to $H_{0\alpha}$, is

$$\begin{aligned}\text{Min}_{\mu, \beta_j, \gamma_{ij}} \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \beta_j - \gamma_{ij})^2 \\ = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \hat{\mu} - \hat{\beta}_j - \hat{\gamma}_{ij})^2 \\ = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \bar{y}_{ijo})^2 + JK \sum_{i=1}^I (\bar{y}_{ioo} - \bar{y}_{ooo})^2 \\ = SSE + JK \sum_{i=1}^I (\bar{y}_{ioo} - \bar{y}_{ooo})^2.\end{aligned}$$

Thus the sum of squares due to deviation from $H_{0\alpha}$ or the sum of squares due to effect A is

$$SSA = \text{Sum of squares due to } H_{0\alpha} - SSE = JK \sum_{i=1}^I (\bar{y}_{ioo} - \bar{y}_{ooo})^2$$

$$\text{with } \frac{SSA}{\sigma^2} \sim \chi^2(I-1).$$

Minimizing the error sum of squares under $H_{0\beta} : \beta_1 = \beta_2 = \dots = \beta_J = 0$, i.e., minimizing

$$E_2 = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \alpha_i - \gamma_{ij})^2,$$

and solving the normal equations

$$\frac{\partial E_2}{\partial \mu} = 0, \quad \frac{\partial E_2}{\partial \alpha_i} = 0 \text{ for all } i \text{ and } \frac{\partial E_2}{\partial \gamma_{ij}} = 0 \text{ for all } i \text{ and } j$$

yields the least-squares estimators as

$$\begin{aligned}\hat{\mu} &= \bar{y}_{ooo} \\ \hat{\alpha}_i &= \bar{y}_{ioo} - \bar{y}_{ooo} \\ \hat{\gamma}_{ij} &= \bar{y}_{ijo} - \bar{y}_{ooo} - \bar{y}_{ioo} + \bar{y}_{ooo} = \bar{y}_{ijo} - \bar{y}_{ioo}.\end{aligned}$$

The minimum error sum of squares is

$$\begin{aligned} & \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\gamma}_{ij})^2 \\ &= SSE + IK \sum_{j=1}^J (\bar{y}_{ojo} - \bar{y}_{ooo})^2 \end{aligned}$$

and the sum of squares due to deviation from $H_{0\beta}$ or the sum of squares due to effect B is

$$SSB = \text{Sum of squares due to } H_{0\beta} - SSE = IK \sum_{j=1}^J (\bar{y}_{ojo} - \bar{y}_{ooo})^2$$

$$\text{with } \frac{SSB}{\sigma^2} \sim \chi^2(J-1).$$

Next, minimizing the error sum of squares under $H_{0\gamma}$: all $\gamma_{ij} = 0$ for all i, j , i.e., minimizing

$$E_3 = \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \alpha_i - \beta_j)^2$$

with respect to μ, α_i and β_j and solving the normal equations

$$\frac{\partial E_3}{\partial \mu} = 0, \quad \frac{\partial E_3}{\partial \alpha_i} = 0 \text{ for all } i \text{ and } \frac{\partial E_3}{\partial \beta_j} = 0 \text{ for all } j$$

yields the least-squares estimators

$$\begin{aligned} \hat{\mu} &= \bar{y}_{ooo} \\ \hat{\alpha}_i &= \bar{y}_{ioo} - \bar{y}_{ooo} \\ \hat{\beta}_j &= \bar{y}_{ojo} - \bar{y}_{ooo}. \end{aligned}$$

The sum of squares due to $H_{0\gamma}$ is

$$\begin{aligned} & \text{Min}_{\mu, \alpha_i, \beta_j} \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \mu - \alpha_i - \beta_j)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j)^2 \\ &= SSE + K \sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{ijo} - \bar{y}_{ioo} - \bar{y}_{ojo} + \bar{y}_{ooo})^2. \end{aligned}$$

Thus the sum of squares due to deviation from $H_{0\gamma}$ or the sum of squares due to the interaction effect AB is

$$SSAB = \text{Sum of squares due to } H_{0\gamma} - SSE = K \sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{ijo} - \bar{y}_{ioo} - \bar{y}_{ojo} + \bar{y}_{ooo})^2$$

$$\text{with } \frac{SSAB}{\sigma^2} \sim \chi^2((I-1)J-1).$$

The total sum of squares can be partitioned as

$$TSS = SSA + SSB + SSAB + SSE$$

where SSA , SSB , $SSAB$ and SSE are mutually orthogonal. So either using the independence of SSA , SSB , $SSAB$ and SSE as well as their respective χ^2 – distributions or using the likelihood ratio test approach, the decision rules for the null hypothesis at α level of significance are based on F -statistic as follows

$$F_1 = \frac{IJ(K-1)}{I-1} \cdot \frac{SSA}{SSE} \sim F[(I-1), IJ(K-1)] \text{ under } H_{0\alpha},$$

$$F_2 = \frac{IJ(K-1)}{J-1} \cdot \frac{SSB}{SSE} \sim F[(J-1), IJ(K-1)] \text{ under } H_{0\beta},$$

and

$$F_3 = \frac{IJ(K-1)}{(I-1)(J-1)} \cdot \frac{SSAB}{SSE} \sim F[(I-1)(J-1), IJ(K-1)] \text{ under } H_{0\gamma}.$$

So

Reject $H_{0\alpha}$ if $F_1 > F_{1-\alpha}[(I-1), IJ(K-1)]$

Reject $H_{0\beta}$ if $F_2 > F_{1-\alpha}[(J-1), IJ(K-1)]$

Reject $H_{0\gamma}$ if $F_3 > F_{1-\alpha}[(I-1)(J-1), IJ(K-1)]$.

If $H_{0\alpha}$ or $H_{0\beta}$ is rejected, one can use t -test or multiple comparison test to find which pairs of α_i 's or β_j 's are significantly different.

If $H_{0\gamma}$ is rejected, one would not usually explore it further but theoretically, t -test or multiple comparison tests can be used.

It can also be shown that

$$E(MSA) = \sigma^2 + \frac{JK}{I-1} \sum_{i=1}^I \alpha_i^2$$

$$E(MSB) = \sigma^2 + \frac{IK}{J-1} \sum_{j=1}^J \beta_j^2$$

$$E(MSAB) = \sigma^2 + \frac{K}{(I-1)(J-1)} \sum_{i=1}^I \sum_{j=1}^J \gamma_{ij}^2$$

$$E(MSE) = \sigma^2.$$

The analysis of variance table is as follows:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F-value
Factor A	$(I - 1)$	SSA	$MSA = \frac{SSA}{I - 1}$	$F_1 = \frac{MSA}{MSE}$
Factor B	$(J - 1)$	SSB	$MSB = \frac{SSB}{J - 1}$	$F_2 = \frac{MSB}{MSE}$
Interaction AB	$(I - 1)(J - 1)$	SSAB	$MSAB = \frac{SSAB}{(I - 1)(J - 1)}$	$F_3 = \frac{MSAB}{MSE}$
Error	$IJ(K - 1)$	SSE	$MSE = \frac{SSE}{IJ(K - 1)}$	
Total	$(IJK - 1)$	TSS		

Tukey's test for nonadditivity:

Consider the set up of two way classification with one observation per cell and interaction as

$$y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ij}, \quad i = 1, 2, \dots, I, j = 1, 2, \dots, J \quad \text{with} \quad \sum_{i=1}^I \alpha_i = 0, \sum_{j=1}^J \beta_j = 0.$$

The distribution of degrees of freedom in this case is as follows:

Source	Degrees of freedom
A	$I - 1$
B	$J - 1$
AB(interaction)	$(I - 1)(J - 1)$
Error	0
Total	$IJ - 1$

There is no degree of freedom for error. The problem is that the two factor interaction effect and random error component are subsumed together and cannot be separated out. There is no estimate for σ^2 .

If no interaction exists, then $H_0 : \gamma_{ij} = 0$ for all i, j is accepted and the additive model

$$y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$$

is well enough to test the hypothesis $H_0 : \alpha_i = 0$ and $H_0 : \beta_j = 0$ with error having $(I-1)(J-1)$ degrees of freedom.

If interaction exists, then $H_0 : \gamma_{ij} = 0$ is rejected. In such a case, if we assume that the structure of the interaction effect is such that it is proportional to the product of individual effects, i.e.,

$$\gamma_{ij} = \lambda \alpha_i \beta_j$$

then a test for testing $H_0 : \lambda = 0$ can be constructed. Such a test will serve as a test for nonadditivity.

It will help in knowing the effect of the presence of interact effect and whether the interaction enters into the model additively. Such a test is given by Tukey's test for nonadditivity which requires one degree of freedom leaving $(I-1)(J-1)-1$ degrees of freedom for error.

Let us assume that departure from additivity can be specified by introducing a product term and writing the model as

$$E(y_{ij}) = \mu + \alpha_i + \beta_j + \lambda \alpha_i \beta_j; i = 1, 2, \dots, I, j = 1, 2, \dots, J \text{ with } \sum_{i=1}^I \alpha_i = 0, \sum_{j=1}^J \beta_j = 0.$$

When $\lambda \neq 0$, the model becomes a nonlinear model and the least-squares theory for linear models is not applicable.

Note that using $\sum_{i=1}^I \alpha_i = 0, \sum_{j=1}^J \beta_j = 0$, we have

$$\begin{aligned} \bar{y}_{oo} &= \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J y_{ij} = \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J [\mu + \alpha_i + \beta_j + \lambda \alpha_i \beta_j + \varepsilon_{ij}] \\ &= \mu + \frac{1}{I} \sum_{i=1}^I \alpha_i + \frac{1}{J} \sum_{j=1}^J \beta_j + \frac{\lambda}{IJ} \left(\sum_{i=1}^I \alpha_i \right) \left(\sum_{j=1}^J \beta_j \right) + \bar{\varepsilon}_{oo} \\ &= \mu + \bar{\varepsilon}_{oo} \end{aligned}$$

$$E(\bar{y}_{oo}) = \mu$$

$$\Rightarrow \hat{\mu} = \bar{y}_{oo}.$$

Next

$$\begin{aligned}
 \bar{y}_{io} &= \frac{1}{J} \sum_{j=1}^J y_{ij} = \frac{1}{J} \sum_{j=1}^J [\mu + \alpha_i + \beta_j + \lambda \alpha_i \beta_j + \varepsilon_{ij}] \\
 &= \mu + \alpha_i + \frac{1}{J} \sum_{j=1}^J \beta_j + \lambda \alpha_i \frac{1}{J} \sum_{j=1}^J \beta_j + \bar{\varepsilon}_{io} \\
 &= \mu + \alpha_i + \bar{\varepsilon}_{io}
 \end{aligned}$$

$$\begin{aligned}
 E(\bar{y}_{io}) &= \mu + \alpha_i \\
 \Rightarrow \hat{\alpha}_i &= \bar{y}_{io} - \hat{\mu} = \bar{y}_{io} - \bar{y}_{oo}.
 \end{aligned}$$

Similarly

$$\begin{aligned}
 \bar{y}_{oj} &= \mu + \beta_j \\
 \Rightarrow \hat{\beta}_j &= \bar{y}_{oj} - \hat{\mu} = \bar{y}_{oj} - \bar{y}_{oo}
 \end{aligned}$$

Thus $\hat{\mu}, \hat{\alpha}_i$ and $\hat{\beta}_j$ remain the unbiased estimators of μ, α_i and β_j , respectively irrespective of whether $\lambda = 0$ or not.

Also

$$\begin{aligned}
 E[y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo}] &= \lambda \alpha_i \beta_j \\
 \text{or} \\
 E[(y_{ij} - \bar{y}_{oo}) - (\bar{y}_{io} - \bar{y}_{oo}) - (\bar{y}_{oj} - \bar{y}_{oo})] &= \lambda \alpha_i \beta_j.
 \end{aligned}$$

Consider the estimation of μ, α_i, β_j and λ based on the minimization of

$$\begin{aligned}
 S &= \sum_i \sum_j (y_{ij} - \mu - \alpha_i - \beta_j - \lambda \alpha_i \beta_j)^2 \\
 &= \sum_i \sum_j S_{ij}^2.
 \end{aligned}$$

The normal equations are solved as

$$\frac{\partial S}{\partial \mu} = 0 \Rightarrow \sum_{i=1}^I \sum_{j=1}^J S_{ij} = 0$$

$$\Rightarrow \hat{\mu} = \bar{y}_{oo}$$

$$\frac{\partial S}{\partial \alpha_i} = 0 \Rightarrow \sum_{j=1}^J (1 + \lambda \beta_j) S_{ij} = 0$$

$$\frac{\partial S}{\partial \beta_j} = 0 \Rightarrow \sum_{i=1}^I (1 + \lambda \alpha_i) S_{ij} = 0$$

$$\frac{\partial S}{\partial \lambda} = 0 \Rightarrow \sum_{i=1}^I \sum_{j=1}^J \alpha_i \beta_j S_{ij} = 0$$

$$\text{or } \sum_{i=1}^I \sum_{j=1}^J \alpha_i \beta_j (y_{ij} - \mu - \alpha_i - \beta_j - \lambda \alpha_i \beta_j) = 0$$

$$\text{or } \lambda = \frac{\sum_{i=1}^I \sum_{j=1}^J \alpha_i \beta_j y_{ij}}{\left(\sum_{i=1}^I \alpha_i^2 \right) \left(\sum_{j=1}^J \beta_j^2 \right)} = \tilde{\lambda} \text{ (say)}$$

which can be estimated provided α_i and β_j are assumed to be known.

Since α_i and β_j can be estimated by $\hat{\alpha}_i = \bar{y}_{io} - \bar{y}_{oo}$ and $\hat{\beta}_j = \bar{y}_{oj} - \bar{y}_{oo}$ irrespective of whether $\lambda = 0$

or not, so we can substitute them in place of α_i and β_j in $\tilde{\lambda}$ which gives

$$\hat{\lambda} = \frac{\sum_{i=1}^I \sum_{j=1}^J \hat{\alpha}_i \hat{\beta}_j y_{ij}}{\left(\sum_{i=1}^I \hat{\alpha}_i^2 \right) \left(\sum_{j=1}^J \hat{\beta}_j^2 \right)} = \frac{(IJ) \sum_{i=1}^I \sum_{j=1}^J \hat{\alpha}_i \hat{\beta}_j y_{ij}}{\left(J \sum_{i=1}^I \hat{\alpha}_i^2 \right) \left(I \sum_{j=1}^J \hat{\beta}_j^2 \right)}$$

$$= \frac{IJ \sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{io} - \bar{y}_{oo})(\bar{y}_{oj} - \bar{y}_{oo}) y_{ij}}{S_A S_B}$$

$$\text{where } S_A = J \sum_{i=1}^I \hat{\alpha}_i^2 = J \sum_{i=1}^I (\bar{y}_{io} - \bar{y}_{oo})^2$$

$$S_B = I \sum_{j=1}^J \hat{\beta}_j^2 = I \sum_{j=1}^J (\bar{y}_{oj} - \bar{y}_{oo})^2.$$

Assuming α_i and β_j to be known

$$\begin{aligned} \text{Var}(\tilde{\lambda}) &= \left(\frac{1}{\sum_{i=1}^I \alpha_i^2 \sum_{j=1}^J \beta_j^2} \right)^2 \left[\sum_{i=1}^I \sum_{j=1}^J \alpha_i^2 \beta_j^2 \text{Var}(y_{ij}) + 0 \right] \\ &= \frac{\sigma^2 \left(\sum_{i=1}^I \alpha_i^2 \right) \left(\sum_{j=1}^J \beta_j^2 \right)}{\left(\sum_{i=1}^I \alpha_i^2 \right)^2 \left(\sum_{j=1}^J \beta_j^2 \right)^2} \\ &= \frac{\sigma^2}{\left(\sum_{i=1}^I \alpha_i^2 \right) \left(\sum_{j=1}^J \beta_j^2 \right)} \end{aligned}$$

using $\text{Var}(y_{ij}) = \sigma^2$, $\text{Cov}(y_{ij}, y_{jk}) = 0$ for all $i \neq k$.

When α_i and β_j are estimated by $\hat{\alpha}_i$ and $\hat{\beta}_j$, then substitute them back in the expression of $\text{Var}(\tilde{\lambda})$ and treating it as $\text{Var}(\hat{\lambda})$ gives

$$\begin{aligned} \text{Var}(\hat{\lambda}) &= \frac{\sigma^2}{\left(\sum_{i=1}^I \hat{\alpha}_i^2 \right) \left(\sum_{j=1}^J \hat{\beta}_j^2 \right)} \\ &= \frac{IJ\sigma^2}{S_A S_B} \end{aligned}$$

for given $\hat{\alpha}_i$ and $\hat{\beta}_j$.

Note that if $\lambda = 0$, then

$$\begin{aligned} E\left[\hat{\lambda} / \hat{\alpha}_i, \hat{\beta}_j \text{ for all } i, j\right] &= E\left[\frac{\sum_{i=1}^I \sum_{j=1}^J \alpha_i \beta_j y_{ij}}{\sum_{i=1}^I \alpha_i^2 \sum_{j=1}^J \beta_j^2} \right] \\ &= E\left[\frac{\sum_{i=1}^I \sum_{j=1}^J \alpha_i \beta_j (\mu + \alpha_i + \beta_j + 0 + \varepsilon_{ij})}{\left(\sum_{i=1}^I \alpha_i^2 \right) \left(\sum_{j=1}^J \beta_j^2 \right)} \right] \\ &= \frac{0}{\left(\sum_{i=1}^I \alpha_i^2 \right) \left(\sum_{j=1}^J \beta_j^2 \right)} = 0. \end{aligned}$$

As $\hat{\alpha}_i$ and $\hat{\beta}_j$ remains valid irrespective of $\lambda = 0$ or not, in this sense $\hat{\lambda}$ is a function of y_{ij} and hence normally distributed as

$$\hat{\lambda} \sim N\left(0, \frac{IJ\sigma^2}{S_A S_B}\right).$$

Thus the statistic

$$\begin{aligned} \frac{(\hat{\lambda})^2}{\text{Var}(\hat{\lambda})} &= \frac{IJ \left[\sum_{i=1}^I \sum_{j=1}^J \hat{\alpha}_i \hat{\beta}_j y_{ij} \right]^2}{\sigma^2 S_A S_B} \\ &= \frac{IJ \left[\sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{io} - \bar{y}_{oo})(\bar{y}_{oj} - \bar{y}_{oo}) y_{ij} \right]^2}{\sigma^2 S_A S_B} \\ &= \frac{IJ \left[\sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{io} - \bar{y}_{oo})(\bar{y}_{oj} - \bar{y}_{oo})(y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo}) \right]^2}{\sigma^2 S_A S_B} \\ &= \frac{S_N}{\sigma^2} \end{aligned}$$

follows a χ^2 - distribution with one degree of freedom where

$$S_N = \frac{IJ \left[\sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{io} - \bar{y}_{oo})(\bar{y}_{oj} - \bar{y}_{oo})(y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo}) \right]^2}{S_A S_B}$$

is the sum of squares due to nonadditivity.

Note that

$$\frac{S_{AB}}{\sigma^2} = \frac{\sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2}{\sigma^2}$$

follows $\chi^2((I-1)(J-1))$.

so $\left(\frac{S_N}{\sigma^2} - \frac{S_{AB}}{\sigma^2}\right)$ is nonnegative and follows $\chi^2[(I-1)(J-1)-1]$.

The reason for this is as follows:

$$y_{ij} = \mu + \alpha_i + \beta_j + \text{non additivity} + \varepsilon_{ij}$$

and so

$$TSS = SSA + SSB + S_N + SSE$$

$$\Rightarrow SSE = TSS - SSA - SSB - S_N$$

has degrees of freedom

$$= (IJ - 1) - (I - 1) - (J - 1) - 1$$

$$= (I - 1)(J - 1) - 1$$

We need to ensure that $SSE > 0$. So using the result

“If Q, Q_1 and Q_2 are quadratic forms such that

$$Q = Q_1 + Q_2 \text{ with } Q \sim \chi^2(a), Q_2 \sim \chi^2(b) \text{ and } Q_2 \text{ is non-negative, then}$$

$$Q_1 \sim \chi^2(a - b)''$$

ensures that the difference

$$\frac{S_N}{\sigma^2} - \frac{SAB}{\sigma^2}$$

is nonnegative.

Moreover S_N (SS due to nonadditivity) and SSE are orthogonal. Thus the F -test for nonadditivity is

$$F = \frac{\left(\frac{S_N / \sigma^2}{1} \right)}{\left(\frac{SSE / \sigma^2}{(I - 1)(J - 1) - 1} \right)}$$

$$= [(I - 1)(J - 1) - 1] \frac{SSN}{SSE}$$

$$\sim F[1, (I - 1)(J - 1) - 1] \text{ under } H_0.$$

So the decision rule is

Reject $H_0 : \lambda = 0$ whenever

$$F > F_{1-\alpha}[1, (I - 1)(J - 1) - 1]$$

The analysis of variance table for the model including a term for nonadditivity is as follows:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	<i>F</i> -value
<i>A</i>	<i>I</i> – 1	<i>S_A</i>	$MS_A = \frac{S_A}{I-1}$	
<i>B</i>	<i>J</i> – 1	<i>S_B</i>	$MS_B = \frac{S_B}{J-1}$	
Nonadditivity	1	<i>S_N</i>	$MS_N = S_N$	$\frac{MS_N}{MSE}$
Error	(<i>I</i> – 1)(<i>J</i> – 1) – 1 (By subtraction)	<i>SSE</i>	$MSE = \frac{SSE}{(I-1)(J-1)-1}$	
Total	<i>IJ</i> – 1	<i>TSS</i>		

Comparison of Variances

One of the basic assumptions in the analysis of variance is that the samples are drawn from different normal populations with different means but the same variances. So before going for analysis of variance, the test of hypothesis about the equality of variance is needed to be done.

We discuss the test of equality of two variances and more than two variances.

Case 1: Equality of two variances

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma^2.$$

Suppose there are two independent random samples

$$A : x_1, x_2, \dots, x_{n_1} ; x_i \sim N(\mu_A, \sigma_A^2)$$

$$B : y_1, y_2, \dots, y_{n_2} ; y_i \sim N(\mu_B, \sigma_B^2)$$

The sample variance corresponding to the two samples are

$$s_x^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_i - \bar{x})^2$$

$$s_y^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - \bar{y})^2.$$

Under $H_0 : \sigma_A^2 = \sigma_B^2 = \sigma^2$,

$$\frac{(n_1-1)s_x^2}{\sigma^2} \sim \chi^2(n_1-1)$$

$$\frac{(n_2-1)s_y^2}{\sigma^2} \sim \chi^2(n_2-1).$$

Moreover, the sample variances s_x^2 and s_y^2 are independent. So

$$\left(\frac{\left(\frac{(n_1-1)s_x^2}{\sigma^2} \right)}{\frac{n_1-1}{\left(\frac{(n_2-1)s_y^2}{\sigma^2} \right)}} \right) = \frac{s_x^2}{s_y^2} \sim F_{n_1-1, n_2-1}.$$

So for testing $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_1 : \sigma_1^2 \neq \sigma_2^2$, the null hypothesis H_0 is rejected if

$$F > F_{1-\frac{\alpha}{2}; n_1-1, n_2-1} \quad \text{or} \quad F < F_{\frac{\alpha}{2}; n_1-1, n_2-1} \quad \text{where} \quad F_{\frac{\alpha}{2}; n_1-1, n_2-1} = \frac{1}{F_{1-\frac{\alpha}{2}; n_2-1, n_1-1}}.$$

If the null hypothesis $H_0 : \sigma_A^2 = \sigma_B^2$ is rejected, then the problem is termed as the Fisher-Behren's problem. The solutions are available for this problem.

Case 2: Equality of more than two variances: Bartlett's test

$H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$ and $H_1 : \sigma_i^2 \neq \sigma_j^2$ for atleast one $i \neq j = 1, 2, \dots, k$.

Let there be k independent normal population $N(\mu_i, \sigma_i^2)$ each of size $n_i, i = 1, 2, \dots, k$. Let

$s_1^2, s_2^2, \dots, s_k^2$ be k independent unbiased estimators of population variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_k^2$ respectively

with $\nu_1, \nu_2, \dots, \nu_k$ degrees of freedom. Under H_0 , all the variances are the same as σ^2 , say and an

unbiased estimate of σ^2 is $s^2 = \sum_{i=1}^k \frac{\nu_i s_i^2}{\nu}$ where $\nu_i = n_i - 1, \nu = \sum_{i=1}^k \nu_i$.

Bartlett has shown that under H_0

$$\frac{\sum_{i=1}^k \left(\nu_i \ln \frac{s_i^2}{s^2} \right)}{\left[1 + \frac{1}{3(k-1)} \left\{ \sum_{i=1}^k \left(\frac{1}{\nu_i} \right) - \frac{1}{\nu} \right\} \right]}$$

is distributed as $\chi^2(k-1)$ based on which H_0 can be tested.

Chapter 4

Experimental Designs and Their Analysis

Design of experiment means how to design an experiment in the sense that how the observations or measurements should be obtained to answer a query in a valid, efficient and economical way. The designing of the experiment and the analysis of obtained data are inseparable. If the experiment is designed properly keeping in mind the question, then the data generated is valid and proper analysis of data provides the valid statistical inferences. If the experiment is not well designed, the validity of the statistical inferences is questionable and may be invalid.

It is important to understand first the basic terminologies used in the experimental design.

Experimental unit:

For conducting an experiment, the experimental material is divided into smaller parts and each part is referred to as an experimental unit. The experimental unit is randomly assigned to treatment is the experimental unit. The phrase “randomly assigned” is very important in this definition.

Experiment:

A way of getting an answer to a question which the experimenter wants to know.

Treatment

Different objects or procedures which are to be compared in an experiment are called treatments.

Sampling unit:

The object that is measured in an experiment is called the sampling unit. This may be different from the experimental unit.

Factor:

A factor is a variable defining a categorization. A factor can be fixed or random in nature. A factor is termed as a fixed factor if all the levels of interest are included in the experiment.

A factor is termed as a random factor if all the levels of interest are not included in the experiment and those that are can be considered to be randomly chosen from all the levels of interest.

Replication:

It is the repetition of the experimental situation by replicating the experimental unit.

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Experimental error:

The unexplained random part of the variation in any experiment is termed as experimental error. An estimate of experimental error can be obtained by replication.

Treatment design:

A treatment design is the manner in which the levels of treatments are arranged in an experiment.

Example: (Ref.: Statistical Design, G. Casella, Chapman and Hall, 2008)

Suppose some varieties of fish food is to be investigated on some species of fishes. The food is placed in the water tanks containing the fishes. The response is the increase in the weight of fish. The experimental unit is the tank, as the treatment is applied to the tank, not to the fish. Note that if the experimenter had taken the fish in hand and placed the food in the mouth of fish, then the fish would have been the experimental unit as long as each of the fish got an independent scoop of food.

Design of experiment:

One of the main objectives of designing an experiment is how to verify the hypothesis in an efficient and economical way. In the context of the null hypothesis of equality of several means of normal populations having the same variances, the analysis of variance technique can be used. Note that such techniques are based on certain statistical assumptions. If these assumptions are violated, the outcome of the test of a hypothesis then may also be faulty and the analysis of data may be meaningless. So the main question is how to obtain the data such that the assumptions are met and the data is readily available for the application of tools like analysis of variance. The designing of such a mechanism to obtain such data is achieved by the design of the experiment. After obtaining the sufficient experimental unit, the treatments are allocated to the experimental units in a random fashion. Design of experiment provides a method by which the treatments are placed at random on the experimental units in such a way that the responses are estimated with the utmost precision possible.

Principles of experimental design:

There are three basic principles of design which were developed by Sir Ronald A. Fisher.

- (i) Randomization
- (ii) Replication
- (iii) Local control

(i) Randomization

The principle of randomization involves the allocation of treatment to experimental units at random to avoid any bias in the experiment resulting from the influence of some extraneous unknown factor that may affect the experiment. In the development of analysis of variance, we assume that the errors are random and independent. In turn, the observations also become random. The principle of randomization ensures this.

The random assignment of experimental units to treatments results in the following outcomes.

- a) It eliminates systematic bias.
- b) It is needed to obtain a representative sample from the population.
- c) It helps in distributing the unknown variation due to confounded variables throughout the experiment and breaks the confounding influence.

Randomization forms a basis of a valid experiment but replication is also needed for the validity of the experiment.

If the randomization process is such that every experimental unit has an equal chance of receiving each treatment, it is called **complete randomization**.

(ii) Replication:

In the replication principle, any treatment is repeated a number of times to obtain a valid and more reliable estimate than which is possible with one observation only. Replication provides an efficient way of increasing the precision of an experiment. The precision increases with the increase in the number of observations. Replication provides more observations when the same treatment is used, so it increases precision. For example, if the variance of x is σ^2 then variance of the sample mean $\bar{x}$ based on n observation is $\frac{\sigma^2}{n}$. So as n increases, $Var(\bar{x})$ decreases.

(ii) Local control (error control)

The replication is used with local control to reduce the experimental error. For example, if the experimental units are divided into different groups such that they are homogeneous within the blocks, then the variation among the blocks is eliminated and ideally, the error component will contain the variation due to the treatments only. This will, in turn, increase the efficiency.

Complete and incomplete block designs:

In most of the experiments, the available experimental units are grouped into blocks having more or less identical characteristics to remove the blocking effect from the experimental error. Such design is termed as **block designs**.

The number of experimental units in a block is called the **block size**.

If

size of block = number of treatments

and

each treatment in each block is randomly allocated,

then it is a **full replication** and the design is called a **complete block design**.

In case, the number of treatments is so large that a full replication in each block makes it too heterogeneous with respect to the characteristic under study, then smaller but homogeneous blocks can be used. In such a case, the blocks do not contain a full replicate of the treatments. Experimental designs with blocks containing an incomplete replication of the treatments are called **incomplete block designs**.

Completely randomized design (CRD)

The CRD is the simplest design. Suppose there are v treatments to be compared.

- All experimental units are considered the same and no division or grouping among them exist.
- In CRD, the v treatments are allocated randomly to the whole set of experimental units, without making any effort to group the experimental units in any way for more homogeneity.
- Design is entirely flexible in the sense that any number of treatments or replications may be used.
- The number of replications for different treatments need not be equal and may vary from treatment to treatment depending on the knowledge (if any) on the variability of the observations on individual treatments as well as on the accuracy required for the estimate of individual treatment effect.

Example: Suppose there are 4 treatments and 20 experimental units, then

- the treatment 1 is replicated, say 3 times and is given to 3 experimental units,
- the treatment 2 is replicated, say 5 times and is given to 5 experimental units,
- the treatment 3 is replicated, say 6 times and is given to 6 experimental units

and

- finally, the treatment 4 is replicated $[20-(6+5+3)]=6$ times and is given to the remaining 6 experimental units.

- All the variability among the experimental units goes into experimented error.
- CRD is used when the experimental material is homogeneous.
- CRD is often inefficient.
- CRD is more useful when the experiments are conducted inside the lab.
- CRD is well suited for the small number of treatments and for the homogeneous experimental material.

Layout of CRD

Following steps are needed to design a CRD:

- Divide the entire experimental material or area into a number of experimental units, say n .
- Fix the number of replications for different treatments in advance (for given total number of available experimental units).
- No local control measure is provided as such except that the error variance can be reduced by choosing a homogeneous set of experimental units.

Procedure

Let the v treatments are numbered from $1, 2, \dots, v$ and n_i be the number of replications required for i^{th} treatment such that $\sum_{i=1}^v n_i = n$.

- Select n_1 units out of n units randomly and apply treatment 1 to these n_1 units.
(**Note:** This is how the randomization principle is utilized in CRD.)
- Select n_2 units out of $(n - n_1)$ units randomly and apply treatment 2 to these n_2 units.
- Continue with this procedure until all the treatments have been utilized.
- Generally, the equal number of treatments are allocated to all the experimental units unless no practical limitation dictates or some treatments are more variable or/and of more interest.

Analysis

There is only one factor which is affecting the outcome – treatment effect. So the set-up of one-way analysis of variance is to be used.

y_{ij} : Individual measurement of j^{th} experimental units for i^{th} treatment $i = 1, 2, \dots, v$, $j = 1, 2, \dots, n_i$.

y_{ij} : Independently distributed following $N(\mu + \alpha_i, \sigma^2)$ with $\sum_{i=1}^v n_i \alpha_i = 0$.

μ : overall mean

α_i : i^{th} treatment effect

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_v = 0$$

$$H_1 : \text{All } \alpha_i\text{'s are not equal.}$$

The data set is arranged as follows:

Treatments				
1	2	...	v	
y_{11}	y_{21}	...	y_{v1}	
y_{12}	y_{22}	...	y_{v2}	
$\vdots$	$\vdots$	$\ddots$	$\vdots$	
y_{1n_1}	y_{2n_2}	...	y_{vn_v}	
T_1	T_2	...	T_v	

where $T_i = \sum_{j=1}^{n_i} y_{ij}$ is the treatment total due to i^{th} effect,

$$G = \sum_{i=1}^v T_i = \sum_{i=1}^v \sum_{j=1}^{n_i} y_{ij} \quad \text{is the grand total of all the observations.}$$

In order to derive the test for H_0 , we can use either the likelihood ratio test or the principle of least squares. Since the likelihood ratio test has already been derived earlier, so we choose to demonstrate the use of the least-squares principle.

The linear model under consideration is

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2, \dots, v, j = 1, 2, \dots, n_i$$

where ε_{ij} 's are identically and independently distributed random errors with mean 0 and variance σ^2 . The normality assumption of ε 's is not needed for the estimation of parameters but will be needed for deriving the distribution of various involved statistics and in deriving the test statistics.

$$\text{Let } S = \sum_{i=1}^v \sum_{j=1}^{n_i} \varepsilon_{ij}^2 = \sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \mu - \alpha_i)^2.$$

Minimizing S with respect to μ and α_i , the normal equations are obtained as

$$\frac{\partial S}{\partial \mu} = 0 \Rightarrow n\mu + \sum_{i=1}^v n_i \alpha_i = 0$$

$$\frac{\partial S}{\partial \alpha_i} = 0 \Rightarrow n_i \mu + n_i \alpha_i = \sum_{j=1}^{n_i} y_{ij} \quad i = 1, 2, \dots, v.$$

Solving them using $\sum_{i=1}^v n_i \alpha_i = 0$, we get

$$\hat{\mu} = \bar{y}_{oo}$$

$$\hat{\alpha}_i = \bar{y}_{io} - \bar{y}_{oo}$$

where $\bar{y}_{io} = \frac{1}{n_i} \sum_{j=1}^{n_i} \bar{y}_{ij}$ is the mean of observation receiving the i^{th} treatment and $\bar{y}_{oo} = \frac{1}{n} \sum_{i=1}^v \sum_{j=1}^{n_i} \bar{y}_{ij}$ is the mean of all the observations.

The fitted model is obtained after substituting the estimate $\hat{\mu}$ and $\hat{\alpha}_i$ in the linear model. Using the fitted model, we can write

$$y_{ij} = \bar{y}_{oo} + (\bar{y}_{io} - \bar{y}_{oo}) + (y_{ij} - \bar{y}_{io})$$

$$\text{or } (y_{ij} - \bar{y}_{oo}) = (\bar{y}_{io} - \bar{y}_{oo}) + (y_{ij} - \bar{y}_{io}).$$

Squaring both sides and summing over all the observation, we have

$$\sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2 = \sum_{i=1}^v n_i (\bar{y}_{io} - \bar{y}_{oo})^2 + \sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2$$

↓

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$$\text{or } \left(\begin{array}{c} \text{Total sum} \\ \text{of squares} \end{array} \right) = \left(\begin{array}{c} \text{Sum of squares} \\ \text{due to treatment} \\ \text{effects} \end{array} \right) + \left(\begin{array}{c} \text{Sum of squares} \\ \text{due to error} \end{array} \right)$$

$$\text{or } TSS = SSTr + SSE$$

- Since $\sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo}) = 0$, so TSS is based on the sum of $(n-1)$ squared quantities. The TSS carries only $(n-1)$ degrees of freedom.

- Since $\sum_{i=1}^v n_i (\bar{y}_{io} - \bar{y}_{oo}) = 0$, so $SSTr$ is based only on the sum of $(v-1)$ squared quantities. The $SSTr$ carries only $(v-1)$ degrees of freedom.

- Since $\sum_{i=1}^{n_i} n_i (\bar{y}_{ij} - \bar{y}_{io}) = 0$ for all $i = 1, 2, \dots, v$, so SSE is based on the sum of squaring n quantities like $(y_{ij} - \bar{y}_{io})$ with v constraints $\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io}) = 0$, So SSE carries $(n-v)$ degrees of freedom.

- Using the Fisher-Cochran theorem,

$$TSS = SSTr + SSE$$

with degrees of freedom partitioned as

$$(n-1) = (v-1) + (n-v).$$

Moreover, equality in $TSS = SStr + SSE$ has to hold exactly. To ensure that the equality holds exactly, we find one of the sums of squares through subtraction. Generally, it is recommended to find SSE by subtraction as

$$SSE = TSS - SStr$$

$$\begin{aligned} TSS &= \sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{io})^2 \\ &= \sum_{i=1}^v \sum_{j=1}^{n_i} y_{ij}^2 - \frac{G^2}{n} \end{aligned}$$

where

$$G = \sum_{i=1}^v \sum_{j=1}^{n_i} y_{ij}.$$

$$\begin{aligned} SStr &= \sum_{i=1}^v n_i (\bar{y}_{io} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^v \left(\frac{T_i^2}{n_i} \right) - \frac{G^2}{n} \end{aligned}$$

$$\text{where } T_i = \sum_{j=1}^{n_i} y_{ij}$$

$$\frac{G^2}{n} : \text{correction factor.}$$

Now under $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_v = 0$, the model become

$$Y_{ij} = \mu + \varepsilon_{ij},$$

$$\text{and minimizing } S = \sum_{i=1}^v \sum_{j=1}^{n_i} \varepsilon_{ij}^2$$

with respect to μ gives

$$\frac{\partial S}{\partial \mu} = 0 \Rightarrow \hat{\mu} = \frac{G}{n} = \bar{y}_{oo}.$$

The SSE under H_0 becomes

$$SSE = \sum_{i=1}^v \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{oo})^2$$

and thus $TSS = SSE$. This TSS under H_0 contains the variation only due to the random error whereas the earlier $TSS = SStr + SSE$ contains the variation due to treatments and errors both. The difference between the two will provides the effect of treatments in terms of the sum of squares as

$$SSTr = \sum_{i=1}^v n_i (\bar{y}_i - \bar{y}_{oo})^2$$

Distributions and decision rules:

Using the normal distribution property of ε_{ij} 's, we find that y_{ij} 's are also normal as they are the linear combination of ε_{ij} 's.

- $-\frac{SSTr}{\sigma^2} \sim \chi^2(v-1)$ under H_0
- $-\frac{SSE}{\sigma^2} \sim \chi^2(n-v)$ under H_0
- $SSTr$ and SSE are independently distributed
- $\frac{MStr}{MSE} \sim F(v-1, n-v)$ under H_0 .
- Reject H_0 at α^* level of significance if $F > F_{\alpha^*; v-1, n-v}$.

[Note: We denote the level of significance here by α^* because α has been used for denoting the factor]

The analysis of variance table is as follows

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F
Between treatments	$v - 1$	$SSTr$	$MStr$	$\frac{MStr}{MSE}$
Errors	$n - v$	SSE	MSE	
Total	$n - 1$	TSS		

Expectations

$$\begin{aligned}
 E(SSE) &= \sum_{i=1}^v \sum_{j=1}^{n_i} E(y_{ij} - y_{io})^2 \\
 &= \sum_{i=1}^v \sum_{j=1}^{n_i} (\varepsilon_{ij} - \bar{\varepsilon}_{io})^2 \\
 &= \sum_{i=1}^v \sum_{j=1}^{n_i} E(\varepsilon_{ij}^2) - \sum_{i=1}^v n_i E(\bar{\varepsilon}_{io}^2) \\
 &= n\sigma^2 - \sum_{i=1}^v n_i \frac{\sigma^2}{n_i} \\
 &= (n-v)\sigma^2
 \end{aligned}$$

$$E(MSE) = E\left(\frac{SSE}{n-v}\right) = \sigma^2$$

$$\begin{aligned}
E(SSTr) &= \sum_{i=1}^v n_i E(\bar{y}_{io} - \bar{y}_{oo})^2 \\
&= \sum_{i=1}^v n_i E(\alpha_i + \bar{\epsilon}_{io} - \bar{\epsilon}_{oo})^2 \\
&= \sum_{i=1}^v n_i \alpha_i^2 + \left[\sum_{i=1}^v n_i \bar{\epsilon}_{io}^2 - n \bar{\epsilon}_{oo}^2 \right] \\
&= \sum_{i=1}^v n_i \alpha_i^2 + \left[\sum_{i=1}^v n_i \frac{\sigma^2}{n_i} - n \frac{\sigma^2}{n} \right] \\
&= \sum_{i=1}^v n_i \alpha_i^2 + (v-1)\sigma^2
\end{aligned}$$

$$E(MSTr) = E\left(\frac{SSTr}{v-1}\right) = \frac{1}{v-1} \sum_{i=1}^v n_i \alpha_i^2 + \sigma^2.$$

In general $E(MSTr) \neq \sigma^2$ but under H_0 , all $\alpha_i = 0$ and so $E(MSTr) = \sigma^2$.

Randomized Block Design

If a large number of treatments are to be compared, then a large number of experimental units are required. This will increase the variation among the responses and CRD may not be appropriate to use. In such a case when the experimental material is not homogeneous and there are v treatments to be compared, then it may be possible to

- group the experimental material into blocks of sizes v units.
- Blocks are constructed such that the experimental units within a block are relatively homogeneous and resemble to each other more closely than the units in the different blocks.
- If there are b such blocks, we say that the blocks are at b levels. Similarly, if there are v treatments, we say that the treatments are at v levels. The responses from the b levels of blocks and v levels of treatments can be arranged in a two-way layout. The observed data set is arranged as follows:

	Treatments (Factor B)							Block totals
		1	2		j		v	
Blocks (Factor A)	1	y_{11}	y_{12}	...	y_{1j}	...	y_{1v}	B_1
	2	y_{21}	y_{22}	...	y_{2j}	...	y_{2v}	B_2
	.	.	.		.		.	.
	.	.	.		.		.	.
	.	.	.		.		.	.
	i	y_{i1}	y_{i2}	...	y_{ij}	...	y_{iv}	B_i
	.		.		.		.	.
	.		.		.		.	.
	.		.		.		.	.
	b	y_{b1}	y_{b2}	...	y_{bj}	...	y_{bv}	B_b
Treatment totals		T_1	T_2	...	T_j	...	T_v	Grand total (G)

Layout:

A two-way layout is called a randomized block design (RBD) or a randomized complete block design (RCB) if, within each block, the v treatments are randomly assigned to v experimental units such that each of the $v!$ ways of assigning the treatments to the units has the same probability of being adopted in the experiment and the assignment in different blocks are statistically independent.

The RBD utilizes the principles of design - randomization, replication and local control - in the following way:

1. Randomization:

- Number the v treatments $1, 2, \dots, v$.
- Number the units in each block as $1, 2, \dots, v$.
- Randomly allocate the v treatments to v experimental units in each block.

2. Replication

Since each treatment is appearing in each block, so every treatment will appear in all the blocks. So each treatment can be considered as if replicated the number of times as the number of blocks. Thus in RBD, the number of blocks and the number of replications are same.

3. Local control

Local control is adopted in RBD in the following way:

- First form the homogeneous blocks of the experimental units.
- Then allocate each treatment randomly in each block.

The error variance now will be smaller because of homogeneous blocks and some variance will be parted away from the error variance due to the difference among the blocks.

Example:

Suppose there are 7 treatments denoted as $T_1, T_2, \dots, T_7$ corresponding to 7 levels of a factor to be included in 4 blocks. So one possible layout of the assignment of 7 treatments to 4 different blocks in an RBD is as follows

Block 1	T_2	T_7	T_3	T_5	T_1	T_4	T_6
Block 2	T_1	T_6	T_7	T_4	T_5	T_3	T_2
Block 3	T_7	T_5	T_1	T_6	T_4	T_2	T_3
Block 4	T_4	T_1	T_5	T_6	T_2	T_7	T_3

Analysis

Let

y_{ij} : Individual measurements of j^{th} treatment in i^{th} block, $i = 1, 2, \dots, b$, $j = 1, 2, \dots, v$.

y_{ij} 's are independently distributed following $N(\mu + \beta_i + \tau_j, \sigma^2)$

where μ : overall mean effect

β_i : i^{th} block effect

τ_j : j^{th} treatment effect

such that $\sum_{i=1}^b \beta_i = 0$, $\sum_{j=1}^v \tau_j = 0$.

There are two null hypotheses to be tested.

- related to the block effects

$$H_{0B} : \beta_1 = \beta_2 = \dots = \beta_b = 0.$$

- related to the treatment effects

$$H_{0T} : \tau_1 = \tau_2 = \dots = \tau_v = 0.$$

The linear model, in this case, is a two-way model as

$$y_{ij} = \mu + \beta_i + \tau_j + \varepsilon_{ij}, i = 1, 2, \dots, b; j = 1, 2, \dots, v$$

where ε_{ij} are identically and independently distributed random errors following a normal distribution with mean 0 and variance σ^2 .

The tests of hypothesis can be derived using the likelihood ratio test or the principle of least squares. The use of likelihood ratio test has already been demonstrated earlier, so we now use the principle of least squares.

$$\text{Minimizing } S = \sum_{i=1}^b \sum_{j=1}^v \varepsilon_{ij}^2 = \sum_{i=1}^b \sum_{j=1}^v (y_{ij} - \mu - \beta_i - \tau_j)^2$$

and solving the normal equation

$$\frac{\partial S}{\partial \mu} = 0, \frac{\partial S}{\partial \beta_i} = 0, \frac{\partial S}{\partial \tau_j} = 0 \text{ for all } i = 1, 2, \dots, b, j = 1, 2, \dots, v.$$

the least squares estimators are obtained as

$$\begin{aligned} \hat{\mu} &= \bar{y}_{oo}, \\ \hat{\beta}_i &= \bar{y}_{io} - \bar{y}_{oo}, \\ \hat{\tau}_j &= \bar{y}_{oj} - \bar{y}_{oo}. \end{aligned}$$

Using the fitted model (obtained after substituting the estimated values of the parameters in the model), we can write

$$y_{ij} = \bar{y}_{oo} + (\bar{y}_{io} - \bar{y}_{oo}) + (\bar{y}_{oj} - \bar{y}_{oo}) + (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo}).$$

Squaring both sides and summing over i and j gives

$$\sum_{i=1}^b \sum_{j=1}^v (\bar{y}_{ij} - \bar{y}_{oo})^2 = v \sum_{i=1}^b (\bar{y}_{io} - \bar{y}_{oo})^2 + b \sum_{j=1}^v (\bar{y}_{oj} - \bar{y}_{oo})^2 + \sum_{i=1}^b \sum_{j=1}^v (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2$$

$$\text{or } TSS = SSBl + SSTr + SSE$$

with degrees of freedom partitioned as

$$bv - 1 = (b - 1) + (v - 1) + (b - 1)(v - 1).$$

The reason for the number of degrees of freedom for different sums of squares is the same as in the case of CRD.

$$\begin{aligned}\text{Here } TSS &= \sum_{i=1}^b \sum_{j=1}^v (y_{ij} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - \frac{G^2}{bv}\end{aligned}$$

$\frac{G^2}{bv}$: correction factor.

$G = \sum_{i=1}^b \sum_{j=1}^v y_{ij}$: Grand total of all the observation.

$$\begin{aligned}SSBl &= v \sum_{i=1}^b (\bar{y}_{io} - \bar{y}_{oo})^2 \\ &= \sum_{i=1}^b \frac{B_i^2}{v} - \frac{G^2}{bv}\end{aligned}$$

$B_i = \sum_{j=1}^v y_{ij}$: i^{th} block total

$$\begin{aligned}SSTr &= b \sum_{j=1}^v (\bar{y}_{oj} - \bar{y}_{oo})^2 \\ &= \sum_{j=1}^v \frac{T_j^2}{b} - \frac{G^2}{bv}\end{aligned}$$

$T_j = \sum_{i=1}^b y_{ij}$: j^{th} treatment total

$$SSE = \sum_{i=1}^b \sum_{j=1}^v (y_{ij} - \bar{y}_{io} - \bar{y}_{oj} + \bar{y}_{oo})^2.$$

The expectations of mean squares are

$$E(MSBl) = E\left(\frac{SSBl}{b-1}\right) = \sigma^2 + \frac{v}{b-1} \sum_{i=1}^b \beta_i^2$$

$$E(MSTr) = E\left(\frac{SSTr}{v-1}\right) = \sigma^2 + \frac{b}{v-1} \sum_{j=1}^v \tau_j^2$$

$$E(MSE) = E\left(\frac{SSE}{(b-1)(v-1)}\right) = \sigma^2.$$

Moreover,

$$(b-1) \frac{SSBl}{\sigma^2} \sim \chi^2(b-1)$$

$$(v-1) \frac{SSTr}{\sigma^2} \sim \chi^2(v-1)$$

$$(b-1)(v-1) \frac{SSE}{\sigma^2} \sim \chi^2(b-1)(v-1).$$

Under $H_{0B} : \beta_1 = \beta_2 = \dots = \beta_b = 0$,

$$E(MSBl) = E(MSE)$$

and $SSBl$ and SSE are independent , so

$$F_{bl} = \frac{MSBl}{MSE} \sim F((b-1), (b-1)(v-1)).$$

Similarly, under $H_{0T} : \tau_1 = \tau_2 = \dots = \tau_v = 0$, so

$$E(MSTr) = E(MSE)$$

and $SSTr$ and SSE are independent, so

$$F_{Tr} = \frac{MSTr}{MSE} \sim F(v-1, (b-1)(v-1)).$$

Reject H_{0B} if $F_{be} > F_{\alpha}((b-1), (b-1)(v-1))$

Reject H_{0T} if $F_{Tr} > F_{\alpha}((v-1), (b-1)(v-1))$

If H_{0B} is accepted, then it indicates that the blocking is not necessary for future experimentation.

If H_{0T} is rejected then it indicates that the treatments are different. Then the multiple comparison tests are used to divide the entire set of treatments into different subgroup such that the treatments in the same subgroup have the same treatment effect and those in the different subgroups have different treatment effects.

The analysis of variance table is as follows

Source of variation	Degrees of freedom	Sum of squares	Mean squares	F
Blocks	$b - 1$	$SSBl$	$MSBl$	F_{Bl}
Treatments	$v - 1$	$SSTr$	$MSTr$	F_{Tr}
Errors	$(b - 1)(v - 1)$	SSE	MSE	
Total	$bv - 1$	TSS		

Latin Square Design

The treatments in the RBD are randomly assigned to b blocks such that each treatment must occur in each block rather than assigning them at random over the entire set of experimental units as in the CRD. There are only two factors – block and treatment effects – which are taken into account and the total number of experimental units needed for complete replication are bv where b and v are the numbers of blocks and treatments respectively.

If there are three factors and suppose there are b , v and k levels of each factor, then the total number of experimental units needed for a complete replication are bvk . This increases the cost of experimentation and the required number of experimental units over RBD.

In Latin square design (LSD), the experimental material is divided into rows and columns, each having the same number of experimental units which is equal to the number of treatments. The treatments are allocated to the rows and the columns such that each treatment occurs once and only once in each row and in each column.

In order to allocate the treatment to the experimental units in rows and columns, we take help from Latin squares.

Latin Square:

A Latin square of order p is an arrangement of p symbols in p^2 cells arranged in p rows and p columns such that each symbol occurs once and only once in each row and in each column. For example, to write a Latin square of order 4, choose four symbols – A, B, C and D. These letters are Latin letters which are used as symbols. Write them in a way such that each of the letters out of A, B, C and D occurs once and only once in each row and each column. For example, as

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

This is a Latin square.

We consider first the following example to illustrate how a Latin square is used to allocate the treatments and in getting the response.

Example:

Suppose different brands of petrol are to be compared with respect to the mileage per litre achieved in motor cars.

Important factors responsible for the variation in mileage are

- the difference between individual cars.
- the difference in the driving habits of drivers.

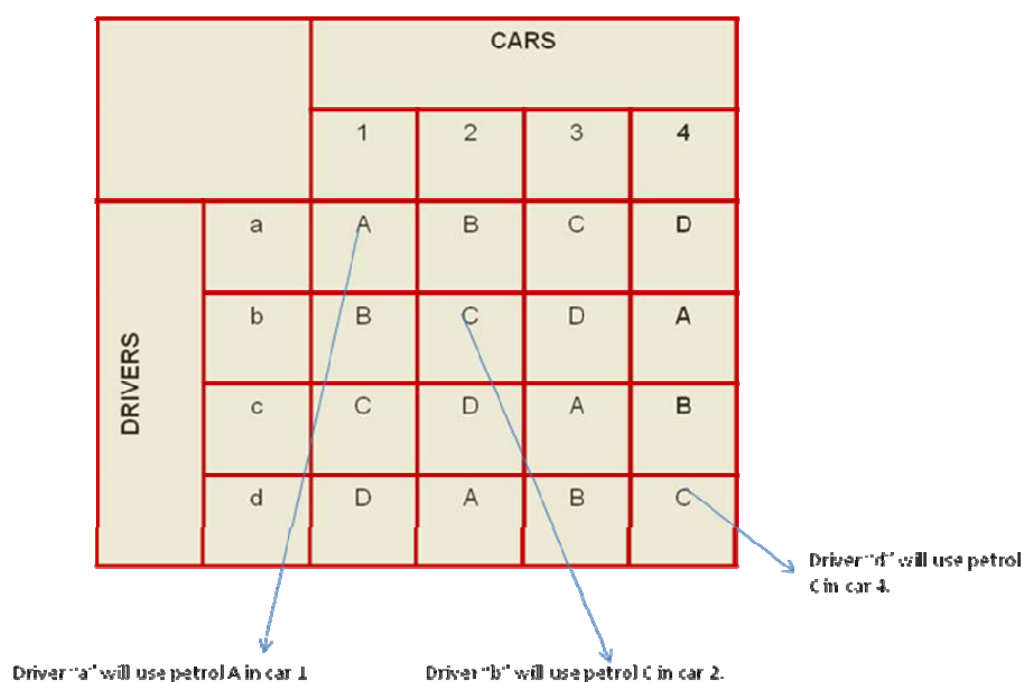
We have three factors – cars, drivers and petrol brands. Suppose we have

- 4 types of cars denoted as 1, 2, 3, 4.
- 4 drivers that are represented by a, b, c, d.
- 4 brands of petrol are indicated as A, B, C, D.

Now the complete replication will require $4 \times 4 \times 4 = 64$ the number of experiments. We choose only 16 experiments. To choose such 16 experiments, we take the help of the Latin square. Suppose we choose the following Latin square:

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

Write them in rows and columns and choose rows for drivers, columns for cars and letter for petrol brands. Thus 16 observations are recorded as per this plan of treatment combination (as shown in the next figure) and further analysis is carried out. Since such design is based on Latin square, so it is called as a Latin square design.



Another choice of a Latin square of order 4 is

C	B	A	D
B	C	D	A
A	D	C	B
D	A	B	C

This will again give a design different from the previous one. The 16 observations will be recorded again but based on different treatment combinations.

Since we use only 16 out of 64 possible observations, so it is an incomplete 3-way layout in which each of the 3 factors – cars, drivers and petrol brands are at 4 levels and the observations are recorded only on 16 of the 64 possible treatment combinations.

Thus in an LSD,

- the treatments are grouped into replication in two-ways
 - once in rows and
 - and in columns,
- rows and columns variations are eliminated from the within treatment variation.
 - In RBD, the experimental units are divided into homogeneous blocks according to the blocking factor. Hence it eliminates the difference among blocks from the experimental error.
 - In LSD, the experimental units are grouped according to two factors. Hence two effects (like as two block effects) are removed from the experimental error.
 - So the error variance can be considerably reduced in LSD.

The LSD is an incomplete three-way layout in which each of the three factors, viz, rows, columns and treatments, is at v levels each and observations only on v^2 of the v^3 possible treatment combinations are taken. Each treatment combination contains one level of each factor.

The analysis of data in an LSD is conditional in the sense it depends on which Latin square is used for allocating the treatments. If the Latin square changes, the conclusions may also change.

We note that Latin squares play an important role in an LSD, so first we study more about these Latin squares before describing the analysis of variance.

Standard form of Latin square

A Latin square is in the standard form if the symbols in the first row and first columns are in the **natural order** (Natural order means the order of alphabets like A, B, C, D,...).

Given a Latin square, it is possible to rearrange the columns so that the first row and first column remain in a natural order.

Example: Four standard forms of 4×4 Latin square are as follows.

A B C D	A B C D	A B C D	A B C D
B A D C	B C D A	B D A C	B A D C
C D B A	C D A B	C A D B	C D A B
D C A B	D A B C	D C B A	D C B A

For each standard Latin square of order p , the p rows can be permuted in $p!$ ways. Keeping a row fixed, vary and permute $(p - 1)$ columns in $(p - 1)!$ ways. So there are $p!(p - 1)!$ different Latin squares.

For illustration

Size of square	Number of Standard squares	Value of $p!(1 - p)!$	Total number of different squares
3 x 3	1	12	12
4 x 4	4	144	576
5 x 5	56	2880	161280
6 x 6	9408	86400	812851250

Conjugate:

Two standard Latin squares are called conjugate if the rows of one are the columns of other.

For example

A B C D		A B C D
B C D A	and	B C D A
C D A B		C D A B
D A B C		D A B C

are conjugate. In fact, they are **self conjugate**.

A Latin square is called **self conjugate** if its arrangement in rows and columns are the same.

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Transformation set:

A set of all Latin squares obtained from a single Latin square by permuting its rows, columns and symbols is called a transformation set.

From a Latin square of order p , $p!(p - 1)!$ different Latin squares can be obtained by making $p!$ permutations of columns and $(p - 1)!$ permutations of rows which leaves the first row in place. Thus

Number of different Latin squares of order p in a transformation set = $p!(p - 1)!$ X number of standard Latin squares in the set

Orthogonal Latin squares

If two Latin squares of the same order but with different symbols are such that when they are superimposed on each other, every ordered pair of symbols (different) occurs exactly once in the Latin square, then they are called orthogonal.

Graeco-Latin square:

A pair of orthogonal Latin squares, one with Latin symbols and the other with Greek symbols form a Graeco-Latin square.

For example

A	B	C	D	α	β	γ	δ
B	A	D	C	δ	γ	β	α
C	D	A	B	β	α	δ	γ
D	C	B	A	γ	δ	α	β

is a Graeco-Latin square of order 4.

Graeco Latin squares design enables to consider one more factor than the factors in Latin square design. For example, in the earlier example, if there are four drivers, four cars, four petrol and each petrol has four varieties, as α, β, γ and δ , then Graeco-Latin square helps in deciding the treatment combination as follows:

Drivers	Cars				
		1	2	3	4
	<i>a</i>	$A\alpha$	$B\beta$	$C\gamma$	$D\delta$
	<i>b</i>	$B\delta$	$A\gamma$	$D\beta$	$C\alpha$
	<i>c</i>	$C\beta$	$D\alpha$	$A\delta$	$B\gamma$
	<i>d</i>	$D\gamma$	$C\delta$	$B\alpha$	$A\beta$

Now

$A\alpha$ means: Driver ‘a’ will use the α variant of petrol A in Car 1.

$B\gamma$ means: Driver ‘c’ will use the γ variant of petrol B in Car 4

and so on.

Mutually orthogonal Latin square

A set of Latin squares of the same order is called a set of mutually orthogonal Latin square (or a hyper Graeco-Latin square) if every pair in the set is orthogonal. The total number of mutually orthogonal Latin squares of order p is at most $(p - 1)$.

Analysis of LSD (one observation per cell)

In designing an LSD of order p ,

- choose one Latin square at random from the set of all possible Latin squares of order p .
- Select a standard Latin square from the set of all standard Latin squares with equal probability.
- Randomize all the rows and columns as follows:
 - Choose a random number, less than p , say n_1 and then 2nd row is the n_1^{th} row.
 - Choose another random number less than p , say n_2 and then 3rd row is the n_2^{th} row and so on.
 - Then do the same for the column.
- For Latin squares of the order less than 5, fix the first row and then randomize rows and then randomize columns. In Latin squares of order 5 or more, need not to fix even the first row. Just randomize all rows and columns.

Example:

Suppose following Latin square is chosen

A	B	C	D	E
B	C	D	E	A
D	E	A	B	C
E	A	B	C	D
C	D	E	A	B

Now randomize rows, e.g., 3rd row becomes 5th row and 5th row becomes 3rd row . The Latin square becomes

A	B	C	D	E
B	C	D	E	A
C	D	E	A	B
E	A	B	C	D
D	E	A	B	C

Now randomize columns, say 5th column becomes 1st column, 1st column becomes 4th column and 4th column becomes 5th column

E	B	C	A	D
A	C	D	B	E
D	A	B	E	C
C	E	A	D	B
B	D	E	C	A

Now use this Latin square for the assignment of treatments.

y_{ijk} : Observation on k^{th} treatment in i^{th} row and j^{th} block, $i = 1, 2, \dots, v$, $j = 1, 2, \dots, v$, $k = 1, 2, \dots, v$.

Triplets (i, j, k) take on only the v^2 values indicated by the chosen particular Latin square selected for the experiment.

y_{ijk} 's are independently distributed as $N(\mu + \alpha_i + \beta_j + \tau_k, \sigma^2)$.

Linear model is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \varepsilon_{ijk}, i = 1, 2, \dots, v; j = 1, 2, \dots, v; k = 1, 2, \dots, v$$

where ε_{ijk} are random errors which are identically and independently distributed following $N(0, \sigma^2)$.

$$\text{with } \sum_{i=1}^v \alpha_i = 0, \sum_{j=1}^v \beta_j = 0, \sum_{k=1}^v \tau_k = 0,$$

α_i : main effect of rows

β_j : main effect of columns

τ_k : main effect of treatments.

The null hypothesis under consideration are

$$H_{0R} : \alpha_1 = \alpha_2 = \dots = \alpha_v = 0$$

$$H_{0C} : \beta_1 = \beta_2 = \dots = \beta_v = 0$$

$$H_{0T} : \tau_1 = \tau_2 = \dots = \tau_v = 0$$

The analysis of variance can be developed on the same lines as earlier.

Minimizing $S = \sum_{i=1}^v \sum_{j=1}^v \sum_{k=1}^v \varepsilon_{ijk}^2$ with respect to μ, α_i, β_j and τ_k given the least-squares estimate as

$$\hat{\mu} = \bar{y}_{ooo}$$

$$\hat{\alpha}_i = \bar{y}_{ioo} - \bar{y}_{ooo} \quad i = 1, 2, \dots, v$$

$$\hat{\beta}_j = \bar{y}_{oj o} - \bar{y}_{ooo} \quad j = 1, 2, \dots, v$$

$$\hat{\tau}_k = \bar{y}_{ook} - \bar{y}_{ooo} \quad k = 1, 2, \dots, v.$$

Using the fitted model based on these estimators, the total sum of squares can be partitioned into the mutually orthogonal sum of squares $SSR, SSC, SStr$ and SSE as

$$TSS = SSR + SSC + SStr + SSE$$

where

$$TSS: \text{Total sum of squares} = \sum_{i=1}^v \sum_{j=1}^v \sum_{k=1}^v (y_{ijk} - \bar{y}_{ooo})^2 = \sum_{i=1}^v \sum_{j=1}^v \sum_{k=1}^v y_{ijk}^2 - \frac{G^2}{v^2}$$

$$SSR: \text{Sum of squares due to rows} = v \sum_{i=1}^v (\bar{y}_{ioo} - \bar{y}_{ooo})^2 = \frac{\sum_{i=1}^v R_i^2}{v} - \frac{G^2}{v^2}; \text{ where } R_i = \sum_{j=1}^v \sum_{k=1}^v y_{ijk}$$

$$SSC: \text{Sum of squares due to column} = v \sum_{j=1}^v (\bar{y}_{oj o} - \bar{y}_{ooo})^2 = \frac{\sum_{j=1}^v C_j^2}{v} - \frac{G^2}{v^2}; \text{ where } C_j = \sum_{i=1}^v \sum_{k=1}^v y_{ijk}$$

$$SSTr : \text{Sum of squares due to treatment} = v \sum_{k=1}^v (\bar{y}_{ook} - \bar{y}_{ooo})^2 = \frac{\sum_{k=1}^v T_k^2}{v} - \frac{G^2}{v^2}; \text{ where } T_k = \sum_{i=1}^v \sum_{j=1}^v y_{ijk}$$

Degrees of freedom carried by SSR , SSC and $SSTr$ are $(v - 1)$ each.

Degrees of freedom carried by TSS is $v^2 - 1$.

Degree of freedom carried by SSE is $(v - 1)(v - 2)$.

The expectations of mean squares are obtained as

$$E(MSR) = E\left(\frac{SSR}{v-1}\right) = \sigma^2 + \frac{v}{v-1} \sum_{i=1}^v \alpha_i^2$$

$$E(MSC) = E\left(\frac{SSC}{v-1}\right) = \sigma^2 + \frac{v}{v-1} \sum_{j=1}^v \beta_j^2$$

$$E(MSTr) = E\left(\frac{SSTr}{v-1}\right) = \sigma^2 + \frac{v}{v-1} \sum_{k=1}^v \tau_k^2$$

$$E(MSE) = E\left(\frac{SSE}{(v-1)(v-2)}\right) = \sigma^2.$$

Thus

$$\text{- under } H_{0R}, F_R = \frac{MSR}{MSE} \sim F((v-1), (v-1)(v-2))$$

$$\text{- under } H_{0C}, F_C = \frac{MSC}{MSE} \sim F((v-1), (v-1)(v-2))$$

$$\text{- under } H_{0T}, F_T = \frac{MSTr}{MSE} \sim F((v-1), (v-1)(v-2)).$$

Decision rules:

Reject H_{0R} at level α if $F_R > F_{1-\alpha; v(-1), (v-1)(v-2)}$

Reject H_{0C} at level α if $F_C > F_{1-\alpha; (v-1), (v-1)(v-2)}$

Reject H_{0T} at level α if $F_T > F_{1-\alpha; (v-1), (v-1)(v-2)}$.

If any null hypothesis is rejected, then use multiple comparison test.

The analysis of variance table is as follows

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F
Rows	$v - 1$	SSR	MSR	F_R
Columns	$v - 1$	SSC	MSC	F_C
Treatments	$v - 1$	$SSTr$	$MSTr$	F_T
Error	$(v - 1)(v - 2)$	SSE	MSE	
Total	$v^2 - 1$	TSS		

Missing plot techniques:

It happens many time in conducting the experiments that some observation are missed. This may happen due to several reasons. For example, in a clinical trial, suppose the readings of blood pressure are to be recorded after three days of giving the medicine to the patients. Suppose the medicine is given to 20 patients and one of the patients doesn't turn up for providing the blood pressure reading. Similarly, in an agricultural experiment, the seeds are sown and yields are to be recorded after few months. Suppose some cattle destroy the crop of any plot or the crop of any plot is destroyed due to storm, insects etc.

In such cases, one option is to

- somehow estimate the missing value on the basis of available data,
- replace it back in the data and make the data set complete.

Now conduct the statistical analysis on the basis of completed data set as if no value was missing by making necessary adjustments in the statistical tools to be applied. Such an area comes under the purview of “missing data models” and a lot of development has taken place. Several books on this issue have appeared, e.g.,

- Little, R.J.A. and Rubin, D.B. (2002). Statistical Analysis with Missing Data, 2nd edition, New York: John Wiley.
- Schafer, J.L. (1997). Analysis of Incomplete Multivariate Data. Chapman & Hall, London etc.

We discuss here the classical missing plot technique proposed by Yates which involve the following steps:

- Estimate the missing observations by the values which makes the error sum of squares to be minimum.
- Substitute the unknown values by the missing observations.
- Express the error sum of squares as a function of these unknown values.
- Minimize the error sum of squares using the principle of maxima/minima, i.e., differentiating it with respect to the missing value and put it to zero and form a linear equation.
- Form as much linear equation as the number of unknown values (i.e., differentiate the error sum of squares with respect to each unknown value).
- Solve all the linear equations simultaneously and solutions will provide the missing values.
- Impute the missing values with the estimated values and complete the data.
- Apply analysis of variance tools.
- The error sum of squares thus obtained is corrected but the treatment sum of squares is not corrected.
- The number of degrees of freedom associated with the total sum of squares is subtracted by the number of missing values and adjusted in the error sum of squares. No change in the degrees of freedom of sum of squares due to treatment is needed.

Missing observations in RBD

One missing observation:

Suppose one observation in $(i, j)^{\text{th}}$ cell is missing and let this be x . The arrangement of observations in RBD then will look like as follows:

	Treatments (Factor B)							Block totals
		1	2	...	j	...	v	
Blocks (Factor A)	1	y_{11}	y_{12}	...	y_{1j}	...	y_{1v}	$B_1 = y_{1o}$
	2	y_{21}	y_{22}	...	y_{2j}	...	y_{2v}	$B_2 = y_{2o}$
	.	.	.		.		.	.
	.	.	.		.		.	.
	.	.	.		.		.	.
	i	y_{i1}	y_{i2}	...	$y_{ij} = x$	...	y_{iv}	$B_i = y'_{io} + x$
	.		.		.		.	.
	.		.		.		.	.
	.		.		.		.	.
	b	y_{b1}	y_{b2}	...	y_{bj}	...	y_{bv}	B_b
Treatment totals		$T_1 = y_{o1}$	$T_2 = y_{o2}$	...	$T_j = y'_{oj} + x$	...	$T_v = y_{ob}$	Grand total (G) = $y'_{oo} + x$

where y'_{oo} : total of known observations

y'_{io} : total of known observations in i^{th} block

y'_{oj} : total of known observations in j^{th} treatment

$$\text{Correction factor (CF)} = \frac{(G')^2}{n} = \frac{(y'_{oo} + x)^2}{bv}$$

$$TSS = \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - CF$$

$$= (x^2 + \text{terms which are constant with respect to } x) - CF$$

$$SSBl = \frac{1}{v} [(y'_{io} + x)^2 + \text{terms which are constant with respect to } x] - CF$$

$$SSTr = \frac{1}{b} [(y'_{oj} + x)^2 + \text{terms which are constant with respect to } x] - CF$$

$$SSE = TSS - SSBl - SSTr$$

$$= x^2 - \frac{1}{v} (y'_{io} + x)^2 - \frac{1}{b} (y'_{oj} + x)^2 + \frac{(y'_{oo} + x)^2}{bv} + (\text{terms which are constant with respect to } x) - CF.$$

Find x such that SSE is minimum

$$\frac{\partial(SSE)}{\partial x} = 0 \Rightarrow 2x - \frac{2(y'_{io} + x)}{v} - \frac{2(y'_{oj} + x)}{b} + \frac{2(y'_{oo} + x)}{bv} = 0$$

$$\text{or } x = \frac{vy'_{oj} + by'_{io} - y'_{oo}}{(b-1)(v-1)}$$

The second-order derivative condition for x to provide minimum SSE can be easily verified.

Two missing observations:

If there are two missing observation, then let they be x and y .

- Let the corresponding row sums (block totals) are $(R_1 + x)$ and $(R_2 + y)$.
- Column sums (treatment totals) are $(C_1 + x)$ and $(C_2 + y)$.
- Total of known observations is S .

Then

$$SSE = x^2 + y^2 - \frac{1}{v}[(R_1 + x)^2 + (R_2 + y)^2] - \frac{1}{b}[(C_1 + x)^2 + (C_2 + y)^2] + \frac{1}{bv}(S + x + y)^2$$

+ terms independent of x and y .

Now differentiate SSE with respect to x and y , as

$$\frac{\partial(SSE)}{\partial x} = 0 \Rightarrow x - \frac{R_1 + x}{v} - \frac{C_1 + x}{b} + \frac{S + x + y}{bv} = 0$$

$$\frac{\partial(SSE)}{\partial y} = 0 \Rightarrow y - \frac{R_2 + y}{v} - \frac{C_2 + y}{b} + \frac{S + x + y}{bv} = 0.$$

Thus solving the following two linear equations in x and y , we obtain the estimated missing values

$$(b-1)(v-1)x = bR_1 + vC_1 - S - y$$

$$(b-1)(v-1)y = bR_2 + vC_2 - S - x.$$

Adjustments to be done in analysis of variance

- (i) Obtain the within block sum of squares from incomplete data.
- (ii) Subtract correct error sum of squares from (i). This gives the correct treatment sum of squares.
- (iii) Reduce the degrees of freedom of error sum of squares by the number of missing observations.
- (iv) No adjustments in other sums of squares are required.

Missing observations in LSD

Let

- x be the missing observation in $(i, j, k)^{\text{th}}$ cell, i.e.,
 $y_{ijk}, i = 1, 2, \dots, v, j = 1, 2, \dots, v, k = 1, 2, \dots, v.$
- R : Total of known observations in i^{th} row
- C : Total of known observations in j^{th} column
- T : Total of known observation receiving the k^{th} treatment.
- S : Total of known observations

Now

$$\text{Correction factor (CF)} = \frac{(S + x)^2}{v^2}$$

Total sum of squares (TSS) = x^2 + term which are constant with respect to x - CF

Row sum of squares (SSR) = $\frac{(R + x)^2}{v}$ + term which are constant with respect to x - CF

Column sum of squares (SSC) = $\frac{(C + x)^2}{v}$ + term which are constant with respect to x - CF

Treatment sum of squares ($SSTr$) = $\frac{(T + x)^2}{v}$ + term which are constant with respect to x - CF

Sum of squares due to error (SSE) = $TSS - SSR - SSC - SSTr$

$$= x^2 - \frac{1}{v} \left[(R + x)^2 + (C + x)^2 + (T + x)^2 \right] + \frac{2(S + x)^2}{v^2}$$

Choose x such that SSE is minimum. So

$$\frac{d(SSE)}{dx} = 0$$

$$\Rightarrow 2x - \frac{2}{v} (R + C + T + 3x) + \frac{4(S + x)}{v^2} = 0$$

$$\text{or } x = \frac{V(R + C + T) - 2S}{(v - 1)(v - 2)}$$

Adjustment to be done in analysis of variance:

Do all the steps as in the case of RBD.

To get the correct treatment sum of squares, proceed as follows:

- Ignore the treatment classification and consider only row and column classification.
- Substitute the estimated values at the place of missing observation.
- Obtain the error sum of squares from complete data, say SSE_1 .
- Let SSE_2 be the error sum of squares based on LSD obtained earlier.
- Find the corrected treatment sum of squares = $SSE_2 - SSE_1$.
- Reduce of degrees of freedom of error sum of squares by the number of missing values.

Chapter 5

Incomplete Block Designs

If the number of treatments to be compared is large, then we need a large number of blocks to accommodate all the treatments. This requires more experimental material and so the cost of experimentation becomes high which may be in terms of money, labour, time etc. The completely randomized design and randomized block design may not be suitable in such situations because they will require a large number of experimental units to accommodate all the treatments. In such cases, when the sufficient number of homogeneous experimental units are not available to accommodate all the treatments in a block, then incomplete block designs can be used. In incomplete block designs, each block receives only some of the selected treatments and not all the treatments. Sometimes it is possible that the available blocks can accommodate only a limited number of treatments due to several reasons. For example, the goodness of a car is judged by different features like fuel efficiency, engine performance, body structure etc. Each of this factor depends on many other factors, e.g., the engine consists of many parts and the performance of every part combined together will result in the final performance of the engine. These factors can be treated as treatment effects. If all these factors are to be compared, then we need a large number of cars to design a complete experiment. This may be an expensive affair. The incomplete block designs overcome such problems. It is possible to use much less number of cars with the set up of incomplete block design and all the treatments need not be assigned to all the cars. Rather some treatments will be implemented in some cars and remaining treatments in other cars. The efficiency of such designs is, in general, not less than the efficiency of a complete block design. In another example, consider a situation of destructive experiments, e.g., testing the life of television sets, LCD panels, etc. If there are a large number of treatments to be compared, then we need a large number of television sets or LCD panels. The incomplete block designs can use a lesser number of television sets or LCD panels to conduct the test of the significance of treatment effects without losing, in general, the efficiency of the design of the experiment. This also results in the reduction of experimental cost. Similarly, in any experiment involving animals like as biological experiments, one would always like to sacrifice fewer animals. Moreover, the government guidelines also restrict the experimenter to use a smaller number of animals. In such cases, either the number of treatments to be compared can be reduced depending upon the number of animals in each block or to reduce the block size. In such cases when the number of treatments to be compared is larger than the number of animals in each block, then the block size is reduced and the setup of incomplete block designs can be used. This will result in a lower cost of experimentation. The incomplete block designs need less number of observations in a block than the

observations in a complete block design to conduct the test of hypothesis without losing the efficiency of the design of experiment, in general.

Complete and incomplete block designs:

The designs in which every block receives all the treatments are called the complete block designs.

The designs in which every block does not receive all the treatments but only some of the treatments are called incomplete block design.

The block size is smaller than the total number of treatments to be compared in the incomplete block designs.

There are three types of analysis in the incomplete block designs

- intrablock analysis,
- interblock analysis and
- recovery of interblock information.

Intrablock analysis:

In intrablock analysis, the treatment effects are estimated after eliminating the block effects and then the analysis and the test of significance of treatment effects are conducted further. If the blocking factor is not marked, then the intrablock analysis is sufficient enough to provide reliable, correct and valid statistical inferences.

Interblock analysis:

There is a possibility that the blocking factor is important and the block totals may carry some important information about the treatment effects. In such situations, one would like to utilize the information on block effects (instead of removing it as in the intrablock analysis) in estimating the treatment effects to conduct the analysis of design. This is achieved through the interblock analysis of an incomplete block design by considering the block effects to be random.

Recovery of interblock information:

When both the intrablock and the interblock analysis have been conducted, then two estimates of treatment effects are available from each of the analysis. A natural question then arises -- Is it possible to pool these two estimates together and obtain an improved estimator of the treatment effects to use it for the construction of test statistic for testing of hypothesis? Since such an estimator comprises of more information to estimate the treatment effects, so this is naturally expected to provide better statistical inferences. This is achieved by combining the intrablock and interblock analysis together through the recovery of interblock information.

Intrablock analysis of incomplete block design:

We start here with the usual approach involving the summations over different subscripts of y 's. Then gradually, we will switch to a matrix-based approach so that the reader can compare both the approaches. They can also learn the one-to-one relationships between the two approaches for better understanding.

Notations and normal equations:

Let

- v treatments have to be compared.
- b blocks are available.
- k_i : Number of plots in i^{th} block ($i = 1, 2, \dots, b$).
- r_j : Number of plots receiving j^{th} treatment ($j = 1, 2, \dots, v$).
- n : Total number of plots.
$$n = r_1 + r_2 + \dots + r_v = k_1 + k_2 + \dots + k_b.$$
- Each treatment may occur more than once in each block

or

may not occur at all.

- n_{ij} denotes the number of times the j^{th} treatment occurs in i^{th} block

For example, $n_{ij} = 1$ or 0 for all i, j means that no treatment occurs more than once in a block and treatment may not occur in some blocks at all. Similarly, $n_{ij} = 1$ means that j^{th} treatment occurs in i^{th} block and $n_{ij} = 0$ means that j^{th} treatment does not occurs in i^{th} block.

It may be noticed that

$$\begin{aligned}\sum_{j=1}^v n_{ij} &= k_i \quad i = 1, \dots, b \\ \sum_i n_{ij} &= r_j \quad j = 1, \dots, v \\ n &= \sum_i \sum_j n_{ij}\end{aligned}$$

Model:

Let y_{ijm} denotes the response (yield) from the m^{th} replicate of j^{th} treatment in i^{th} block and

$$y_{ijm} = \beta_i + \tau_j + \varepsilon_{ijm} \quad i = 1, 2, \dots, b, \quad j = 1, 2, \dots, v, \quad m = 1, 2, \dots, n_{ij}$$

[**Note:** We are not considering here the general mean effect in this model for better understanding of the issues in the estimation of parameters. Later, we will consider it in the analysis.]

Following notations are used in further description.

Block totals : $B_1, B_2, \dots, B_b$ where $B_i = \sum_j \sum_m y_{ijm}$.

Treatment totals: $V_1, V_2, \dots, V_v$ where $V_j = \sum_i \sum_m y_{ijm}$

Grand total : $Y = \sum_i \sum_j \sum_m y_{ijm}$

Generally, a design is denoted by $D(v, b, r, k, n)$ where v, b, r, k and n are the parameters of the design.

Example:

Let us consider an example to understand the meaning of these notations. Suppose there are 3 blocks (Block 1, Block 2 and Block 3) and 5 treatments ($\tau_1, \tau_2, \tau_3, \tau_4, \tau_5$). So $b = 3$ and $v = 5$. These treatments are arranged in different plots in blocks as follows:

Block 1: 5 plots

Plot 1	τ_1	Plot 2	τ_1	Plot 3	τ_2	Plot 4	τ_2	Plot 5	τ_3
--------	----------	--------	----------	--------	----------	--------	----------	--------	----------

Block 2: 4 plots

Plot 1	τ_2	Plot 2	τ_4	Plot 3	τ_5	Plot 4	τ_5
--------	----------	--------	----------	--------	----------	--------	----------

Block 3: 2 plots

Plot 1	τ_2	Plot 2	τ_2
--------	----------	--------	----------

k_i

Number of plots in Block 1: $k_1 = 5$

Number of plots in Block 2: $k_2 = 3$

Number of plots in Block 3: $k_3 = 2$

r_j

Number of times τ_1 appears in

- Block 1 = 2
- Block 2 = 0
- Block 3 = 0.

Total number of times τ_1 appears in the entire design is $r_1 = 2 + 0 + 0 = 2$.

Number of times τ_2 appears in

- Block 1 = 2
- Block 2 = 1
- Block 3 = 2.

Total number of times τ_2 appears in the entire design is $r_2 = 2 + 1 + 2 = 5$.

Number of times τ_3 appears in

- Block 1 = 1
- Block 2 = 0
- Block 3 = 0.

Total number of times τ_3 appears in the entire design is $r_3 = 1 + 0 + 0 = 1$.

Number of times τ_4 appears in

- Block 1 = 0
- Block 2 = 1
- Block 3 = 0.

Total number of times τ_4 appears in the entire design is $r_4 = 0 + 1 + 0 = 1$.

Number of times τ_5 appears in

- Block 1 = 0
- Block 2 = 2
- Block 3 = 0.

Total number of times τ_5 appears in the entire design is $r_5 = 0 + 2 + 0 = 2$.

n_{ij}

Total number of times τ_1 appears in Block 1: $n_{11} = 2$

Total number of times τ_2 appears in Block 1: $n_{12} = 2$

Total number of times τ_3 appears in Block 1: $n_{13} = 1$

Total number of times τ_4 appears in Block 1: $n_{14} = 0$

Total number of times τ_5 appears in Block 1: $n_{15} = 0$

Total number of times τ_1 appears in Block 2: $n_{21} = 0$

Total number of times τ_2 appears in Block 2: $n_{22} = 1$

Total number of times τ_3 appears in Block 2: $n_{23} = 0$

Total number of times τ_4 appears in Block 2: $n_{24} = 1$

Total number of times τ_5 appears in Block 2: $n_{25} = 2$

Total number of times τ_1 appears in Block 3: $n_{31} = 0$

Total number of times τ_2 appears in Block 3: $n_{32} = 2$

Total number of times τ_3 appears in Block 3: $n_{33} = 0$

Total number of times τ_4 appears in Block 3: $n_{34} = 0$

Total number of times τ_5 appears in Block 3: $n_{35} = 0$

 y_{ijm}

y_{ijm} : response from the m^{th} replicate of j^{th} treatment in i^{th} block, $i = 1, 2, 3$; $j = 1, 2, 3, 4, 5$; $m = 1, 2, \dots, n_{ij}$

Following are the notations for for y_{ijm} different treatments in the blocks

Block 1:	τ_1	y_{111}	τ_1	y_{112}	τ_2	y_{121}	τ_2	y_{122}	τ_3	y_{131}
----------	----------	-----------	----------	-----------	----------	-----------	----------	-----------	----------	-----------

Block 2:	τ_2	y_{221}	τ_4	y_{241}	τ_5	y_{251}	τ_5	y_{252}
----------	----------	-----------	----------	-----------	----------	-----------	----------	-----------

Block 3:	τ_2	y_{321}	τ_2	y_{322}
----------	----------	-----------	----------	-----------

B_i	V_j
$B_1 = y_{111} + y_{112} + y_{121} + y_{122} + y_{131}$	$V_1 = y_{111} + y_{112}$
$B_2 = y_{221} + y_{241} + y_{251} + y_{252}$	$V_2 = y_{121} + y_{122} + y_{221} + y_{321} + y_{322}$
$B_3 = y_{321} + y_{322}$	$V_3 = y_{131}$
	$V_4 = y_{241}$
	$V_5 = y_{251} + y_{252}$

Normal equations:

Minimizing $S = \sum_i \sum_j \sum_m \varepsilon_{ijm}^2$ with respect to β_i and τ_j , we obtain the least-squares estimators of the parameters as follows:

$$S = \sum_i \sum_j \sum_m (y_{ijm} - \beta_i - \tau_j)^2$$

$$\frac{\partial S}{\partial \beta_i} = 0$$

$$\Rightarrow \sum_j \sum_m (y_{ijm} - \beta_i - \tau_j) = 0$$

or

$$B_i - \beta_i \sum_j \sum_m 1 - \sum_j \tau_j \sum_m 1 = 0 \quad (1)$$

or

$$B_i = \beta_i k_i + n_{i1} \tau_1 + n_{i2} \tau_2 + \dots + n_{iv} \tau_v, \quad i = 1, \dots, b$$

$$B_i = \beta_i k_i + \sum_j \tau_j n_{ij} \quad [b \text{ equations}]$$

$$\frac{\partial S}{\partial \tau_j} = 0$$

$$\Rightarrow \sum_i \sum_m (y_{ijm} - \beta_i - \tau_j) = 0$$

$$\text{or } \sum_i \sum_m y_{ijm} - \sum_i \beta_i \sum_m 1 - \tau_j \sum_i \sum_m 1 = 0$$

$$V_j - \sum_i \beta_i n_{ij} - \tau_j \sum_i n_{ij} = 0 \quad (2)$$

$$\text{or } V_j = n_{ij} \beta_1 + n_{2j} \beta_2 + \dots + n_{bj} \beta_b + r_j \tau_j, \quad j = 1, 2, \dots, v$$

$$\text{or } V_j = \sum_i \beta_i n_{ij} + r_j \tau_j \quad [v \text{ equations}]$$

Equations (1) and (2) constitute $(b + v)$ equations.

Note that

$$\sum_i \text{equation (1)} = \sum_j \text{equation (2)}$$

$$\sum_i B_i = \sum_j V_j$$

$$\sum_i \left(\sum_j \sum_m y_{ijm} \right) = \sum_j \left(\sum_i \sum_m y_{ijm} \right).$$

Thus there are at most $(b + v - 1)$ degrees of freedom for estimates. So the estimates of only $(b + v - 1)$ parameters can be obtained out of all $(b + v)$ parameters.

[**Note:** We will see later that degrees of freedom may be less than or equal to $(b + v - 1)$ in special cases. Also, note that we have not assumed any side conditions like $\sum_i \alpha_i = \sum_j \beta_j = 0$ as in the case of complete block designs.]

To obtain the estimates of the parameters, there are two options-

1. Using equation (1), eliminate β_i from equation (2) to estimate τ_j or
2. Using equation (2), eliminate τ_j from equation (1) to estimate β_i .

We consider first the approach 1., i.e., using equation (1), eliminate β_i from equation (2).

From equation (1),

$$\beta_i = \frac{1}{k_i} \left[B_i - \sum_{j=1}^v n_{ij} \tau_j \right].$$

Use it in (2) as follows.

$$\begin{aligned} V_j &= n_{1j} \beta_1 + \dots + n_{bj} \beta_b + r_j \tau_j \\ &= n_{1j} \left[\frac{1}{k_1} (B_1 - n_{11} \tau_1 - \dots - n_{1v} \tau_v) \right] \\ &\quad + n_{2j} \left[\frac{1}{k_2} (B_2 - n_{21} \tau_1 - \dots - n_{2v} \tau_v) \right] + \dots \\ &\quad + n_{bj} \left[\frac{1}{k_b} (B_b - n_{b1} \tau_1 - \dots - n_{bv} \tau_v) \right] + r_j \tau_j \end{aligned}$$

or

$$V_j = \frac{n_{1j}B_1}{k_1} - \frac{n_{2j}B_2}{k_2} - \dots - \frac{n_{bj}B_b}{k_b}$$

$$= \tau_1 \left[-\frac{n_{11}n_{1j}}{k_1} - \frac{n_{21}n_{2j}}{k_2} - \dots - \frac{n_{b1}n_{bj}}{k_b} \right] + \dots$$

$$+ \tau_v \left[-\frac{n_{1v}n_{1j}}{k_1} - \frac{n_{2v}n_{2j}}{k_2} - \dots - \frac{n_{bv}n_{bj}}{k_b} \right] + r_j \tau_j, \quad j = 1, \dots, v$$

or

$$V_j - \sum_{i=1}^b \frac{n_{ij}B_i}{k_i} = \tau_1 \left[-\frac{n_{11}n_{1j}}{k_1} \dots - \frac{n_{b1}n_{bj}}{k_b} \right] + \dots + \tau_v \left[-\frac{n_{1v}n_{1j}}{k_1} \dots - \frac{n_{bv}n_{bj}}{k_b} \right] + r_j \tau_j$$

or

$$Q_j = \tau_1 \left[-\frac{n_{11}n_{1j}}{k_1} \dots - \frac{n_{b1}n_{bj}}{k_b} \right] + \dots + \tau_v \left[-\frac{n_{1v}n_{1j}}{k_1} \dots - \frac{n_{bv}n_{bj}}{k_b} \right] + r_j \tau_j, \quad j = 1 \dots v$$

where

$$Q_j = V_j - \left[\frac{n_{1j}B_1}{k_1} + \dots + \frac{n_{bj}B_b}{k_b} \right], \quad j = 1, 2, \dots, v$$

are called **adjusted treatment totals**.

[**Note:** Compared to the earlier case, the j^{th} treatment total V_j is adjusted by a factor $\sum_{i=1}^b \frac{n_{ij}B_i}{k_i}$, that is why

it is called “adjusted”. The adjustment is being made for the block effects because they were eliminated to estimate the treatment effects.]

Note that

k_i : Number of plots in i^{th} block.

$\frac{B_i}{k_i}$ is called the average (response) yield per plot from i^{th} block.

$\frac{n_{ij}B_i}{k_i}$ is considered as an average contribution to the j^{th} treatment total from the i^{th} block.

Q_j is obtained by removing the sum of the average contributions of the b blocks from the j^{th} treatment total V_j .

Write

$$Q_j = \tau_1 \left[-\frac{n_{11}n_{1j}}{k_1} \dots - \frac{n_{b1}n_{bj}}{k_b} \right] + \dots + \tau_v \left[-\frac{n_{1v}n_{1j}}{k_1} \dots - \frac{n_{bv}n_{bj}}{k_b} \right] + r_j \tau_j, \quad j = 1, 2, \dots, v.$$

as

$$Q_j = C_{j1}\tau_1 + C_{j2}\tau_2 + \dots + C_{jv}\tau_v$$

where

$$C_{jj} = r_j - \frac{n_{1j}^2}{k_1} - \frac{n_{2j}^2}{k_2} - \dots - \frac{n_{bj}^2}{k_b}$$

$$C_{jj'} = -\frac{n_{1j}n_{1j'}}{k_1} - \frac{n_{2j}n_{2j'}}{k_2} - \dots - \frac{n_{bj}n_{bj'}}{k_b}; \quad j \neq j', \quad j = 1, 2, \dots, v.$$

The $v \times v$ matrix $C = ((C_{jj'}))$, $j = 1, 2, \dots, v$; $j' = 1, 2, \dots, v$ with C_{jj} as diagonal elements and $C_{jj'}$ as off-diagonal elements is called the **C-matrix** of the incomplete block design.

C matrix is symmetric. Its row sum and column sum are zero. (proved later)

Rewrite

$$Q_j = \tau_1 \left[-\frac{n_{11}n_{1j}}{k_1} \dots - \frac{n_{b1}n_{bj}}{k_b} \right] + \dots + \tau_v \left[-\frac{n_{1v}n_{1j}}{k_1} \dots - \frac{n_{bv}n_{bj}}{k_b} \right] + r_j \tau_j, \quad j = 1, 2, \dots, v.$$

as

$$Q = C\tau.$$

This equation is called as **reduced normal equations** where $Q' = (Q_1, Q_2, \dots, Q_v)$, $\tau' = (\tau_1, \tau_2, \dots, \tau_v)$

Equations (1) and (2) are EQUIVALENT.

Alternative presentation in matrix notations:

Now let us try to represent and translate the same algebra in matrix notations.

Let

E_{mn} : $m \times n$ matrix whose all elements are unity.

$N = (n_{ij})$ is $b \times v$ matrix called as **incidence matrix**.

$$k_i = \sum_{j=1}^v n_{ij}$$

$$r_j = \sum_{i=1}^b n_{ij}$$

$$n = \sum_i \sum_j n_{ij}$$

$$E_{1b}N = (r_1, r_2, \dots, r_v) = r'$$

$$NE_{v1} = (k_1, k_2, \dots, k_b)' = k.$$

For illustration, we verify one of the relationships as follows.

$$\begin{aligned}
 E_{1b} &= (1, 1, \dots, 1)_{1 \times b} \\
 N &= \begin{pmatrix} n_{11} & n_{12} & \dots & n_{1v} \\ n_{21} & n_{22} & \dots & n_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ n_{b1} & n_{b2} & \dots & n_{bv} \end{pmatrix} \\
 E_{1b} N &= (1, 1, \dots, 1)_{1 \times b} \begin{pmatrix} n_{11} & n_{12} & \dots & n_{1v} \\ n_{21} & n_{22} & \dots & n_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ n_{b1} & n_{b2} & \dots & n_{bv} \end{pmatrix}_{b \times v} \\
 &= \left(\sum_{i=1}^b n_{i1}, \sum_{i=1}^b n_{i2}, \dots, \sum_{i=1}^b n_{iv} \right) \\
 &= (r_1, r_2, \dots, r_v) \\
 &= r'.
 \end{aligned}$$

It is now clear that the treatment and blocks are not estimable as such as in the case of complete block designs. Note that we have not made any assumption like $\sum_i \alpha_i = \sum_j \beta_j = 0$ also.

Now we introduce the general mean effect (denoted by μ) in the linear model and carry out further analysis on the same lines as earlier.

Consider the model

$$y_{ijm} = \mu + \beta_i + \tau_j + \varepsilon_{ijm}, \quad i = 1, 2, \dots, b; \quad j = 1, 2, \dots, v; \quad m = 0, 1, \dots, n_{ij}.$$

The normal equations are obtained by minimizing $S = \sum_i \sum_j \sum_m \varepsilon_{ijm}^2$ with respect to the parameters

μ, β_i and τ_j and solving them, we can obtain the least-squares estimators of the parameters.

Minimizing $S = \sum_i \sum_j \sum_m \varepsilon_{ijm}^2$ with respect to the parameters μ, β_i and τ_j , the normal equations are

obtained as

$$\begin{aligned}
 n\hat{\mu} + \sum_i n_{i0}\hat{\beta}_i + \sum_j n_{0j}\hat{\tau}_j &= G \\
 n_i\hat{\mu} + n_{i0}\hat{\beta}_i + \sum_j n_{ij}\hat{\tau}_j &= B_i \quad i = 1, \dots, b \\
 n_{0j}\hat{\mu} + n_{0j}\hat{\tau}_j + \sum_i n_{ij}\hat{\beta}_i &= V_j \quad j = 1, \dots, v.
 \end{aligned}$$

Now we write these normal equations in matrix notations. Denote

$$\begin{aligned}\beta &= Col(\beta_1, \beta_2, \dots, \beta_b) \\ \tau &= Col(\tau_1, \tau_2, \dots, \tau_v) \\ B &= Col(B_1, B_2, \dots, B_b) \\ V &= Col(V_1, V_2, \dots, V_v) \\ N &= ((n_{ij})) : \text{incidence matrix of order } b \times v\end{aligned}$$

where $Col(.)$ denotes the column vector.

Let

$$K = diag(k_1, \dots, k_b) : b \times b \text{ diagonal matrix}$$

$$R = diag(r_1, \dots, r_v) : v \times v \text{ diagonal matrix.}$$

Then the $(b + v + 1)$ normal equations can be written as

$$\begin{pmatrix} G \\ B \\ V \end{pmatrix} = \begin{pmatrix} n & E_{1b}K & E_{1v}R \\ KE_{b1} & K & N \\ RE_{v1} & N' & R \end{pmatrix} \begin{pmatrix} \hat{\mu} \\ \hat{\beta} \\ \hat{\tau} \end{pmatrix}. \quad (*)$$

Since we are presently interested in the testing of hypothesis related to the treatment effects, so we eliminate the block effects $\hat{\beta}$ to estimate the treatment effects. For doing so, multiply both sides on the left of equation (*) by

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & I_b & -NR^{-1} \\ 0 & -N'K^{-1} & I_v \end{bmatrix}$$

where

$$R^{-1} = diag\left(\frac{1}{r_1}, \frac{1}{r_2}, \dots, \frac{1}{r_v}\right), \quad K^{-1} = diag\left(\frac{1}{k_1}, \frac{1}{k_2}, \dots, \frac{1}{k_b}\right).$$

Solving it further, we get set of three equations as follows:

$$\begin{aligned}G &= n\hat{\mu} + E_{1b}K\hat{\beta} + E_{1v}R\hat{\tau} \\ B - NR^{-1}V &= [K - NR^{-1}N']\hat{\beta} \\ V - N'K^{-1}B &= [R - N'K^{-1}N]\hat{\tau}\end{aligned}$$

These are called as ‘**reduced normal equations**’ or ‘**reduced intrablock equations**’.

The same reduced intrablock equations can also be obtained as follows. Rewrite equation (*) as

$$G = n\hat{\mu} + E_{1b}K\hat{\beta} + E_{1v}R\hat{\tau} \quad (i)$$

$$B = KE_{b1}\hat{\mu} + K\hat{\beta} + N\hat{\tau} \quad (ii)$$

$$V = RE_{v1}\hat{\mu} + N'\hat{\beta} + R\hat{\tau} \quad (iii)$$

Pre-multiply equation (ii) by $N'K^{-1}$ as

$$N'K^{-1}B = N'K^{-1}KE_{b1}\hat{\mu} + N'K^{-1}K\hat{\beta} + N'K^{-1}N\hat{\tau}$$

and subtract it from equation (iii) as

$$V - N'K^{-1}B = (RE_{v1} - N'K^{-1}KE_{b1})\hat{\mu} + (N' - N'K^{-1}K)\hat{\beta} + (R - N'K^{-1}N)\hat{\tau}$$

$$\text{or } V - N'K^{-1}B = [R - N'K^{-1}N]\hat{\tau}.$$

Next, pre-multiply equation (ii) by NR^{-1} as

$$NR^{-1}V = NR^{-1}RE_{v1}\hat{\mu} + NR^{-1}N'\hat{\beta} + NR^{-1}R\hat{\tau}$$

and subtract it from equation (ii) as

$$B - NR^{-1}V = (KE_{b1} - NR^{-1}RE_{v1})\hat{\mu} + (K - NR^{-1}N')\hat{\beta} + (N - NR^{-1}R)\hat{\tau}$$

$$\text{or } B - NR^{-1}V = [K - NR^{-1}N']\hat{\beta}$$

The reduced normal equation in the treatment effects can be written as

$$Q = C\hat{\tau}$$

where

$$Q = V - N'K^{-1}B$$

$$C = R - N'K^{-1}N.$$

The vector Q is called as the vector of **adjusted treatment totals** since it contains the treatment totals adjusted for the block effects, the matrix C is called as **C-matrix**.

The C matrix is symmetric and its row sums and columns sums are zero.

To show that row sum is zero in C matrix, we proceed as follows:

Row sum:

$$\begin{aligned} CE_{v1} &= RE_{v1} - N'K^{-1}NE_{v1} \\ &= (r_1, r_2, \dots, r_v)' - N'K^{-1}k^{-1} = (r_1, r_2, \dots, r_v)' - N'E_{b1} \\ &= r - r = 0 \end{aligned}$$

Similarly, the column sum can also be shown to be zero.

In order to obtain the reduced normal equation for treatment effects, we first estimated the block effects from one of the normal equation and substituted it into another normal equation related to the treatment effects. This way the adjusted treatment total vector Q (which is adjusted for block effects) is obtained.

Similarly, the reduced normal equations for the block effects can be found as follows. First, estimate the treatment effects from one of the normal equations and substitute it into another normal equation related to the block effects. Then we get the **adjusted block totals** (adjusted for treatment totals).

So, similar to $Q = C\hat{\tau}$, we can obtain another equation which can be represented as

$$D\hat{\beta} = P$$

where

$$D = \text{diag}(k_1, k_2, \dots, k_b) - N \text{diag}\left(\frac{1}{r_1}, \frac{1}{r_2}, \dots, \frac{1}{r_v}\right) N' = K - NR^{-1}N'$$

$$P = B - N \text{diag}\left(\frac{1}{r_1}, \frac{1}{r_2}, \dots, \frac{1}{r_v}\right) V = B - NR^{-1}V$$

$$\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_b)'$$

and P is the adjusted block totals which are obtained after removing the treatment effects

Analysis of variance table:

Under the null hypothesis $H_0 : \tau = 0$, the design is one-way analysis of variance set up with blocks as classifications. In this set up, we have the following:

$$\begin{aligned} \text{sum of squares due to blocks} &= \sum_{i=1}^b \frac{B_i^2}{k_i} - \frac{G^2}{n} \\ &= (B_1, B_2, \dots, B_b)' \text{diag}\left(\frac{1}{k_1}, \frac{1}{k_2}, \dots, \frac{1}{k_b}\right) \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_b \end{pmatrix} - \frac{G^2}{n} \\ &= B'K^{-1}B - \frac{G^2}{n} \end{aligned}$$

If y is the vector of all the observations, then

$$\begin{aligned} \text{Error sum of squares } (S_e) &= \sum_i \sum_j \sum_m (y_{ijm} - \hat{\mu} - \hat{\beta}_i - \hat{\tau}_j)^2 \\ &= \sum_i \sum_j \sum_m y_{ijm} (y_{ijm} - \hat{\mu} - \hat{\beta}_i - \hat{\tau}_j) \quad [\text{Using normal equations, other terms will be zero}] \\ &= \sum_i \sum_j \sum_m y_{ijm}^2 - \hat{\mu}G - \sum_j \hat{\tau}_j V_j - \sum_i \hat{\beta}_i B_i \\ &= y'y - \hat{\mu}G - V'\hat{\tau} - B'\hat{\beta}. \end{aligned}$$

Using original normal equations given by

$$B = KE_{b1}\hat{\mu} + K\hat{\beta} + N\hat{\tau},$$

we have

$$\hat{\beta} = K^{-1}B - E_{b1}\hat{\mu} - K^{-1}N\hat{\tau}.$$

Since $G = V'E_{v1} = B'E_{b1}$, substituting $\hat{\beta}$ in S_e gives

$$\begin{aligned} S_e &= y'y - G\hat{\mu} - B'[K^{-1}B - E_{b1}\hat{\mu} - K^{-1}N\hat{\tau}] - V'\hat{\tau} \\ &= y'y - G\hat{\mu} - B'[K^{-1}B - E_{b1}\hat{\mu} - K^{-1}N\hat{\tau}] - V'\hat{\tau} \\ &= y'y - G\hat{\mu} - B'K^{-1}B + G\hat{\mu} + B'K^{-1}N\hat{\tau} - V'\hat{\tau} \\ &= y'y - B'K^{-1}B + (B'K^{-1}N - V')\hat{\tau} \\ &= \left(y'y - \frac{G^2}{n}\right) - \left(B'K^{-1}B - \frac{G^2}{n}\right) - (V - N'K^{-1}B)'\hat{\tau} \end{aligned}$$

$$S_e = \left(y'y - \frac{G^2}{n}\right) - \left(B'K^{-1}B - \frac{G^2}{n}\right) - Q'\hat{\tau}$$

$$\begin{array}{ccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{Error SS} & = & \text{Total SS} & & \text{Block SS} & & \text{Adjusted treatment SS} \\ & & & & \text{(unadjusted)} & & \text{(adjusted for blocks)} \end{array}$$

The degrees of freedom associated with the different sum of squares are as follows:

Block SS (unadjusted) : $b - 1$

Treatment SS (adjusted) : $v - 1$

Error SS : $n - b - v + 1$

Total SS : $n - 1$

The adjusted treatment sum of squares and the sum of squares due to error are independently distributed and follow a Chi-square distribution with $(v-1)$ and $(n-b-v+1)$ degrees of freedom, respectively.

The analysis of variance table for $H_0 : \tau = 0$ is as follows:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F
Treat	$v - 1$	$Q' \hat{\tau}$ (Adjusted)	$\frac{Q' \hat{\tau}}{v - 1}$	$F = \frac{Q' \hat{\tau} / (v - 1)}{S_e / (n - b - v + 1)}$
Blocks	$b - 1$	$B' K^{-1} B - \frac{G^2}{n}$ (Unadjusted)		
Error	$n - b - v + 1$	S_e	$\frac{S_e}{n - b - v + 1}$	
Total	$n - 1$	$y' y - \frac{G^2}{n}$		

Under H_0 , $\frac{Q' \hat{\tau} / (v - 1)}{S_e / (n - b - v + 1)} \sim F(v - 1, n - b - v + 1)$.

Thus in an incomplete block design, it matters whether we are estimating the block effects first and then the treatment effects are estimated

or

first estimate the treatment effects and then the block effects are estimated.

In complete block designs, it doesn't matter at all. So the testing of hypothesis related to the block and treatment effects can be done from the same estimates.

A reason for this is as follows: In an incomplete block design, either the

- Adjusted sum of squares due to treatments, the unadjusted sum of squares due to blocks and the corresponding sum of squares due to errors are orthogonal

or

- Adjusted sum of squares due to blocks, the unadjusted sum of squares due to treatments and the corresponding sum of squares due to errors are orthogonal.

Note that the adjusted sum of squares due to treatment and the adjusted sum of squares due to blocks are not orthogonal. So

either

$$\text{Error S.S} = \text{Total SS} - \text{SS block (Unadjusted)} - \text{SS treat (Adjusted)}$$

holds true

or

$$\text{Error S.S} = \text{Total SS} - \text{SS block (Adjusted)} - \text{SS treat (Unadjusted)}$$

holds true due to Fisher Cochran theorem.

Since $CE_{v1} = 0$, so C is a rank deficient matrix. Also, since

$$\begin{aligned} Q'E_{v1} &= V'E_{v1} - (N'K^{-1}B)'E_{v1} \\ &= (V_1, \dots, V_v)E_{v1} - B'K^{-1}NE_{v1} \\ &= \left(\sum_i V_i\right) - B'K^{-1}k' \\ &= \sum_i V_i - (B_1, \dots, B_b) \text{diag}\left(\frac{1}{k_1}, \frac{1}{k_2}, \dots, \frac{1}{k_b}\right) \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_b \end{pmatrix} \\ &= \sum_i V_i - (B_1 \dots B_b)E_{b1} \\ &= \sum_i V_i - \sum_j B_j \\ &= G - G \\ &= 0 \end{aligned}$$

so the intrablock equations are consistent.

We will confine our attention to those designs for which $\text{rank}(C) = v - 1$. These are called **connected designs** and for which all contrasts in the treatments, i.e., all linear combinations $l'\tau$ where $l'E_{v1} = 0$ have unique least-squares solutions. This we prove now as follows.

Let G^* and H^* be any two generalized inverses of C by which we mean that they are the square matrix of order v such that G^*Q and H^*Q are both the solution vectors to the intrablock equation, i.e., $\hat{\tau} = G^*Q$ and $\hat{\tau} = H^*Q$, respectively.

Then

$$Q = C\hat{\tau}$$

$$\Rightarrow Q = CG^*Q$$

$$\text{and } Q = CH^*Q \text{ for all } Q$$

$$\text{so that } C(G^* - H^*)Q = 0.$$

It follows that $(G^* - H^*)Q$ can be written as a^*E_{v1} where a is any scalar which may be zero.

Let ℓ be a vector such that $\ell'E_{v1} = 0$. The two estimates of $\ell'\tau$ are $\ell'G^*Q$ and $\ell'H^*Q$ but

$$\begin{aligned}\ell'G^*Q - \ell'H^*Q &= \ell'(G^* - H^*)Q \\ &= \ell'a^*E_{v1} \\ &= a^*\ell'E_{v1} \\ &= 0\end{aligned}$$

$$\Rightarrow \ell'\tau \text{ is unique.}$$

Theorem: The adjusted treatment totals are orthogonal to the block totals.

Proof: It is enough to prove that

$$\text{Cov}(B_i, Q_j) = 0 \text{ for all } i, j.$$

Now

$$\begin{aligned}\text{Cov}(B_i, Q_j) &= \text{Cov}\left[B_i, V_j - \sum_i \left(\frac{n_{ij}}{k_i}\right) B_i\right] \\ &= \text{Cov}(B_i, V_j) - \frac{n_{ij}}{k_i} \text{Var}(B_i)\end{aligned}$$

because the block totals are mutually orthogonal, see how:

For $y_{11}, y_{12}, \dots, y_{1v}$, the block total $B_1 = \sum_{j=1}^v y_{1j}$.

For $y_{21}, y_{22}, \dots, y_{2v}$, the block total $B_2 = \sum_{j=1}^v y_{2j}$.

$$\text{Var}(B_1) = \sum_{j=1}^v \text{Var}(y_{1j}) = v\sigma^2 \text{ as } \text{Cov}(y_{1j}, y_{1k}) = 0 \text{ for } j \neq k$$

$$\text{Var}(B_2) = \sum_{j=1}^v \text{Var}(y_{2j}) = v\sigma^2 \text{ as } \text{Cov}(y_{2j}, y_{2k}) = 0 \text{ for } j \neq k$$

$$\text{Var}(B_1) + \text{Var}(B_2) = 2v\sigma^2 \text{ as } \text{Cov}(y_{1j}, y_{2k}) = 0 \text{ for } j \neq k$$

$$\Rightarrow B_1 \text{ and } B_2 \text{ are mutually orthogonal as all } y_{ij} \text{'s are independent.}$$

As B_i and V_j have n_{ij} observations in common and the observations are mutually independent, so

$$\text{Cov}(B_i, V_j) = n_{ij} \sigma^2$$

$$\text{Var}(B_i) = k_i \sigma^2$$

$$\text{Thus } \text{Cov}(B_i, Q_j) = n_{ij} \sigma^2 - \frac{n_{ij}}{k_i} k_i \sigma^2 = 0$$

Hence proved.

Theorem:

$$E(Q) = C\tau$$

$$\text{Var}(Q) = \sigma^2 C$$

Proof:

$$Q_j = V_j - \left[\frac{n_{1j} B_1}{k_1} + \dots + \frac{n_{bj} B_b}{k_b} \right]$$

$$= V_j - \sum_{i=1}^b \frac{n_{ij} B_i}{k_i}$$

$$E(Q_j) = E(V_j) - \sum_{i=1}^b \frac{n_{ij}}{k_i} E(B_i)$$

$$\begin{aligned} E(V_j) &= \sum_i \sum_m E(y_{ijm}) \\ &= \sum_i \sum_m E(\mu + \beta_i + \tau_j + \varepsilon_{ijm}) \\ &= \mu \sum_i n_{ij} + \sum_i \beta_i n_{ij} + \tau_j \sum_i n_{ij} \\ &= \mu r_j + \sum_i \beta_i n_{ij} + \tau_j r_j \end{aligned}$$

$$\begin{aligned} E(B_i) &= \sum_j \sum_m E(y_{ijm}) \\ &= \sum_j \sum_m E(\mu + \beta_i + \tau_j + \varepsilon_{ijm}) \\ &= \sum_j \sum_m (\mu + \beta_i + \tau_j) \\ &= \mu k_i + \beta_i k_i + \sum_j \tau_j n_{ij} \end{aligned}$$

$$\begin{aligned} \sum_{i=1}^b \frac{n_{ij}}{k_i} E(B_i) &= \sum_{i=1}^b \frac{n_{ij}}{k_i} \left[\mu k_i + \beta_i k_i + \sum_j \tau_j n_{ij} \right] \\ &= \mu r_j + \sum_i \beta_i n_{ij} + \sum_i \frac{n_{ij}}{k_i} \left(\sum_j \tau_j n_{ij} \right). \end{aligned}$$

Thus substituting these expressions in $E(Q_j)$, we have

$$\begin{aligned} E(Q_j) &= r_j \tau_j - \sum_i \frac{n_{ij}}{k_i} (\sum_j \tau_j n_{ij}) \\ &= \left(r_j - \sum_i \frac{n_{ij}^2}{k_i} \right) \tau_j - \sum_i \frac{n_{ij}}{k_i} \sum_{j(\neq \ell)} n_{i\ell} \tau_\ell = c_{jj} \tau_j + \sum_{j(\neq \ell)} c_{\ell j} \tau_j. \end{aligned}$$

Further, substituting $E(Q_j)$ in $E(Q) = (E(Q_1), E(Q_2), \dots, E(Q_b))'$, we get

$$E(Q) = C\tau$$

Next

$$\text{Var}(Q) = \begin{pmatrix} \text{Var}(Q_1) & \text{Cov}(Q_1, Q_2) & \dots & \text{Cov}(Q_1, Q_v) \\ \text{Cov}(Q_2, Q_1) & \text{Var}(Q_2) & \dots & \text{Cov}(Q_2, Q_v) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(Q_v, Q_1) & \text{Cov}(Q_v, Q_2) & \dots & \text{Var}(Q_v) \end{pmatrix}$$

$$\begin{aligned} \text{Var}(Q_j) &= \text{Var}\left[V_j - \sum_i \frac{n_{ij}}{k_i} B_i\right] \\ &= \text{Var}(V_j) + \sum_i \left(\frac{n_{ij}}{k_i}\right)^2 \text{Var}(B_i) - 2 \sum_i \frac{n_{ij}}{k_i} \text{Cov}(V_j, B_i), \end{aligned}$$

Note that

$$\text{Var}(V_j) = \text{Var}\left(\sum_i \sum_m y_{ijm}\right) = r_j \sigma^2$$

$$\text{Var}(B_i) = \text{Var}\left(\sum_j \sum_m y_{ijm}\right) = k_i \sigma^2$$

$$\text{Cov}(V_j, B_i) = \text{Cov}\left(\sum_i \sum_m y_{ijm}, \sum_j \sum_m y_{ijm}\right) = n_{ij} \sigma^2$$

$$\begin{aligned} \text{Var}(Q_j) &= r_j \sigma^2 + \sum_i \left(\frac{n_{ij}}{k_i}\right)^2 k_i \sigma^2 - 2 \sum_i \left(\frac{n_{ij}}{k_i}\right) n_{ij} \sigma^2 \\ &= r_j \sigma^2 - \sum_i \frac{n_{ij}^2}{k_i} \sigma^2 - 2 \sum_i \left(\frac{n_{ij}^2}{k_i}\right) \sigma^2 \\ &= r_j \sigma^2 - \sum_i \left(\frac{n_{ij}^2}{k_i}\right) \sigma^2 \\ &= c_{jj} \sigma^2. \end{aligned}$$

$$\begin{aligned}
\text{Cov}(Q_j, Q_\ell) &= \text{Cov}\left[V_j - \sum_i \frac{n_{ij}}{k_i} B_i, V_\ell - \sum_i \frac{n_{i\ell}}{k_i} B_i\right] \\
&= \text{Cov}(V_j, V_\ell) - \sum_i \frac{n_{i\ell}}{k_i} \text{Cov}(V_j, B_i) - \sum_i \frac{n_{ij}}{k_i} \text{Cov}(B_i, V_\ell) + \sum_i \frac{n_{ij} n_{i\ell}}{k_i^2} \text{Cov}(B_i, B_i) \\
&= 0 - \sum_i \frac{n_{i\ell}}{k_i} \text{Cov}(V_j, B_i) - \sum_i \frac{n_{ij}}{k_i} \text{Cov}(B_i, V_\ell) + \sum_i \frac{n_{ij} n_{i\ell}}{k_i^2} \text{Var}(B_i) \\
&= -\sum_i \left(\frac{n_{i\ell} n_{ij}}{k_i}\right) \sigma^2 - \sum_i \left(\frac{n_{ij} n_{i\ell}}{k_i}\right) \sigma^2 + \sum_i \frac{n_{ij} n_{i\ell}}{k_i^2} k_i \sigma^2 \\
&= c_{j\ell} \sigma^2
\end{aligned}$$

Substituting the terms of $\text{Var}(Q_j) = c_{jj} \sigma^2$ and $\text{Cov}(Q_j, Q_\ell) = c_{j\ell} \sigma^2$ in $\text{Var}(Q)$, we get $\text{Var}(Q) = C \sigma^2$.

Hence proved.

[Note: We will prove this result using the matrix approach later].

Covariance matrix of adjusted treatment totals:

Consider $Z = \begin{pmatrix} V \\ B \end{pmatrix}$ with $b + v$ variables.

We can express

$$\begin{aligned}
Q &= V - N' K^{-1} B \\
&= [I \quad -N' K^{-1}] \begin{bmatrix} V \\ B \end{bmatrix} \\
&= [I \quad -N' K^{-1}] Z.
\end{aligned}$$

So

$$\text{Cov}(Q) = [I \quad -N' K^{-1}] \text{Cov}(Z) \begin{bmatrix} I' \\ (-N' K^{-1})' \end{bmatrix}.$$

Now we find that

$$\text{Cov}(Z) = \begin{pmatrix} \text{Var}(V) & \text{Cov}(V, B) \\ \text{Cov}(B, V) & \text{Var}(B) \end{pmatrix}$$

Since B_i and V_j have n_{ij} observations in common and the observations are mutually independent, so

$$\text{Cov}(B_i, V_j) = n_{ij} \sigma^2$$

$$\text{Var}(B_i) = k_i \sigma^2$$

$$\text{Var}(V_j) = r_j \sigma^2.$$

Thus

$$\begin{aligned}
Cov(Z) &= \begin{pmatrix} R & N' \\ N & K \end{pmatrix} \sigma^2 \\
Cov(Q) &= \begin{bmatrix} I & -N'K^{-1} \end{bmatrix} \begin{bmatrix} R & N' \\ N & K \end{bmatrix} \begin{bmatrix} I \\ -K^{-1}N \end{bmatrix} \sigma^2 \\
&= \begin{bmatrix} R - N'K^{-1}N & N' - N' \end{bmatrix} \begin{bmatrix} I \\ -K^{-1}N \end{bmatrix} \sigma^2 \\
&= (R - N'K^{-1}N) \sigma^2 \\
&= C\sigma^2.
\end{aligned}$$

Next, we show that $Cov(B, Q) = 0$

$$\begin{aligned}
Cov(B, Q) &= Cov(B, V) - Cov(B, V - N'K^{-1}B) \\
&= Cov(B, V) - Var(B)K^{-1}N \\
&= N\sigma^2 - KK^{-1}N\sigma^2 \\
&= 0.
\end{aligned}$$

An alternative approach to find/ prove $E(Q) = C\tau$, $D(Q) = C\sigma^2$

Now we illustrate another approach to find the expectations etc in the set up of an incomplete block design. We have now learnt three approaches- the classical approach based on summations, the approach based on matrix theory and this new approach which is also based on the matrix theory. We can choose any of the approaches. The objective here is to let the reader know these different approaches.

Rewrite the linear model

$$y_{ijm} = \mu + \beta_i + \tau_j + \varepsilon_{ijm}, \quad i = 1, 2, \dots, b; \quad j = 1, 2, \dots, v; \quad m = 0, 1, \dots, n_{ij}.$$

as

$$y = \mu E_{n1} + D_1' \tau + D_2' \beta + \varepsilon$$

where

$$\begin{aligned}
\tau &= (\tau_1, \tau_2, \dots, \tau_v)' \\
\beta &= (\beta_1, \beta_2, \dots, \beta_b)' .
\end{aligned}$$

Since B_i and V_j have n_{ij} observations in common and the observations are mutually independent, so denote

$D_1 : v \times n$ matrix of treatment effect versus N , i.e., $(i, j)^{\text{th}}$ element of this matrix is given by

$$D_1 = \begin{cases} 1 & \text{if } j^{\text{th}} \text{ observation comes from } i^{\text{th}} \text{ treatment} \\ 0 & \text{otherwise} \end{cases}$$

$D_2 : b \times n$ matrix of block effects versus N , i.e., $(i, j)^{\text{th}}$ element of this matrix is given by

$$D_2 = \begin{cases} 1 & \text{if } j^{\text{th}} \text{ observation comes from } i^{\text{th}} \text{ block} \\ 0 & \text{otherwise.} \end{cases}$$

Following results can be verified:

$$D_1 D_1' = R = \text{diag}(r_1, r_2, \dots, r_v),$$

$$D_2 D_2' = K = \text{diag}(k_1, k_2, \dots, k_b),$$

$$D_2 D_1' = N \text{ or } D_1 D_2' = N'$$

$$D_1 E_{n1} = (r_1, r_2, \dots, r_v)'$$

$$D_2 E_{n1} = (k_1, k_2, \dots, k_b)'$$

$$D_1' E_{v1} = E_{n1} = D_2' E_{b1}.$$

In earlier notations,

$$V = (V_1, V_2, \dots, V_v)' = D_1 y$$

$$B = (B_1, B_2, \dots, B_b)' = D_2 y$$

Express Q in terms of D_1 and D_2 as

$$\begin{aligned} Q &= V - N' K^{-1} B \\ &= [D_1 - D_1 D_2' (D_2 D_2')^{-1} D_2] y. \end{aligned}$$

Then

$$\begin{aligned} E(Q) &= [D_1 - D_1 D_2' (D_2 D_2')^{-1} D_2] E(y) \\ &= [D_1 - D_1 D_2' (D_2 D_2')^{-1} D_2] (\mu E_{n1} + D_1' \tau + D_2' \beta) \\ &= [D_1 E_{n1} - D_1 D_2' (D_2 D_2')^{-1} D_2 E_{n1}] \mu + [D_1 D_1' - D_1 D_2' (D_2 D_2')^{-1} D_2 D_1'] \tau \\ &\quad + [D_1 D_2' - D_1 D_2' (D_2 D_2')^{-1} D_2 D_2'] \beta \\ &= [(r_1, r_2, \dots, r_v)' - N' K^{-1} (k_1, \dots, k_b)'] \mu + [R - N' K^{-1} N] \tau + [N' - N' K^{-1} K] \beta. \end{aligned}$$

$$\begin{aligned}
\text{Since } N'K^{-1}(k_1, \dots, k_b)' &= N' \text{diag} \left(\frac{1}{k_1}, \frac{1}{k_2}, \dots, \frac{1}{k_b} \right) \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_b \end{pmatrix} \\
&= N' \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} n_{11} & n_{21} & \dots & n_{b1} \\ n_{12} & n_{22} & \dots & n_{b2} \\ \vdots & \vdots & \ddots & \vdots \\ n_{1v} & n_{2v} & \dots & n_{bv} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\
&= \left(\sum_{i=1}^b n_{i1}, \sum_{i=1}^b n_{i2}, \dots, \sum_{i=1}^b n_{iv} \right)' = (r_1, r_2, \dots, r_v)'.
\end{aligned}$$

Thus

$$N' - N'K^{-1}K = (r_1, r_2, \dots, r_v)' - N'K^{-1}(k_1, \dots, k_b) = 0$$

and so

$$\begin{aligned}
E(Q) &= [R - N'K^{-1}N]\tau \\
&= C\tau.
\end{aligned}$$

Next

$$\begin{aligned}
\text{Var}(Q) &= D_1 [I - D_2'(D_2 D_2')^{-1} D_2] \text{Var}(y) [I - D_2'(D_2 D_2')^{-1} D_2] D_1' \\
&= \sigma^2 D_1 [I - D_2'(D_2 D_2')^{-1} D_2] D_1' \\
&= \sigma^2 [D_1 D_1' - D_1 D_2' (D_2 D_2')^{-1} D_2 D_1'] \\
&= \sigma^2 [R - N'K^{-1}N] \\
&= \sigma^2 C.
\end{aligned}$$

Note that $[I - D_2'(D_2 D_2')^{-1} D_2]$ is an idempotent matrix.

Similarly, we can also express

$$\begin{aligned}
P &= B - NR^{-1}V \\
&= [D_2 - D_2 D_1' R^{-1} D_1] y.
\end{aligned}$$

Theorem: $E(P) = D\beta$, $\text{Var}(P) = \sigma^2 D$

Proof:

$$\begin{aligned}
D &= K - NR^{-1}N' \\
P &= B - NR^{-1}V \\
&= D_2 [I - D_1 R^{-1} D_1'] y \\
&= D_2 [I - D_1' (D_1 D_1')^{-1} D_1] y
\end{aligned}$$

$$\begin{aligned}
E(P) &= D_2[I_1 - D_1'(D_1D_1')^{-1}D_1](\mu E_{n_1} + D_1'\tau + D_2'\beta) \\
&= [D_2E_{n_1} - D_2D_1'R^{-1}D_1E_{n_1}]\mu + [D_2D_1' - D_2D_1'R^{-1}D_1D_1']\tau \\
&\quad + [D_2D_1' - D_2D_1'R^{-1}D_1D_2']\beta \\
&= [(k_1, k_2, \dots, k_b)' - NR^{-1}(r_1, r_2, \dots, r_v)']\mu + [N - NR^{-1}R]\tau + [K - NR^{-1}N']\beta \\
&= [(k_1, k_2, \dots, k_b)' - NE_{v_1}]\mu + 0 + D\beta \\
&= [(k_1, k_2, \dots, k_b)' - (k_1, k_2, \dots, k_b)']\mu + D\beta \\
&= D\beta
\end{aligned}$$

Next

$$\begin{aligned}
\text{Var}(P) &= \sigma^2 D_2[I - D_1'(D_1D_1')^{-1}D_1]D_2' \\
&= \sigma^2 [D_2D_2' - D_2D_1'(D_1D_1')^{-1}D_1D_2'] \\
&= \sigma^2 [K - NR^{-1}N'] = \sigma^2 D
\end{aligned}$$

Note that $[I - D_1'(D_1D_1')^{-1}D_1]$ is an idempotent matrix.

Alternatively, we can also find $\text{Var}(P)$ as follows:

$$\begin{aligned}
P &= (I \quad -NR^{-1}) \begin{pmatrix} B \\ V \end{pmatrix} = (I \quad -NR^{-1})Z \quad \text{where } Z = (B, \quad V)' \\
\text{Var}(P) &= (I \quad -NR^{-1}) \text{Cov}(Z) \begin{pmatrix} I \\ -R^{-1}N' \end{pmatrix} \\
&= (I \quad -NR^{-1}) \begin{pmatrix} K & N \\ N' & R \end{pmatrix} \begin{pmatrix} I \\ -R^{-1}N' \end{pmatrix} \sigma^2 \\
&= (K - NR^{-1}N' \quad N - NR^{-1}R) \begin{pmatrix} I \\ -R^{-1}N' \end{pmatrix} \sigma^2 \\
&= (K - NR^{-1}N') \sigma^2 \\
&= D\sigma^2
\end{aligned}$$

Now we consider some properties of incomplete block designs.

Lemma: $b + \text{rank}(C) = v + \text{rank}(D)$.

Proof: Consider $(b + v) \times (b + v)$ matrix

$$A = \begin{bmatrix} K & N \\ N' & R \end{bmatrix}$$

Note that A is a submatrix of C .

Using the result that the rank of a matrix does not change by the pre-multiplication of the nonsingular matrix, consider the following matrices:

$$M = \begin{bmatrix} I_b & 0 \\ -N'K^{-1} & I_v \end{bmatrix}$$

and

$$S = \begin{bmatrix} I_b & 0 \\ -R^{-1}N' & I_v \end{bmatrix}.$$

M and S are nonsingular, so we have

$$\text{rank}(A) = \text{rank}(MA) = \text{rank}(AS).$$

Now

$$MA = \begin{bmatrix} I_b & 0 \\ -N'K^{-1} & I_v \end{bmatrix} \begin{bmatrix} K & N \\ N' & R \end{bmatrix} = \begin{bmatrix} K & N \\ 0 & C \end{bmatrix}$$

$$AS = \begin{bmatrix} D & N \\ 0 & R \end{bmatrix}.$$

Thus

$$\text{rank} \begin{bmatrix} K & N \\ 0 & C \end{bmatrix} = \text{rank} \begin{bmatrix} D & N \\ 0 & R \end{bmatrix}$$

or

$$\text{rank}(K) + \text{rank}(C) = \text{rank}(D) + \text{rank}(R)$$

or

$$b + \text{rank}(C) = v + \text{rank}(D)$$

Remark:

$C : v \times v$ and $D : b \times b$ are symmetric matrices.

One can verify that

$$CE_{v1} = 0$$

$$\text{and } DE_{b1} = 0$$

$$\text{Thus } \text{rank}(C) \leq v - 1$$

$$\text{and } \text{rank}(D) \leq b - 1.$$

Lemma:

If $\text{rank}(C) = v - 1$, then all blocks and treatment contrasts are estimable.

Proof:

If $\text{rank}(C) = v - 1$, it is obvious that all the treatment contrasts are estimable.

Using the result from the lemma $b + \text{rank}(C) = v + \text{rank}(D)$, we have

$$\begin{aligned}\text{rank}(D) + v &= \text{rank}(C) + b \\ &= v - 1 + b\end{aligned}$$

Thus

$$\text{rank}(D) = b - 1.$$

Thus all the block contrasts are also estimable.

Orthogonality of Q and P:

Now we explore the conditions under which Q and P can be orthogonal.

$$\begin{aligned}Q &= V - N'K^{-1}B \\ &= (D_1 - D_1D_2'K^{-1}D_2)y\end{aligned}$$

$$\begin{aligned}P &= B - NR^{-1}V \\ &= (D_2 - D_2D_1'R^{-1}D_1)y\end{aligned}$$

$$\begin{aligned}\text{Cov}(Q, P) &= (D_1 - D_1D_2'K^{-1}D_2)(D_2 - D_2D_1'R^{-1}D_1)'\sigma^2 \\ &= (D_1D_2' - D_1D_2'R^{-1}D_1D_2' - D_1D_2'K^{-1}D_2D_2' + D_1D_2'K^{-1}D_2D_1'R^{-1}D_1D_2')\sigma^2 \\ &= (N' - RR^{-1}N' - N'K^{-1}K + N'K^{-1}NR^{-1}N')\sigma^2 \\ &= (N'K^{-1}NR^{-1}N' - N')\sigma^2\end{aligned}$$

Q and P (or equivalently Q_i and P_j) are orthogonal when

$$\text{Cov}(Q, P) = 0$$

$$\text{or } N'K^{-1}NR^{-1}N' - N' = 0 \quad (i)$$

$$\Rightarrow (R - C)R^{-1}N' - N' = 0 \quad (\text{Using } C = R - N'K^{-1}N)$$

$$\Rightarrow CR^{-1}N' = 0 \quad (ii)$$

or equivalently

$$N'K^{-1}NR^{-1}N' - N' = 0$$

$$\Rightarrow N'K^{-1}(K - D) - N' = 0 \quad (\text{Using } D = K - NR^{-1}N')$$

$$\Rightarrow N'K^{-1}D = 0 \quad (iii)$$

Thus Q_i and P_j are orthogonal if

$$NR^{-1}N'K^{-1}N = N'$$

or equivalently

$$NR^{-1}C = 0$$

or equivalently

$$DK^{-1}N = 0.$$

Orthogonal block design:

A block design is said to be orthogonal if Q_i 's and P_j 's are orthogonal for all i and j . Thus the condition for the orthogonality of design is $NR^{-1}N'K^{-1}N = N$, $NR^{-1}C = 0$ or $DK^{-1}N = 0$.

Lemma: If $\frac{n_{ij}}{r_j}$ is constant for all j , then $\frac{n_{ij}}{k_i}$ is constant for all i and vice versa. In this case, we have

$$n_{ij} = \frac{k_i r_j}{n}.$$

Proof. If $\frac{n_{ij}}{r_j}$ is constant for all j then $\frac{n_{ij}}{r_j} = a_i$, say.

$$\Rightarrow n_{ij} = a_i r_j$$

$$\text{or } \sum_j n_{ij} = \sum_j a_i r_j = a_i \sum_j r_j = a_i n$$

$$\text{or } k_i = a_i n$$

$$\text{or } a_i = \frac{k_i}{n}$$

Thus

$$\frac{n_{ij}}{r_j} = \frac{k_i}{n}$$

$$\text{or } n_{ij} = \frac{k_i r_j}{n}$$

$$\text{So } \frac{n_{ij}}{k_j} = \frac{r_j}{n} : \text{ independent of } i.$$

Hence proved.

Contrast:

A linear function $\sum_{j=1}^v c_j \tau_j = C' \tau$ where $c_1, c_2, \dots, c_v$ are given number such that $\sum_{j=1}^v c_j = 0$ is called a contrast of τ_j 's.

Elementary contrast:

A contrast $\sum_{j=1}^v c_j \tau_j = C' \tau$ with $C = (c_1, c_2, \dots, c_v)'$ in treatment effects $\tau = (\tau_1, \tau_2, \dots, \tau_v)'$ is called an elementary contrast if C has only two non-zero components 1 and -1.

Elementary contrasts in the treatment effects involve all the differences in the form $\tau_i - \tau_j$, $i \neq j$.

It is desirable to design experiments where all the elementary contrasts are estimable.

Connected Design:

A design where all the elementary contrasts are estimable is called a connected design otherwise it is called a **disconnected design**.

The physical meaning of connectedness of a design is as follows:

Given any two treatment effects τ_{i1} and τ_{i2} , it is possible to have a chain of treatment effects like $\tau_{i1}, \tau_{1j}, \tau_{2j}, \dots, \tau_{nj}, \tau_{i2}$, such that two adjoining treatments in this chain occur in the same block.

Example of connected design:

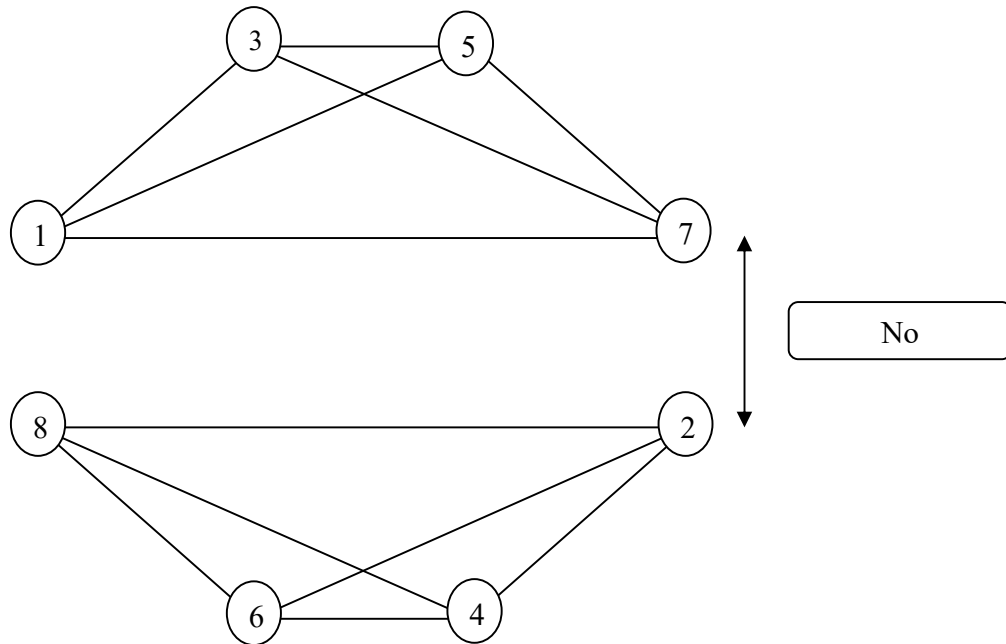
In a connected design, within every block, all the treatment contrasts are estimable and pair-wise comparison of estimators have similar variances.

Consider a disconnected incomplete block design as follows:

$b = 8$ (Block numbers: I, II, ..., VIII), $k = 3$, $v = 8$ (treatment numbers: 1, 2, ..., 8), $r = 3$

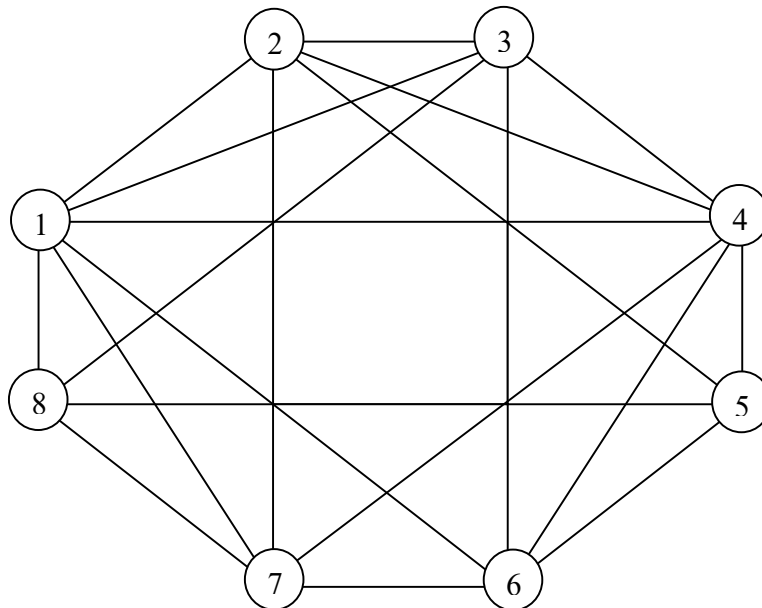
Blocks	Treatments
I	1 3 5
II	2 4 6
III	3 5 7
IV	4 6 8
V	5 7 1
VII	6 8 2
VII	7 1 3
VIII	8 2 4

The blocks of this design can be represented graphically as follows:



Note that it is not possible to reach the treatment, e.g., 7 from 2, 3 from 4 etc. So the design is not connected.

Moreover, if the blocks of the design are given like in the following figure, then any treatment can be reached from any treatment. So the design, in this case, is connected. For example, treatment 2 can be reached from treatment 6 through different routes like $6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2$, $6 \rightarrow 3 \rightarrow 2$, $6 \rightarrow 7 \rightarrow 8 \rightarrow 1 \rightarrow 2$, $6 \rightarrow 7 \rightarrow 2$ etc.



A design is connected if every treatment can be reached from every treatment via lines in the connectivity graph.

Theorem: An incomplete block design with v treatments is connected if and only if $\text{rank}(C) = v - 1$.

Proof: Let the design be connected. Consider a set of $(v - 1)$ linearly independent contrasts $\tau_i - \tau_j$ ($j = 2, 3, \dots, v$). Let these contrasts be $C_2'\tau, C_3'\tau, \dots, C_v'\tau$ where $\tau = (\tau_1, \tau_2, \dots, \tau_v)'$. Obviously, vectors $C_2, C_3, \dots, C_v$ form the basis of vector space of dimension $(v - 1)$. Thus any contrast $p'\tau$ is expressible as a linear combination of the contrasts $C_i'\tau$ ($i = 2, 3, \dots, v$). Also $p'\tau$ is estimable if and only if p belongs to the column space of C -matrix of the design.

Therefore, the dimension of column space of C must be the same as that of the vector space spanned by the vectors C_i ($i = 2, 3, \dots, v$), i.e., equal to $(v - 1)$.

Thus $\text{rank}(C) = v - 1$.

Conversely, let $\text{rank}(C) = v - 1$ and let $\xi_1, \xi_2, \dots, \xi_{v-1}$ be a set of orthonormal eigenvectors corresponding to the (not necessarily distinct) non-zero eigenvalues $\theta_1, \theta_2, \dots, \theta_{v-1}$ of C .

Then

$$\begin{aligned} E(\xi_i'Q) &= \xi_i' C \tau \\ &= \theta_i \xi_i' \tau. \end{aligned}$$

Thus an unbiased estimator of $\xi_i' \tau$ is $\frac{\xi_i' Q}{\theta_i}$.

Also, since each ξ_i is orthogonal to E_{v1} and ξ_i 's are mutually orthogonal, so any contrast $p'\tau$ belongs to the vector space spanned by $\{\xi_i, i = 1 \dots v\}$, i.e., $p = \sum_{i=1}^{v-1} a_i \xi_i$.

$$\text{So } E \left[\sum_{i=1}^{v-1} a_i \frac{\xi_i' Q}{\theta_i} \right] = p' \tau.$$

Thus $p'\tau$ is estimable and this completes the proof.

Lemma: For a connected block design $Cov(Q, P) = 0$ if and only if $N' = \frac{rk'}{n}$.

Proof: “if” part

When $N' = \frac{rk'}{n}$, we have

$$\begin{aligned} \frac{1}{\sigma^2} Cov(Q, P) &= N' K^{-1} N R^{-1} N' - N' \\ &= \frac{rk' K^{-1} k r' R^{-1} N'}{n^2} - N' \end{aligned}$$

$$\begin{aligned} \text{Since } k' K^{-1} k &= (k_1, k_2, \dots, k_b) \text{diag} \left(\frac{1}{k_1}, \frac{1}{k_2}, \dots, \frac{1}{k_b} \right) \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_b \end{pmatrix} \\ &= (1, \dots, 1) \begin{pmatrix} k_1 \\ \vdots \\ k_b \end{pmatrix} = \sum_{i=1}^b k_i = n \end{aligned}$$

and

$$r' R^{-1} = (r_1, r_2, \dots, r_v) \text{diag} \left(\frac{1}{r_1}, \frac{1}{r_2}, \dots, \frac{1}{r_v} \right) = E_{1v}.$$

Then

$$\begin{aligned} \frac{1}{\sigma^2} Cov(Q, P) &= \frac{rn E_{1v} N'}{n^2} - N' \\ &= \frac{r E_{1v} N'}{n} - N' = \frac{rk'}{n} - N' \\ &= N' - N' = 0. \end{aligned}$$

“Only if part”

Let $Cov(Q, P) = 0$

$$\Rightarrow N' K^{-1} N R^{-1} N' - N' = 0 \quad (\text{Since } C = R - N' K^{-1} N)$$

$$\text{or } (R - C) R^{-1} N' - N' = 0$$

$$\text{or } C R^{-1} N' = 0.$$

Let

$$R^{-1} N' = A = (\underline{a}_1, \underline{a}_2, \dots, \underline{a}_b)$$

where $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_b$ are the columns of A.

Since the design is connected, so the columns of A are proportional to E_{v1} . Also, all row/column sums of C are zero.

$$\text{So } (CE_{v1}, CE_{v1}, \dots, CE_{v1}) = 0$$

and

$$CA = 0$$

$$\text{or } C(\underline{a}_1, \underline{a}_2, \dots, \underline{a}_b) = 0$$

$$\Rightarrow a_i \propto E_{v1}$$

$$\text{or } a_i = \alpha_i E_{v1}; i = 1, 2, \dots, b$$

where α_i are some scalars.

This gives

$$A = R^{-1}N' = E_{v1}\alpha' \text{ where } \alpha = (\alpha_1, \dots, \alpha_b)'$$

So we have

$$\begin{aligned} N' &= RE_{v1}\alpha = (r_1, r_2, \dots, r_v)'\alpha' \\ &= r\alpha' \text{ where } r = (r_1, r_2, \dots, r_v)' \end{aligned}$$

Pre-multiply by E_{1v} gives

$$E_{1v}N' = (k_1, k_2, \dots, k_b)' = E_{1v}r\alpha' = n\alpha'$$

$$\text{or } k = n\alpha'$$

$$\Rightarrow \alpha' = \frac{k'}{n} \text{ where } k = (k_1, k_2, \dots, k_b)'$$

Thus

$$N' = r\alpha' = \frac{rk'}{n}.$$

Hence proved.

Definition: A connected block design is said to be **orthogonal** if and only if the incidence matrix of the design satisfies the condition $N' = \frac{rk'}{n}$.

Designs which do not satisfy this condition are called **non-orthogonal**. It is clear from this result that if at least one entry of N is zero, the design cannot be orthogonal.

A block design with at least one zero-entry in its incidence matrix is called an incomplete block design.

Theorem: A sufficient condition for the orthogonality of design is that $\frac{n_{ij}}{r_j}$ is constant for all j .

Conclusion: It is obvious from the condition of orthogonality of a design that a design which is not connected and an incomplete design even though it may be connected cannot have an orthogonal structure.

Now we illustrate the general nature of the incomplete block design. We try to obtain the results for a randomized block design through the results of an incomplete block design.

Randomized block design:

The randomized block design is an arrangement of v treatment in b blocks of v plots each, such that every treatment occurs in every block, one treatment in each plot.

The arrangement of treatment within a block is random and in terms of incidence matrix,

$$n_{ij} = 1 \quad \text{for all } i = 1, 2, \dots, v; \quad j = 1, 2, \dots, b.$$

Thus we have

$$k_i = \sum_j n_{ij} = v \quad \text{for all } i$$

$$r_j = \sum_i n_{ij} = v \quad \text{for all } j.$$

We have $\frac{n_{ij}}{r_j} = \frac{1}{v}$ constant for all j .

$$C_{jj} = v - \frac{b}{v}$$

$$C_{jj'} = -\frac{b}{v}$$

$$Q_j = V_j - \frac{G}{v}.$$

Normal equations for τ 's are

$$\left(v - \frac{b}{v}\right)\tau_j - \frac{b}{v} \sum_{i \neq j} \tau_i = V_j - \frac{G}{v}; \quad j = 1, \dots, v$$

$$\tau_1 + \tau_2 + \dots + \tau_v = 0.$$

Thus

$$b\tau_j - \frac{b}{v} \sum_j \tau_j = V_j - \frac{G}{v}$$

$$\text{or } \hat{\tau}_j = \frac{1}{b} \left(V_j - \frac{G}{v} \right) = \bar{y}_{oj} - \bar{y}_{oo}.$$

The sum of squares due to treatments adjusted for blocks is

$$= \sum_j \hat{\tau}_j Q_j$$

$$= \frac{1}{b} \sum_j \left(V_j - \frac{G}{v} \right)^2$$

$$= \frac{\sum_j V_j^2}{b} - \frac{G^2}{bv},$$

which is also the sum of squares due to treatments which are unadjusted for blocks because the design is orthogonal.

$$\text{Sum of squares due to blocks} = \frac{\sum_i B_i^2}{v} - \frac{G^2}{bv}$$

$$\text{Sum of squares due to error} = \sum_i \sum_j \left(y_{ij} - \frac{B_i}{v} - \frac{V_j}{b} + \frac{G}{bv} \right)^2.$$

These expressions are the same as obtained under the analysis of variance in the set up of a randomized block design.

Interblock analysis of incomplete block design

The purpose of block designs is to reduce the variability of response by removing the part of the variability as block numbers. If in fact, this removal is illusory, the block effects being all equal, then the estimates are less accurate than those obtained by ignoring the block effects and using the estimates of treatment effects. On the other hand, if the block effect is very marked, the reduction in the basic variability may be sufficient to ensure a reduction of the actual variances for the block analysis.

In the intrablock analysis related to treatments, the treatment effects are estimated after eliminating the block effects. If the block effects are marked, then the block comparisons may also provide information about the treatment comparison. So a question arises how to utilize the block information additionally to develop an analysis of variance to test the hypothesis about the significance of treatment effects.

Such an analysis can be derived regarding the block effects as random variables. This assumption involves the random allocation of different blocks of the design to be the blocks of the material selected (at random from the population of possible blocks) in addition to the random allocation of treatments occurring in a block to the units of the block selected to contain them. Now the two responses from the same block are correlated because the error associated with each contains the block number in common. Such an analysis of incomplete block design is termed as interblock analysis.

To illustrate the idea behind the interblock analysis and how the block comparisons also contain information about the treatment comparisons, consider an allocation of four selected treatments in two blocks each. The outputs (y_{ij}) are recorded as follows:

Block 1: $y_{14} \ y_{16} \ y_{17} \ y_{19}$

Block 2: $y_{23} \ y_{25} \ y_{26} \ y_{27}$.

The block totals are

$$B_1 = y_{14} + y_{16} + y_{17} + y_{19},$$

$$B_2 = y_{23} + y_{25} + y_{26} + y_{27}.$$

Following the model $y_{ij} = \mu + \beta_i + \tau_j + \varepsilon_{ij}$, $i = 1, 2, j = 1, 2, \dots, 9$, we have

$$y_{14} = \mu + \beta_1 + \tau_4 + \varepsilon_{14},$$

$$y_{16} = \mu + \beta_1 + \tau_6 + \varepsilon_{16},$$

$$y_{17} = \mu + \beta_1 + \tau_7 + \varepsilon_{17},$$

$$y_{19} = \mu + \beta_1 + \tau_9 + \varepsilon_{19},$$

$$y_{23} = \mu + \beta_2 + \tau_3 + \varepsilon_{23},$$

$$y_{25} = \mu + \beta_2 + \tau_5 + \varepsilon_{25},$$

$$y_{26} = \mu + \beta_2 + \tau_6 + \varepsilon_{26},$$

$$y_{27} = \mu + \beta_2 + \tau_7 + \varepsilon_{27},$$

and thus

$$\begin{aligned} B_1 - B_2 &= 4(\beta_1 - \beta_2) + (\tau_4 + \tau_6 + \tau_7 + \tau_9) - (\tau_3 + \tau_5 + \tau_6 + \tau_7) \\ &\quad + (\varepsilon_{14} + \varepsilon_{16} + \varepsilon_{17} + \varepsilon_{19}) - (\varepsilon_{23} + \varepsilon_{25} + \varepsilon_{26} + \varepsilon_{27}). \end{aligned}$$

If we assume additionally that the block effects β_1 and β_2 are random with mean zero, then

$$E(B_1 - B_2) = (\tau_4 + \tau_9) - (\tau_3 + \tau_5)$$

which reflects that the block comparisons can also provide information about the treatment comparisons.

The intrablock analysis of an incomplete block design is based on estimating the treatment effects (or their contrasts) by eliminating the block effects. Since different treatments occur in different blocks, so one may expect that the block totals may also provide some information on the treatments. The interblock analysis utilizes the information on block totals to estimate the treatment differences. The block effects are assumed to be random and so we consider the set up of mixed effect model in which the treatment effects are fixed but the block effects are random. This approach is applicable only when the number of blocks is more than the number of treatments. We consider here the interblock analysis of binary proper designs for which $n_{ij} = 0$ or 1 and $k_1 = k_2 = \dots = k_b = k$ in connection with the intrablock analysis.

Model and Normal Equations

Let y_{ij} denotes the response from the j^{th} treatment in i^{th} block from the model

$$y_{ij} = \mu^* + \beta_i^* + \tau_j + \varepsilon_{ij}, i = 1, 2, \dots, b; j = 1, 2, \dots, v$$

where

μ^* is the general mean effect;

β_i^* is the random additive i^{th} block effect;

τ_j is the fixed additive j^{th} treatment effect; and

ε_{ij} is the i.i.d. random error with $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Since the block effect is now considered to be random, so we additionally assume that $\beta_i^* (i = 1, 2, \dots, b)$ are independently distributed following $N(0, \sigma_{\beta}^2)$ and are uncorrelated with ε_{ij} . One may note that we can not assume here $\sum_i \beta_i^* = 0$ as in other cases of fixed-effect models. In place of this, we take $E(\beta_i^*) = 0$. Also, y_{ij} 's are no longer independently distributed but

$$\begin{aligned} \text{Var}(y_{ij}) &= \sigma_{\beta}^2 + \sigma^2, \\ \text{Cov}(y_{ij}, y_{i'j'}) &= \begin{cases} \sigma_{\beta}^2 & \text{if } i = i', j \neq j' \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

In the case of interblock analysis, we work with the block totals B_i in place of y_{ij} where

$$\begin{aligned} B_i &= \sum_{j=1}^v n_{ij} y_{ij} \\ &= \sum_{j=1}^v n_{ij} (\mu^* + \beta_i^* + \tau_j + \varepsilon_{ij}) \\ &= k\mu^* + \sum_j n_{ij} \tau_j + f_i \end{aligned}$$

where $f_i = \beta_i^* k + \sum_j n_{ij} \varepsilon_{ij}$, ($i = 1, 2, \dots, b$) are independent and normally distributed with mean 0 and

$$\text{Var}(f_i) = k^2 \sigma_\beta^2 + k \sigma^2 = \sigma_f^2.$$

Thus

$$\begin{aligned} E(B_i) &= k\mu^* + \sum_j n_{ij} \tau_j, \\ \text{Var}(B_i) &= \sigma_f^2; \quad i = 1, 2, \dots, b, \\ \text{Cov}(B_i, B_{i'}) &= 0; \quad i \neq i'; i, i' = 1, 2, \dots, b. \end{aligned}$$

In matrix notations, the model under consideration can be written as

$$B = k\mu^* E_{b1} + N\tau + f$$

where $f = (f_1, f_2, \dots, f_b)'$.

Estimates of μ^* and τ in interblock analysis:

In order to obtain the estimates of μ^* and τ , we minimise the sum of squares due to error

$f = (f_1, f_2, \dots, f_b)'$, i.e., minimize $(B - k\mu^* E_{b1} - N\tau)'(B - k\mu^* E_{b1} - N\tau)$ with respect to μ^* and τ .

The estimates of μ^* and τ are the solutions of following normal equations:

$$\begin{pmatrix} kE_{b1}' \\ N' \end{pmatrix} \begin{pmatrix} kE_{b1} & N \end{pmatrix} \begin{pmatrix} \tilde{\mu} \\ \tilde{\tau} \end{pmatrix} = \begin{pmatrix} kE_{b1}' \\ N' \end{pmatrix} B$$

$$\text{or} \begin{pmatrix} k^2 E_{b1}' E_{b1} & kE_{b1}' N \\ kN' E_{b1} & N' N \end{pmatrix} \begin{pmatrix} \tilde{\mu} \\ \tilde{\tau} \end{pmatrix} = \begin{pmatrix} kG \\ N' B \end{pmatrix}$$

$$\text{or} \begin{pmatrix} k^2 b & kE_{v1}' R \\ kRE_{v1} & N' N \end{pmatrix} \begin{pmatrix} \tilde{\mu} \\ \tilde{\tau} \end{pmatrix} = \begin{pmatrix} kG \\ N' B \end{pmatrix} \quad (\text{using } N' E_{b1} = r = RE_{v1})$$

Premultiplying both sides of the equation by

$$\begin{pmatrix} 1 & 0 \\ -\frac{RE_{v1}}{b} & I_v \end{pmatrix},$$

we get

$$\begin{pmatrix} bk & E'_{v1} \\ 0 & N'N - \frac{RE_{v1}E'_{v1}R}{b} \end{pmatrix} \begin{pmatrix} \tilde{\mu} \\ \tilde{\tau} \end{pmatrix} = \begin{pmatrix} G \\ N'B - \frac{RE_{v1}G}{b} \end{pmatrix}.$$

Using the side condition $E'_{v1}R\tilde{\tau}=0$ and assuming $N'N$ to be nonsingular, we get the estimates of μ^* and τ as $\tilde{\mu}$ and $\tilde{\tau}$ given by

$$\begin{aligned} \tilde{\mu} &= \frac{G}{bk}, \\ \tilde{\tau} &= (N'N)^{-1} \left(N'B - \frac{RE_{v1}G}{b} \right) \\ &= (N'N)^{-1} \left(N'B - \frac{kGN'E_{b1}}{bk} \right) \quad (\text{using } RE_{v1} = r = N'E_{v1}) \\ &= (N'N)^{-1} \left(N'B - \frac{G}{bk} N'NE_{v1} \right) \\ &= (N'N)^{-1} N'B - \frac{GE_{v1}}{bk}. \end{aligned}$$

The normal equations can also be solved in an alternative way also as follows.

The normal equations $\begin{pmatrix} k^2b & kE'_{v1}R \\ kRE_{v1} & N'N \end{pmatrix} \begin{pmatrix} \tilde{\mu} \\ \tilde{\tau} \end{pmatrix} = \begin{pmatrix} kG \\ N'\beta \end{pmatrix}$ can be written as

$$\begin{aligned} k^2b\tilde{\mu} + kE'_{v1}R\tilde{\tau} &= kG \\ kRE_{v1}\tilde{\mu} + N'N\tilde{\tau} &= N'B. \end{aligned}$$

Using the side condition $E'_{v1}R\tilde{\tau}=0$ (or equivalently $\sum_j r_j \tilde{\tau}_j = 0$) and assuming $N'N$ to be nonsingular,

the first equation gives $\tilde{\mu} = \frac{G}{bk}$. Substituting $\tilde{\mu}$ in the second equation gives $\tilde{\tau}$.

$$\begin{aligned} \tilde{\tau} &= (N'N)^{-1} \left(N'B - \frac{RE_{v1}G}{b} \right) \\ &= (N'N)^{-1} N'B - \frac{GE_{v1}}{bk}. \end{aligned}$$

Generally, we are not interested merely in the interblock analysis of variance but we want to utilize the information from interblock analysis along with the intrablock information to improve upon the statistical inferences.

After obtaining the interblock estimate of treatment effects, the next question that arises is how to use this information for improved estimation of treatment effects and use it further for the testing of significance of treatment effects. Such an estimate will be based on the use of more information, so it is expected to provide better statistical inferences.

We now have two different estimates of the treatment effect as

- based on intrablock analysis $\hat{\tau} = C^{-1}Q$ and
- based on interblock analysis $\tilde{\tau} = (N'N)^{-1}N'B - \frac{GE_{v1}}{bk}$.

Let us consider the estimation of linear contrast of treatment effects $L = l'\tau$. Since the intrablock and interblock estimates of τ are based on Gauss-Markov model and least-squares principle, so the best estimate of L based on intrablock estimation is

$$\begin{aligned} L_1 &= l'\hat{\tau} \\ &= l'C^{-1}Q \end{aligned}$$

and the best estimate of L based on interblock estimation is

$$\begin{aligned} L_2 &= l'\tilde{\tau} \\ &= l'\left[(N'N)^{-1}N'B - \frac{GE_{v1}}{bk}\right] \\ &= l'(N'N)^{-1}N'B \quad (\text{since } l'E_{v1} = 0 \text{ being contrast.}) \end{aligned}$$

The variances of L_1 and L_2 are

$$\text{Var}(L_1) = \sigma^2 l'C^{-1}l$$

and

$$\text{Var}(L_2) = \sigma_f^2 l'(N'N)^{-1}l,$$

respectively. The covariance between Q (from intrablock) and B (from interblock) is

$$\begin{aligned} \text{Cov}(Q, B) &= \text{Cov}(V - N'K^{-1}B^*, B) \\ &= \text{Cov}(V, B) - \text{Cov}(N'K^{-1}B^*, B) \\ &= N'\sigma_f^2 - N'K^{-1}K\sigma_f^2 \\ &= 0. \end{aligned}$$

Note that B^* denotes the block total based on intrablock analysis and B denotes the block totals based on interblock analysis. We are using two notations B and B^* just to indicate that the two block totals are different. The reader should not misunderstand that it follows from the result $Cov(Q, B) = 0$ in case of intrablock analysis.

Thus

$$Cov(L_1, L_2) = 0$$

irrespective of the values of l .

The question now arises that given the two estimators $\hat{\tau}$ and $\tilde{\tau}$ of τ , how to combine them and obtain a minimum variance unbiased estimator of τ . It is illustrated with the following example:

Example:

Let $\hat{\phi}_1$ and $\hat{\phi}_2$ be any two unbiased estimators of a parameter ϕ with $Var(\hat{\phi}_1) = \sigma_1^2$ and $Var(\hat{\phi}_2) = \sigma_2^2$.

Consider a linear combination $\hat{\phi} = \theta_1 \hat{\phi}_1 + \theta_2 \hat{\phi}_2$ with weights θ_1 and θ_2 . In order that $\hat{\phi}$ is an unbiased estimator of ϕ , we need

$$\begin{aligned} E(\hat{\phi}) &= \phi \\ \text{or } \theta_1 E(\hat{\phi}_1) + \theta_2 E(\hat{\phi}_2) &= \phi \\ \text{or } \theta_1 \phi + \theta_2 \phi &= \phi \\ \text{or } \theta_1 + \theta_2 &= 1. \end{aligned}$$

So modify $\hat{\phi}$ as $\frac{\theta_1 \hat{\phi}_1 + \theta_2 \hat{\phi}_2}{\theta_1 + \theta_2}$ which is the weighted mean of $\hat{\phi}_1$ and $\hat{\phi}_2$.

Further, if $\hat{\phi}_1$ and $\hat{\phi}_2$ are independent, then

$$Var(\hat{\phi}) = \theta_1^2 \sigma_1^2 + \theta_2^2 \sigma_2^2.$$

Now we find θ_1 and θ_2 such that $Var(\hat{\phi})$ is minimum such that $\theta_1 + \theta_2 = 1$.

$$\frac{\partial Var(\hat{\phi})}{\partial \theta_1} = 0 \Rightarrow 2\theta_1 \sigma_1^2 - 2(1 - \theta_1) \sigma_2^2 = 0$$

$$\text{or } \theta_1 \sigma_1^2 - \theta_2 \sigma_2^2 = 0$$

$$\text{or } \frac{\theta_1}{\theta_2} = \frac{\sigma_2^2}{\sigma_1^2}$$

$$\text{or } \text{weight} \propto \frac{1}{\text{variance}}.$$

Alternatively, the Lagrangian function approach can be used to obtain such a result as follows. The Lagrangian function with λ^* as Lagrangian multiplier is given by

$$\phi = \text{Var}(\hat{\phi}) - \lambda^*(\theta_1 + \theta_2 - 1)$$

and solving $\frac{\partial \phi}{\partial \theta_1} = 0$, $\frac{\partial \phi}{\partial \theta_2} = 0$ and $\frac{\partial \phi}{\partial \lambda^*} = 0$ also gives the same result that $\frac{\theta_1}{\theta_2} = \frac{\sigma_2^2}{\sigma_1^2}$.

We note that a pooled estimator of τ in the form of weighted arithmetic mean of uncorrelated L_1 and L_2 is the minimum variance unbiased estimator of τ when the weights θ_1 and θ_2 of L_1 and L_2 , respectively are chosen such that

$$\frac{\theta_1}{\theta_2} = \frac{\text{Var}(L_2)}{\text{Var}(L_1)},$$

i.e., the chosen weights are reciprocal to the variance of respective estimators, irrespective of the values of l . So consider the weighted average of L_1 and L_2 with weights θ_1 and θ_2 , respectively as

$$\begin{aligned} \tau^* &= \frac{\theta_1 L_1 + \theta_2 L_2}{\theta_1 + \theta_2} \\ &= \frac{l'(\theta_1 \hat{\tau} + \theta_2 \tilde{\tau})}{\theta_1 + \theta_2} \end{aligned}$$

with

$$\begin{aligned} \theta_1^{-1} &= l' C^{-1} l \sigma^2 \\ \theta_2^{-1} &= l' (N' N)^{-1} l \sigma_f^2. \end{aligned}$$

The linear contrast of τ^* is

$$L^* = l' \tau^*$$

and its variance is

$$\begin{aligned} \text{Var}(L^*) &= \frac{\theta_1^2 \text{Var}(L_1) + \theta_2^2 \text{Var}(L_2)}{(\theta_1 + \theta_2)^2} l' l \quad (\text{since } \text{Cov}(L_1, L_2) = 0) \\ &= \frac{l' l}{(\theta_1 + \theta_2)} \end{aligned}$$

because the weights of estimators are chosen to be inversely proportional to the variance of the respective estimators. We note that τ^* can be obtained provided θ_1 and θ_2 are known. But θ_1 and θ_2 are known only if σ^2 and σ_β^2 are known. So τ^* can be obtained when σ^2 and σ_β^2 are known. In case, if σ^2 and σ_β^2 are unknown, then their estimates can be used. A question arises how to obtain such

estimators? One such approach to obtain the estimates of σ^2 and σ_β^2 is based on utilizing the results from intrablock and interblock analysis both and is as follows.

From intrablock analysis

$$E(SS_{Error(t)}) = (n - b - v + 1)\sigma^2,$$

so an unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_{Error(t)}}{n - b - v + 1}.$$

An unbiased estimator of σ_β^2 is obtained by using the following results based on the intrablock analysis:

$$SS_{Treat(unadj)} = \sum_{j=1}^v \frac{V_j^2}{\tau_j} - \frac{G^2}{n},$$

$$SS_{Block(unadj)} = \sum_{i=1}^b \frac{B_i^2}{k_i} - \frac{G^2}{n},$$

$$SS_{Treat(adj)} = \sum_{j=1}^v Q_j \hat{\tau}_j,$$

$$SS_{Total} = \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - \frac{G^2}{n},$$

where

$$\begin{aligned} SS_{Total} &= SS_{Treat(adj)} + SS_{Block(unadj)} + SS_{Error(t)} \\ &= SS_{Treat(unadj)} + SS_{Block(adj)} + SS_{Error(t)}. \end{aligned}$$

Hence

$$SS_{Block(adj)} = SS_{Treat(adj)} + SS_{Block(unadj)} - SS_{Treat(unadj)}.$$

Under the interblock analysis model

$$E[SS_{Block(adj)}] = E[SS_{Treat(adj)}] + E[SS_{Block(unadj)}] - E[SS_{Treat(unadj)}]$$

which is obtained as follows:

$$E[SS_{Block(adj)}] = (b-1)\sigma^2 + (n-v)\sigma_\beta^2$$

or

$$E\left[SS_{Block(adj)} - \frac{b-1}{n-b-v+1} SS_{Error(t)}\right] = (n-v)\sigma_\beta^2.$$

Thus an unbiased estimator of σ_β^2 is

$$\hat{\sigma}_\beta^2 = \frac{1}{n-v} \left[SS_{Block(adj)} - \frac{b-1}{n-b-v+1} SS_{Error(t)} \right].$$

Now the estimates of weights θ_1 and θ_2 can be obtained by replacing σ^2 and σ_β^2 by $\hat{\sigma}^2$ and $\hat{\sigma}_\beta^2$ respectively. Then the estimate of τ^* can be obtained by replacing θ_1 and θ_2 by their estimates and can be used in place of τ^* . It may be noted that the exact distribution of the associated sum of squares due to treatments is difficult to find when σ^2 and σ_β^2 are replaced by $\hat{\sigma}^2$ and $\hat{\sigma}_\beta^2$, respectively in τ^* . Some approximate results are possible which we will present while dealing with the balanced incomplete block design. An increase in the precision using interblock analysis as compared to intrablock analysis is measured by

$$\frac{1/\text{variance of pooled estimate}}{1/\text{variance of intrablock estimate}} - 1.$$

In the interblock analysis, the block effects are treated as a random variable which is appropriate if the blocks can be regarded as a random sample from a large population of blocks. The best estimate of the treatment effect from the intrablock analysis is further improved by utilizing the information on block totals. Since the treatments in different blocks are not all the same, so the difference between block totals is expected to provide some information about the differences between the treatments. So the interblock estimates are obtained and pooled with intrablock estimates to obtain the combined estimate of τ . The procedure of obtaining the interblock estimates and then obtaining the pooled estimates is called the **recovery of interblock information**.

Chapter 6

Balanced Incomplete Block Design (BIBD)

The designs like CRD and RBD are the complete block designs. We now discuss the balanced incomplete block design (BIBD) and the partially balanced incomplete block design (PBIBD) which are the incomplete block designs.

A balanced incomplete block design (BIBD) is an incomplete block design in which

- b blocks have the same number k of plots each and
- every treatment is replicated r times in the design.
- Each treatment occurs at most once in a block, i.e., $n_{ij} = 0$ or 1 where n_{ij} is the number of times the j^{th} treatment occurs in i^{th} block, $i = 1, 2, \dots, b; j = 1, 2, \dots, v$.
- Every pair of treatments occurs together in λ of the b blocks.

Five parameters denote such design as $D(b, k, v, r; \lambda)$.

The parameters b, k, v, r and λ are not chosen arbitrarily.

They satisfy the following relations:

$$(I) \quad bk = vr$$

$$(II) \quad \lambda(v-1) = r(k-1)$$

$$(III) \quad b \geq v \text{ (and hence } r > k).$$

$$\begin{aligned} \text{Hence } \sum_j n_{ij} &= k \quad \text{for all } i \\ \sum_j n_{ij} &= r \quad \text{for all } j \end{aligned}$$

and $n_{1j}n_{ij'} + n_{2j}n_{ij'} + \dots + n_{bj}n_{ij'} = \lambda$ for all $j \neq j' = 1, 2, \dots, v$. Obviously $\frac{n_{ij}}{r}$ cannot be a constant for all j . So the design is not orthogonal.

Example of BIBD

In the design $D(b, k; v, r; \lambda)$: consider $b = 10$ (say, $B_1, \dots, B_{10}$), $v = 6$ (say, $T_1, \dots, T_6$), $k = 3, r = 5, \lambda = 2$

Blocks Treatments

B_1 T_1 T_2 T_5

B_2 T_1 T_2 T_6

B_3 T_1 T_3 T_4

B_4 T_1 T_3 T_6

B_5 T_1 T_4 T_5

B_6 T_2 T_3 T_4

B_7 T_2 T_3 T_5

B_8 T_2 T_4 T_6

B_9 T_3 T_5 T_6

B_{10} T_4 T_5 T_6

Now we see how the conditions of BIBD are satisfied.

- (i) $bk = 10 \times 3 = 30$ and $vr = 6 \times 5 = 30$
 $\Rightarrow bk = vr$
- (ii) $\lambda(v-1) = 2 \times 5 = 10$ and $r(k-1) = 5 \times 2 = 10$
 $\Rightarrow \lambda(v-1) = r(k-1)$
- (iii) $b = 10 \geq 6$

Even if the parameters satisfy the relations, it is not always possible to arrange the treatments in blocks to get the corresponding design.

The necessary and sufficient conditions to be satisfied by the parameters for the existence of a BIBD are not known.

The conditions (I)-(III) are some necessary condition only. The construction of such design depends on the actual arrangement of the treatments into blocks and this problem is handled in combinatorial mathematics. Tables are available, giving all the designs involving at most 20 replication and their method of construction.

Theorem:

- (I) $bk = vr$
- (II) $\lambda(v-1) = r(k-1)$
- (III) $b \geq v$.

Proof: (I)

Let $N = (n_{ij}) : b \times v$ the incidence matrix

Observing that the quantities $E_{1b}NE_{v1}$ and $E_{1v}N'E_{b1}$ are the scalars and the transpose of each other, we find their values.

Consider

$$\begin{aligned}
 E_{1b}NE_{v1} &= (1, 1, \dots, 1) \begin{pmatrix} n_{11} & n_{21} & \dots & n_{b1} \\ n_{12} & n_{22} & \dots & n_{b2} \\ \vdots & \vdots & \ddots & \vdots \\ n_{1v} & n_{2v} & \dots & n_{bv} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\
 &= (1, 1, \dots, 1) \begin{pmatrix} \sum_j n_{1j} \\ \sum_j n_{2j} \\ \vdots \\ \sum_j n_{bj} \end{pmatrix} \\
 &= (1, 1, \dots, 1)_{1 \times b} \begin{pmatrix} k \\ k \\ \vdots \\ k \end{pmatrix} = bk.
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 E_{1v}N'E_{b1} &= (1, \dots, 1) \begin{pmatrix} n_{11} & n_{21} & \dots & n_{b1} \\ n_{12} & n_{22} & \dots & n_{b2} \\ \vdots & \vdots & \ddots & \vdots \\ n_{1v} & n_{2v} & \dots & n_{bv} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ \vdots \\ 1 \end{pmatrix} \\
 &= (1, 1, \dots, 1) \begin{pmatrix} \sum_i n_i \\ \vdots \\ \sum_i n_{iv} \end{pmatrix} = (1, 1, \dots, 1)_{1 \times v} \begin{pmatrix} r \\ \vdots \\ r \end{pmatrix} = vr
 \end{aligned}$$

But

$$E_{1b}NE_{v1} = E_{1v}N'E_{b1} \text{ as both are scalars.}$$

Thus $bk = vr$.

Proof: (II)

Consider

$$\begin{aligned}
 N'N &= \begin{pmatrix} n_{11} & n_{21} & \cdots & n_{b1} \\ n_{12} & n_{22} & \cdots & n_{b2} \\ \vdots & \vdots & \ddots & \vdots \\ n_{1v} & n_{2v} & \cdots & n_{bv} \end{pmatrix} \begin{pmatrix} n_{11} & n_{12} & \cdots & n_{1v} \\ n_{21} & n_{22} & \cdots & n_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ n_{b1} & n_{b2} & \cdots & n_{bv} \end{pmatrix} \\
 &= \begin{pmatrix} \sum_i n_{i1}^2 & \sum_i n_{i1}n_{i2} & \cdots & \sum_i n_{i1}n_{iv} \\ \sum_i n_{i1}n_{i2} & \sum_i n_{i2}^2 & \cdots & \sum_i n_{i2}n_{iv} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_i n_{iv}n_{i1} & \sum_i n_{iv}n_{i2} & \cdots & \sum_i n_{iv}^2 \end{pmatrix} \\
 &= \begin{pmatrix} r & \lambda & \cdots & \lambda \\ \lambda & r & \cdots & \lambda \\ \vdots & \vdots & \ddots & \vdots \\ r & \lambda & \cdots & \lambda \end{pmatrix}. \tag{1}
 \end{aligned}$$

Since $n_{ij}^2 = 1$ or 0 as $n_{ij} = 1$ or 0 ,

so $\sum_i n_{ij}^2 = \text{Number of times } \tau_j \text{ occurs in the design}$

$= r$ for all $j = 1, 2, \dots, v$ of times occurs in the design

and $\sum_i n_{ij}n_{ij'} = \text{Number of blocks in which } \tau_j \text{ and } \tau_{j'} \text{ occurs together}$

$= \lambda$ for all $j \neq j'$.

$$\begin{aligned}
 N'NE_{v1} &= \begin{pmatrix} r & \lambda & \cdots & \lambda \\ \lambda & r & \cdots & \lambda \\ \vdots & \vdots & \ddots & \vdots \\ \lambda & \lambda & \cdots & r \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \\
 &= \begin{pmatrix} r + \lambda(v-1) \\ r + \lambda(v-1) \\ \vdots \\ r + \lambda(v-1) \end{pmatrix} = [r + \lambda(v-1)]E_{v1}. \tag{2}
 \end{aligned}$$

Also

$$\begin{aligned}
N'NE_{v1} &= N' \left[\begin{pmatrix} n_{11} & n_{12} & \cdots & n_{1v} \\ n_{21} & n_{22} & \cdots & n_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ n_{b1} & n_{b2} & \cdots & n_{bv} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \right] \\
&= N' \begin{pmatrix} \sum_j n_{1j} \\ \sum_j n_{2j} \\ \vdots \\ \sum_j n_{bj} \end{pmatrix} \\
&= \begin{pmatrix} n_{11} & n_{12} & \cdots & n_{1v} \\ n_{21} & n_{22} & \cdots & n_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ n_{b1} & n_{b2} & \cdots & n_{bv} \end{pmatrix} \begin{pmatrix} k \\ k \\ \vdots \\ k \end{pmatrix}_{b \times 1} \\
&= k \begin{pmatrix} \sum_i n_{i1} \\ \sum_i n_{i2} \\ \vdots \\ \sum_i n_{iv} \end{pmatrix} \\
&= k \begin{pmatrix} r \\ r \\ \vdots \\ r \end{pmatrix} \\
&= krE_{v1} \tag{3}
\end{aligned}$$

From (2) and (3)

$$[r + \lambda(v-1)]E_{v1} = krE_{v1}$$

$$\text{or } r + \lambda(v-1) = kr$$

$$\text{or } \lambda(v-1) = r(k-1)$$

Proof: (III)

From (I), the determinant of $N'N$ is

$$\begin{aligned}
\det |N'N| &= [r + \lambda(v-1)](r - \lambda)^{v-1} \\
&= [r + r(k-1)](r - \lambda)^{v-1} \\
&= rk(r - \lambda)^{v-1} \\
&\neq 0
\end{aligned}$$

because since if $r = \lambda \Rightarrow$ from (II) that $k = v$. This contradicts the incompleteness of the design.

Thus $N'N$ is a $v \times v$ nonsingular matrix.

Thus $\text{rank}(N'N) = v$.

We know from matrix theory result

$$\text{rank}(N) = \text{rank}(N'N)$$

so $\text{rank}(N) = v$

But $\text{rank}(N) \leq b$, there being b rows in N .

Thus $v \leq b$.

Interpretation of conditions of BIBD

Interpretation of (I) $bk = vr$

This condition is related to the total number of plots in an experiment. In our settings, there are k plots in each block and there are b blocks. So the total number of plots are bk .

Further, there are v treatments and each treatment is replicated r times such that each treatment occurs atmost in one block. So total number of plots containing all the treatments is vr . Since both the statements counts the total number of plots, hence $bk = vr$.

Interpretation of (II)

Each block has k plots. Thus the total pairs of plots in a block $= \binom{k}{2} = \frac{k(k-1)}{2}$.

There are b blocks. Thus the total pairs of plots such that each pair consists of plots within a block $= b \frac{k(k-1)}{2}$.

There are v treatments, thus the total number of pairs of treatment $= \binom{v}{2} = \frac{v(v-1)}{2}$.

Each pair of treatment is replicated λ times, i.e., each pair of treatment occurs in λ blocks.

Thus the total number of pairs of plots within blocks must be $= \lambda \frac{v(v-1)}{2}$.

$$\text{Hence } b \frac{k(k-1)}{2} = \lambda \frac{v(v-1)}{2}$$

Using $bk = vr$ in this relation, we get $r(k-1) = \lambda(v-1)$.

Proof of (III) was given by Fisher but quite long, so not needed here.

Balancing in designs:

There are two types of balancing – Variance balanced and efficiency balanced. We discuss the variance balancing now and the efficiency balancing later.

Balanced Design (Variance Balanced):

A connected design is said to be balanced (variance balanced) if all the elementary contrasts of the treatment effects can be estimated with the same precision. This definition does not hold for the disconnected design, as all the elementary contrasts are not estimable in this design.

Proper Design:

An incomplete block design with $k_1 = k_2 = \dots = k_b = k$ is called a proper design.

Symmetric BIBD:

A BIBD is called symmetrical if the number of blocks = number of treatments, i.e., $b = v$.

Since $b = v$, so from $bk = vr$

$$\Rightarrow k = r.$$

Thus the number of pairs of treatments common between any two blocks = λ .

The determinant of $N'N$ is

$$\begin{aligned} |N'N| &= [r + \lambda(v-1)](r - \lambda)^{v-1} \\ &= [r + r(k-1)](r - \lambda)^{v-1} \\ &= rk(r - \lambda)^{v-1}. \end{aligned}$$

When BIBD is symmetric, $b = v$ and then using $bk = vr$, we have $k = r$. Thus

$$|N'N| = |N|^2 = r^2(r - \lambda)^{v-1},$$

so

$$|N| = \pm r(r - \lambda)^{\frac{v-1}{2}}.$$

Since $|N|$ is an integer, hence when v is an even number, $(r - \lambda)$ must be a perfect square. So

$$\begin{aligned} |N'N| &= (r - \lambda)I + \lambda E_{v1}E_{v1}', \\ (N'N)^{-1} &= N^{-1}N'^{-1} \\ &= \frac{1}{r - \lambda} \left[I - \frac{\lambda}{r^2} E_{v1}E_{v1}' \right], \\ N'^{-1} &= \frac{1}{r - \lambda} \left[I - \frac{\lambda}{r^2} E_{v1}E_{v1}' \right]. \end{aligned}$$

Post-multiplying both sides by N' , we get

$$NN' = (r - \lambda)I + \lambda E_{v1} E_{v1}' = N'N.$$

Hence in the case of a symmetric BIBD, any two blocks have λ treatment in common.

Since BIBD is an incomplete block design. So every pair of treatment can occur at most once in a block, we must have $v \geq k$.

If $v = k$, then it means that each treatment occurs once in every block which occurs in case of RBD.

So in BIBD, always assume $v > k$.

Similarly $\lambda < r$.

[If $\lambda = r$ then $\lambda(v-1) = r(k-1) \Rightarrow v = k \Rightarrow$ which means that the design is RBD]

Resolvable design:

A block design of

- b blocks in which
- each of v treatments is replicated r times

is said to be resolvable if b blocks can be divided into r sets of b/r blocks each, such that every treatment appears in each set precisely once. Obviously, in a resolvable design, b is a multiple of r .

Theorem: If in a BIBD $D(v, b, r, k, \lambda)$, b is divisible by r , then

$$b \geq v + r - 1.$$

Proof: Let $b = nr$ (where $n > 1$ is a positive integer).

For a BIBD, $\lambda(v-1) = r(k-1)$

$$\begin{aligned} \text{or } r &= \frac{\lambda(v-1)}{(k-1)} && \left[\begin{array}{l} \text{because } vr = bk \\ \text{or } vr = nrk \\ \text{or } v = nk \end{array} \right] \\ &= \frac{\lambda(nk-1)}{(k-1)} \\ &= \lambda \left(\frac{n-1}{k-1} \right) + \lambda n. \end{aligned}$$

Since $n > 1$ and $k > 1$, so $\lambda n > 1$ is an integer. Since r has to be an integer.

$$\Rightarrow \lambda \frac{(n-1)}{k-1} \text{ is also a positive integer.}$$

Now, if possible, let

$$\begin{aligned}
b &< v + r - 1 \\
\Rightarrow nr &< v + r - 1 \\
\text{or } r(n-1) &< v - 1 \\
\text{or } r(n-1) &< \frac{r(k-1)}{\lambda} \quad \left(\text{because } v-1 = \frac{r(k-1)}{\lambda} \right) \\
\Rightarrow \frac{\lambda(n-1)}{k-1} &< 1 \text{ which is a contradiction as integer can not be less than one} \\
\Rightarrow b &< v + r - 1 \text{ is impossible. Thus the opposite is true.} \\
\Rightarrow b &\geq v + r - 1 \text{ holds correct.}
\end{aligned}$$

Intrablock analysis of BIBD:

Consider the model

$$y_{ij} = \mu + \beta_i + \tau_j + \varepsilon_{ij}; \quad i = 1, 2, \dots, b; \quad j = 1, 2, \dots, v,$$

where

- μ is the general mean effect;
- β_i is the fixed additive i^{th} block effect;
- τ_j is the fixed additive j^{th} treatment effect and
- ε_{ij} is the i.i.d. random error with $\varepsilon_{ij} \sim N(0, \sigma^2)$.

We don't need to develop the analysis of BIBD from starting. Since BIBD is also an incomplete block design and the analysis of incomplete block design has already been presented in the earlier module, so we implement those derived expressions directly under the setup and conditions of BIBD. Using

the same notations, we represent the blocks totals by $B_i = \sum_{j=1}^v y_{ij}$, treatment totals by $V_j = \sum_{i=1}^b y_{ij}$,

adjusted treatment totals by Q_j and grand total by $G = \sum_{i=1}^b \sum_{j=1}^v y_{ij}$. The normal equations are obtained

by differentiating the error sum of squares. Then the block effects are eliminated from the normal equations and the normal equations are solved for the treatment effects. The resulting intrablock equations of treatment effects in matrix notations are expressible as

$$Q = C\hat{\tau}.$$

Now we obtain the forms of C and Q in the case of BIBD. The diagonal elements of C are given by

$$\begin{aligned}
c_{jj} &= r - \frac{\sum_{i=1}^b n_{ij}^2}{k} \quad (j = 1, 2, \dots, v) \\
&= r - \frac{r}{k}.
\end{aligned}$$

The off-diagonal elements of C are given by

$$\begin{aligned} c_{jj'} &= -\frac{1}{k} \sum_{i=1}^b n_{ij} n_{ij'} \quad (j \neq j'; j, j' = 1, 2, \dots, v) \\ &= -\frac{\lambda}{k}. \end{aligned}$$

The adjusted treatment totals are obtained as

$$\begin{aligned} Q_j &= V_j - \frac{1}{k} \sum_{i=1}^b n_{ij} B_i \quad (j = 1, 2, \dots, v) \\ &= V_j - \frac{1}{k} \sum_{i(j)} B_i \end{aligned}$$

where $\sum_{i(j)}$ denotes the sum over those blocks containing j^{th} treatment. Denote

$$T_j = \sum_{i(j)} B_i, \text{ then}$$

$$Q_j = V_j - \frac{T_j}{k}.$$

The C matrix is simplified as follows:

$$\begin{aligned} C &= rI - \frac{N'N}{k} \\ &= rI - \frac{1}{k} \left[(r - \lambda)I + \lambda E_{v1} E_{v1}' \right] \\ &= r \left(\frac{k-1}{k} \right) I + \frac{\lambda}{k} (I - E_{v1} E_{v1}') \\ &= \lambda \left(\frac{v-1}{k} \right) I + \frac{\lambda}{k} (I - E_{v1} E_{v1}') \\ &= \frac{\lambda v}{k} \left(I - \frac{E_{v1} E_{v1}'}{v} \right). \end{aligned}$$

Since C is not as a full rank matrix, so its unique inverse does not exist. The generalized inverse of C is denoted as C^- which is obtained as

$$C^- = \left(C + \frac{E_{v1} E_{v1}'}{v} \right)^{-1}.$$

Since

$$\begin{aligned} C &= \frac{\lambda v}{k} \left(I_v - \frac{E_{v1} E_{v1}'}{v} \right) \\ \text{or } \frac{kC}{\lambda v} &= I_v - \frac{E_{v1} E_{v1}'}{v}, \end{aligned}$$

the generalized inverse of $\frac{k}{\lambda v}C$ is

$$\begin{aligned}\left(\frac{k}{\lambda v}\right)^{-1} C^{-} &= \left[C + \frac{E_{v1}E'_{v1}}{v} \right]^{-1} \\ &= \left[I_v - \frac{E_{v1}E'_{v1}}{v} + \frac{E_{v1}E'_{v1}}{v} \right]^{-1} \\ &= I_v.\end{aligned}$$

Thus $C^{-} = \frac{\lambda v}{k} I_v$.

Thus an estimate of τ is obtained from $Q = C\tau$ as

$$\begin{aligned}\hat{\tau} &= C^{-}Q \\ &= \frac{\lambda v}{k}Q.\end{aligned}$$

The null hypothesis of our interest is $H_0 : \tau_1 = \tau_2 = \dots = \tau_v$ against the alternative hypothesis H_1 : at least one pair of τ_j 's is different. Now we obtain the various sum of squares involved in the development of analysis of variance as follows.

The adjusted treatment sum of squares is

$$\begin{aligned}SS_{Treat(adj)} &= \hat{\tau}'Q \\ &= \frac{k}{\lambda v}Q'Q \\ &= \frac{k}{\lambda v} \sum_{j=1}^v Q_j^2,\end{aligned}$$

The unadjusted block sum of squares is

$$SS_{Block(unadj)} = \sum_{i=1}^b \frac{B_i^2}{k} - \frac{G^2}{bk}.$$

The total sum of squares is

$$SS_{Total} = \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - \frac{G^2}{bk}$$

The residual sum of squares is obtained by

$$SS_{Error(t)} = SS_{Total} - SS_{Block(unadj)} - SS_{Treat(adj)}.$$

A test for $H_0 : \tau_1 = \tau_2 = \dots = \tau_v$ is then based on the statistic

$$F_{Tr} = \frac{SS_{Treat(adj)} / (v-1)}{SS_{Error(t)} / (bk - b - v + 1)}$$

$$= \frac{k}{\lambda v} \cdot \frac{bk - b - v + 1}{v - 1} \cdot \frac{\sum_{j=1}^v Q_j^2}{SS_{Error(t)}}$$

If $F_{Tr} > F_{1-\alpha, v-1, bk-b-v+1}$; then $H_{0(t)}$ is rejected.

This completes the analysis of variance test and is termed as intrablock analysis of variance. This analysis can be compiled into the intrablock analysis of variance table for testing the significance of the treatment effect given as follows.

Intrablock analysis of variance table of BIBD for $H_0 : \tau_1 = \tau_2 = \dots = \tau_v$

Source	Sum of squares	Degrees of freedom	Mean squares	F
Between treatment (adjusted)	$SS_{Treat(adj)}$	$v - 1$	$MS_{treat} = \frac{SS_{Treat(adj)}}{v - 1}$	$\frac{MS_{Treat}}{MS_E}$
Between blocks (unadjusted)	$SS_{Block(unadj)}$	$b - 1$		
Intrablock error	$SS_{Error(t)}$ (by subtraction)	$bk - b - v + 1$	$MS_E = \frac{SS_{Error(t)}}{bk - b - v + 1}$	
Total	$SS_{Total} =$ $\sum_i \sum_j y_{ij}^2 - \frac{G^2}{bk}$	$bk - 1$		

In case, the null hypothesis is rejected, then we go for a pairwise comparison of the treatments. For that, we need an expression for the variance of the difference of two treatment effects.

The variance of an elementary contrast $(\tau_j - \tau_{j'}, j \neq j')$ under the intrablock analysis is

$$\begin{aligned}
 V^* &= \text{Var}(\hat{\tau}_j - \hat{\tau}_{j'}) \\
 &= \text{Var}\left(\frac{k}{\lambda v}(Q_j - Q_{j'})\right) \\
 &= \frac{k^2}{\lambda^2 v^2} [\text{Var}(Q_j) + \text{Var}(Q_{j'}) - 2\text{Cov}(Q_j, Q_{j'})] \\
 &= \frac{k^2}{\lambda^2 v^2} (c_{jj} + c_{j'j'} - 2c_{jj'}) \sigma^2 \\
 &= \frac{k^2}{\lambda^2 v^2} \left[2r \left(1 - \frac{1}{k} \right) + \frac{2\lambda}{k} \right] \sigma^2 \\
 &= \frac{2k}{\lambda v} \sigma^2.
 \end{aligned}$$

This expression depends on σ^2 which is unknown. So it is unfit for use in the real data applications. One solution is to estimate σ^2 from the given data and use it in the place of σ^2 .

An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_{\text{Error}(t)}}{bk - b - v + 1}.$$

Thus an unbiased estimator of V^* can be obtained by substituting $\hat{\sigma}^2$ in it as

$$\hat{V}_* = \frac{2k}{\lambda v} \cdot \frac{SS_{\text{Error}(t)}}{bk - b - v + 1}.$$

If H_0 is rejected, then we make pairwise comparison and use the multiple comparison test. To test

$H_0 : \tau_j = \tau_{j'} (j \neq j')$, a suitable statistic is

$$t = \frac{k(bk - b - v + 1)}{\lambda v} \cdot \frac{Q_j - Q_{j'}}{\sqrt{SS_{\text{Error}(t)}}}$$

which follows a t -distribution with $(bk - b - v + 1)$ degrees of freedom under H_0 .

A question arises that how a BIBD compares to an RBD. Note that BIBD is an incomplete block design whereas RBD is a complete block design. This point should be kept in mind while making such restrictive comparison.

We now compare the efficiency of BIBD with a randomized block (complete) design with r replicates. The variance of an elementary contrast under a randomized block design (RBD) is

$$V_R^* = \text{Var}(\hat{\tau}_j^2 - \hat{\tau}_{j'})_{\text{RBD}} = \frac{2\sigma_*^2}{r}$$

where $\text{Var}(y_{ij}) = \sigma_*^2$ under RBD.

Thus the relative efficiency of BIBD relative to RBD is

$$\begin{aligned}\frac{Var(\hat{\tau}_j - \hat{\tau}_{j'})_{RBD}}{Var(\hat{\tau}_j - \hat{\tau}_{j'})_{BIBD}} &= \frac{\left(\frac{2\sigma_*^2}{r}\right)}{\left(\frac{2k\sigma^2}{\lambda v}\right)} \\ &= \frac{\lambda v}{rk} \left(\frac{\sigma_*^2}{\sigma^2}\right).\end{aligned}$$

The factor $\frac{\lambda v}{rk} = E$ (say) is termed as the **efficiency factor** of BIBD and

$$\begin{aligned}E = \frac{\lambda v}{rk} &= \frac{v}{k} \left(\frac{k-1}{v-1}\right) \\ &= \left(1 - \frac{1}{k}\right) \left(1 - \frac{1}{v}\right)^{-1} \\ &< 1 \text{ (since } v > k\text{)}.\end{aligned}$$

The actual efficiency of BIBD over RBD not only depends on the efficiency factor but also on the ratio of variances σ_*^2 / σ^2 . So BIBD can be more efficient than RBD as σ_*^2 can be more than σ^2 because $k < v$.

Efficiency balanced design:

A block design is said to be efficiency balanced if every contrast of the treatment effects is estimated through the design with the same efficiency factor.

If a block design satisfies any two of the following properties:

- (i) efficiency balanced,
- (ii) variance balanced and
- (iii) an equal number of replications,

then the third property also holds true.

Missing observations in BIBD:

The intrablock estimate of missing $(i, j)^{th}$ observation y_{ij} is

$$y_{ij} = \frac{vr(k-1)B_i - k(v-1)Q_j - (v-1)Q'_j}{k(k-1)(bk - b - v + 1)}$$

Q'_j : the sum of Q value for all other treatment (but not the j^{th} one) which are present in the i^{th} block.

All other procedures remain the same.

Interblock analysis and recovery of interblock information in BIBD

In the intrablock analysis of variance of an incomplete block design or BIBD, the treatment effects were estimated after eliminating the block effects from the normal equations. In a way, the block effects were assumed to be not marked enough and so they were eliminated. It is possible in many situations that the block effects are influential and marked. In such situations, the block totals may carry information about the treatment combinations also. This information can be used in estimating the treatment effects which may provide more efficient results. This is accomplished by an interblock analysis of BIBD and used further through the recovery of interblock information. So we first conduct the interblock analysis of BIBD. We do not derive the expressions a fresh but we use the assumptions and results from the interblock analysis of an incomplete block design. We additionally assume that the block effects are random with variance σ_β^2 .

After estimating the treatment effects under interblock analysis, we use the results for the pooled estimation and recovery of interblock information in a BIBD.

In case of BIBD,

$$\begin{aligned}
 N'N &= \begin{pmatrix} \sum_i n_{i1}^2 & \sum_i n_{i1}n_{i2} & \cdots & \sum_i n_{i1}n_{iv} \\ \sum_i n_{i1}n_{i2} & \sum_i n_{i2}^2 & \cdots & \sum_i n_{i2}n_{iv} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_i n_{iv}n_{i1} & \sum_i n_{iv}n_{i2} & \cdots & \sum_i n_{iv}^2 \end{pmatrix} \\
 &= \begin{pmatrix} r & \lambda & \cdots & \lambda \\ \lambda & r & \cdots & \lambda \\ \vdots & \vdots & \ddots & \vdots \\ \lambda & \lambda & \cdots & r \end{pmatrix} \\
 &= (r - \lambda)I_v + \lambda E_{v1}E_{v1}'
 \end{aligned}$$

$$(N'N)^{-1} = \frac{1}{r - \lambda} \left[I_v - \frac{\lambda E_{v1}E_{v1}'}{rk} \right]$$

The interblock estimate of τ can be obtained by substituting the expression on $(N'N)^{-1}$ in the earlier obtained interblock estimate.

$$\tilde{\tau} = (N'N)^{-1} N'B - \frac{GE_{v1}}{bk}.$$

Our next objective is to use the intrablock and interblock estimates of treatment effects together to find an improved estimate of treatment effects.

In order to use the interblock and intrablock estimates of τ together through pooled estimate, we consider the interblock and intrablock estimates of the treatment contrast.

The intrablock estimate of treatment contrast $l'\tau$ is

$$\begin{aligned} l'\hat{\tau} &= l'C^{-1}Q \\ &= \frac{k}{\lambda v} l'Q \\ &= \frac{k}{\lambda v} \sum_j l_j Q_j \\ &= \sum_j l_j \hat{\tau}_j, \text{ say.} \end{aligned}$$

The interblock estimate of treatment contrast $l'\tau$ is

$$\begin{aligned} l'\tilde{\tau} &= \frac{l'N'B}{r-\lambda} \quad (\text{since } l'E_{v1} = 0) \\ &= \frac{1}{r-\lambda} \sum_{j=1}^v l_j \left(\sum_{i=1}^b n_{ij} B_i \right) \\ &= \frac{1}{r-\lambda} \sum_{j=1}^v l_j T_j \\ &= \sum_{j=1}^v l_j \tilde{\tau}_j. \end{aligned}$$

The variance of $l'\hat{\tau}$ is obtained as

$$\begin{aligned} \text{Var}(l'\hat{\tau}) &= \left(\frac{k}{\lambda v} \right) \text{Var} \left(\sum_j l_j Q_j \right) \\ &= \left(\frac{k}{\lambda v} \right)^2 \left[\sum_j l_j^2 \text{Var}(Q_j) + 2 \sum_j \sum_{j'(\neq j)} l_j l_{j'} \text{Cov}(Q_j, Q_{j'}) \right]. \end{aligned}$$

Since

$$\text{Var}(Q_j) = r \left(1 - \frac{1}{k} \right) \sigma^2,$$

$$\text{Cov}(Q_j, Q_{j'}) = -\frac{\lambda}{k} \sigma^2, \quad (j \neq j'),$$

so

$$\begin{aligned}
\text{Var}(l'\hat{\tau}) &= \left(\frac{k}{\lambda v}\right)^2 \left[r \left(1 - \frac{1}{k}\right) \sigma^2 \sum_j l_j^2 - \frac{\lambda}{k} \left\{ \left(\sum_j l_j \right)^2 - \sum_j l_j^2 \right\} \sigma^2 \right] \\
&= \left(\frac{k}{\lambda v}\right)^2 \left[\frac{r(k-1)}{k} \sum_j l_j^2 + \frac{\lambda}{k} \sum_j l_j^2 \right] \sigma^2 \quad (\text{since } \sum_j \ell_j = 0 \text{ being contrast}) \\
&= \left(\frac{k}{\lambda v}\right)^2 \frac{1}{k} [\lambda(v-1) + \lambda] \sum_j l_j^2 \quad (\text{using } r(k-1) = \lambda(v-1)) \\
&= \left(\frac{k}{\lambda v}\right) \sigma^2 \sum_j l_j^2.
\end{aligned}$$

Similarly, the variance of $\ell'\hat{\tau}$ is obtained as

$$\begin{aligned}
\text{Var}(l'\tilde{\tau}) &= \left(\frac{1}{r-\lambda}\right)^2 \left[\sum_j l_j^2 \text{Var}(T_j) + 2 \sum_j \sum_{j' \neq j} l_j l_{j'} \text{Cov}(T_j, T_{j'}) \right] \\
&= \left(\frac{1}{r-\lambda}\right)^2 \left[r \sigma_f^2 \sum_j l_j^2 + \lambda \sigma_f^2 \left\{ \left(\sum_j l_j \right)^2 - \sum_j l_j^2 \right\} \right] \\
&= \frac{\sigma_f^2}{r-\lambda} \sum_j l_j^2.
\end{aligned}$$

The information on $\ell'\hat{\tau}$ and $\ell'\tilde{\tau}$ can be used together to obtain a more efficient estimator of $\ell'\tau$ by considering the weighted arithmetic mean of $\ell'\hat{\tau}$ and $\ell'\tilde{\tau}$. This will be the minimum variance unbiased estimator of $\ell'\tau$ when the weights of the corresponding estimates are chosen such that they are inversely proportional to the respective variances of the estimators. Thus the weights to be assigned to intrablock and interblock estimates are reciprocal to their variances as $\lambda v/(k\sigma^2)$ and $(r-\lambda)/\sigma_f^2$, respectively. Then the pooled mean of these two estimators is

$$\begin{aligned}
L^* &= \frac{\frac{\lambda v}{k\sigma^2} l'\hat{\tau} + \frac{r-\lambda}{\sigma_f^2} l'\tilde{\tau}}{\frac{\lambda v}{k\sigma^2} + \frac{r-\lambda}{\sigma_f^2}} = \frac{\frac{\lambda v}{k\sigma^2} \sum_j l_j \hat{\tau}_j + \frac{r-\lambda}{\sigma_f^2} \sum_j l_j \tilde{\tau}_j}{\frac{\lambda v}{k\sigma^2} + \frac{r-\lambda}{\sigma_f^2}} \\
&= \frac{\frac{\lambda v \omega_1}{k} \sum_j l_j \hat{\tau}_j + (r-\lambda) \omega_2 \sum_j l_j \tilde{\tau}_j}{\frac{\lambda v}{k} \omega_1 + (r-\lambda) \omega_2} \\
&= \frac{\lambda v \omega_1 \sum_j l_j \hat{\tau}_j + k(r-\lambda) \omega_2 \sum_j l_j \tilde{\tau}_j}{\lambda v \omega_1 + k(r-\lambda) \omega_2} \\
&= \sum_j l_j \left[\frac{\lambda v \omega_1 \hat{\tau}_j + k(r-\lambda) \omega_2 \tilde{\tau}_j}{\lambda v \omega_1 + k(r-\lambda) \omega_2} \right] \\
&= \sum_j l_j \tau_j^*
\end{aligned}$$

where $\tau_j^* = \frac{\lambda v \omega_1 \hat{\tau}_j + k(r - \lambda) \omega_2 \hat{\tau}_j}{\lambda v \omega_1 + k(r - \lambda) \omega_2}$, $\omega_1 = \frac{1}{\sigma^2}$, $\omega_2 = \frac{1}{\sigma_f^2}$.

Now we simplify the expression of τ_j^* so that it becomes more compatible in further analysis.

Since $\hat{\tau}_j = (k / \lambda v) Q_j$ and $\tilde{\tau}_j = T_j / (r - \lambda)$, so the numerator of τ_j^* can be expressed as

$$\omega_1 \lambda v \hat{\tau}_j + \omega_2 k(r - \lambda) \tilde{\tau}_j = \omega_1 k Q_j + \omega_2 k T_j$$

Similarly, the denominator of τ_j^* can be expressed as

$$\begin{aligned} & \omega_1 \lambda v + \omega_2 k(r - \lambda) \\ &= \omega_1 \left[\frac{vr(k-1)}{v-1} \right] + \omega_2 \left[k \left(r - \frac{r(k-1)}{v-1} \right) \right] \quad (\text{using } \lambda(v-1) = r(k-1)) \\ &= \frac{1}{v-1} [\omega_1 vr(k-1) + \omega_2 kr(v-k)]. \end{aligned}$$

Let

$$W_j^* = (v-k)V_j - (v-1)T_j + (k-1)G$$

where $\sum_j W_j^* = 0$. Using these results we have

$$\begin{aligned} \tau_j^* &= \frac{(v-1) [\omega_1 k Q_j + \omega_2 k T_j]}{\omega_1 rv(k-1) + \omega_2 kr(v-k)} \\ &= \frac{(v-1) [\omega_1 (kV_j - T_j) + \omega_2 k T_j]}{r [\omega_1 v(k-1) + \omega_2 k(v-k)]} \quad (\text{using } Q_j = V_j - \frac{T_j}{k}) \\ &= \frac{\omega_1 k(v-1)V_j + (k\omega_2 - \omega_1)(v-1)T_j}{r [\omega_1 v(k-1) + \omega_2 k(v-k)]} \\ &= \frac{\omega_1 k(v-1)V_j + (\omega_1 - k\omega_2) [W_j^* - (v-k)V_j - (k-1)G]}{r [\omega_1 v(k-1) + \omega_2 k(v-k)]} \\ &= \frac{[\omega_1 k(v-1) - (\omega_1 - k\omega_2)(v-k)]V_j + (\omega_1 - k\omega_2) [W_j^* - (k-1)G]}{r [\omega_1 v(k-1) + \omega_2 k(v-k)]} \\ &= \frac{1}{r} \left[V_j + \frac{\omega_1 - k\omega_2}{\omega_1 v(k-1) + \omega_2 k(v-k)} \{W_j^* - (k-1)G\} \right] \\ &= \frac{1}{r} \left[V_j + \xi \{W_j^* - (k-1)G\} \right] \end{aligned}$$

where

$$\xi = \frac{\omega_1 - k\omega_2}{\omega_1 v(k-1) + \omega_2 k(v-k)}, \omega_1 = \frac{1}{\sigma^2}, \omega_2 = \frac{1}{\sigma_f^2}.$$

Thus the pooled estimate of the contrast $l'\tau$ is

$$\begin{aligned} l'\tau^* &= \sum_j l_j \tau_j^* \\ &= \frac{1}{r} \sum_j l_j (V_j + \xi W_j^*) \quad (\text{since } \sum_j l_j = 0 \text{ being contrast}) \end{aligned}$$

The variance of $l'\tau^*$ is

$$\begin{aligned} \text{Var}(l'\tau^*) &= \frac{k}{\lambda v \omega_1 + k(r - \lambda) \omega_2} \sum_j l_j^2 \\ &= \frac{k(v-1)}{r[v(k-1)\omega_1 + k(v-k)\omega_2]} \sum_j l_j^2 \quad (\text{using } \lambda(v-1) = r(k-1)) \\ &= \sigma_E^2 \frac{\sum_j l_j^2}{r} \end{aligned}$$

where

$$\sigma_E^2 = \frac{k(v-1)}{v(k-1)\omega_1 + k(v-k)\omega_2}$$

is called as the **effective variance**.

Note that the variance of any elementary contrast based on the pooled estimates of the treatment effects is

$$\text{Var}(\tau_i^* - \tau_j^*) = \frac{2}{r} \sigma_E^2.$$

The effective variance can be approximately estimated by

$$\hat{\sigma}_E^2 = \text{MSE} [1 + (v-k)\omega^*]$$

where MSE is the mean square due to error obtained from the intrablock analysis as

$$\text{MSE} = \frac{SS_{\text{Error}(t)}}{bk - b - v + 1}$$

and

$$\omega^* = \frac{\omega_1 - \omega_2}{v(k-1)\omega_1 + k(v-k)\omega_2}.$$

The quantity ω^* depends upon the unknown σ^2 and σ_β^2 . To obtain an estimate of ω^* , we can obtain the unbiased estimates of σ^2 and σ_β^2 and then substitute them back in place of σ^2 and σ_β^2 in ω^* . To do this, we proceed as follows.

An estimate of ω_1 can be obtained by estimating σ^2 from the intrablock analysis of variance as

$$\hat{\omega}_1 = \frac{1}{\hat{\sigma}_2} = [MSE]^{-1}.$$

The estimate of ω_2 depends on $\hat{\sigma}^2$ and $\hat{\sigma}_\beta^2$. To obtain an unbiased estimator of σ_β^2 , consider

$$SS_{Block(adj)} = SS_{Treat(adj)} + SS_{Block(unadj)} - SS_{Treat(unadj)}$$

for which

$$E(SS_{Block(adj)}) = (bk - v)\sigma_\beta^2 + (b - 1)\sigma^2.$$

Thus an unbiased estimator of σ_β^2 is

$$\begin{aligned}\hat{\sigma}_\beta^2 &= \frac{1}{bk - v} [SS_{Block(adj)} - (b - 1)\hat{\sigma}^2] \\ &= \frac{1}{bk - v} [SS_{Block(adj)} - (b - 1)MSE] \\ &= \frac{b - 1}{bk - v} [MS_{Block(adj)} - MSE] \\ &= \frac{b - 1}{v(r - 1)} [MS_{Block(adj)} - MSE]\end{aligned}$$

where

$$MS_{Block(adj)} = \frac{SS_{Block(adj)}}{b - 1}.$$

Thus

$$\begin{aligned}\hat{\omega}_2 &= \frac{1}{k\hat{\sigma}^2 + \hat{\sigma}_\beta^2} \\ &= \frac{1}{v(r - 1) [k(b - 1)MS_{Block(adj)} - (v - k)MSE]}.\end{aligned}$$

Recall that our main objective is to develop a test of hypothesis for $H_0 : \tau_1 = \tau_2 = \dots = \tau_v$ and we now want to develop it using the information based on both interblock and intrablock analysis.

To test the hypothesis related to treatment effects based on the pooled estimate, we proceed as follows.

Consider the adjusted treatment totals based on the intrablock and the interblock estimates as

$$T_j^* = T_j + \omega^* W_j^*; j = 1, 2, \dots, v$$

and use it as usual treatment total as in earlier cases.

The sum of squares due to T_j^* is

$$S_{T^*}^2 = \sum_{j=1}^v T_j^{*2} - \frac{\left(\sum_{j=1}^v T_j^* \right)^2}{v}.$$

Note that in the usual analysis of variance technique, the test statistic for such null hypothesis is developed by taking the ratio of the sum of squares due to treatment divided by its degrees of freedom and the sum of squares due to error divided by its degrees of freedom. Following the same idea, we define the statistics

$$F^* = \frac{S_{T^*}^2 / [(v-1)r]}{MSE[1 + (v-k)\hat{\omega}^*]}$$

where $\hat{\omega}^*$ is an estimator of ω^* . It may be noted that F^* depends on $\hat{\omega}^*$. The value of $\hat{\omega}^*$ itself depends on the estimated variances $\hat{\sigma}^2$ and $\hat{\sigma}_f^2$. So it cannot be ascertained that the statistic F^* necessary follow the F distribution. Since the construction of F^* is based on the earlier approaches where the statistic was found to follow the exact F -distribution, so based on this idea, the distribution of F^* can be considered to be approximately F distributed. Thus the approximate distribution of F^* is considered as F distribution with $(v-1)$ and $(bk-b-v+1)$ degrees of freedom. Also, $\hat{\omega}^*$ is an estimator of ω^* which is obtained by substituting the unbiased estimators of ω_1 and ω_2 .

An approximate best pooled estimator of $\sum_{j=1}^v l_j \tau_j$ is

$$\sum_{j=1}^v l_j \frac{V_j + \xi W_j}{r}$$

and its variance is approximately estimated by

$$\frac{k \sum_j l_j^2}{\lambda v \hat{\omega}_1 + (r - \lambda) k \hat{\omega}_2}.$$

In case of the resolvable BIBD, $\hat{\sigma}_\beta^2$ can be obtained by using the adjusted block with replications sum of squares from the intrablock analysis of variance. If sum of squares due to such block total is SS_{Block}^* and corresponding mean square is

$$MS_{Block}^* = \frac{SS_{Block}^*}{b-r}$$

then

$$\begin{aligned}
E(MS_{Block}^*) &= \sigma^2 + \frac{(v-k)(r-1)}{b-r} \sigma_\beta^2 \\
&= \sigma^2 + \frac{(r-1)k}{r} \sigma_\beta^2
\end{aligned}$$

and $k(b-r) = r(v-k)$ for a resolvable design. Thus

$$E[rMS_{Block}^* - MSE] = (r-1)(\sigma^2 + k\sigma_\beta^2)$$

and hence

$$\begin{aligned}
\hat{\omega}_2 &= \left[\frac{rMS_{block}^* - MSE}{r-1} \right]^{-1}, \\
\hat{\omega}_1 &= [MSE]^{-1}.
\end{aligned}$$

The analysis of variance table for the recovery of interblock information in BIBD is described in the following table:

Source	Sum of squares	Degrees of freedom	Mean square	F^*
Between treatment (unadjusted)	$S_{T^*}^2$	$v - 1$		$F^* = \frac{MS_{Blocks(adj)}}{MSE}$
Between blocks (adjusted)	$SS_{Block(adj)} =$	$b - 1$	$MS_{Blocks(adj)} = \frac{SS_{Block(adj)}}{b-1}$	
Intrablock error	$SS_{Treat(adj)} +$ $SS_{Block(unadj)} -$ $SS_{Treat(unadj)}$ $SS_{Error(t)}$ (by subtraction)	$bk - b - v + 1$	$MSE = \frac{SS_{Error(t)}}{bk - b - v + 1}$	
Total	SS_{Total}	$bk - 1$		

The increase in the precision using interblock analysis as compared to intrablock analysis is

$$\begin{aligned} & \frac{Var(\hat{\tau})}{Var(\tau^*)} - 1 \\ &= \frac{\lambda v \omega_1 + \omega_2 k(r - \lambda)}{\lambda v \omega_1} - 1 \\ &= \frac{\omega_2 (r - \lambda) k}{\lambda v \omega_1}. \end{aligned}$$

Such an increase may be estimated by

$$\frac{\hat{\omega}_2 (r - \lambda) k}{\lambda v \hat{\omega}_1}.$$

Although $\omega_1 > \omega_2$ but this may not hold true for $\hat{\omega}_1$ and $\hat{\omega}_2$. The estimates $\hat{\omega}_1$ and $\hat{\omega}_2$ may be negative also and in that case we take $\hat{\omega}_1 = \hat{\omega}_2$.

Chapter 7

Partially Balanced Incomplete Block Design (PBIBD)

The balanced incomplete block designs have several advantages. They are connected designs as well as the block sizes are also equal. A restriction on using the BIBD is that they are not available for all parameter combinations. They exist only for certain parameters. Sometimes, they require a large number of replications also. This hampers the utility of the BIBDs. For example, if there are $v = 8$ treatments and block size is $k = 3$ (i.e., 3 plots in each block) then the total number of required blocks are

are $b = \binom{8}{3} = 56$ and so using the relationship $bk = vr$, the total number of required replicates is

$$r = \frac{bk}{v} = 21.$$

Another important property of the BIBD is that it is efficiency balanced. This means that all the treatment differences are estimated with the same accuracy. The partially balanced incomplete block designs (PBIBD) compromise on this property up to some extent and help in reducing the number of replications. In simple words, the pairs of treatments can be arranged in different sets such that the difference between the treatment effects of a pair, for all pairs in a set, is estimated with the same accuracy. The partially balanced incomplete block designs remain connected like BIBD but no more balanced. Rather they are partially balanced in the sense that some pairs of treatments have the same efficiency whereas some other pairs of treatments have the same efficiency but different from the efficiency of earlier pairs of treatments. This will be illustrated more clearly in the further discussion.

Before describing the set up of PBIBD, first, we need to understand the concept of “**Association Scheme**”. Instead of explaining the theory related to the association schemes, we consider here some examples and then understand the concept of an association scheme. Let there be a set of v treatments. These treatments are denoted by the symbols $1, 2, \dots, v$.

Partially Balanced Association Schemes

A relationship satisfying the following three conditions is called a partially balanced association scheme with m -associate classes.

- (i) Any two symbols are either first, second, ..., or m^{th} associates and the relation of associations is symmetrical, i.e., if the treatment A is the i^{th} associate of treatment B , then B is also the i^{th} associate of treatment A .

- (ii) Each treatment A in the set has exactly n_i treatments in the set which are the i^{th} associate and the number n_i ($i = 1, 2, \dots, m$) does not depend on the treatment A .
- (iii) If any two treatments A and B are the i^{th} associates, then the number of treatments which are both j^{th} associate A and k^{th} associate of B is p_{jk}^i and is independent of the pair of i^{th} associates A and B .

The numbers $v, n_1, n_2, \dots, n_m, p_{jk}^i$ ($i, j, k = 1, 2, \dots, m$) are called the **parameters of m -associate partially balanced scheme**.

We consider now the examples based on rectangular and triangular association schemes to understand the conditions stated in the partially balanced association scheme.

Rectangular Association Scheme

Consider an example of $m = 3$ associate classes. Let there be six treatments denoted as 1, 2, 3, 4, 5 and 6. Suppose these treatments are arranged as follows:

1		2		3
4		5		6

Under this arrangement, with respect to each symbol, the

- two other symbols in the same row are the first associates.
- One another symbol in the same column is the second associate and
- remaining two symbols are in the other row are the third associates.

For example, with respect to treatment 1,

- treatments 2 and 3 occur in the same row, so they are the first associates of treatment 1,
- treatment 4 occurs in the same column, so it is the second associate of treatment 1 and
- the remaining treatments 5 and 6 are the third associates of treatment 1 as they occur in the other (second) row.

Similarly, for treatment 5,

- treatments 4 and 6 occur in the same row, so they are the first associates of treatment 5,
- treatment 2 occurs in the same column, so it is the second associate of treatment 5 and
- remaining treatments 1 and 3 are in the other (second) row, so they are the third associates of treatment 5.

The table below describes the first, second and third associates of all the six treatments.

Treatment number	First associates	Second associates	Third associates
1	2, 3	4	5, 6
2	1, 3	5	4, 6
3	1, 2	6	4, 5
4	5, 6	1	2, 3
5	4, 6	2	1, 3
6	4, 5	3	1, 2

Further, we observe that for treatment 1, the

- o number of first associates (n_1) = 2,
- o number of second associates (n_2) = 1, and
- o the number of third associates (n_3) = 2.

The same values of n_1, n_2 and n_3 hold true for other treatments also.

Now we understand the condition (iii) of the definition of partially balanced association schemes related to p_{jk}^i .

Consider treatments 1 and 2. They are the first associates (which means $i = 1$), i.e. treatments 1 and 2 are the first associate of each other; treatment 6 is the third associate (which means $j = 3$) of treatment 1 and also the third associate (which means $k = 3$) of treatment 2. Thus the number of treatments which are both, i.e., the j^{th} ($j = 3$) associate of treatment A (here $A \equiv 1$) and k^{th} associate of treatment B (here $B \equiv 2$) are i^{th} (i.e., $i = 1$) associate is $p_{jk}^i = p_{33}^1 = 1$.

Similarly, consider the treatments 2 and 3 which are the first associate (which means $i = 1$); treatment 4 is the third (which means $j = 3$) associate of treatment 2 and treatment 4 is also the third (which means $k = 3$) associate of treatment 3. Thus $p_{33}^1 = 1$.

Other values of p_{jk}^i ($i, j, k = 1, 2, 3$) can also be obtained similarly.

Remark: This method can be used to generate a 3-class association scheme in general for $m \times n$ treatments (symbols) by arranging them in m -row and n -columns.

Triangular Association Scheme

The triangular association scheme gives rise to a 2-class association scheme. Let there be a set of v treatments which are denoted as $1, 2, \dots, v$. The treatments in this scheme are arranged in q rows and q columns where.

$$v = \binom{q}{2} = \frac{q(q-1)}{2}.$$

These symbols are arranged as follows:

- Positions in the leading diagonals are left blank (or crossed).
- The $\binom{q}{2}$ positions are filled up in the positions above the principal diagonal by the treatment numbers $1, 2, \dots, v$ corresponding to the symbols.
- Fill the positions below the principal diagonal symmetrically.

This assignment is shown in the following table:

Rows $\rightarrow$	1	2	3	4	...	$q - 1$	q
Columns $\downarrow$							
1	$\times$	1	2	3	...	$q - 2$	$q - 1$
2	1	$\times$	q	$q + 1$	...	$2q - 2$	$2q - 1$
3	2	q	$\times$	...	...	...	...
4	3	$q + 1$	...	...	...	...	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$q - 1$	$q - 2$	$2q - 2$	...	...	...	$\times$	$q(q - 1)/2$
q	$q - 1$	$2q - 1$	...	...	...	$q(q - 1)/2$	$\times$

Now based on this arrangement, we define the first and second associates of the treatments as follows. The symbols entering in the same column i ($i = 1, 2, \dots, q$) are the first associates of i and rest are the second associates. Thus two treatments in the same row or in the same column are the first associates of treatment i . Two treatments which do not occur in the same row or in the same column are the second associates of treatment i .

Now we illustrate this arrangement by the following example:

Let there be 10 treatments. Then $q = 5$ as $v = \binom{5}{2} = 10$. The ten treatments denoted as $1, 2, \dots, 10$ are arranged under the triangular association scheme as follows:

Rows →	1	2	3	4	5
Columns ↓					
1	×	1	2	3	4
2	1	×	5	6	7
3	2	5	×	8	9
4	3	6	8	×	10
5	4	7	9	10	×

For example,

- for treatment 1,
 - the treatments 2, 3 and 4 occur in the same row (or same column) and
 - treatments 5, 6 and 7 occur in the same column (or same row).

So the treatments 2, 3, 4, 5, 6 and 7 are the first associates of treatment.

- Then rest of the treatments 8, 9 and 10 are the second associates of treatment 1.

The first and second associates of the other treatments are stated in the following table.

Treatment number	First associates	Second associates
1	2, 3, 4 5, 6, 7	8, 9, 10
2	1, 3, 4 5, 8, 9	6, 7, 10
3	1, 2, 4 6, 8, 10	5, 7, 9
4	1, 2, 3 7, 9, 10	5, 6, 8
5	1, 6, 7 2, 8, 9	3, 4, 10
6	1, 5, 7 3, 8, 10	2, 4, 9
7	1, 5, 6 4, 9, 10	2, 3, 8
8	2, 5, 9 3, 6, 10	1, 4, 7
9	2, 5, 8 4, 7, 10	1, 3, 6
10	3, 6, 8 4, 7, 9	1, 2, 5

We observe from this table that the number of first and second associates of each of the 10 treatments ($v = 10$) is same with $n_1 = 6$, $n_2 = 3$ and $n_1 + n_2 = 9 = v - 1$. For example, the treatment 2 in the column of first associates occurs six times, viz., in first, third, fourth, fifth, eighth and ninth rows. Similarly, the treatment 2 in the column of second associates occurs three times, viz., in the sixth, seventh and tenth rows. Similar conclusions can be verified for other treatments.

There are six parameters, viz., $p_{11}^1, p_{22}^1, p_{12}^1$ (or p_{21}^1), p_{11}^2, p_{22}^2 and p_{12}^2 (or p_{21}^2) which can be arranged in symmetric matrices P_1 and P_2 as follows:

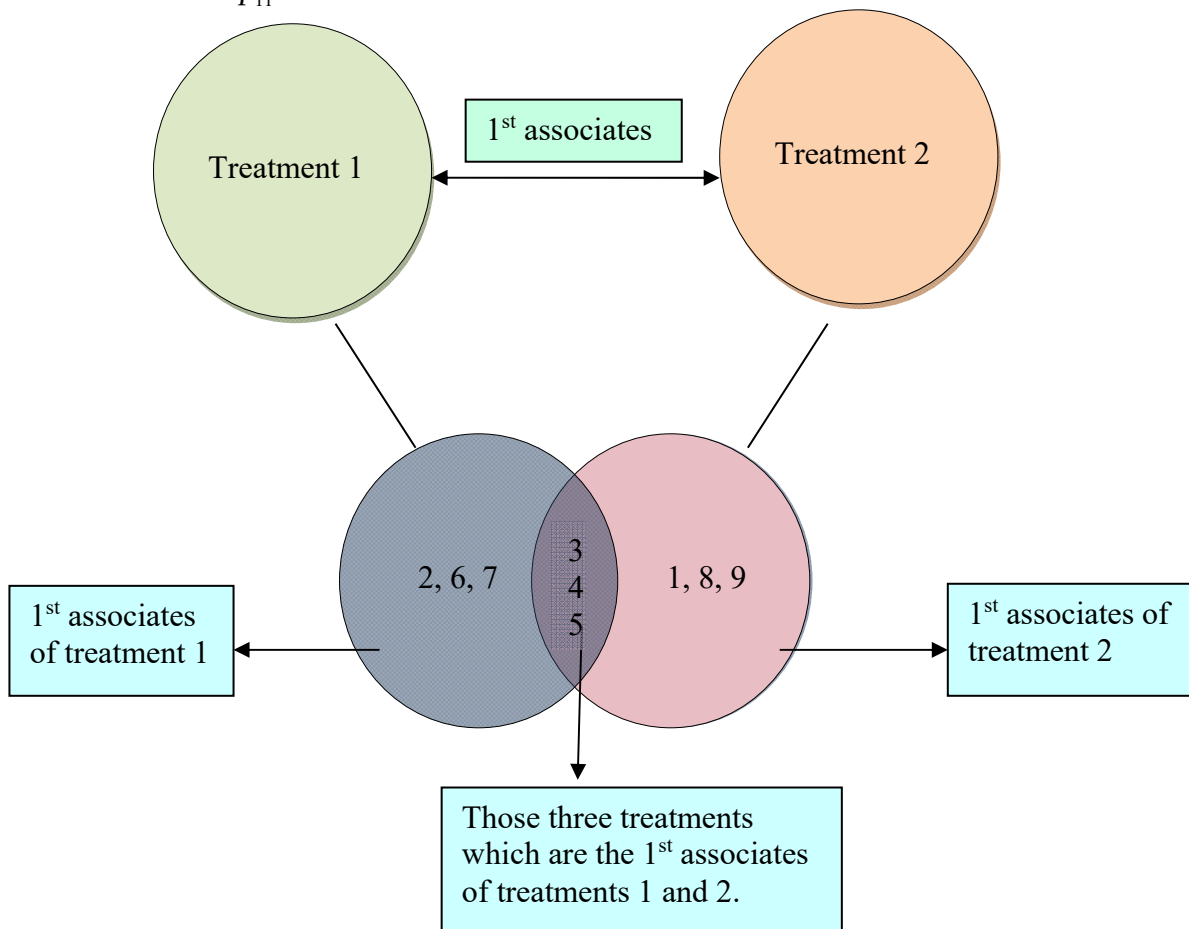
$$P_1 = \begin{bmatrix} p_{11}^1 & p_{12}^1 \\ p_{21}^1 & p_{22}^1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} p_{11}^2 & p_{12}^2 \\ p_{21}^2 & p_{22}^2 \end{bmatrix}.$$

[Note: We would like to caution the reader not to read p_{11}^2 as squares of p_{11} but 2 in p_{11}^2 is only a superscript.]

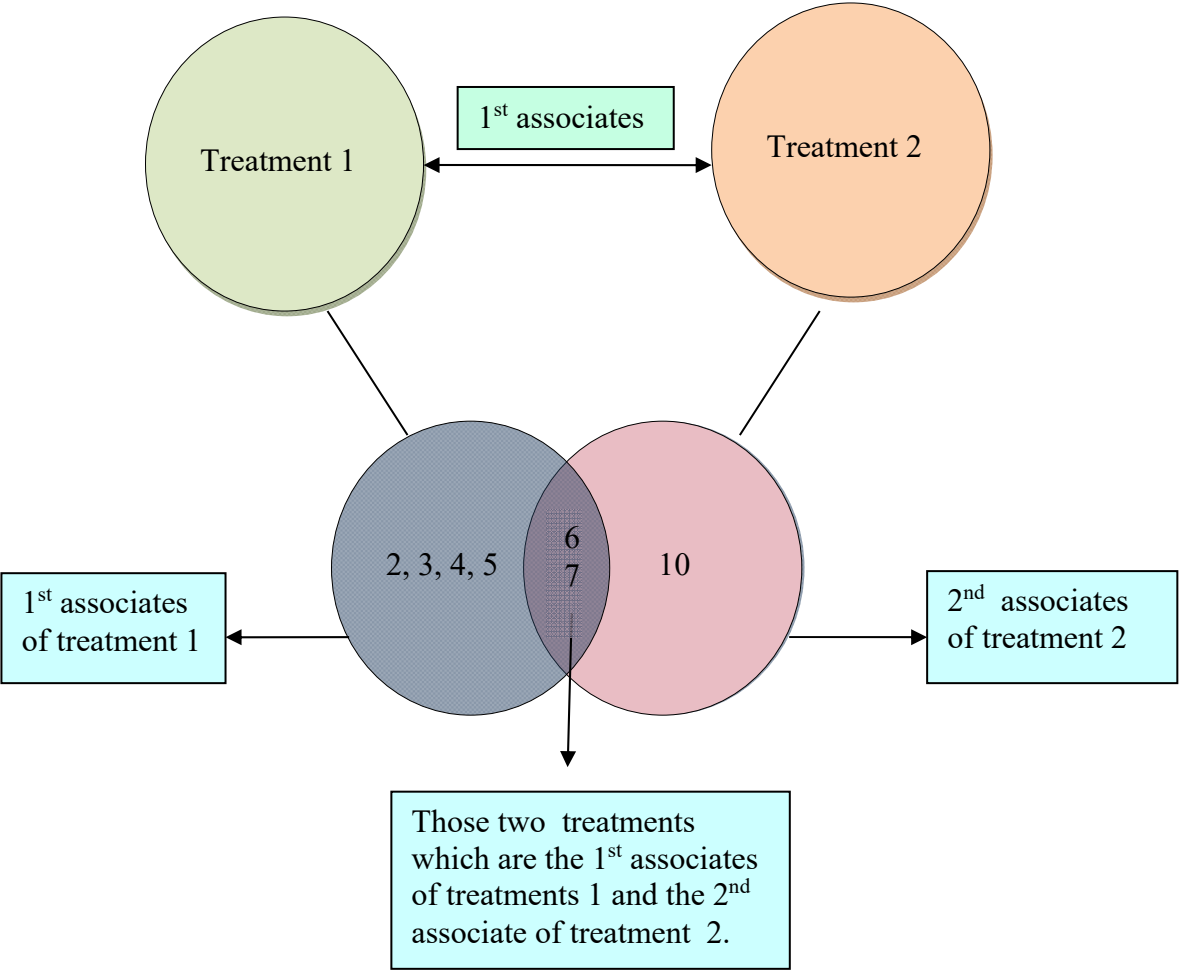
$$P_1 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 4 & 2 \\ 2 & 0 \end{bmatrix}.$$

In order to learn how to write these matrices P_1 and P_2 , we consider treatments 1, 2 and 8. Note that the treatment 8 is the second associate of treatment 1. Consider only the rows corresponding to treatments 1, 2 and 8 and obtain the elements of P_1 and P_2 as follows:

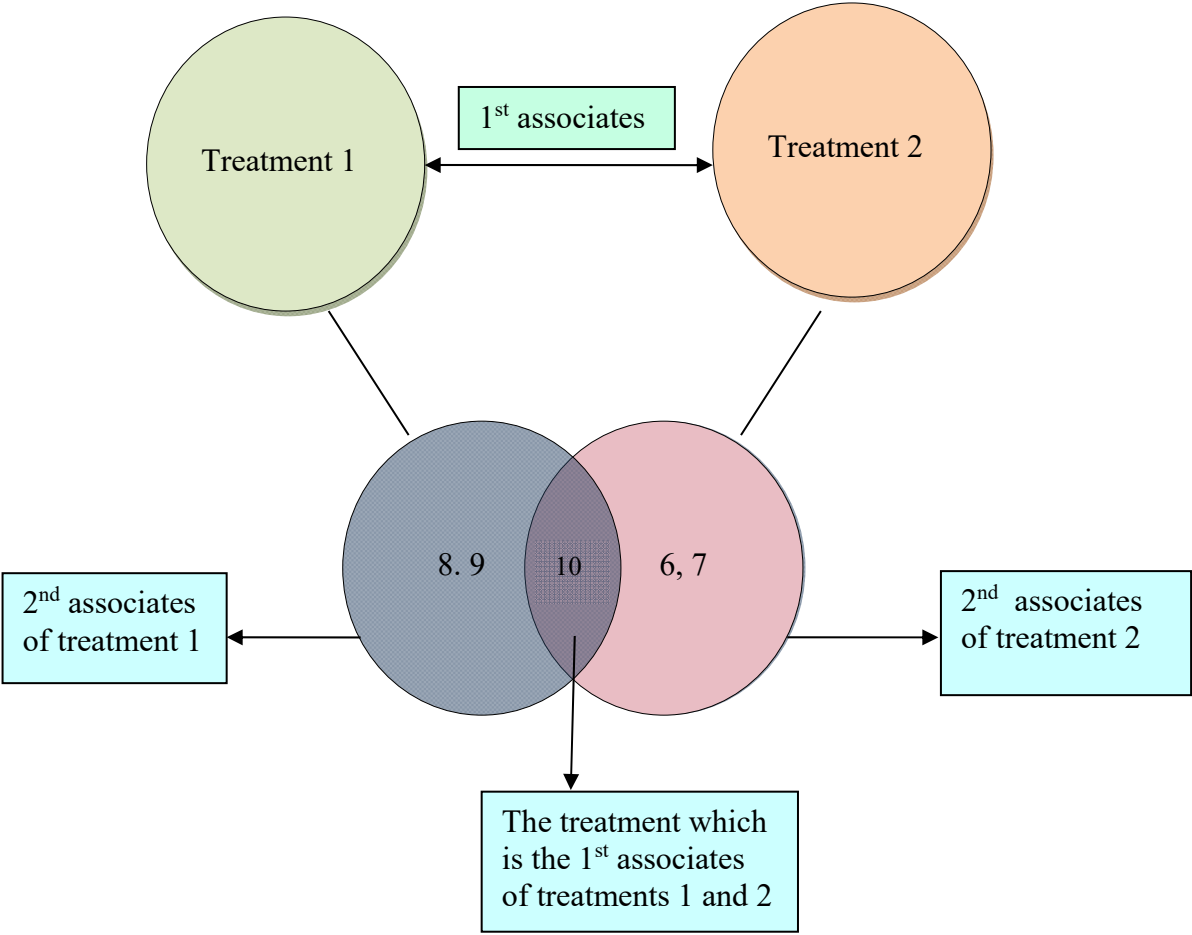
p_{11}^1 : Treatments 1 and 2 are the first associates of each other. There are three common treatments (viz., 3, 4 and 5) between the first associates of treatment 1 and the first associates of treatment 2. So $p_{11}^1 = 3$.



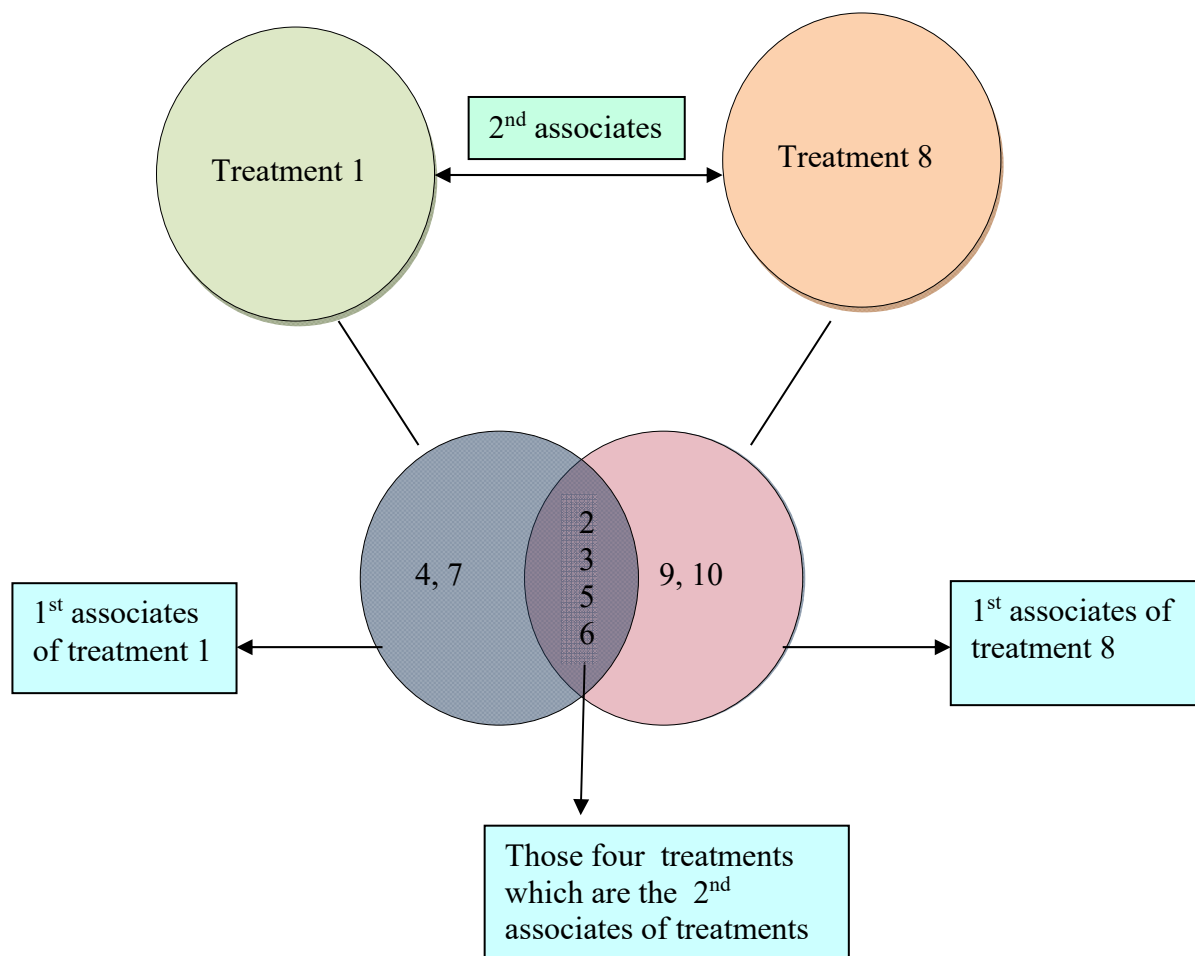
p_{12}^1 and p_{21}^1 : Treatments 1 and 2 are the first associates of each other. There are two treatments (viz., 6 and 7) which are common between the first associates of treatment 1 and the second associates of treatment 2. So $p_{12}^1 = 2 = p_{21}^1$.



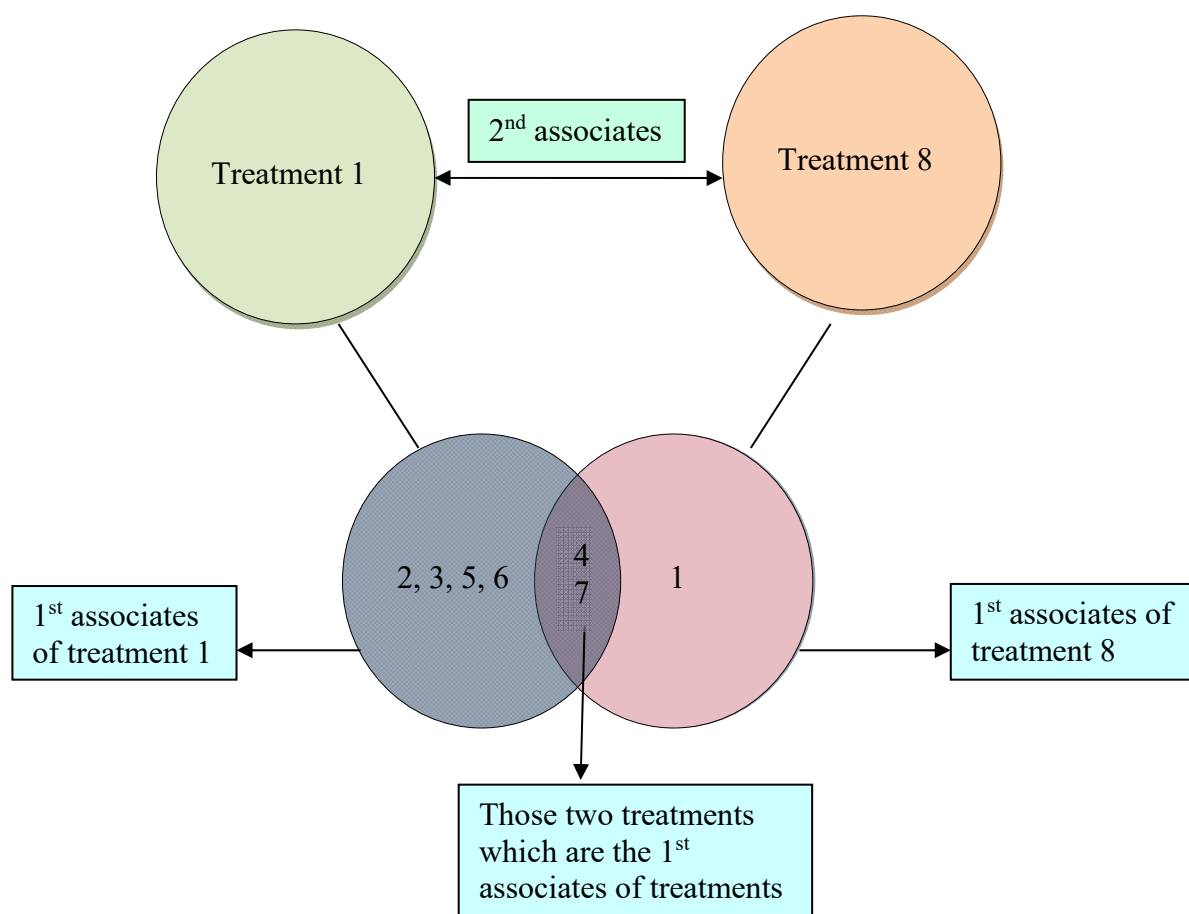
p_{22}^1 : Treatments 1 and 2 are the first associates of each other. There is only one treatment (viz., treatment 10) which is common between the second associates of treatment 1 and the second associates of treatment 2. So $p_{22}^1 = 1$.



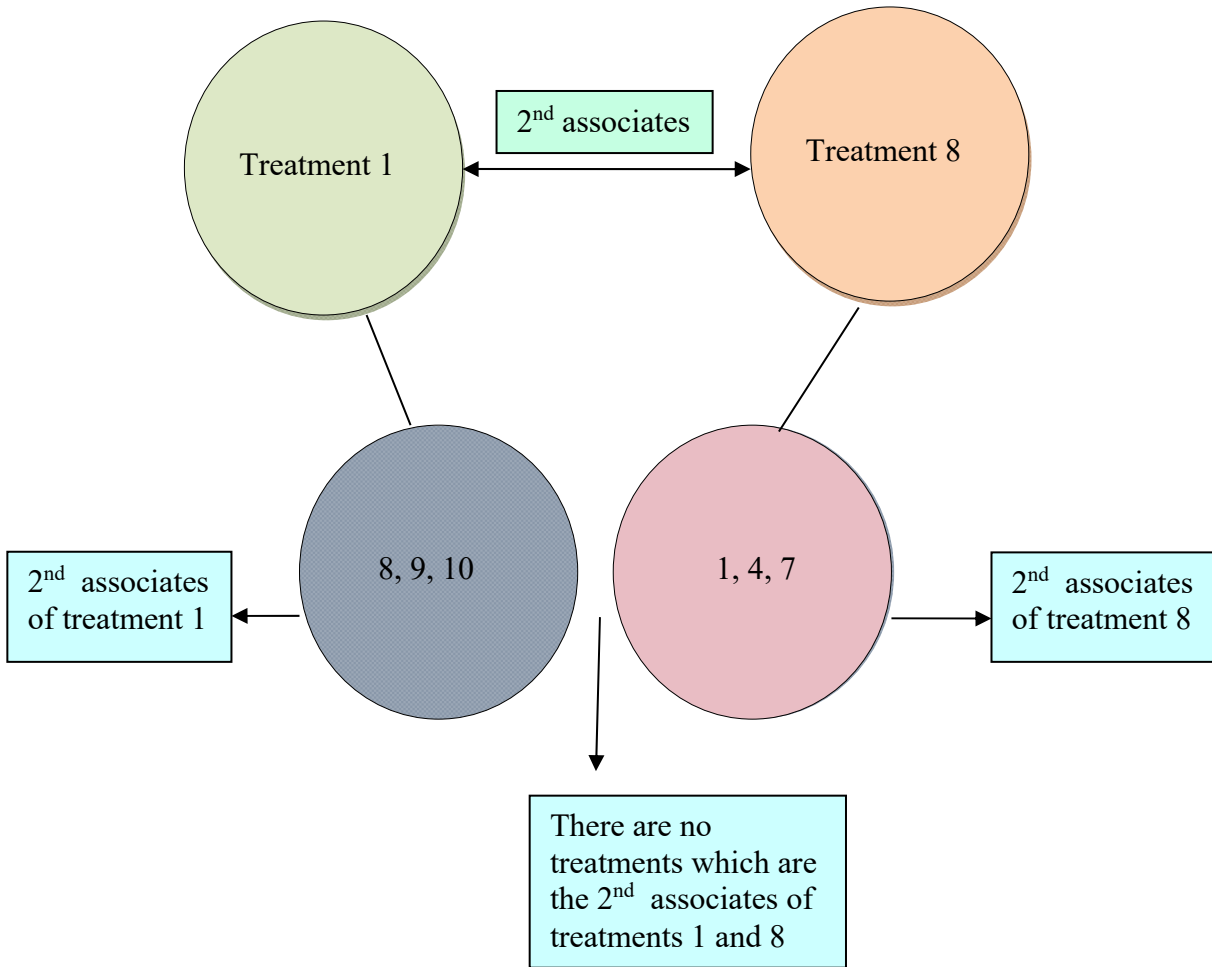
p_{11}^2 : Treatments 1 and 8 are the second associates of each other. There are four treatment (viz., 2,3,5 and 6) which are common between the first associates of treatment 1 and the first associates of treatment 8. So $p_{11}^1 = 4$.



p_{12}^2 and p_{21}^2 : Treatments 1 and 8 are the second associates of each other. There are two treatments (viz., 4 and 7) which are common between the first associates of treatment 1 and the second associates of treatment 8. So $p_{12}^1 = 2 = p_{21}^1$.



p_{22}^2 : Treatments 1 and 8 are the second associates of each other. There is no treatment which is common between the second associates of treatment 1 and the second associates of treatment 8. So $p_{22}^2 = 0$.



In general, if q rows and q columns of a square are used, then for $q > 3$

$$v = \binom{q}{2} = \frac{q(q-1)}{2},$$

$$n_1 = 2q - 4,$$

$$n_2 = \frac{(q-2)(q-3)}{2},$$

$$P_1 = \begin{bmatrix} q-2 & q-3 \\ q-3 & \frac{(q-3)(q-4)}{2} \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 4 & 2q-8 \\ 2q-8 & \frac{(q-4)(q-5)}{2} \end{bmatrix}.$$

For $q = 3$, there are no second associates. This is a degenerate case where second associates do not exist and hence P_2 cannot be defined.

The graph theory techniques can be used for counting p_{jk}^i .

Further, it is easy to see that all the parameters in P_1, P_2 , etc. are not independent.

Construction of Blocks of PBIBD under Triangular Association Scheme

The blocks of a PBIBD can be obtained in different ways through an association scheme. Even different block structure can be obtained using the same association scheme. We now illustrate this by obtaining different block structures using the same triangular scheme. Consider the earlier illustration

where $v = \binom{q}{2} = 10$ with $q = 5$ was considered and the first, second and third associates of treatments were obtained.

Approach 1: One way to obtain the treatments in a block is to consider the treatments in each row. This constitutes the set of treatments to be assigned in a block. When $q = 5$, the blocks of PBIBD are constructed by considering the rows of the following table.

Rows →	1	2	3	4	5
Columns ↓					
1	×	1	2	3	4
2	1	×	5	6	7
3	2	5	×	8	9
4	3	6	8	×	10
5	4	7	9	10	×

From this arrangement, the treatments are assigned in different blocks and following blocks are obtained.

Blocks	Treatments
Block 1	1, 2, 3, 4
Block 2	1, 5, 6, 7
Block 3	2, 5, 8, 9
Block 4	3, 6, 8, 10
Block 5	4, 7, 9, 10

The parameters of such a design are $b = 5, v = 10, r = 2, k = 4, \lambda_1 = 1$ and $\lambda_2 = 0$.

Approach 2: Another approach to obtain the blocks of PBIBD from a triangular association scheme is as follows:

- Consider the pair-wise columns of the triangular scheme.
- Then delete the common treatments between the chosen columns and retain others.
- The retained treatments will constitute the blocks.

Consider, e.g., the triangular association scheme for $q = 5$. Then the first block under this approach is obtained by deleting the common treatments between columns 1 and 2 which results in a block containing the treatments 2, 3, 4, 5, 6 and 7. Similarly, considering the other pairs of columns, the other blocks can be obtained which are presented in the following table:

Blocks	Columns of an association scheme	Treatments
Block 1	(1, 2)	2, 3, 4, 5, 6, 7
Block 2	(1, 3)	1, 3, 4, 5, 8, 9
Block 3	(1, 4)	1, 2, 4, 6, 8, 10
Block 4	(1, 5)	1, 2, 3, 7, 9, 10
Block 5	(2, 3)	1, 2, 6, 7, 8, 9
Block 6	(2, 4)	1, 3, 5, 7, 8, 10
Block 7	(2, 5)	1, 4, 5, 6, 9, 10
Block 8	(3, 4)	2, 3, 5, 6, 9, 10
Block 9	(3, 5)	2, 4, 5, 7, 8, 10
Block 10	(4, 5)	3, 4, 6, 7, 8, 9

The parameters of the PBIBD are $b = 10, v = 10, r = 6, k = 6, \lambda_1 = 3$ and $\lambda_2 = 4$.

Since both these PBIBDs are arising from the same association scheme, so the values of n_1, n_2, P_1 and P_2 remains the same for both the designs. In this case, we have $n_1 = 6, n_2 = 3$,

$$P_1 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 4 & 2 \\ 2 & 0 \end{bmatrix}.$$

Approach 3: Another approach to derive the blocks of PBIBD is to consider all the first associates of a given treatment in a block. For example, in the case of $q = 5$, the first associates of treatment 1 are the treatments 2, 3, 4, 5, 6 and 7. So these treatments constitute one block. Similarly, other blocks can also be found. This results in the same arrangement of treatments in blocks as considered by deleting the common treatments between the pair of columns.

The PBIBD with two associate classes are popular in practical applications and can be classified into the following types depending on the association scheme,

(Reference: *Classification and analysis of partially balanced incomplete block designs with two associate classes*, R. Bose and Shimamoto, 1952, *Journal of American Statistical Association*, 47, pp. 151-184).

1. Triangular
2. Group divisible
3. Latin square with i constraints (L_i)
4. Cyclic and
5. Singly-linked blocks.

The triangular association scheme has already been discussed. We now briefly present other types of association schemes.

Group divisible association scheme:

Let there be v treatments which can be represented as $v = pq$. Now divide the v treatments into p groups with each group having q treatments such that any two treatments in the same group are the first associates and two treatments in the different groups are the second associates. This is called the group divisible type scheme. The scheme simply amounts to arrange the $v = pq$ treatments in a $(p \times q)$ rectangle and then the association scheme can be exhibited. The columns in the $(p \times q)$ rectangle will form the groups.

Under this association scheme,

$$\begin{aligned} n_1 &= q-1 \\ n_2 &= q(p-1), \end{aligned}$$

hence

$$(q-1)\lambda_1 + q(p-1)\lambda_2 = r(k-1)$$

and the parameters of the second kind are uniquely determined by p and q . In this case

$$P_1 = \begin{pmatrix} q-2 & 0 \\ 0 & q(p-1) \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0 & q-1 \\ q-1 & q(p-2) \end{pmatrix},$$

For every group divisible design,

$$\begin{aligned} r &\geq \lambda_1, \\ rk - v\lambda_2 &\geq 0. \end{aligned}$$

If $r = \lambda_1$, then the group divisible design is said to be **singular**. Such a singular group divisible design can always be derived from a corresponding BIBD. To obtain this, just replace each treatment by a group of q treatments. In general, if a BIBD has the following parameters $b^*, v^*, r^*, k^*, \lambda^*$, then a group divisible design is obtained from this BIBD which has parameters

$$b = b^*, \quad v = qv^*, \quad r = r^*, \quad k = qk^*, \quad \lambda_1 = r, \quad \lambda_2 = \lambda^*, \quad n_1 = p, \quad n_2 = q.$$

A group divisible design is **nonsingular** if $r \neq \lambda_1$. Nonsingular group divisible designs can be divided into two classes – semi-regular and regular.

A group divisible design is said to be **semi-regular** if $r > \lambda_1$ and $rk - v\lambda_2 = 0$. For this design

$$b \geq v - p + 1.$$

Also, each block contains the same number of treatments from each group so that k must be divisible by p .

A group divisible design is said to be **regular** if $r > \lambda_1$ and $rk - v\lambda_2 > 0$. For this design

$$b \geq v.$$

Latin Square Type Association Scheme

Let L_i denotes the Latin square type PBIBD with i constraints. In this PBIBD, the number of treatments are expressible as $v = q^2$. The treatments may be arranged in a $(q \times q)$ square. For the case of $i = 2$, two treatments are the first associates if they occur in the same row or the same column, and second associates otherwise. For the general case, we take a set of $(i = 2)$ mutually orthogonal Latin squares, provided it exists. Then two treatments are the first associates if they occur in the same row or the same column, or corresponding to the same letter of one of the Latin squares. Otherwise, they are second associates.

Under this association scheme,

$$\begin{aligned} v &= q^2, \\ n_1 &= i(q-1), \\ n_2 &= (q-1)(q-i+1) \\ P_1 &= \begin{pmatrix} (i-1)(i-2) + q - 2 & (q-i+1)(i-1) \\ (q-i+1)(i-1) & (q-i+1)(q-i) \end{pmatrix}, \\ P_2 &= \begin{pmatrix} i(i-1) & i(q-i) \\ i(q-i) & (q-i)(q-i-1) + q - 2 \end{pmatrix}. \end{aligned}$$

Cyclic Type Association Scheme

Let there be v treatments and they are denoted by integers $1, 2, \dots, v$. In a cyclic type PBIBD, the first associates of treatment i are

$$i + d_1, i + d_2, \dots, i + d_{n_1} \pmod{v},$$

where the d 's satisfy the following conditions:

- (i) the d 's are all different and $0 < d_j < v$ ($j = 1, 2, \dots, n_1$);
- (ii) among the $n_1(n_1 - 1)$ differences $d_j - d_{j'}$, ($j, j' = 1, 2, \dots, n_1, j \neq j'$) reduced $\pmod{v}$, each of numbers $d_1, d_2, \dots, d_{n_1}$ occurs α times, whereas each of the numbers $e_1, e_2, \dots, e_{n_2}$ occurs β times, where $d_1, d_2, \dots, d_{n_1}, e_1, e_2, \dots, e_{n_2}$ are all the different $v - 1$ numbers $1, 2, \dots, v - 1$.

[**Note:** To reduce an integer mod v , we have to subtract from it a suitable multiple of v , so that the reduced integer lies between 1 and v . For example, 17 when reduced mod 13 is 4.]

For this scheme,

$$\begin{aligned} n_1\alpha + n_2\beta &= n_1(n_1 - 1), \\ P_1 &= \begin{pmatrix} \alpha & n_1 - \alpha - 1 \\ n_1 - \alpha - 1 & n_2 - n_1 + \alpha + 1 \end{pmatrix}, \\ P_2 &= \begin{pmatrix} \beta & n_1 - \beta \\ n_1 - \beta & n_2 - n_1 + \beta - 1 \end{pmatrix}. \end{aligned}$$

Singly Linked Block Association Scheme

Consider a BIBD D with parameters $b^{**}, v^{**}, r^{**}, k^{**}, \lambda^{**} = 1$ and $b^{**} > v^{**}$.

Let the block numbers of this design be treated as treatments, i.e., $v = b^{**}$. The first and second associates in the singly linked block association scheme with two classes are determined as follows: Define two block numbers of D to be the first associates if they have exactly one treatment in common and second associates otherwise.

Under this association scheme,

$$\begin{aligned} v &= b^{**}, \\ n_1 &= k^{**}(r^{**} - 1), \\ n_2 &= b^{**} - 1 - n_1, \\ P_1 &= \begin{pmatrix} r^{**} - 2 + (k^{**} - 1)^2 & n_1 - r^{**} - (k^{**} - 1)^2 + 1 \\ n_1 - r^{**} - (k^{**} - 1)^2 + 1 & n_2 - n_1 + r^{**} + (k^{**} - 1)^2 - 1 \end{pmatrix}, \\ P_2 &= \begin{pmatrix} k^{**2} & n_1 - k^{**2} \\ n_1 - k^{**2} & n_2 - n_1 + k^{**2} - 1 \end{pmatrix}. \end{aligned}$$

General Theory of PBIBD

A PBIBD with m -associate classes is defined as follows. Let there be v treatments. Let there are b blocks and the size of each block is k , i.e., there are k plots in each block. Then the v treatments are arranged in b blocks according to an m -associate partially balanced association scheme such that

- (a) every treatment occurs at most once in a block,
- (b) every treatment occurs exactly in r blocks and
- (c) if two treatments are the i^{th} associates of each other then they occur together exactly in $\lambda_i (i = 1, 2, \dots, m)$ blocks.

The number λ_i is independent of the particular pair of i^{th} associate chosen. It is not necessary that λ_i should all be different and some of them λ_i 's may be zero.

If v treatments can be arranged in such a scheme then we have a PBIBD. Note that here two treatments which are the i^{th} associates, occur together in λ_i blocks.

The parameters $b, v, r, k, \lambda_1, \lambda_2, \dots, \lambda_m, n_1, n_2, \dots, n_m$ are termed as the **parameters of the first kind** and p_{jk}^i are termed as the **parameters of the second kind**. It may be noted that $n_1, n_2, \dots, n_m$ and all p_{jk}^i of the design is obtained from the association scheme under consideration. Only $\lambda_1, \lambda_2, \dots, \lambda_m$, occur in the definition of PBIBD.

If $\lambda_i = \lambda$ for all $i = 1, 2, \dots, m$ then PBIBD reduces to BIBD. So BIBD is essentially a PBIBD with one associate class.

Conditions for PBIBD

The parameters of a PBIBD are chosen such that they satisfy the following relations:

- (i) $bk = vr$
- (ii) $\sum_{i=1}^m n_i = v - 1$
- (iii) $\sum_{i=1}^m n_i \lambda_i = r(k - 1)$
- (iv) $n_k p_{ij}^k = n_i p_{jk}^i = n_j p_{ki}^j$
- (v) $\sum_{k=1}^m p_{jk}^i = \begin{cases} n_j - 1 & \text{if } i = j \\ n_j & \text{if } i \neq j. \end{cases}$

It follows from these conditions that there are only $m(m^2 - 1)/6$ independent parameters of the second kind.

Interpretations of Conditions of PBIBD

The interpretations of conditions (i)-(v) are as follows.

(i) $bk = vr$

This condition is related to the total number of plots in an experiment. In our settings, there are k plots in each block and there are b blocks. So the total number of plots are bk . Further, there are v treatments and each treatment is replicated r times such that each treatment occurs atmost in one block. So the total number of plots containing all the treatments is vr . Since both, the statements counts the total number of blocks, hence $bk = vr$.

(ii) $\sum_{i=1}^m n_i = v - 1$

This condition interprets as follows:

Since with respect to each treatment, the remaining $(v - 1)$ treatments are classified as first, second, ..., or m^{th} associates and each treatment has n_i associates, so the sum of all n_i 's is the same as the total number of treatments except the treatment under consideration.

(iii) $\sum_{i=1}^m n_i \lambda_i = r(k - 1)$

Consider r blocks in which a particular treatment A occurs. In these r blocks, $r(k - 1)$ pairs of treatments can be found, each having A as one of its members. Among these pairs, the i^{th} associate of A must occur λ_i times and there are n_i associates, so $\sum_i n_i \lambda_i = r(k - 1)$.

(iv) $n_i p_{jk}^i = n_j p_{ki}^j = n_k p_{ij}^k$

Let G_i be the set of i^{th} associates, $i = 1, 2, \dots, m$ of a treatment A . For $i \neq j$, each treatment in G_i has exactly p_{jk}^i numbers of k^{th} associates in G_j . Thus the number of pairs of k^{th} associates that can be obtained by taking one treatment from G_i and another treatment from G_j on the one hand is $n_i p_{jk}^i$ and $n_j p_{ki}^j$, respectively.

$$(v) \sum_{k=1}^m p_{jk}^i = n_j - 1 \text{ if } i = j \text{ and } \sum_{k=1}^m p_{jk}^i = n_j \text{ if } i \neq j.$$

Let the treatments A and B be i^{th} associates. The k^{th} associate of $A (k = 1, 2, \dots, m)$ should contain all the n_j number of j^{th} associates of $B (j \neq i)$. When $j = i$, A itself will be one of the j^{th} associates of B . Hence k^{th} associate of $A, (k = 1, 2, \dots, m)$ should contain all the $(n_j - 1)$ numbers of j^{th} associate of B . Thus the condition holds.

Intrablock Analysis of PBIBD With Two Associates

Consider a PBIBD under two associates scheme. The parameters of this scheme are $b, v, r, k, \lambda_1, \lambda_2, n_1, n_2, p_{11}^1, p_{22}^1, p_{12}^1, p_{11}^2, p_{22}^2$ and p_{12}^2 . The linear model involving block and treatment effects is

$$y_{ij} = \mu + \beta_i + \tau_j + \varepsilon_{ij}; i = 1, 2, \dots, b, j = 1, 2, \dots, v,$$

where

μ is the general mean effect;

β_i is the fixed additive i^{th} block effect satisfying $\sum_i \beta_i = 0$;

τ_j is the fixed additive j^{th} treatment effect satisfying $\sum_{j=1}^r \tau_j = 0$ and

ε_{ijm} is the i.i.d. random error with $\varepsilon_{ijm} \sim N(0, \sigma^2)$.

The PBIBD is a binary, proper and equi-replicate design. So in this case, the values of the parameters become $n_{ij} = 0$ or $1, k_i = k$ for all $i = 1, 2, \dots, b$ and $r_j = r$ for all $j = 1, 2, \dots, v$. There can be two types of null hypothesis which can be considered – one for the equality of treatment effects and another for the equality of block effects. We are considering here the intrablock analysis, so we consider the null hypothesis related to the treatment effects only. As done earlier in the case of BIBD, the block effects can be considered under the interblock analysis of PBIBD and the recovery of interblock information.

The null hypothesis of interest is

$H_0 : \tau_1 = \tau_2 = \dots = \tau_v$ against alternative hypothesis

$H_1 : \text{at least one pair of } \tau_j \text{ is different.}$

The null hypothesis related to the block effects is of not much practical relevance and can be treated similarly. This was illustrated earlier in case of BIBD. In order to obtain the least-squares estimates of μ, β_i and ρ_j , we minimize of the sum of squares due to residuals

$$\sum_{i=1}^b \sum_{j=1}^v (y_{ij} - \mu - \beta_i - \tau_j)^2$$

with respect to μ, β_i and τ_j . This results in the three normal equations which can be unified using the matrix notations. The set of reduced normal equations in matrix notation after eliminating the block effects are expressed as

$$Q = C\tau$$

where

$$C = R - N'K^{-1}N,$$

$$Q = V - N'K^{-1}B,$$

$$R = rI_v,$$

$$K = kI_b.$$

Then the diagonal elements of C are

$$c_{jj} = r - \frac{\sum_{i=1}^b n_{ij}^2}{k} = \frac{r(k-1)}{k}, \quad (j = 1, 2, \dots, v),$$

the off-diagonal elements of C are obtained as

$$c_{jj'} = -\frac{1}{k} \sum_{i=1}^b n_{ij} n_{ij'} = \begin{cases} -\frac{\lambda_1}{k} & \text{if treatment } j \text{ and } j' \text{ are the first associates} \\ -\frac{\lambda_2}{k} & \text{if treatment } j \text{ and } j' \text{ are the second associates } (j \neq j' = 1, 2, \dots, v) \end{cases}$$

and the j^{th} value in Q is

$$\begin{aligned} Q_j &= V_j - \frac{1}{k} [\text{Sum of block totals in which } j^{th} \text{ treatment occurs}] \\ &= \frac{1}{k} \left[r(k-1)\tau_j - \sum_i \sum_{j'(j \neq j')} n_{ij} n_{ij'} \tau_j \right]. \end{aligned}$$

Next, we attempt to simplify the expression of Q_j .

Let S_{j1} be the sum of all treatments which are the first associates of j^{th} treatment and S_{j2} be the sum of all treatments which are the second associates of j^{th} treatment. Then

$$\tau_j + S_{j1} + S_{j2} = \sum_{j=1}^v \tau_j.$$

Thus the equation in Q_j using this relationship for $j = 1, 2, \dots, v$, becomes

$$\begin{aligned} kQ_j &= [r(k-1)\tau_j - (\lambda_1 S_{j1} + \lambda_2 S_{j2})] \\ &= r(k-1)\tau_j - \lambda_1 S_{j1} - \lambda_2 \left[\sum_{j=1}^v \tau_j - \tau_j - S_{j1} \right] \\ &= [r(k-1) + \lambda_2] \tau_j + (\lambda_2 - \lambda_1) S_{j1} - \lambda_2 \sum_{j=1}^v \tau_j. \end{aligned}$$

Imposing the side condition $\sum_{j=1}^v \tau_j = 0$, we have

$$\begin{aligned} kQ_j &= [r(k-1)\tau_j + \lambda_2] \tau_j + (\lambda_2 - \lambda_1) S_{j1} \\ &= a_{12}^* \tau_j + b_{12}^* S_{j1}, \quad j = 1, 2, \dots, v \end{aligned}$$

where $a_{12}^* = r(k-1) + \lambda_2$ and $a_{12}^* = \lambda_2 - \lambda_1$.

These equations are used to obtain the adjusted treatment sum of squares.

Let Q_{j1} denotes the adjusted sum of Q_j 's over the set of those treatments which are the first associate of j^{th} treatment. We note that when we add the terms S_{j1} for all j , then j occurs n_1 times in the sum, every first associate of j occurs p_{11}^1 times in the sum and every second associate of j occurs p_{11}^2 times in the sum with $p_{11}^1 + p_{11}^2 = n_1$. Then using $K = kI_b$ and $\sum_{j=1}^v \tau_j = 0$, we have

$$\begin{aligned} kQ_{j1} &= k \sum_{j \in S_{j1}} Q_j = \text{Sum of } Q_j \text{'s which are the first associates of treatment } j \\ &= [r(k-1) + \lambda_2] S_{j1} + (\lambda_2 - \lambda_1) [n_1 \tau_j + p_{11}^1 S_{j1} + p_{11}^2 S_{j2}] \\ &= [r(k-1) + \lambda_2 + (\lambda_2 - \lambda_1)(p_{11}^1 - p_{11}^2)] S_{j1} + (\lambda_2 - \lambda_1) p_{11}^2 \tau_j \\ &= b_{22}^* S_{j1} + a_{22}^* \tau_j \end{aligned}$$

where

$$\begin{aligned} a_{22}^* &= (\lambda_2 - \lambda_1) p_{11}^2 \\ b_{22}^* &= r(k-1) + \lambda_2 + (\lambda_2 - \lambda_1)(p_{11}^1 - p_{11}^2). \end{aligned}$$

Now we have following two equations in kQ_j and kQ_{j1} :

$$\begin{aligned} kQ_j &= a_{12}^* \tau_j + b_{12}^* S_{j1} \\ kQ_{j1} &= a_{22}^* \tau_j + b_{22}^* S_{j1} \end{aligned}$$

Now solving these two equations, the estimate of τ_j is obtained as

$$\hat{\tau}_j = \frac{k[b_{22}^* Q_j - b_{12}^* Q_{j1}]}{a_{12}^* b_{12}^* - a_{22}^* b_{12}^*}, \quad (j = 1, \dots, v).$$

We see that

$$\sum_{j=1}^v Q_j = \sum_{j=1}^v Q_{j1} = 0,$$

so

$$\sum_{j=1}^v \hat{\tau}_j = 0.$$

Thus $\hat{\tau}_j$ is a solution of the reduced normal equation.

The analysis of variance can be carried out by obtaining the unadjusted block sum of squares as

$$SS_{Block(unadj)} = \sum_{i=1}^b \frac{B_i^2}{k} - \frac{G^2}{bk},$$

the adjusted sum of squares due to treatment as

$$SS_{Treat(adj)} = \sum_{j=1}^v \hat{\tau}_j Q_j$$

where $G = \sum_{i=1}^b \sum_{j=1}^v y_{ij}$. The sum of squares due to error as

$$SS_{Error(t)} = SS_{Total} - SS_{Block(unadj)} - SS_{Treat(adj)}$$

where

$$SS_{Total} = \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - \frac{G^2}{bk}.$$

A test for $H_0 : \tau_1 = \tau_2 = \dots = \tau_v$ is then based on the statistic

$$F_{Tr} = \frac{SS_{Treat(adj)} / (v-1)}{SS_{Error(unadj)} / (bk - b - v + 1)}.$$

If $F_{Tr} > F_{1-\alpha; v-1, bk-v-b+1}$ then H_0 is rejected.

The intrablock analysis of variance for testing the significance of treatment effects is given in the following table:

Source	Sum of squares	Degrees of freedom	Mean squares	F
Between treatments (adjusted)	$SS_{Treat(adj)} = \sum_{j=1}^v \hat{\tau}_j Q_j$	$df_{Treat} = v - 1$	$MS_{Treat} = \frac{SS_{Treat(adj)}}{df_{Treat}}$	$\frac{MS_{Treat}}{MSE}$
Between blocks (unadjusted)	$SS_{Block(unadj)} = \sum_{i=1}^b \frac{B^2}{k} - \frac{G^2}{bk}$	$df_{Block} = b - 1$		
Intrablock Error	$SS_{Error(t)}$ (By subtraction)	$df_{ET} = bk - b - v + 1$	$MSE = \frac{SS_{Error}}{df_{ET}}$	
Total	$SS_{Total} = \sum_{i=1}^b \sum_{j=1}^v y_{ij}^2 - \frac{G^2}{bk}$	$df_T = bk - 1$		

Note that in the intrablock analysis of PBIBD analysis, at the step where we obtained S_{j1} and eliminated S_{j2} to obtain the following equation

$$kQ_j = [r(k-1) + \lambda_2] \tau_j + (\lambda_2 - \lambda_1) S_{j1} - \lambda_2 \sum_{j=1}^v \tau_j,$$

another possibility is to eliminate S_{j1} instead of S_{j2} . If we eliminate S_{j2} instead of S_{j1} (as we approached), then the solution has less work involved in the summing of Q_{j1} if $n_1 < n_2$. If $n_1 > n_2$, then eliminating S_{j1} will involve less work in obtaining Q_{j2} where Q_{j2} denotes the adjusted sum of Q_j 's is over the set of those treatments which are the second associate of j^{th} treatment. When we do so, obtain the following estimate of the treatment effect is obtained as:

$$\hat{\tau}_j^* = \frac{k[b_{21}^* Q_j - b_{11}^* Q_{j2}]}{a_{11}^* b_{21}^* - a_{21}^* b_{11}^*}$$

where

$$\begin{aligned}
a_{11}^* &= r(k-1) + \lambda_1 \\
b_{11}^* &= \lambda_1 - \lambda_2 \\
a_{21}^* &= (\lambda_1 - \lambda_2) p_{12}^1 \\
b_{21}^* &= r(k-1) + \lambda_1 + (\lambda_1 - \lambda_2)(p_{22}^2 - p_{22}^1).
\end{aligned}$$

The analysis of variance is then based on $\hat{\tau}_j^*$ and can be carried out similarly.

The variance of the elementary contrasts of estimates of treatments (in case of $n_1 < n_2$)

$$\hat{\tau}_j - \hat{\tau}_{j'} = \frac{b_{22}^*(kQ_j - kQ_{j'}) - b_{12}^*(kQ_{j1} - kQ_{j'1})}{a_{12}^*b_{22}^* - a_{22}^*b_{12}^*}$$

is

$$\text{Var}(\hat{\tau}_j - \hat{\tau}_{j'}) = \begin{cases} \frac{2k(b_{22}^* + b_{12}^*)}{a_{12}^*b_{22}^* - a_{22}^*b_{12}^*} & \text{if treatment } j \text{ and } j' \text{ are the first associates} \\ \frac{2kb_{12}^*}{a_{12}^*b_{22}^* - a_{22}^*b_{12}^*} & \text{if treatment } j \text{ and } j' \text{ are the second associates.} \end{cases}$$

We observe that the variance of $\hat{\tau}_j - \hat{\tau}_{j'}$ depends on the nature of j and j' in the sense that whether they are the first or second associates. So design is not (variance) balanced. But variance of any elementary contrast are equal under a given order of association, viz., first or second. That is why the design is said to be partially balanced in this sense.

Chapter 8

Factorial Experiments

Factorial experiments involve simultaneously more than one factor and each factor is at two or more levels. Several factors affect simultaneously the characteristic under study in factorial experiments and the experimenter is interested in the main effects and the interaction effects among different factors.

First, we consider an example to understand the utility of factorial experiments.

Example: Suppose the yield from different plots in an agricultural experiment depends upon

1. (i) variety of crop and
- (ii) type of fertilizer.

Both the factors are in the control of the experimenter.

2. (iii) Soil fertility. This factor is not in the control of the experimenter.

In order to compare different crop varieties

- assign it to different plots keeping other factors like irrigation, fertilizer, etc. fixed and the same for all the plots.
- The conclusions for this will be valid only for the crops grown under similar conditions with respect to the factors like fertilizer, irrigation etc.

In order to compare different fertilizers (or different dosage of fertilizers)

- sow single crop on all the plots and vary the quantity of fertilizer from plot to plot.
- The conclusions will become invalid if different varieties of the crop are sown.
- It is quite possible that one variety may respond differently than another to a particular type of fertilizer.

Suppose we wish to compare

- two crop varieties – a and b , keeping the fertilizer fixed and
- three varieties of fertilizers – A , B and C .

This can be accomplished with two randomized block designs (RBD) by assigning the treatments at random to three plots in any block and two crop varieties at random.

The possible arrangement of the treatments may appear can be as follows.

bB	bA	bC	and	aA	aB	aC
bC	bB	bA		aC	aA	aB
bA	bC	bB		aB	aC	aA

With these two *RBDs*,

- the difference among two fertilizers can be estimated
- but the difference among the crop varieties cannot be estimated. The difference among the crop varieties is entangled with the difference in blocks.

On the other hand, if we use three sets of three blocks each and each block having two plots, then

- randomize the varieties inside each block and
- assign treatments at random to three sets.

The possible arrangement of treatment combinations in blocks can be as follows:

bB	aB	,	aC	bC	and	aA	bA
aB	bB		bC	aC		bA	aA
bB	aB		aC	bC		bA	aA

Here the difference between crop varieties is estimable but the difference between fertilizer treatment is not estimable.

Factorial experiments overcome this difficulty and combine each crop with each fertilizer treatment. There are six treatment combinations as

aA, aB, aC, bA, bB, bC .

Keeping the total number of observations to be 18 (as earlier), we can use *RBD* with three blocks with six plots each, e.g.

bA	aC	aB	bB	aA	bC
aA	aC	bC	aB	bB	bA
bB	aB	bA	aC	aA	bC

Now we can estimate

- the difference between crop varieties and
- the difference between fertilizer treatments. Factorial experiments involve simultaneously more than one factor each at two or more levels.

If the number of levels for each factor is the same, we call it is a **symmetrical factorial experiment**.

If the number of levels of each factor is not the same, then we call it as a **symmetrical or mixed factorial experiment**.

We consider only symmetrical factorial experiments.

Through the factorial experiments, we can study

- the individual effect of each factor and
- interaction effect.

Now we consider a 2^2 factorial experiment with an example and try to develop and understand the theory and notations through this example.

General notation for representing the factors is to use capital letters, e.g., A , B , C etc. and levels of a factor are represented in small letters. For example, if there are two levels of A , they are denoted as a_0 and a_1 . Similarly, the two levels of B are represented as b_0 and b_1 . Another alternative representation to indicate the two levels of A is 0 (for a_0) and 1 (for a_1). The factors of B are then 0 (for b_0) and 1 (for b_1).

Note: An important point to remember is that the factorial experiments are conducted in the design of an experiment. For example, the factorial experiment is conducted as an RBD.

Factorial experiments with factors at two levels (2^2 factorial experiment):

Suppose in an experiment, the values of current and voltage in an experiment affect the rotation per minutes (*rpm*) of fan speed. Suppose there are two levels of current.

- 5 Ampere, call it as level 1 (C_1) and denote it as a_0
- 10 Ampere, call it as level 2 (C_1) and denote it as a_1 .

Similarly, the two levels of voltage are

- 200 volts, call it as level 1 (V_0) and denote it as b_0
- 220 volts, call it as level 2 (V_1) and denote it as b_1 .

The two factors are denoted as A , say for current and B , say for voltage.

In order to make an experiment, there are 4 different combinations of values of current and voltage.

1. Current = 5 Ampere and Voltage = 200 Volts, denoted as $C_0V_0 \equiv a_0b_0$
2. Current = 5 Ampere and Voltage = 220 Volts, denoted as $C_0V_1 \equiv a_0b_1$.
3. Current = 10 Ampere and Voltage = 200 Volts, denoted as $C_1V_0 \equiv a_1b_0$
4. Current = 10 Ampere and Voltage = 220 Volts, denoted as $C_1V_1 \equiv a_1b_1$

The responses from those treatment combinations are represented by $a_0b_0 \equiv (1), (a_0b_1) \equiv (b), (a_1b_0) \equiv (a)$ and $(a_1b_1) \equiv (ab)$, respectively.

Now consider the following:

$$\begin{aligned} \text{I. } \frac{(C_0V_0) + (C_0V_1)}{2} &: \text{Average effect of voltage for the current level } C_0 \\ &: \frac{(a_0b_0) + (a_0b_1)}{2} \equiv \frac{(1) + (b)}{2} \\ \text{II. } \frac{(C_1V_0) + (C_1V_1)}{2} &: \text{Average effect of voltage for the current level } C_1 \\ &: \frac{(a_1b_0) + (a_1b_1)}{2} \equiv \frac{(a) + (ab)}{2} \end{aligned}$$

Compare these two group means (or totals) as follows:

Average effect of V_1 level – Average effect at V_0 level

$$\begin{aligned} &= \frac{(b) + (ab)}{2} - \frac{(1) + (a)}{2} \\ &= \text{Main effect of voltage} \\ &= \text{Main effect of } B. \end{aligned}$$

Comparison like

$$(C_0V_1) - (C_0V_0) \equiv (a) - (1): \text{ indicate the effect of voltage at current level } C_0$$

and

$$(C_1V_1) - (C_1V_0) \equiv (ab) - (b): \text{ indicate the effect of voltage at current level } C_1.$$

The average interaction effect of voltage and current can be obtained as

$$\begin{aligned}
 & \left(\text{Average effect of voltage} \right)_{\text{at current level } I_0} - \left(\text{Average effect of voltage} \right)_{\text{at current level } I_1} \\
 &= \text{Average effect of voltage at different levels of current.} \\
 &= \frac{(C_1V_1) - (C_1V_0)}{2} - \frac{(C_0V_1) - (C_0V_0)}{2} \\
 &= \frac{(ab) - (b)}{2} - \frac{(a) - (1)}{2} \\
 &= \text{Average interaction effect.}
 \end{aligned}$$

Similarly

$$\begin{aligned}
 \frac{(C_0V_0) + (C_1V_0)}{2} &= \frac{(1) + (b)}{2} : \text{Average effect of current at voltage level } V_0. \\
 \frac{(C_0V_1) + (C_1V_1)}{2} &= \frac{(a) + (ab)}{2} : \text{Average effect of current at voltage level } V_1
 \end{aligned}$$

Comparison of these two as

$$\begin{aligned}
 & \left(\text{Average effect of current} \right)_{\text{at voltage level } V_0} - \left(\text{Average effect of current} \right)_{\text{at voltage level } V_1} \\
 &= \frac{(C_0V_1) + (C_1V_1)}{2} - \frac{(C_0V_0) + (C_1V_0)}{2} \\
 &= \frac{(a) + (ab)}{2} - \frac{(1) + (b)}{2} \\
 &= \text{Main effect of current} \\
 &= \text{Main effect of } A.
 \end{aligned}$$

Comparison like

$$\begin{aligned}
 (C_1V_0) - (C_0V_0) &= (b) - (1) : \text{Effect of current at voltage level } V_0 \\
 (C_1V_1) - (C_0V_1) &= (ab) - (a) : \text{Effect of current at voltage level } V_1
 \end{aligned}$$

The average interact effect of current and voltage can be obtained as

$$\begin{aligned}
 & \left(\text{Average effect of current} \right)_{\text{at voltage level } V_0} - \left(\text{Average effect of current} \right)_{\text{at voltage level } V_1} \\
 &= \text{Average effect of current at different levels of voltage} \\
 &= \frac{(C_1V_1) - (C_0V_1)}{2} - \frac{(C_1V_0) - (C_0V_0)}{2} \\
 &= \frac{(ab) - (a)}{2} - \frac{(b) - (1)}{2} \\
 &= \text{Average interaction effect} \\
 &= \text{Same as average effects of voltage at different levels of current.}
 \end{aligned}$$

(It is expected *that* the interaction effect of current and voltage is same as the interaction effect of voltage and current).

The quantity

$$\frac{(C_0V_0) + (C_1V_0) + (C_0V_1) + (C_1V_1)}{4} = \frac{(1) + (a) + (b) + (ab)}{4}$$

gives the **general mean effect** of all the treatment combination.

Treating (ab) as $(a)(b)$ symbolically (mathematically and conceptually, it is incorrect), we can now express all the main effects, interaction effect and general mean effect as follows:

$$\text{Main effect of } A = \frac{(a) + (ab)}{2} - \frac{(1) + (b)}{2} = \frac{1}{2}[(ab) - (b) + (a) - (1)] = \frac{(a-1)(b+1)}{2}$$

$$\text{Main effect of } B = \frac{(b) + (ab)}{2} - \frac{(1) + (a)}{2} = \frac{1}{2}[(ab) - (a) + (b) - (1)] = \frac{(a+1)(b-1)}{2}$$

$$\text{Interaction effect of } A \text{ and } B = \frac{(ab) - (b)}{2} - \frac{(a) - (1)}{2} = \frac{1}{2}[(ab) - (a) + (1) - (b)] = \frac{(a-1)(b-1)}{2}$$

$$\text{General mean effect } (M) = \frac{(1) + (a) + (b) + (ab)}{4} = \frac{1}{4}[(1) + (a) + (b) + (ab)] = \frac{(a+1)(b+1)}{4}$$

Notice the roles of + and – signs as well as the divisor.

- There are two effects related to A and B .
- To obtain the effect of a factor, write the corresponding factor with – sign and others with + sign. For example, in the main effect of A , a occurs with – sign as in $(a - 1)$ and b occurs with + sign as in $(b + 1)$.
- In AB , both the effects are present so a and b both occur with + signs as in $(a + 1)(b + 1)$.
- Also note that the main and interaction effects are obtained by considering the typical differences of averages, so they have divisor 2 whereas the general mean effect is based on all the treatment combinations and so it has divisor 4.
- There is a well defined statistical theory behind this logic but this logic helps in writing the final treatment combination easily. This is demonstrated later with appropriate reasoning.

Other popular notations of treatment combinations are as follows:

$$a_0b_0 \equiv 0 \quad 0 \equiv I$$

$$a_0b_1 \equiv 0 \quad 1 \equiv a$$

$$a_1b_0 \equiv 1 \quad 0 \equiv b$$

$$a_1b_1 \equiv 1 \quad 1 \equiv ab.$$

Sometimes 0 is referred to as ‘low level’ and 1 is referred to ‘high level’.

Here I denote that both factors are at lower levels (a_0b_0 or 00). This is called as the **control treatment**.

These effects can be represented in the following table

Factorial effects	Treatment combinations				Divisor
	(1)	(a)	(b)	(ab)	
<i>M</i>	+	+	+	+	4
<i>A</i>	-	+	-	+	2
<i>B</i>	-	-	+	+	2
<i>AB</i>	+	-	-	+	2

The model corresponding to 2^2 factorial experiment is

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}, \quad i = 1, 2, \quad j = 1, 2, \quad k = 1, 2, \dots, n$$

where n observations are obtained for each treatment combinations.

When the experiments are conducted factor by factor, then much more resources are required in comparison to the factorial experiment. For example, if we conduct *RBD* for three-levels of voltage V_0, V_1 and V_2 and two levels of current I_0 and I_1 , then to have 10 degrees of freedom for the error variance, we need

- 6 replications on voltage
- 11 replications on current.

So the total number of fans needed is 40.

For the factorial experiment with 6 combinations of 2 factors, the total number of fans needed are 18 for the same precision.

We have considered the situation up to now by assuming only one observation for each treatment combination, i.e., no replication. If r replicated observations for each of the treatment combinations are obtained, then the expressions for the main and interaction effects can be expressed as

$$A = \frac{1}{2r}[(ab) + (a) - b - (1)]$$

$$B = \frac{1}{2r}[(ab) + (b) - a - (1)]$$

$$AB = \frac{1}{2r}[(ab) + (1) - a - (b)]$$

$$M = \frac{1}{4r}[(ab) + (a) + (b) + (1)].$$

Now we detail the statistical theory and concepts related to these expressions.

Let $Y_* = ((1), a, b, ab)'$ be the vector of total response values. Then

$$A = \frac{1}{2r} \ell'_A Y_* = \frac{1}{2r} (-1 \ 1 \ -1 \ 1) Y_*$$

$$B = \frac{1}{2r} \ell'_B Y_* = \frac{1}{2r} (-1 \ -1 \ 1 \ 1) Y_*$$

$$AB = \frac{1}{2r} \ell'_{AB} Y_* = \frac{1}{2r} (1 \ -1 \ -1 \ 1) Y_*.$$

Note that A, B and AB are the linear contrasts. Recall that a linear parametric function is estimable only when it is in the form of linear contrast. Moreover, A, B and AB are the linear orthogonal contrasts in the total response values $(1), a, b, ab$ except for the factor $1/2r$.

The sum of squares of a linear parametric function $\ell' y$ is given by $\frac{(\ell' y)^2}{\ell' \ell}$. If there are r replicates, then the sum of squares is $\frac{(\ell' y)^2}{r \ell' \ell}$. It may also be recalled under the normality of y 's, this sum of squares has a Chi-square distribution with one degree of freedom (χ_1^2). Thus the various associated sum of squares due to A, B and AB are given by the following:

$$SSA = \frac{(\ell'_A Y_*)^2}{r \ell'_A \ell_A} = \frac{1}{4r} (ab + a - b - (1))^2$$

$$SSB = \frac{(\ell'_B Y_*)^2}{r \ell'_B \ell_B} = \frac{1}{4r} (ab + b - a - (1))^2$$

$$SSAB = \frac{(\ell'_{AB} Y_*)^2}{r \ell'_{AB} \ell_{AB}} = \frac{1}{4r} (ab + (1) - a - b)^2.$$

Each of SSA, SSB and $SSAB$ has χ_1^2 under normality of Y_* .

The sum of squares due to total is computed as usual

$$TSS = \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^r y_{ijk}^2 - \frac{G^2}{4r}$$

where

$$G = \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^r y_{ijk}$$

is the grand total of all the observations.

The TSS has χ^2 distribution with $(2^2 r - 1)$ degrees of freedom.

The sum of squares due to error is also computed as usual as

$$SSE = TSS - SSA - SSB - SSAB$$

which has χ^2 distribution with

$$(4r - 1) - 1 - 1 - 1 = 4(r - 1)$$

degrees of freedom.

The mean squares are

$$MSA = \frac{SSA}{1},$$

$$MSB = \frac{SSB}{1},$$

$$MSAB = \frac{SSAB}{1},$$

$$MSE = \frac{SSE}{4(r - 1)}.$$

The F -statistic corresponding to A, B and AB are

$$F_A = \frac{MSA}{MSE} \sim F(1, 4(r - 1)) \text{ under } H_0,$$

$$F_B = \frac{MSB}{MSE} \sim F(1, 4(r - 1)) \text{ under } H_0,$$

$$F_{AB} = \frac{MSAB}{MSE} \sim F(1, 4(r - 1)) \text{ under } H_0.$$

The ANOVA table is case of 2^2 factorial experiment is given as follows:

Source	Sum of squares	Degrees of freedom	Mean squares	F
A	SSA	1	MSA	$F_A = \frac{MSA}{MSE}$ $F_B = \frac{MSB}{MSE}$ $F_{AB} = \frac{MSAB}{MSE}$
B	SSB	1	MSB	
AB	$SSAB$	1	$MSAB$	
$Error$	SSE	$4(r - 1)$	MSE	
Total	TSS	$4r - 1$		

The decision rule is to reject the concerned null hypothesis when the value of the concerned F statistic

$$F_{\text{effect}} > F_{1-\alpha}(1, 4(r - 1)).$$

2³ Factorial experiment:

Suppose that in a complete factorial experiment, there are three factors - A, B and C , each at two levels, viz., $a_0, a_1; b_0, b_1$ and c_0, c_1 respectively. There are a total of eight number of combinations:

$$a_0b_0c_0, a_0b_0c_1, a_0b_1c_0, a_0b_1c_1, \\ a_1b_0c_0, a_1b_0c_1, a_1b_1c_0, a_1b_1c_1.$$

Each treatment combination has r replicates, so the total number of observations are $N = 2^3 r = 8r$ that are to be analyzed for their influence on the response.

Assume the total response values are

$$Y_* = [(1), a, b, ab, c, ac, bc, abc]'$$

The response values can be arranged in a three-dimensional contingency table. The effects are determined by the linear contrasts

$$\ell'_{effect} Y_* = \ell'_{effect} ((1), a, b, ab, c, ac, bc, abc)$$

using the following table:

Factorial effect	Treatment combinations							
	(1)	a	b	ab	c	ac	bc	abc
I	+	+	+	+	+	+	+	+
A	-	+	-	+	-	+	-	+
B	-	-	+	+	-	-	+	+
AB	+	-	-	+	+	-	-	+
C	-	-	-	-	+	+	+	+
AC	+	-	+	-	-	+	-	+
BC	+	+	-	-	-	-	+	+
ABC	-	+	+	-	+	-	-	+

Note that once a few rows have been determined in this table, rest can be obtained by simple multiplication of the symbols. For example, consider the column corresponding to a , we note that A has + sign, B has – sign, so AB has – sign (sign of $A \times$ sign of B). Once AB has - sign, C has – sign then ABC has (sign of $AB \times$ sign of C) which is + sign and so on.

The first row is a basic element. With this $a = 1'Y_*$ can be computed where 1 is a column vector of all elements unity. If other rows are multiplied with the first row, they stay unchanged (therefore we call it as identity and denoted as I). Every other row has the same number of + and – signs. If + is replaced by 1 and – is replaced by -1, we obtain the vectors of orthogonal contrasts with the norm $8 (= 2^3)$.

If each row is multiplied by itself, we obtain I (first row). The product of any two rows leads to a different row in the table. For example

$$A.B = AB$$

$$AB.B = AB^2 = A$$

$$AC.BC = A.C^2.BB = AB.$$

The structure in the table helps in estimating the average effect.

For example, the average effect of A is

$$A = \frac{1}{4r} [(a) - (1) + (ab) - (b) + (ac) - (c) + (abc) - (bc)]$$

which has the following explanation.

- (i) Average effect of A at low level of B and $C \equiv (a_1 b_0 c_0) - (a_0 b_0 c_0) \equiv \frac{[(a) - (1)]}{r}$.
- (ii) Average effect of A at high level of B and low level of $C \equiv (a_1 b_1 c_0) - (a_0 b_1 c_0) \equiv \frac{[(ab) - (b)]}{r}$.
- (iii) Average effect of A at low level of B and high level of $C \equiv (a_1 b_0 c_1) - (a_0 b_0 c_1) \equiv \frac{[(ac) - (c)]}{r}$.
- (iv) Average effect of A at high level of B and $C \equiv (a_1 b_1 c_1) - (a_0 b_1 c_1) \equiv \frac{[(abc) - (bc)]}{r}$.

Hence for all combinations of B and C , the average effect of A is the average of all the average effects in (i)-(iv).

Similarly, other main and interaction effects are as follows:

$$B = \frac{1}{4r} [(b) + (ab) + (bc) + (abc) - (1) - (a) - (c) - (ac)] = \frac{(a+1)(b-1)(c+1)}{4r}$$

$$C = \frac{1}{4r} [c + (ac) + (bc) + (abc) - (1) - (a) - (b) - (ab)] = \frac{(a+1)(b+1)(c-1)}{4r}$$

$$AB = \frac{1}{4r} [(1) + (ab) + (c) + (abc) - (a) - (b) - (ac) - (bc)] = \frac{(a-1)(b-1)(c+1)}{4r}$$

$$AC = \frac{1}{4r} [(1) + (b) + (ac) + (abc) - (a) - (ab) - (c) - (bc)] = \frac{(a-1)(b+1)(c-1)}{4r}$$

$$BC = \frac{1}{4r} [(1) + (a) + (bc) + (abc) - (b) - (ab) - (c) - (ac)] = \frac{(a+1)(b-1)(c-1)}{4r}$$

$$ABC = \frac{1}{4r} [(abc) + (ab) + (b) + (c) - (ab) - (ac) - (bc) - (1)] = \frac{(a-1)(b-1)(c-1)}{4r}.$$

Various sum of squares in the 2^3 factorial experiment are obtained as

$$SS(\text{Effect}) = \frac{(\text{linear contrast})^2}{8r} = \frac{(\ell'_{\text{effect}} Y_*)^2}{r \ell'_{\text{effect}} \ell_{\text{effect}}}$$

which follow a Chi-square distribution with one degree of freedom under normality of Y_* . The corresponding mean sum of squares is obtained as

$$MS(\text{Effect}) = \frac{SS(\text{Effect})}{\text{Degrees of freedom}}.$$

The corresponding F -statistics are obtained by

$$F_{\text{Effect}} = \frac{MS(\text{Effect})}{MS(\text{Error})}$$

which follows an F – distribution with degrees of freedoms 1 and error degrees of freedom under the respective null hypothesis. The decision rule is to reject the corresponding null hypothesis at the α level of significance whenever

$$F_{\text{effect}} > F_{1-\alpha}(1, df_{\text{error}}).$$

These outcomes are presented in the following ANOVA table

Sources	Sum of squares	Degrees of freedom	Mean sum of squares	F
A	SSA	1	$MSA = SSA / 1$	F_A
B	SSB	1	$MSB = SSB / 1$	F_B
AB	$SSAB$	1	$MSAB = SSAB / 1$	F_{AB}
C	SSC	1	$MSC = SSC / 1$	F_C
AC	$SSAC$	1	$MSAC = SSAC / 1$	F_{AC}
BC	$SSBC$	1	$MSBC = SSBC / 1$	F_{BC}
ABC	$SSABC$	1	$MSABC = SSABC / 1$	F_{ABC}
Error	$SS(\text{Error})$	$8(r-1)$	$MS(\text{Error}) = SS(\text{Error}) / \{8(r-1)\}$	
Total	TSS	$8r-1$		

2^n Factorial experiment:

Based on the theory developed for 2^2 and 2^3 factorial experiments, we now extend them for the 2^n factorial experiment.

- Capital letters $A, B, C, \dots$ denote the **factors**. They are the main effect contrast for the factors $A, B, C, \dots$
- $AB, AC, BC, \dots$ denote the first order or **2-factor interactions**
- $ABC, ABD, BCD, \dots$ denote the **second-order** or **3-factor interactions** and so on.
- Each of the main effect and interaction effect carries one degree of freedom.
- Total number of main effects $= \binom{n}{1} = n$.
- Total number of first-order interactions $= \binom{n}{2}$.
- Total number of second-order interactions $= \binom{n}{3}$

and so on.

Standard order for treatment combinations:

The list of treatments can be expressed in a standard order.

- For one factor A , the standard order is $(1), a$.
- For two factors A and B , the standard order is obtained by adding b and ab in the standard order of one factor A . This is derived by multiplying (1) and a by b , i.e.
 $b \times \{(1), a\} = (1), a, b, ab$.
- For three factors, add c, ac, bc and abc which are derived by multiplying the standard order of A and B by c , i.e.
 $c \times \{(1), a, b, ab\} = (1), a, b, ab, c, ac, bc, abc$.

Thus the standard order of any factor is obtained step by step by multiplying it with an additional letter to preceding standard order.

For example, the standard order of A, B, C and D is 2^4 factorial experiment is

$$\begin{aligned} & (1), a, b, ab, c, ac, bc, abc, d \times \{(1), a, b, ab, c, ac, bc, abc\} \\ & = (1), a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd. \end{aligned}$$

How to find the contrasts for main effects and interaction effect:

Recall that earlier, we had illustrated the concept in writing the contrasts for main and interaction effects.

For example, in a 2^2 factorial experiment, we had expressed

$$A = \frac{1}{2}(a-1)(b+1) = \frac{1}{2}[(-1) + (a) - (b) + (ab)]$$
$$AB = \frac{1}{2}(a-1)(b-1) = \frac{1}{2}[(1) - (a) - (b) + (ab)].$$

Note that each effect has two component - divisor and contrast. When the order of factorial increases, it is cumbersome to derive such expressions. Some methods have been suggested to write the expressions for factorial effects. First, we detail how to write divisor and then illustrate the methods for obtaining the contrasts.

How to write divisor:

In a 2^n factorial experiment,

- the general mean effect has divisor 2^n and
- any effect (main or interaction) has divisor 2^{n-1} .

For example, in a 2^6 factorial experiment, the general mean effect has divisor 2^6 and any main effect or interaction effect of any order has divisor $2^{6-1} = 2^5$.

If r replicates of each effect are available, then

- the general mean effect has divisor $r2^n$ and
- any main effect or interaction effect of any order has a divisor $r2^{n-1}$.

How to write contrasts:

Method 1:

Contrast belonging to the main effects and the interaction effects are written as follows:

$$\begin{aligned} A &= (a-1)(b+1)(c+1)\dots(z+1) \\ B &= (a+1)(b-1)(c+1)\dots(z+1) \\ C &= (a+1)(b+1)(c-1)\dots(z+1) \\ &\vdots \\ AB &= (a-1)(b-1)(c+1)\dots(z+1) \\ BC &= (a+1)(b-1)(c-1)\dots(z+1) \\ &\vdots \quad \quad \quad \vdots \\ ABC &= (a-1)(b-1)(c-1)\dots(z+1). \\ &\vdots \\ ABC\dots Z &= (a-1)(b-1)(c-1)\dots(z-1) \end{aligned}$$

Look at the pattern of assigning + and – signs on the right-hand side. The letters common on left and right hand sides of the equality (=) sign (irrespective of small or capital letters) contain – sign and rest contain + sign.

The expression on right-hand side when simplified algebraically give the contrasts in terms of treatment combination. For example, in a 2^3 factorial

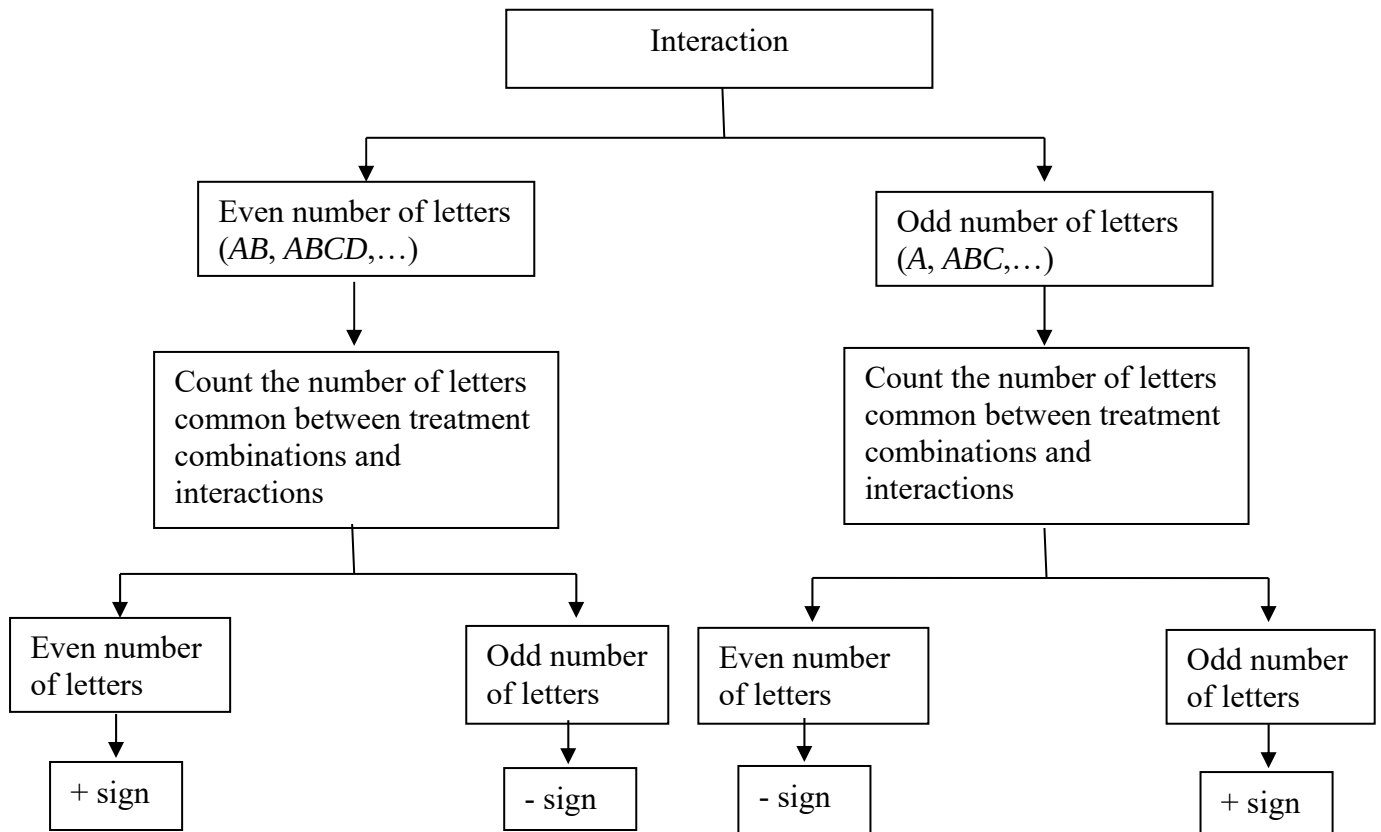
$$\begin{aligned}
 A &= \frac{1}{2^{3-1}}(a-1)(b+1)(c+1) \\
 &= \frac{1}{4}[-(1) + (a) - (b) + (ab) - (c) + (ac) - (bc) + (abc)] \\
 M &= \frac{1}{2^3}(a+1)(b+1)(c+1) \\
 &= \frac{1}{8}[(1) + (a) + (b) + (ab) + (c) + (ac) + (bc) + (abc)]
 \end{aligned}$$

Method 2

- Form a table such that
 - rows correspond to the main or interaction effect and
 - columns correspond to treatment combinations (or another way round)
- + and – signs in the table indicate the sign of the treatment combinations of main and interaction effects.
- Signs are determined by the “rule of odds and evens” given as follows:
 - if the interaction has an even number of letters ($AB, ABCD, \dots$), a treatment combination having an even number of letters common with the interaction enters with a + sign and one with an odd number of letters common enters with a – sign.
 - if the interaction has an odd number of letters ($A, ABC, \dots$), the rule is reversed.
- Once few rows are filled up, others can be obtained through multiplication rule. For example, the sign of $ABCD$ is obtained as

$$(\text{sign of } A \times \text{sign of } BCD) \quad \text{or} \quad (\text{sign of } AB \times \text{sign of } CD).$$
- Treatment combination (1) is taken to have an even number (zero) of letters common with every interaction.

This rule of assignment of + or – is illustrated in the following flow diagram:



For example, in a 2^3 factorial experiment, write

- rows for main and interaction effects and
- columns for treatment combinations in standard order.
- Take treatment combination (1) to have an even number (zero) of letter common with every interaction.

This gives the following table

Factorial effect	Treatment combinations							
	(1)	<i>a</i>	<i>b</i>	<i>ab</i>	<i>c</i>	<i>ac</i>	<i>bc</i>	<i>abc</i>
<i>I</i>	+	+	+	+	+	+	+	+
<i>A</i>	-	+	-	+	-	+	-	+
<i>B</i>	-	-	+	+	-	-	+	+
<i>AB</i>	+	-	-	+	+	-	-	+
<i>C</i>	-	-	-	-	+	+	+	+
<i>AC</i>	+	-	+	-	-	+	-	+
<i>BC</i>	+	+	-	-	-	-	+	+
<i>ABC</i>	-	+	+	-	+	-	-	+

Sums of squares:

Suppose 2^n factorial experiment is carried out in a randomized block design with r replicates.

Denote the total yield (output) from r plots (experimental units) receiving a particular treatment combination by the same symbol within a square bracket. For example, $[ab]$ denotes the total yield from the plots receiving the treatment combination (ab) .

In a 2^2 factorial experiment, the factorial effect totals are

$$[A] = [ab] - [b] + [a] - [1]$$

$[ab]$ = treatment total, i.e. the sum of r observations in which both the factors A and B are at the second level.

$[a]$ = treatment total, i.e., the sum of r observations in which factor A is at the second level and factor B is at the first level

$[b]$ = treatment total, i.e., the sum of r observations in which factor A is at the first level and factor B is at the second level.

$[1]$ = treatment total, i.e. the sum of r observations in which both the factors A and B are at the first level.

$$[A] = \sum_{i=1}^r [y_{i(ab)} - y_{i(b)} + y_{i(a)} - y_{i(1)}]$$
$$= \ell'_A y_A \text{ (say).}$$

where ℓ_A is a vector of $+1$ and $=1$ and y_A is a vector denoting the responses from ab, b, a and 1 .

Similarly, other effects can also be found.

Similarly, other effects can also be found.

The sum of squares due to a particular effect is obtained as

$$\frac{[\text{Total yield}]^2}{\text{Total number of observations}}.$$

In a 2^2 factorial experiment in an RBD, the sum of squares due to A is

$$SSA = \frac{(\ell'_A y_A)^2}{r2^2}.$$

In a 2^n factorial experiment in an RBD, the divisor will be $r.2^n$. If Latin square design is used based on $2^n \times 2^n$ Latin square, then r is replaced by 2^n .

Yates method of computation of the sum of squares:

Yates method gives a systematic approach to find the sum of squares. We are not presenting here the complete method. Only the part which is used for computing only the sum of squares is presented and the method to verify them is not presented.

It has the following steps

1. First, write the treatment combinations in the standard order in the column at the beginning of the table, called a **treatment column**.
2. Find the total yield for each treatment. Write this as the second column of the table, called a **yield column**.
3. Obtain columns (1),(2),...,(n) successively
 - (i) obtain column (1) from yield column
 - (a) upper half is obtained by adding yields in pairs.
 - (b) the second half is obtained by taking differences in pairs, the difference obtained by subtracting the first term of pairs from the second term.
 - (ii) The columns (2),(3),...,(n) are obtained from preceding ones in the same manner as used for getting (1) from the yield columns.
4. This process of finding columns is repeated n times in 2^n factorial experiment.
5. Sum of squares due to interaction = $\frac{[\text{column}(n)]^2}{\text{Total number of observations}}$

Example: Yates procedure for a 2^2 factorial experiment

Treatment Combination	Yield (total from all r replicates)	(1)	(2)
(1)	(1)	(1) + (a)	(1) + (a) + (b) + (ab) = [M]
a	(a)	(b) + (ab)	-(1) + (a) - (b) + (ab) = [A]
b	(b)	(a) - (1)	-(1) - (a) + (b) + (ab) = [B]
ab	(ab)	(ab) - (b)	(1) - (a) - (b) + (ab) = [AB]

Note: The columns are repeatedly obtained 2 times due to 2^2 factorial experiment.

$$\text{Now } SSA = \frac{[A]^2}{4r}; \quad SSB = \frac{[B]^2}{4r}; \quad SSAB = \frac{[AB]^2}{4r}$$

Example: Yates procedure for 2^3 factorial experiment.

Treatment	Yield (total from all r replicates)				
(1)	(2)	(3)	(4)	(5)	(6)
1	(1)	$u_1 = (1) + (a)$	$v_1 = u_1 + u_2$	$w_1 = v_1 + v_2$	$[M]$
a	(a)	$u_2 = (b) + (ab)$	$v_2 = u_3 + u_4$	$w_2 = v_3 + v_4$	$[A]$
b	(b)	$u_3 = (c) + (ac)$	$v_3 = u_5 + u_6$	$w_3 = v_5 + v_6$	$[B]$
ab	(ab)	$u_4 = (bc) + (abc)$	$v_4 = u_7 + u_8$	$w_4 = v_7 + v_8$	$[AB]$
c	(ac)	$u_5 = (a) - (1)$	$v_5 = u_2 - u_7$	$w_5 = v_2 - v_1$	$[C]$
ac	(ac)	$u_6 = (ab) - (b)$	$v_6 = u_4 - u_3$	$w_6 = v_4 - v_3$	$[AC]$
bc	(bc)	$u_7 = (ac) - (c)$	$v_7 = v_6 - u_5$	$w_7 = v_6 - v_5$	$[BC]$
abc	(abc)	$u_8 = (abc) - (bc)$	$v_8 = u_8 - u_7$	$w_8 = v_8 - v_7$	$[ABC]$

The sum of squares are obtained as follows when the design is RBD.

$$SS(Effect) = \frac{[Effect]^2}{r \cdot 2^3}.$$

For analysis of 2^n factorial experiment, the analysis of variance involves the partitioning of treatment sum of squares so as to obtain sum of squares due to main and interaction effects of factors. These sum of squares are mutually orthogonal, so

Treatment SS = Total of SS due to main and interaction effects.

For example:

In 2^2 factorial experiment in an RBD with r replications, the division of degrees of freedom and the treatment sum of squares are as follows:

Source	Degree of freedom	Sum of squares
Replications	$r - 1$	
Treatments	$4 - 1 = 3$	
- A	1	$[A]^2 / 4r$
- B	1	$[B]^2 / 4r$
- AB	1	$[AB]^2 / 4r$
Error	$3(r - 1)$	
Total	$4r - 1$	

The decision rule is to reject the concerned null hypothesis when the related F -statistic $F_{effect} > F_{1-\alpha}(1, 3(r-1))$.

Chapter 9

Confounding

If the number of factors or levels increase in a factorial experiment, then the number of treatment combinations increases rapidly. When the number of treatment combinations is large, then it may be difficult to get the blocks of sufficiently large size to accommodate all the treatment combinations. Under such situations, one may use either connected incomplete block designs, e.g., balanced incomplete block designs (BIBD) where all the main effects and interaction contrasts can be estimated or use unconnected designs where not all these contrasts can be estimated.

Non-estimable contrasts are said to be confounded.

Note that a linear function $\lambda'\beta$ is said to be estimable if there exist a linear function $l'y$ of the observations on random variable y such that $E(l'y) = \lambda'\beta$. Now there arise two questions. Firstly, what does confounding means and secondly, how does it compares to using BIBD.

In order to understand the confounding, let us consider a simple example of 2^2 factorial with factors a and b . The four treatment combinations are (1) , a, b and ab . Suppose each batch of raw material to be used in the experiment is enough only for two treatment combinations to be tested. So two batches of raw material are required. Thus two out of four treatment combinations must be assigned to each block. Suppose this 2^2 factorial experiment is being conducted in a randomized block design. Then the corresponding model is

$$E(y_{ij}) = \mu + \beta_i + \tau_j,$$

then

$$A = \frac{1}{2r} [ab + a - b - (1)],$$

$$B = \frac{1}{2r} [ab + b - a - (1)],$$

$$AB = \frac{1}{2r} [ab + (1) - a - b].$$

Suppose the following block arrangement is opted:

Block 1	Block 2
(1) ab	a b

The block effects of blocks 1 and 2 are β_1 and β_2 , respectively, then the average responses corresponding to treatment combinations a, b, ab and (1) are

$$\begin{aligned}
E[y(a)] &= \mu + \beta_2 + \tau(a), \\
E[y(b)] &= \mu + \beta_2 + \tau(b), \\
E[y(ab)] &= \mu + \beta_1 + \tau(ab), \\
E[y(1)] &= \mu + \beta_1 + \tau(1),
\end{aligned}$$

respectively. Here $y(a)$, $y(b)$, $y(ab)$, $y(1)$ and $\tau(a)$, $\tau(b)$, $\tau(ab)$, $\tau(1)$ denote the responses and treatments corresponding to a, b, ab and (1) , respectively. Ignoring the factor $1/2r$ in A, B, AB and using $E[y(a)]$, $E[y(b)]$, $E[y(ab)]$, $E[y(1)]$, the effect A is expressible as follows :

$$\begin{aligned}
A &= [\mu + \beta_1 + \tau(ab)] + [\mu + \beta_2 + \tau(a)] - [\mu + \beta_2 + \tau(b)] - [\mu + \beta_1 + \tau(1)] \\
&= \tau(ab) + \tau(a) - \tau(b) - \tau(1).
\end{aligned}$$

So the block effect is not present in A and it is not mixed up with the treatment effects. In this case, we say that the main effect A is not confounded. Similarly, for the main effect B , we have

$$\begin{aligned}
B &= [\mu + \beta_1 + \tau(ab)] + [\mu + \beta_2 + \tau(b)] - [\mu + \beta_2 + \tau(a)] - [\mu + \beta_1 + \tau(1)] \\
&= \tau(ab) + \tau(b) - \tau(a) - \tau(1).
\end{aligned}$$

So there is no block effect present in B and thus B is not confounded. For the interaction effect AB , we have

$$\begin{aligned}
AB &= [\mu + \beta_1 + \tau(ab)] + [\mu + \beta_1 + \tau(1)] - [\mu + \beta_2 + \tau(a)] - [\mu + \beta_2 + \tau(b)] \\
&= 2(\beta_1 - \beta_2) + \tau(ab) + \tau(1) - \tau(a) - \tau(b).
\end{aligned}$$

Here the block effects are present in AB . In fact, the block effects are β_1 and β_2 are mixed up with the treatment effects and cannot be separated individually from the treatment effects in AB . So AB is said to be confounded (or mixed up) with the blocks.

Alternatively, if the arrangement of treatments in blocks are as follows:

Block 1	Block 2
$\begin{array}{c} ab \\ a \end{array}$	$\begin{array}{c} (1) \\ b \end{array}$

then the main effect A is expressible as

$$\begin{aligned} A &= [\mu + \beta_1 + \tau(ab)] + [\mu + \beta_1 + \tau(a)] - [\mu + \beta_2 + \tau(b)] - [\mu + \beta_2 + \tau(1)] \\ &= 2(\beta_1 - \beta_2) + \tau(ab) + \tau(a) - \tau(b) - \tau(1) \end{aligned}$$

Observe that the block effects β_1 and β_2 are present in this expression. So the main effect A is confounded with the blocks in this arrangement of treatments.

So the main effect A is confounded with the blocks in this arrangement of treatments.

We notice that it is in our control to decide that which of the effect is to be confounded. The order in which treatments are run in a block is determined randomly. The choice of block to be run first is also randomly decided.

The following observation emerges from the allocation of treatments in blocks:

“For a given effect, when two treatment combinations with the same signs are assigned to one block and the other two treatment combinations with the same but opposite signs are assigned to another block, then the effect gets confounded”.

For example, in case AB is confounded, then

- ab and (1) with $+$ signs are assigned to block 1 whereas
- a and b with $-$ signs are assigned to block 2.

Similarly, when A is confounded, then

- a and ab with $+$ signs are assigned to block 1 whereas
- (1) and b with $-$ signs are assigned to block 2.

The reason behind this observation is that if every block has treatment combinations in the form of linear contrast, then effects are estimable and thus unconfounded. This is also evident from the theory of linear estimation that a linear parametric function is estimable if it is in the form of a linear contrast.

The contrasts which are not estimable are said to be **confounded with the differences between blocks** (or block effects). The contrasts which are estimable are said to be unconfounded with blocks or free from block effects.

Comparison of balanced Incomplete Block design (BIBD versus factorial:

Now we explain how confounding and BIBD compares together. Consider a 2^3 factorial experiment which needs the block size to be 8. Suppose the raw material available to conduct the experiment is sufficient only for a block of size 4. One can use BIBD in this case with parameters $b = 14, k = 4, v = 8, r = 7$ and $\lambda = 3$ (such BIBD exists). For this BIBD, the efficiency factor is

$$E = \frac{\lambda v}{kr} = \frac{6}{8}$$

and

$$\text{Var}(\hat{\tau}_j - \hat{\tau}_{j'})_{\text{BIBD}} = \frac{2k}{\lambda v} \sigma^2 = \frac{2}{6} \sigma^2 \quad (j \neq j').$$

Consider now an unconnected design in which 7 out of 14 blocks get treatment combination in block 1 as

$$\boxed{a \ b \ c \ abc}$$

and remaining 7 blocks get treatment combination in block 2 as

$$\boxed{(1) \ ab \ bc \ ac}$$

In this case, all the effects A, B, C, AB, BC and AC are estimable but ABC is not estimable because the treatment combinations with all + and all – signs in

$$\begin{aligned} ABC &= (a-1)(b-1)(c-1) \\ &= \underbrace{(a+b+c+abc)}_{\text{in block 1}} - \underbrace{((1)+ab+bc+ac)}_{\text{in block 2}} \end{aligned}$$

are contained in same blocks. In this case, the variance of estimates of unconfounded main effects and interactions is $8\sigma^2/7$. Note that in case of RBD,

$$\text{Var}(\hat{\tau}_j - \hat{\tau}_{j'})_{\text{RBD}} = \frac{2\sigma^2}{r} = \frac{2\sigma^2}{7} \quad (j \neq j')$$

and there are four linear contrasts, so the total variance is $4 \times (2\sigma^2/7)$ which gives the factor $8\sigma^2/7$ and which is smaller than the variance under BIBD.

We observe that at the cost of not being able to estimate ABC , we have better estimates of A, B, C, AB, BC and AC with the same number of replicates as in BIBD. Since higher order interactions are difficult to interpret and are usually not large, so it is much better to use confounding arrangements which provide better estimates of the interactions in which we are more interested.

Note that this example is for understanding only. As such the concepts behind incomplete block design and confounding are different.

Confounding arrangement:

The arrangement of treatment combinations in different blocks, whereby some pre-determined effect (either main or interaction) contrasts are confounded is called a confounding arrangement.

For example, when the interaction ABC is confounded in a 2^3 factorial experiment, then the confounding arrangement consists of dividing the eight treatment combinations into following two sets:

$$\boxed{a \ b \ c \ abc}$$

and

$$\boxed{(1) \ ab \ bc \ ac}$$

With the treatments of each set being assigned to the same block and each of these sets being replicated same number of times in the experiment, we say that we have a confounding arrangement of a 2^3 factorial in two blocks. It may be noted that any confounding arrangement has to be such that only predetermined interactions are confounded and the estimates of interactions which are not confounded are orthogonal whenever the interactions are orthogonal.

Defining contrast:

The interactions which are confounded are called the defining contrasts of the confounding arrangement.

A confounded contrast will have treatment combinations with the same signs in each block of the confounding arrangement. For example, if effect $AB = (a-1)(b-1)(c+1)$ is to be confounded, then put all factor combinations with + sign, i.e., (1), ab , c and abc in one block and all other factor combinations with – sign, i.e., a , b , ac and bc in another block. So the **block size reduces to 4 from 8** when one effect is confounded in 2^3 factorial experiment.

Suppose if along with ABC confounded, we want to confound C also,. To obtain such blocks, consider the blocks where ABC is confounded and divide them into further halves. So the block

$$\boxed{a \ b \ c \ abc}$$

is divided into following two blocks: $\boxed{a \ b}$ and $\boxed{c \ abc}$

and the block

$$(1) \begin{array}{|c|c|c|c|} \hline ab & bc & ac & \\ \hline \end{array}$$

is divided into following two blocks:

$$(1) \begin{array}{|c|c|} \hline ab & \\ \hline \end{array} \quad \text{and} \quad \begin{array}{|c|c|} \hline bc & ac \\ \hline \end{array}$$

These blocks of 4 treatments are divided into 2 blocks with each having 2 treatments and they are obtained in the following way. If only C is confounded then the block with + sign of treatment combinations in C is

$$\begin{array}{|c|c|c|c|} \hline c & ac & bc & abc \\ \hline \end{array}$$

and block with – sign of treatment combinations in C is

$$(1) \begin{array}{|c|c|c|} \hline a & b & ab \\ \hline \end{array}.$$

Now look into the

(i) following block with + sign when $ABC = (a-1)(b-1)(c-1)$ is confounded,

$$\begin{array}{|c|c|c|c|} \hline a & b & c & abc \\ \hline \end{array}$$

(ii) following block with + sign when $C = (a+1)(b+1)(c-1)$ is confounded and

$$\begin{array}{|c|c|c|c|} \hline c & ab & ac & abc \\ \hline \end{array}$$

(iii) table of + and – signs in case of 2^3 factorial experiment.

Identify the treatment combinations having common - signs in these two blocks in (i) and (ii). These treatment combinations are c and abc . So assign them into one block. The remaining treatment combinations out of a , b , c and abc are a and b which go into another block.

Similarly look into the

(a) following block with – sign when ABC is confounded,

$$(1) \begin{array}{|c|c|c|c|} \hline ab & bc & ac & \\ \hline \end{array}$$

(b) following block with – sign when C is confounded and

$$(1) \begin{array}{|c|c|c|} \hline a & b & ab \\ \hline \end{array}$$

(c) table of + and – signs in case of 2^3 factorial experiment.

Identify the treatment combinations having common – sign in these two blocks in (a) and (b). These treatment combinations are (1) and ab which go into one block and the remaining two treatment combinations ac and bc out of c , ac , bc and abc go into another block. So the blocks where both ABC and C are confounded together are

$$\begin{array}{|c|} \hline (1) \quad ab \\ \hline \end{array}, \begin{array}{|c|} \hline a \quad b \\ \hline \end{array}, \begin{array}{|c|c|} \hline ac & bc \\ \hline \end{array} \quad \text{and} \quad \begin{array}{|c|c|} \hline c & abc \\ \hline \end{array}.$$

While making these assignments of treatment combinations into four blocks, each of size two, we notice that another effect, viz., AB also gets confounded automatically. Thus we see that when we confound two factors, a third factor is automatically getting confounded. This situation is quite general. The defining contrasts for a confounding arrangement cannot be chosen arbitrarily. If some defining contrasts are selected then some other will also get confounded.

Now we present some definitions which are useful in describing the confounding arrangements.

Generalized interaction:

Given any two interactions, the generalized interaction is obtained by multiplying the factors (in capital letters) and ignoring all the terms with an even exponent.

For example, the generalized interaction of the factors ABC and BCD is $ABC \times BCD = AB^2C^2D = AD$ and the generalized interaction of the factors AB, BC and ABC is $AB \times BC \times ABC = A^2B^3C^2 = B$.

Independent set :

A set of main effects and interaction contrasts is called independent if no member of the set can be obtained as a generalized interaction of the other members of the set.

For example, the set of factors AB, BC and AD is an independent set but the set of factors AB, BC, CD and AD is not an independent set because $AB \times BC \times CD = AB^2C^2D = AD$ which is already contained in the set.

Orthogonal treatment combinations:

The treatment combination $a^p b^q c^r \dots$ is said to be orthogonal to the interaction $A^x B^y C^z \dots$ if $(px + qy + rz + \dots)$ is divisible by 2. Since $p, q, r, \dots, x, y, z, \dots$ are either 0 or 1, so a treatment combination is orthogonal to an interaction if they have an even number of letters in common. Treatment combination (1) is orthogonal to every interaction.

If $a^{p_1} b^{q_1} c^{r_1} \dots$ and $a^{p_2} b^{q_2} c^{r_2} \dots$ are both orthogonal to $A^x B^y C^z \dots$, then the product $a^{p_1+p_2} b^{q_1+q_2} c^{r_1+r_2} \dots$ is also orthogonal to $A^x B^y C^z \dots$. Similarly, if two interactions are orthogonal to a treatment combination, then their generalized interaction is also orthogonal to it.

Now we give some general results for a confounding arrangement. Suppose we wish to have a confounding arrangement in 2^p blocks of a 2^k factorial experiment. Then we have the following observations:

1. The size of each block is 2^{k-p} .
2. The number of elements in defining contrasts is $(2^p - 1)$, i.e., $(2^p - 1)$ interactions have to be confounded.

Proof: If p factors are to be confounded, then the number of m th order interaction

with p factors is $\binom{p}{m}$, $(m = 1, 2, \dots, p)$. So the total number of factors to be

confounded are $\sum_{m=1}^p \binom{p}{m} = 2^{p-1}$.

3. If any two interactions are confounded, then their generalized interactions are also confounded.
4. The number of independent contrasts out of $(2^p - 1)$ defining contrasts is p and rest are obtained as generalized interactions.
5. Number of effects getting confounded automatically is $(2^p - p - 1)$.

To illustrate this, consider a 2^5 factorial ($k = 5$) with 5 factors, viz., A, B, C, D and E . The factors are to be confounded in 2^3 blocks ($p = 3$). So the size of each block is $2^{5-3} = 4$. The number of defining contrasts is $2^3 - 1 = 7$. The number of independent contrasts which can be chosen arbitrarily is 3 (i.e., p) out of 7 defining contrasts. Suppose we choose $p = 3$ following independent contrasts as

- (i) ACE
- (ii) CDE
- (iii) $ABDE$

and then the remaining 4 out of 7 defining contrasts are obtained as

- (iv) $(ACE) \times (CDE) = AC^2DE^2 = AD$
- (v) $(ACE) \times (ABDE) = A^2BCDE^2 = BCD$
- (vi) $(CDE) \times (ABDE) = ABCD^2E^2 = ABC$
- (vii) $(ACE) \times (CDE) \times (ABDE) = A^2BC^2D^2E^3 = BE$.

Alternatively, if we choose another set of $p = 3$ independent contrast as

- (i) $ABCD$,
- (ii) $ACDE$,
- (iii) $ABCDE$,

then the defining contrasts are obtained as

- (iv) $(ABCD) \times (ACDE) = A^2BC^2D^2E = BE$
- (v) $(ABCD) \times (ABCDE) = A^2B^2C^2D^2E = E$
- (vi) $(ACDE) \times (ABCDE) = A^2BC^2D^2E^2 = B$
- (vii) $(ABCD) \times (ACDE) \times (ABCDE) = A^3B^2C^3D^3E^2 = ACD$.

In this case, the main effects B and E also get confounded.

As a rule, try to confound, as far as possible, higher order interactions only because they are difficult to interpret.

After selecting p independent defining contrasts, divide the 2^k treatment combinations into 2^p groups of 2^{k-p} combinations each, and each group going into one block.

Principal (key) block:

Group containing the combination (1) is called the principal block or key block. It contains all the treatment combinations which are orthogonal to the chosen independent defining contrasts.

If there are p independent defining contrasts, then any treatment combination in principal block is orthogonal to p independent defining contrasts. In order to obtain the principal block,

- write the treatment combinations in standard order.
- check each one of them for orthogonality.
- If two treatment combinations belongs to the principal block, their product also belongs to the principal block.
- When few treatment combinations of the principal block have been determined, other treatment combinations can be obtained by multiplication rule.

Now we illustrate these steps in the following example.

Example:

Consider the set up of a 2^5 factorial experiment in which we want to divide the total treatment effects into 2^3 groups by confounding three effects AD , BE and ABC . The generalized interactions in this case are $ADBE$, BCD , ACE and CDE .

In order to find the principal block, first write the treatment combinations in standard order as follows:

(1) a b ab c ac bc abc
 d ad bd abd cd acd bcd $abcd$
 e ae be abe ce ace bce $abce$
 de ade bde $abde$ cde $acde$ $bcde$ $abcde$.

Place a treatment combination in the principal block if it has an even number of letters in common with the confounded effects AD , BE and ABC . The principal block has (1), acd , bce and $abde (= acd \times bce)$. Obtain other blocks of confounding arrangement from principal block by multiplying the treatment combinations of the principal block by a treatment combination not occurring in it or in any other block already obtained. In other words, choose treatment combinations not occurring in it and multiply with them in the principal block. Choose only distinct blocks. In this case, obtain other blocks by multiplying a , b , ab , c , ac , bc , abc like as in the following.

Arrangement of the treatments in blocks when AD , BE and ABC are confounded

Principal Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7	Block 8
(1)	a	b	ab	c	ac	bc	abc
acd bce $abde$	cd $abce$ bde	$abcd$ ce ade	bcd ace de	ad be $abcde$	d abe $bcde$	abd e $acde$	bd ae cde

For example, block 2 is obtained by multiplying a with each factor combination in principal block as $(1) \times a = a$, $acd \times a = a^2cd = cd$, $bce \times a = abce$, $abde \times a = a^2bde = bde$; block 3 is obtained by multiplying b with (1), acd , bce and $abde$ and similarly other blocks are obtained. If any other treatment combination is chosen to be multiplied with the treatments in principal block, then we get

a block which will be one among the blocks 1 to 8. For example, if ae is multiplied with the treatments in principal block, then the blocks obtained consists of $(1) \times ae = ae$, $acd \times ae = cde$, $bce \times ae = abc$ and $abde \times ae = bd$ which is same as the block 8.

Alternatively, if ACD , $ABCD$ and $ABCDE$ are to be confounded, then independent defining contrasts are ACD , $ABCD$, $ABCDE$ and the principal block has (1) , ac , ad and $cd (= ac \times ad)$.

Analysis of variance in case of confounded effects

When an effect is confounded, it means that it is not estimable. The following steps are followed to conduct the analysis of variance in case of factorial experiments with confounded effects:

- Obtain the sum of squares due to main and interaction effects in the usual way as if no effect is confounded.
- Drop the sum of squares corresponding to confounded effects and retain only the sum of squares due to unconfounded effects.
- Find the total sum of squares.
- Obtain the sum of squares due to error and associated degrees of freedom by subtraction.
- Conduct the test of hypothesis in the usual way.